

Appendix C

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I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

151145

OR

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

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☒ The address associated with the above-mentioned Customer Number

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

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Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Netlist, Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

12/5/17

Name

Noel B. Whitley

Title

VP, IP & Licensing

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

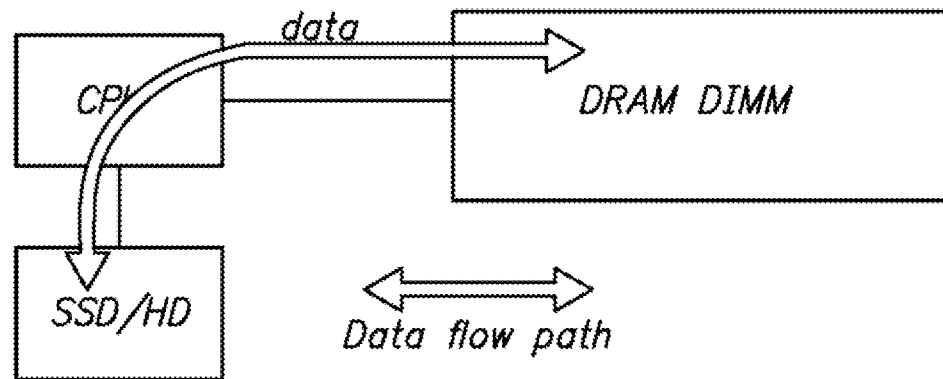


FIG. 1
(PRIOR ART)

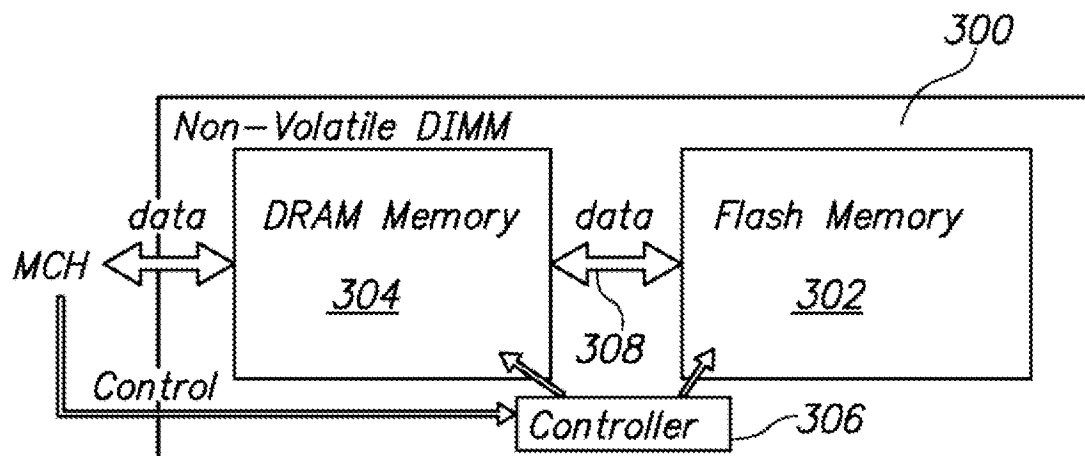
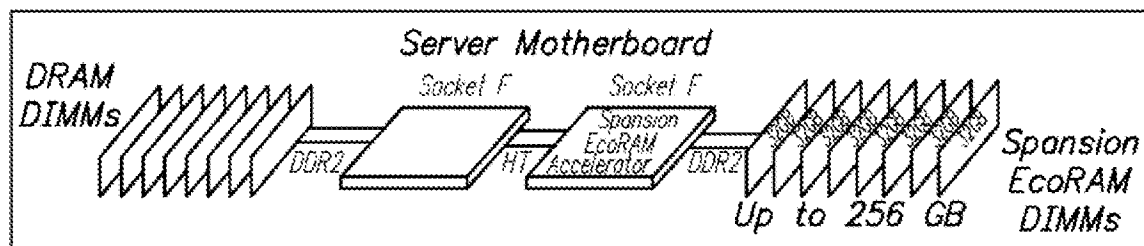


FIG. 3A

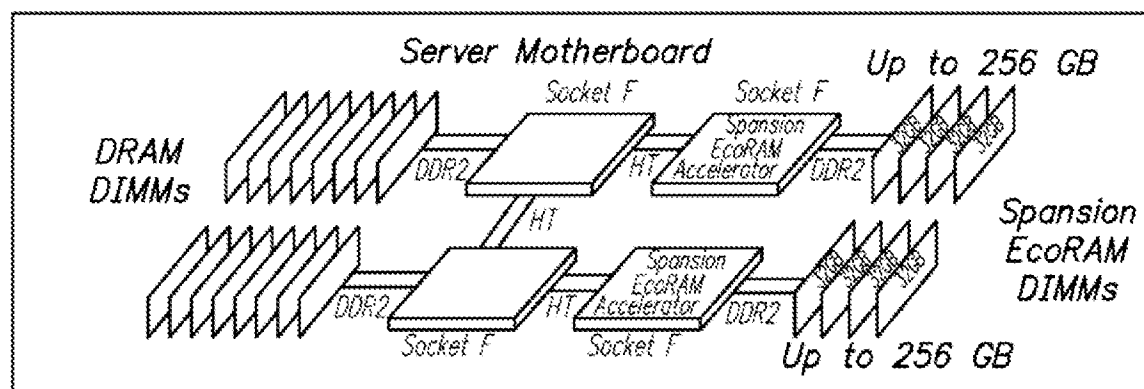
Spancion EcoRAM Configurations

256GB Spancion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

256GB Spancion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

FIG. 2
(PRIOR ART)

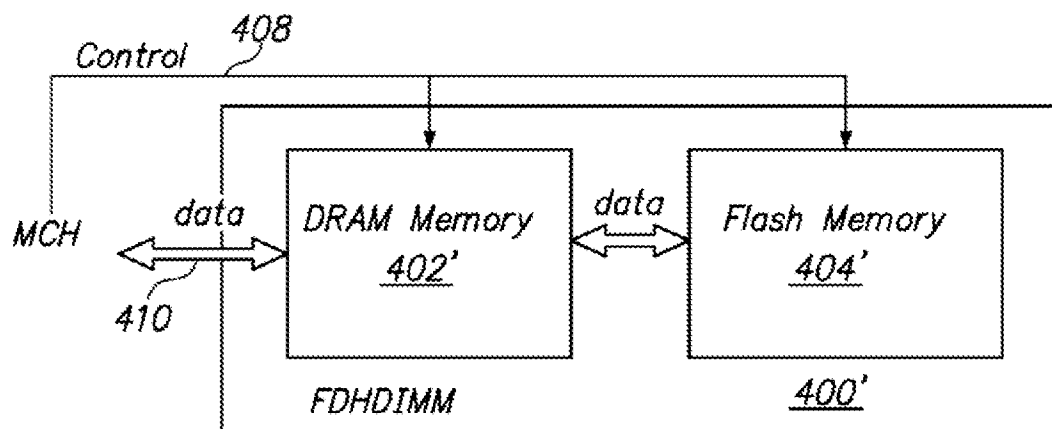


FIG. 4B

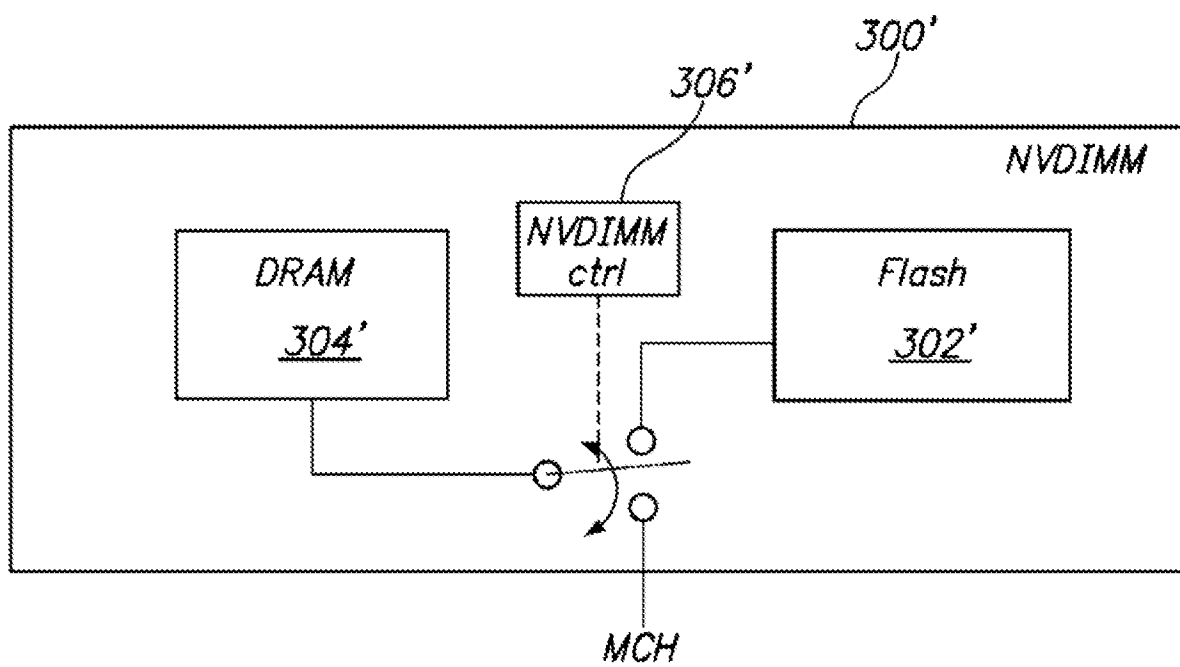


FIG. 3B

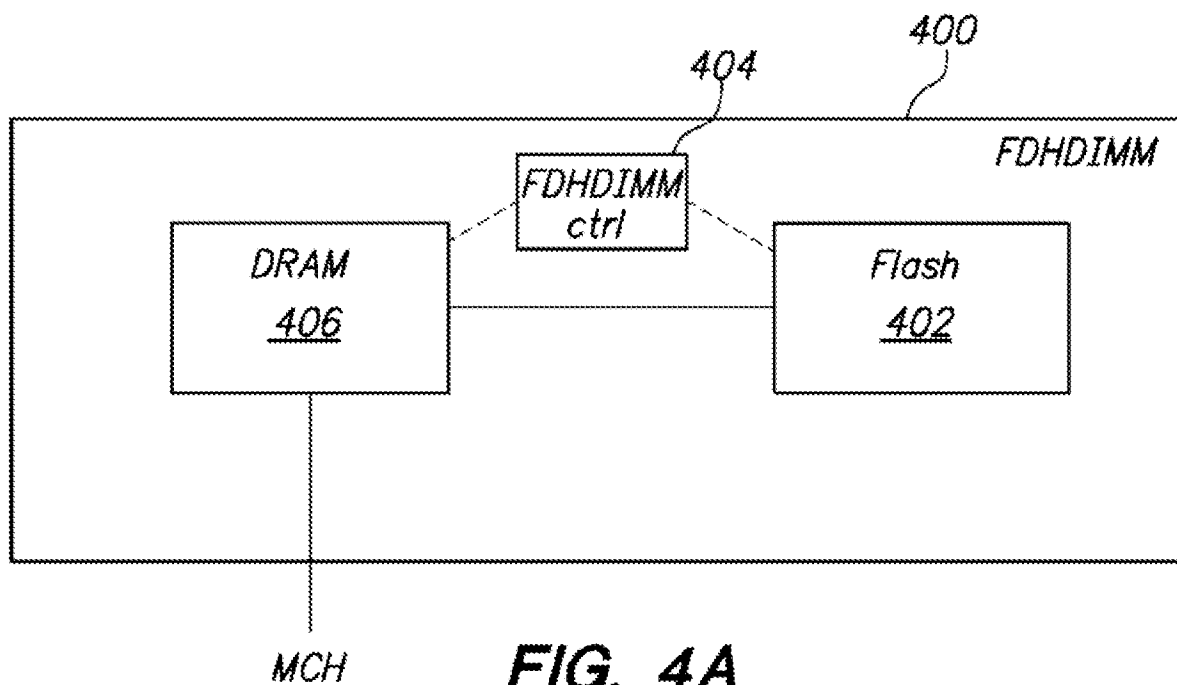


FIG. 4A

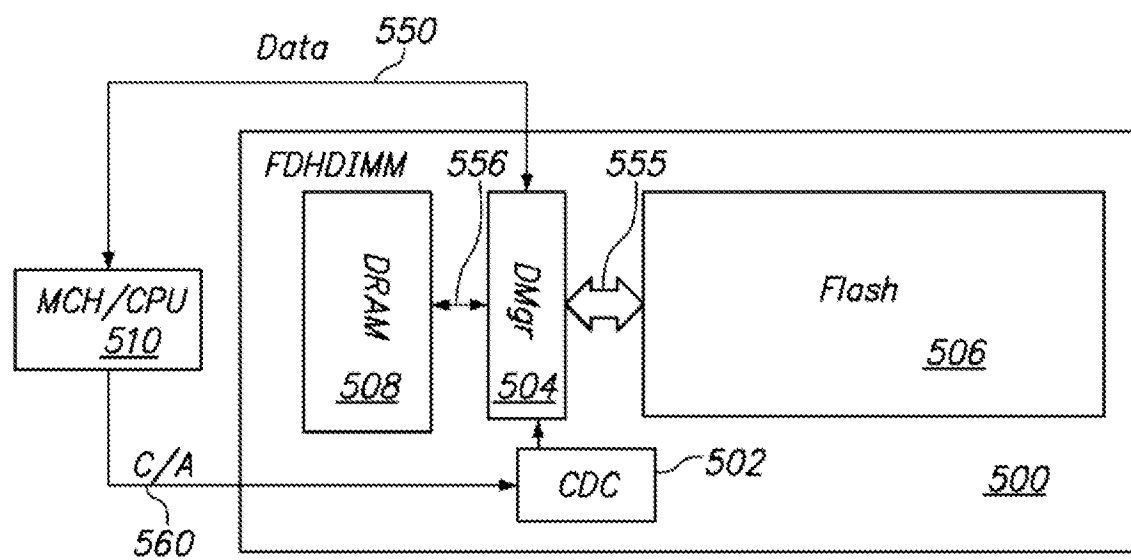


FIG. 5A

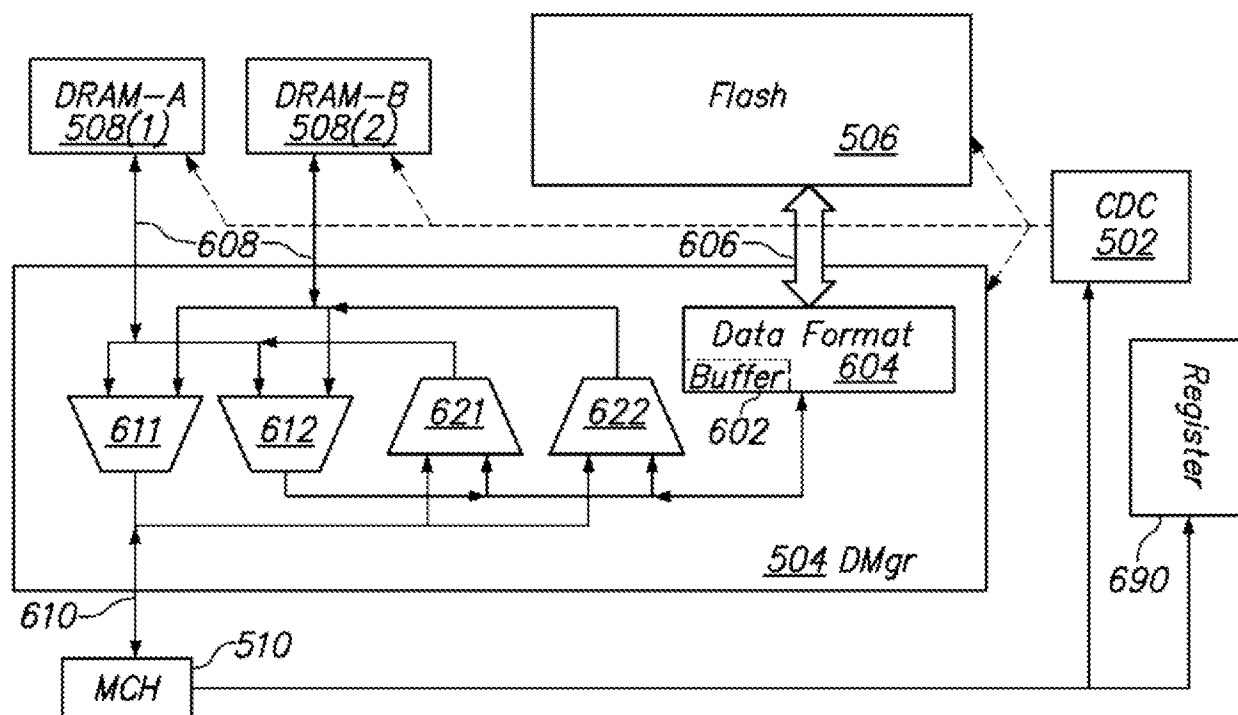


FIG. 6

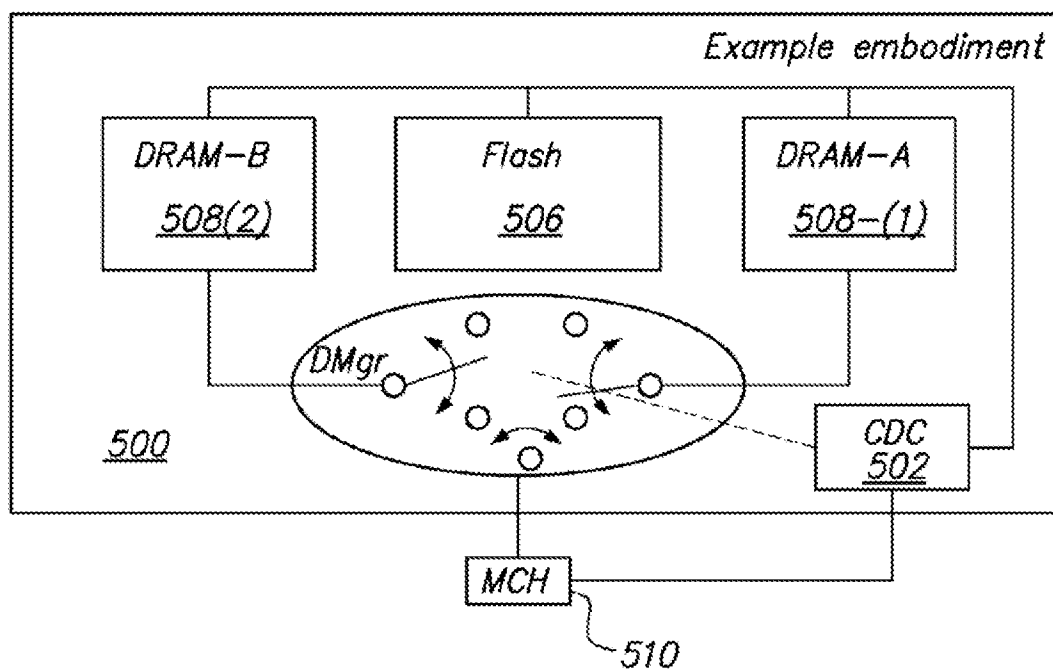


FIG. 5B

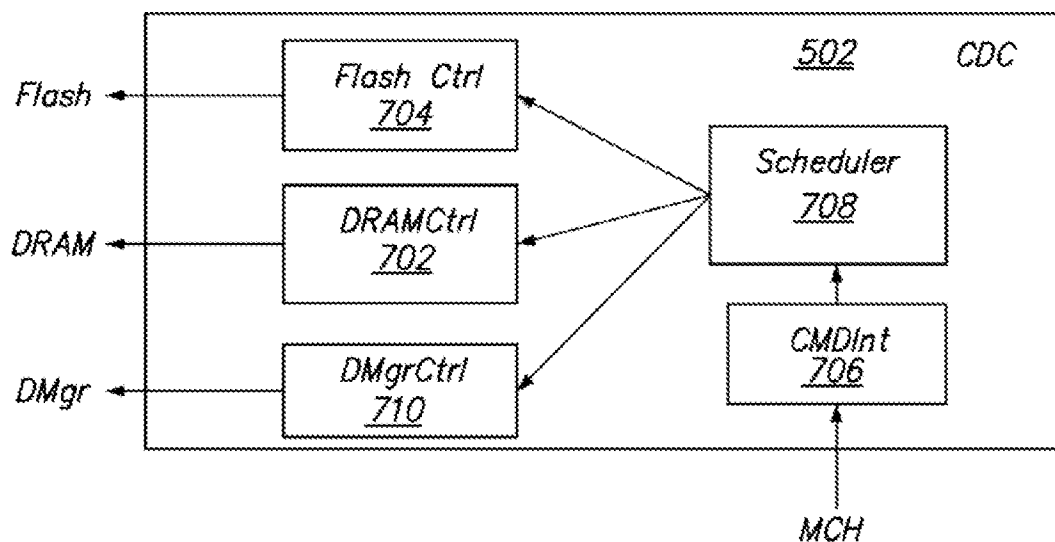


FIG. 7

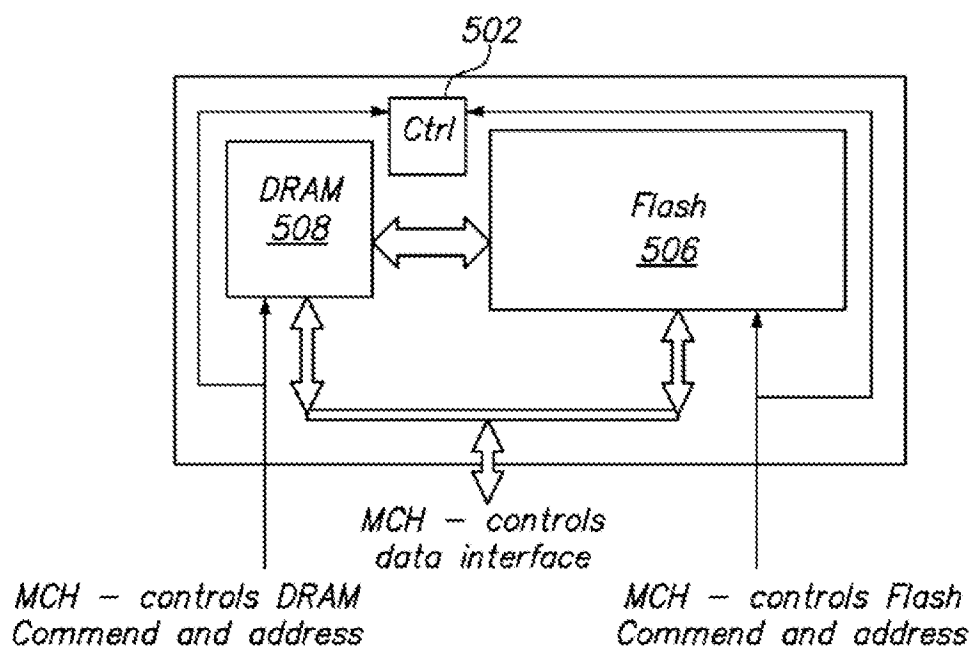


FIG. 8A

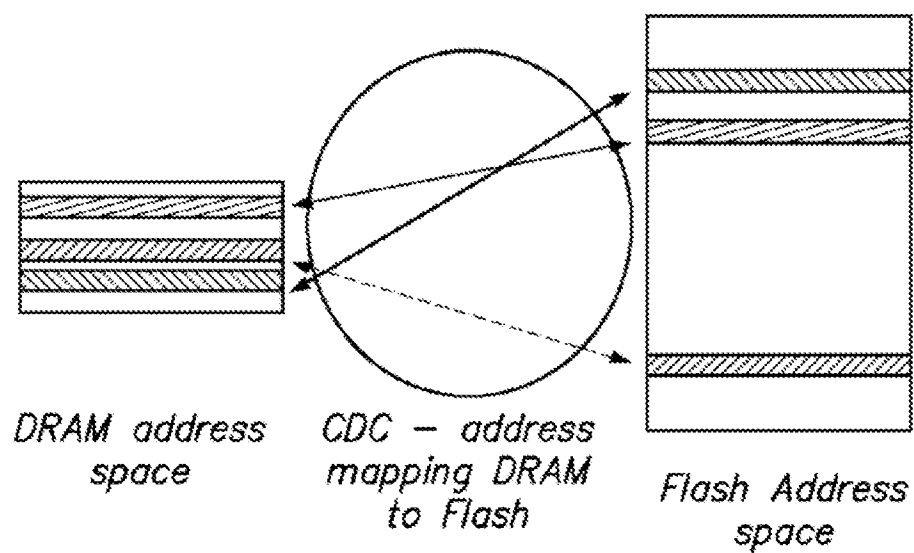
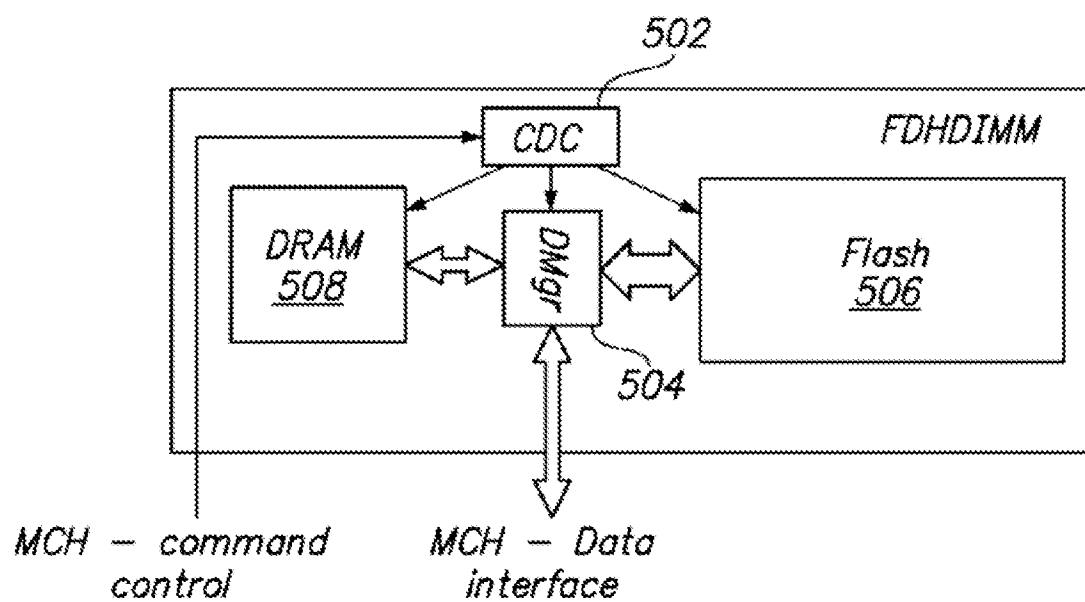


FIG. 8B

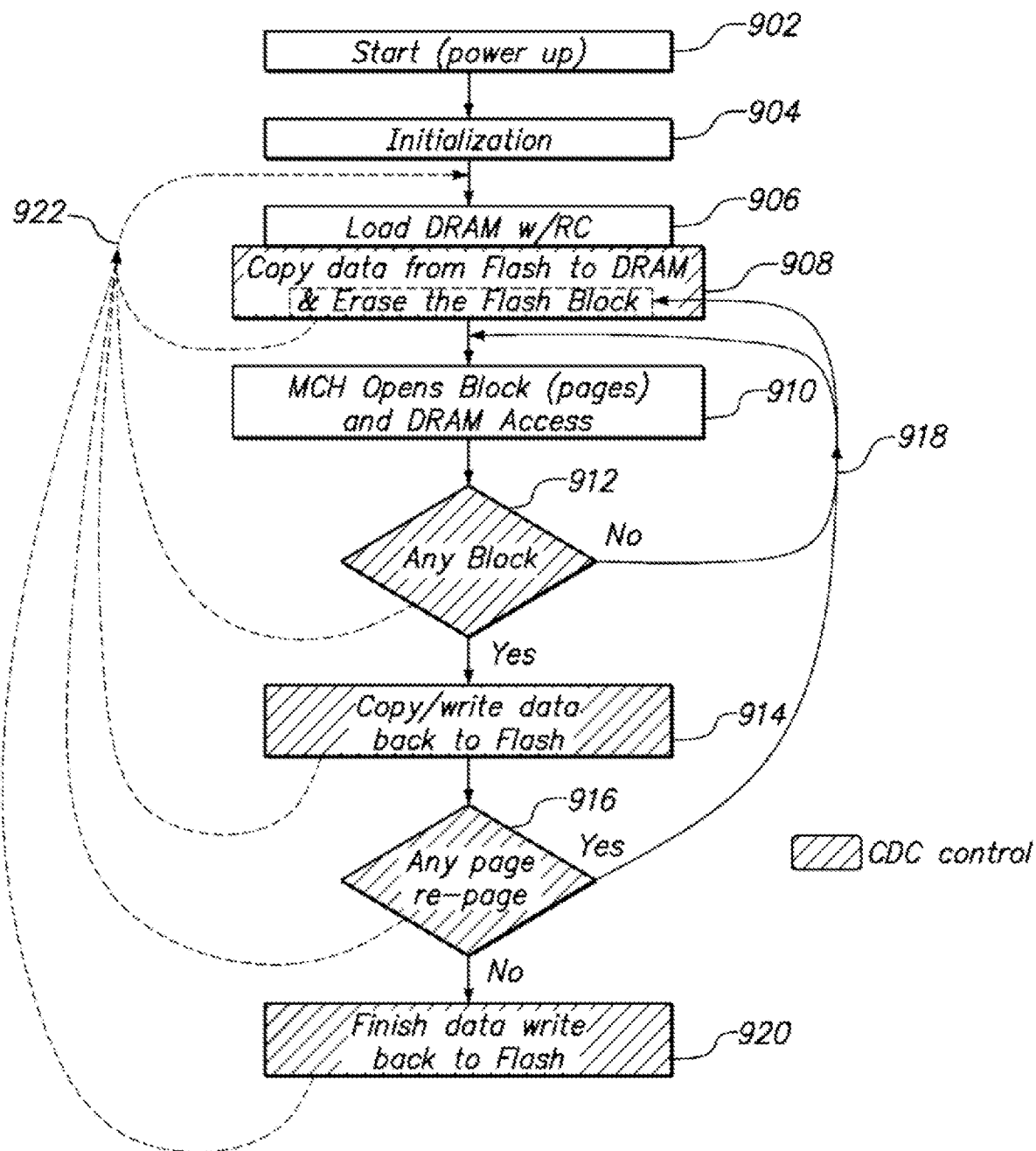


FIG. 9

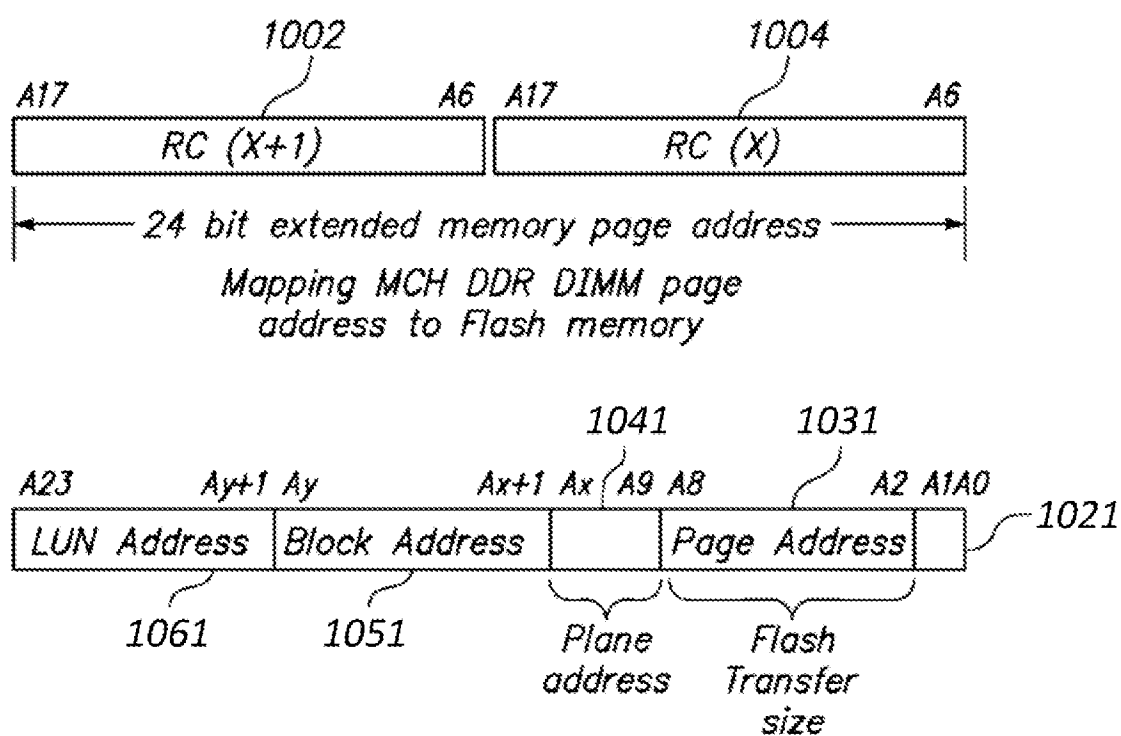


FIG. 10

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

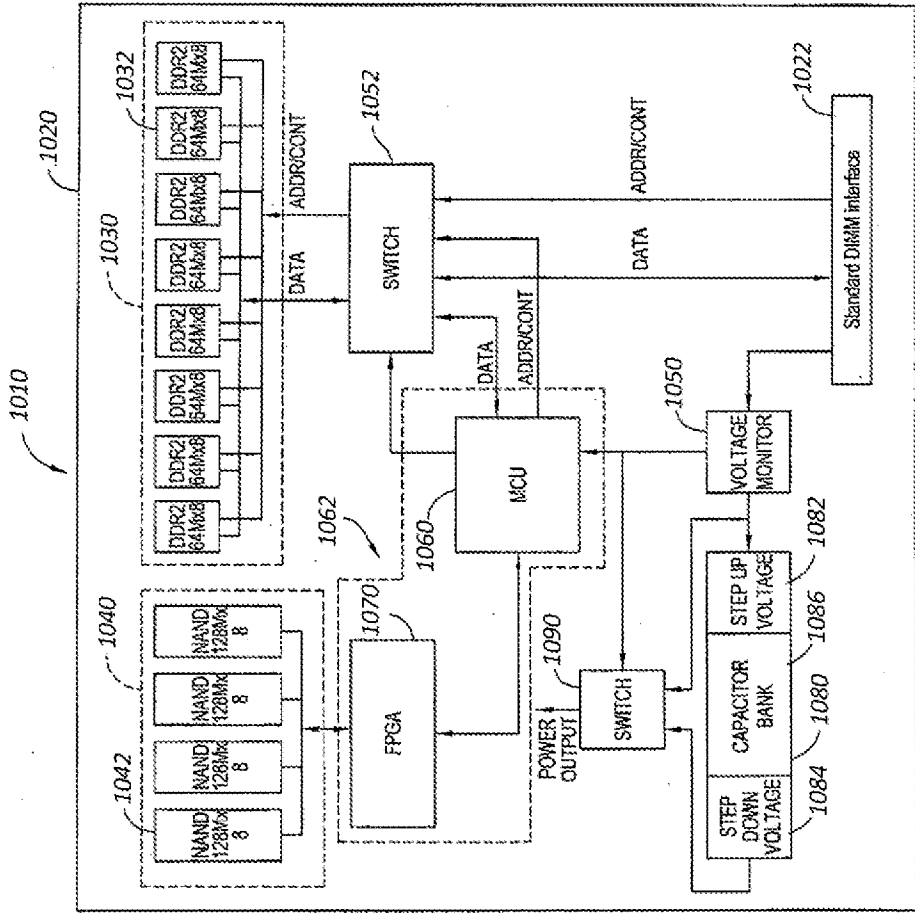


FIG. 12

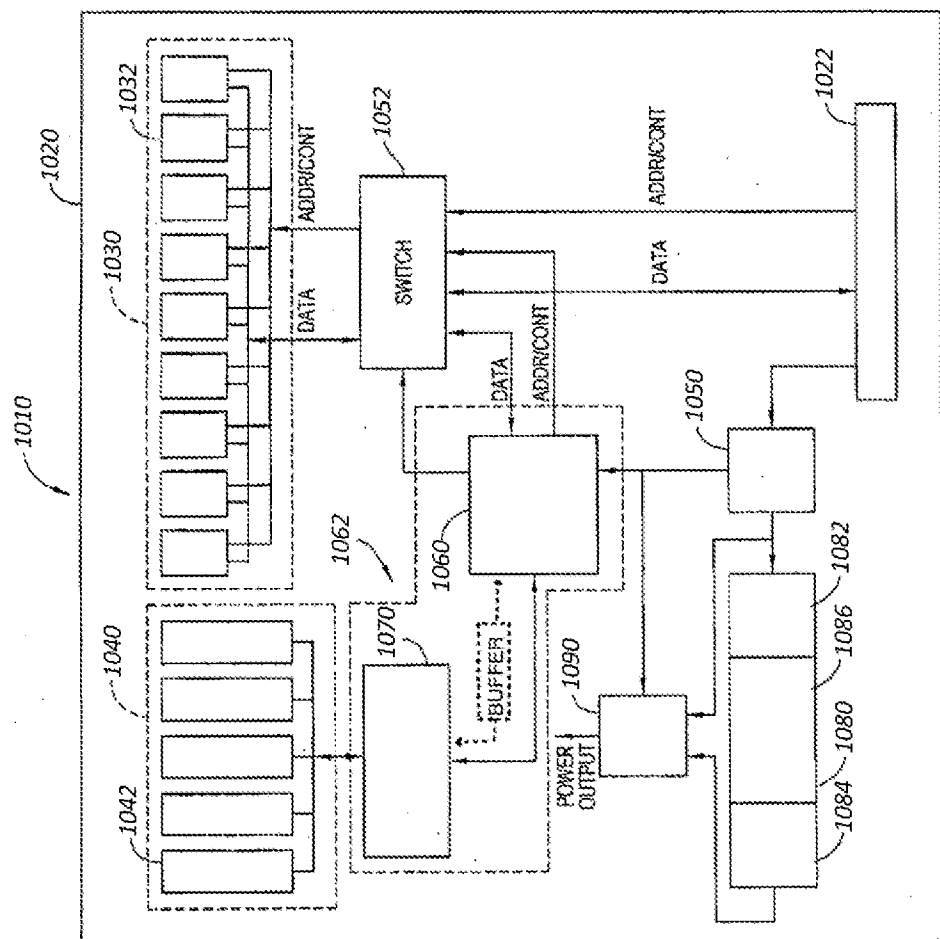


FIG. 13

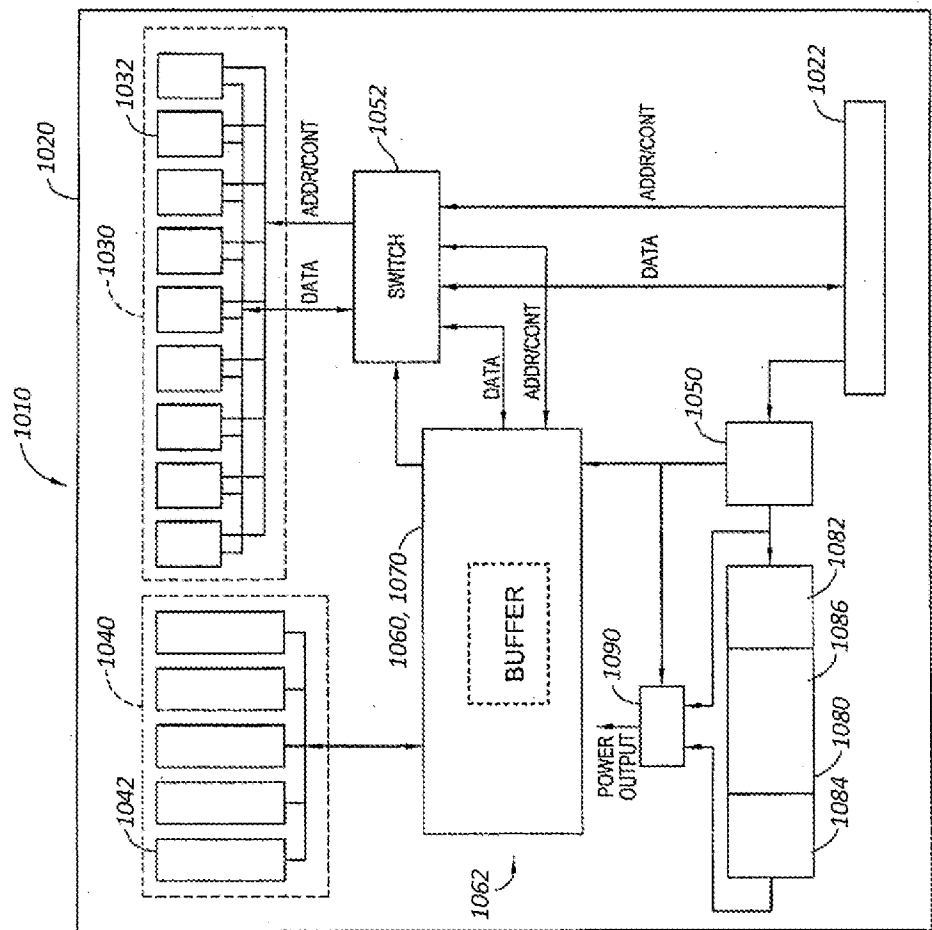


FIG. 14

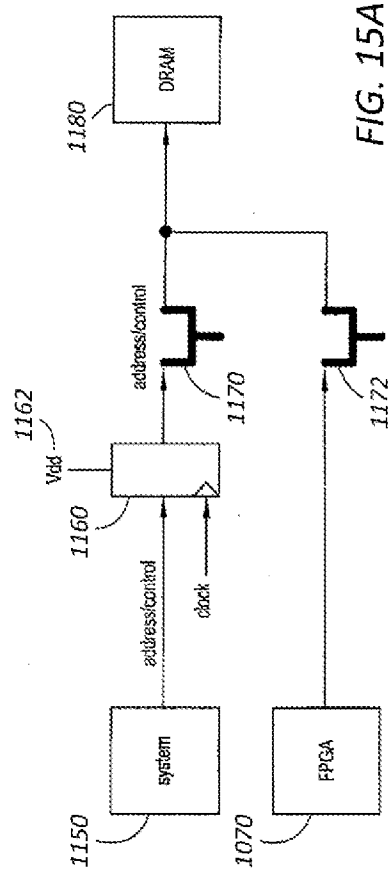


FIG. 15A

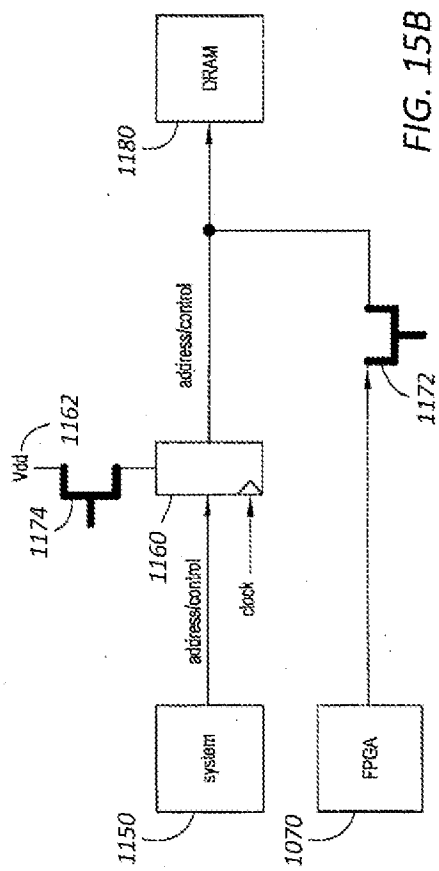


FIG. 15B

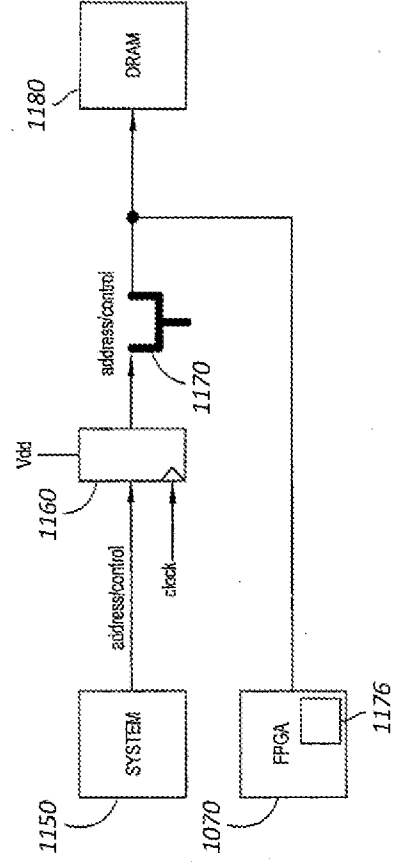


FIG. 15C

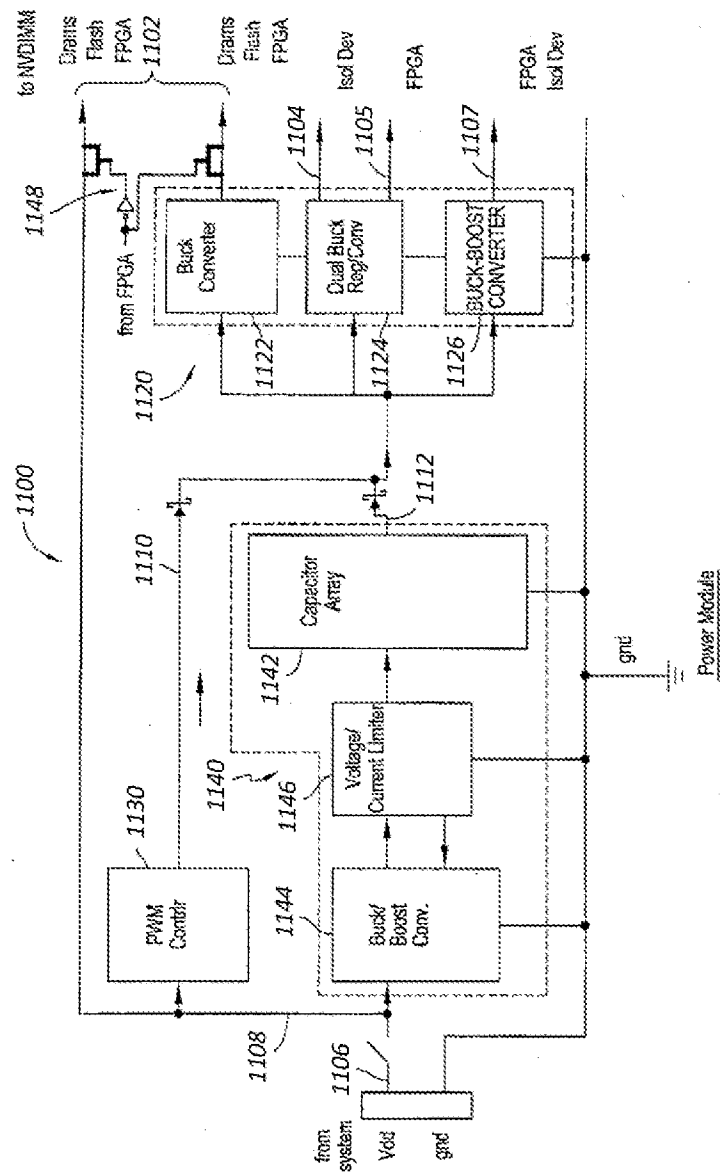


FIG. 16

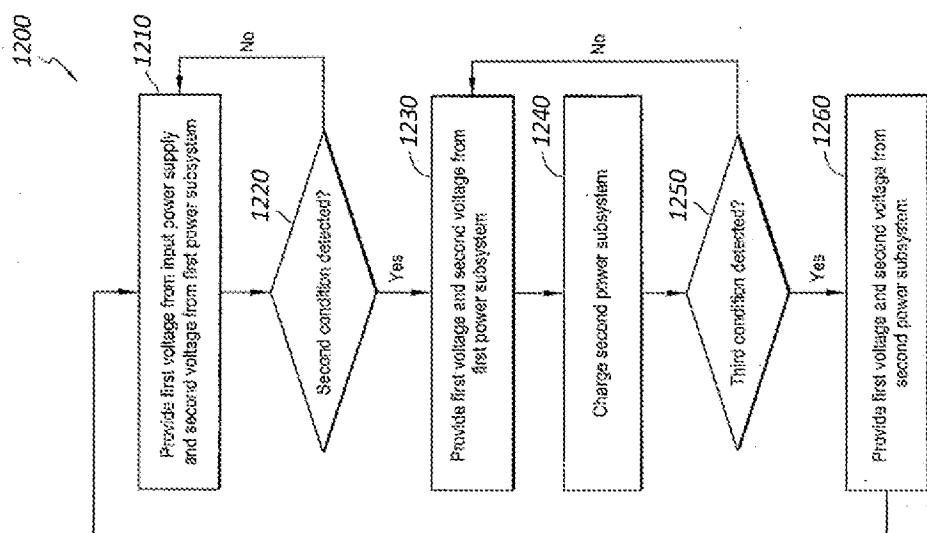


FIG. 17

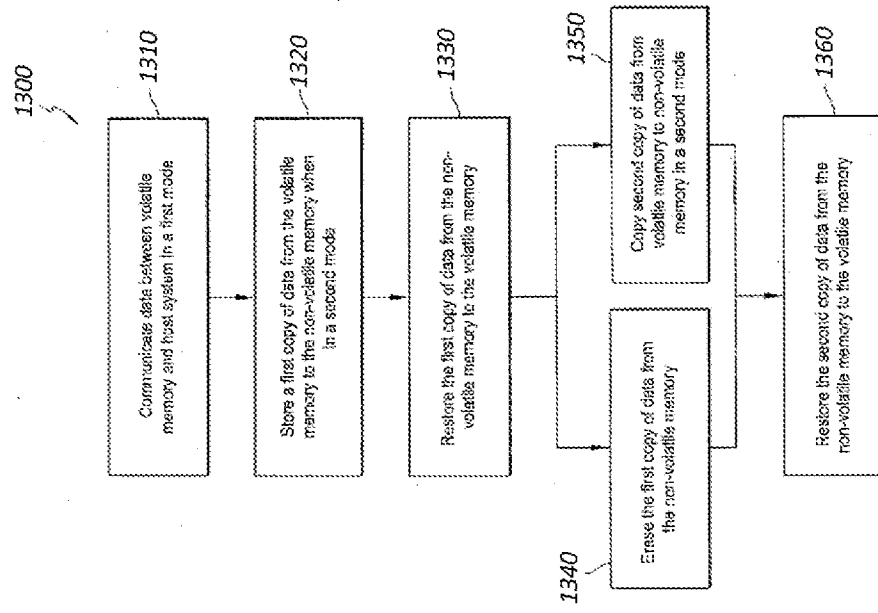


FIG. 18

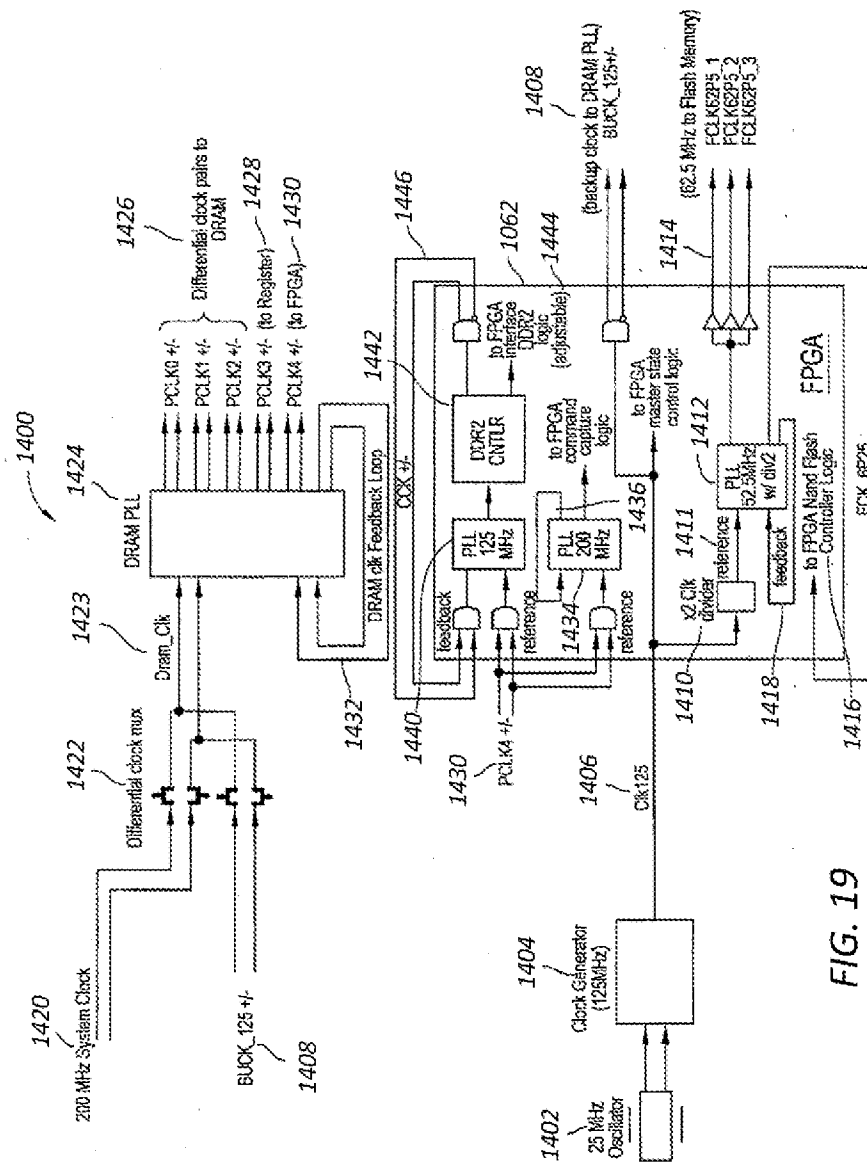


FIG. 19

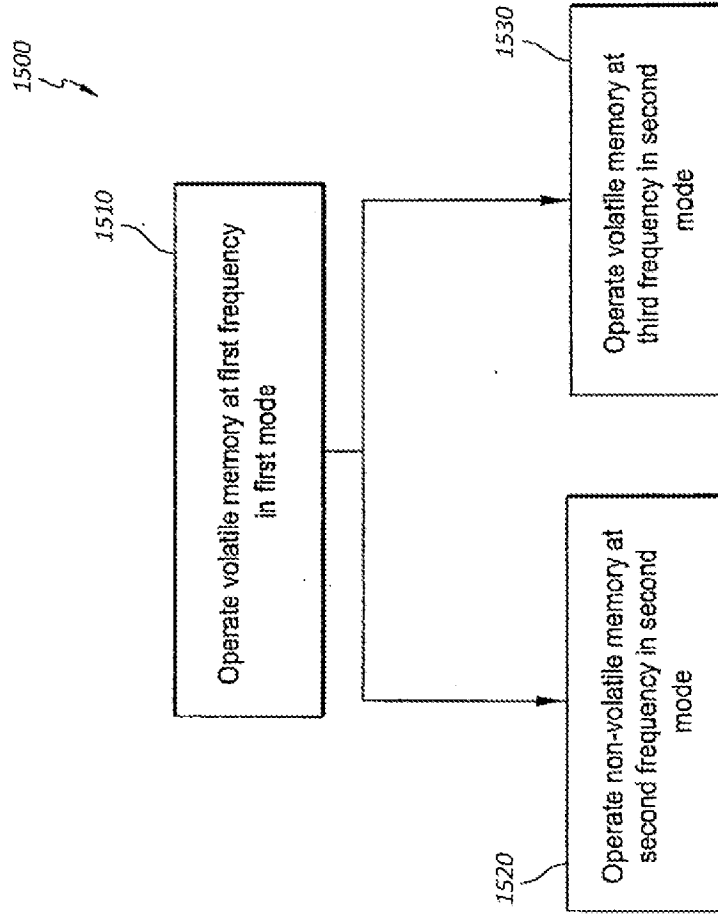


FIG. 20

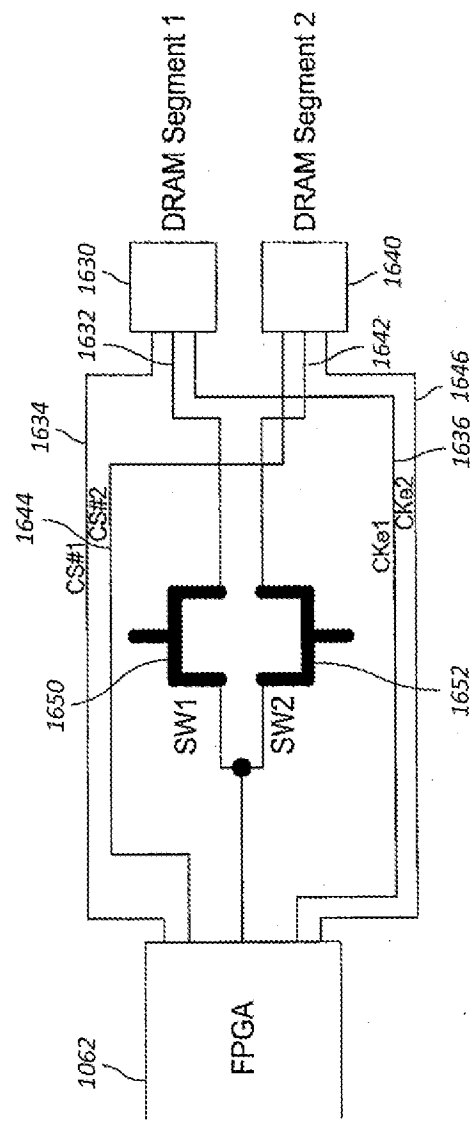


FIG. 21

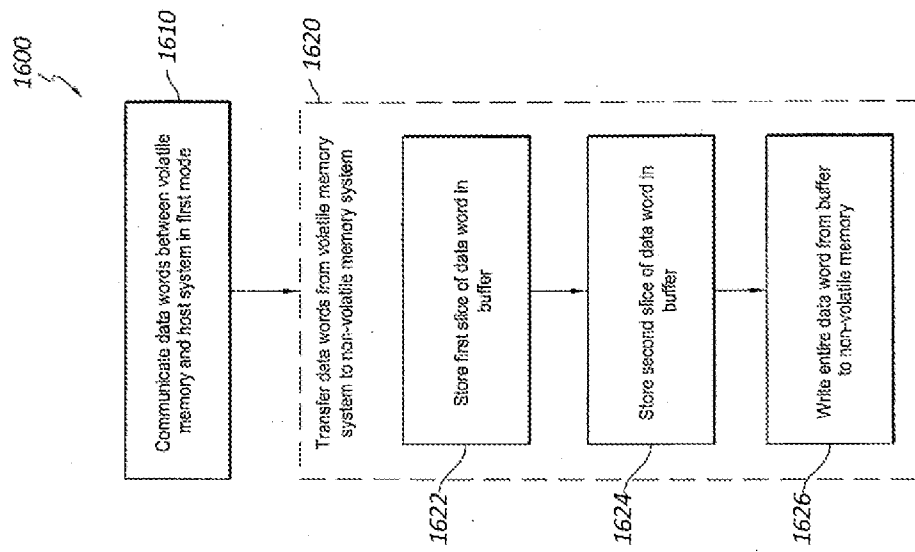


FIG. 22

Under the Paperwork Reduction Act of 1995 no persons are required to respond to a collection of information unless it displays a valid OMB control number.

UTILITY PATENT APPLICATION TRANSMITTAL <i>(Only for new nonprovisional applications under 37 CFR 1.53(b))</i>		Attorney Docket No. 022955-000090	
		First Named Inventor Chi-She Chen, et al.	
		Title FLASH-DRAM HYBRID MEMORY MODULE	
		Priority Mail Express® Label No.	

APPLICATION ELEMENTS <i>See MPEP chapter 600 concerning utility patent application contents.</i>	ADDRESS TO: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450
--	---

1. ☐ Fee Transmittal Form (PTO/SB/17 or equivalent) 2. ☐ Applicant asserts small entity status. See 37 CFR 1.27 3. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Applicant must attach form PTO/SB/15A or B or equivalent. 4. ☒ Specification [Total Pages <u>77</u>] Both the claims and abstract must start on a new page. (See MPEP § 608.01(a) for information on the preferred arrangement) 5. ☒ Drawing(s) (35 U.S.C. 113) [Total Sheets <u>22</u>] 6. Inventor's Oath or Declaration [Total Pages <u>4</u>] (including substitute statements under 37 CFR 1.64 and assignments serving as an oath or declaration under 37 CFR 1.63(e)) a. ☐ Newly executed (original or copy) b. ☒ A copy from a prior application (37 CFR 1.63(d)) 7. ☒ Application Data Sheet * See note below. See 37 CFR 1.76 (PTO/AIA/14 or equivalent) 8. CD-ROM or CD-R in duplicate, large table, or Computer Program (Appendix) ☐ Landscape Table on CD 9. Nucleotide and/or Amino Acid Sequence Submission (if applicable, items a. – c. are required) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on: i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statements verifying identity of above copies	ACCOMPANYING APPLICATION PAPERS 10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee _____ 11. ☐ 37 CFR 3.73(c) Statement ☒ Power of Attorney (when there is an assignee) 12. ☐ English Translation Document (if applicable) 13. ☐ Information Disclosure Statement (PTO/SB/08 or PTO-1449) ☐ Copies of citations attached 14. ☒ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP § 503) (Should be specifically itemized) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Nonpublication Request Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent. 18. ☐ Other: _____ _____ _____ _____
--	--

***Note:** (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 **must** be included in an Application Data Sheet (ADS).
 (2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).

19. CORRESPONDENCE ADDRESS				
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Address				
City	State	Zip Code		
Country	Telephone	Email		

Signature	/Khaled Shami/	Date	January 24, 2022
Name (Print/Type)	Khaled Shami	Registration No. (Attorney/Agent)	38745

This collection of information is required by 37 CFR 1.53(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	44819531
Application Number:	17582797
International Application Number:	
Confirmation Number:	3546
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000090
Receipt Date:	24-JAN-2022
Filing Date:	
Time Stamp:	17:11:04
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	WebADS.pdf	261304	no	9
			30ebf733d33f0f76d66b665610463e482bcd26ec		

Warnings:

Information:					
2		022955-000090_Appl_Text.pdf	638118	yes	77
			25e3644bd94b899a1ab5a2ca6849c07204c048bb		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	75	
	Claims		76	76	
	Abstract		77	77	
Warnings:					
Information:					
3	Oath or Declaration filed	022955-000090_Inventors_Oath.PDF	372159	no	4
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Warnings:					
Information:					
4		022955-000090_PA_ConApp.pdf	369499	yes	3
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	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Specification		2	2	
	Applicant Arguments/Remarks Made in an Amendment		3	3	
Warnings:					
Information:					
5	Power of Attorney	022955-000090_SignedPOA.PDF	144453	no	1
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Warnings:					
Information:					

6	Drawings-only black and white line drawings	022955-000090_Appl_Drawings_ForFiling.pdf	6452231	no	22
			f25d28133d2a7a6c4efa22b0fe87e19f4c69aaf2		

Warnings:

Information:

7	Transmittal of New Application	022955-000090_NewAppTransmittal_aia0015.pdf	282121	no	2
			393d79d68ded9f915027833d3bb026297133b0b3c		

Warnings:

Information:

Total Files Size (in bytes):			8519885
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Chi-She		Chen		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Walnut	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1		944 Crystal Water Lane			
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country ⁱ	US		
Inventor 2					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1		6 Silver Fir			
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country ⁱ	US		
Inventor 3					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Scott	H.	Milton		

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	49 Statehouse Place				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92602	Country ⁱ	US		

Inventor 4

Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country ⁱ	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

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Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	022955-000090	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
	Continuation of		17328019	2021-05-24	

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17328019	Continuation of	17138766	2020-12-30	11016918	2021-05-25

Prior Application Status		Abandoned		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
17138766	Continuation of		15934416	2018-03-23	

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
15934416	Continuation of	14840865	2015-08-31	9928186	2018-03-27

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14840865	Continuation of	14489269	2014-09-17	9158684	2015-10-13

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14489269	Continuation of	13559476	2012-07-26	8874831	2014-10-28

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
13559476	Claims benefit of provisional		61512871	2011-07-28	

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Prior Application Status		Patented	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30

Prior Application Status		Abandoned	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		

Prior Application Status		Pending	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12131873	Claims benefit of provisional	60941586	2007-06-01		

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This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

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Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

☐ NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

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The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

- ☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.								
Applicant 1								
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.								
			Clear					
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If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:								
Name of the Deceased or Legally Incapacitated Inventor:								
If the Applicant is an Organization check here. ☒								
Organization Name	Netlist, Inc.							
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Address 2	Suite 100							
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

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Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.						
If the Assignee or Non-Applicant Assignee is an Organization check here.			☒			
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Mailing Address Information For Assignee including Non-Applicant Assignee:						
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Address 2		Suite 100				
City	Irvine	State/Province	CA			
Country i	US	Postal Code	92617			
Phone Number	US	Fax Number				
Email Address						
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Khaled Shami/			Date (YYYY-MM-DD)	
First Name	Khaled	Last Name	Shami	Registration Number	38745
Additional Signature may be generated within this form by selecting the Add button.					

FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 17/138,766, filed December 30, 2020, titled "Flash-Dram Hybrid Memory", now U.S. Patent No. 11,016,918, which is a continuation of U.S. Patent Application No. 15/934,416, filed March 23, 2018, titled "Flash-Dram Hybrid Memory Module," which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled "Non-Volatile Memory Module," now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may be considered related to U.S. Patent Application No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed February 5, 2014, which is a continuation of U.S. Patent Application 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Patent 8,677,060, issued

March 18, 2014, which is a continuation of U.S. Patent Application 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed June 28, 2012, now U.S. Patent 8,516,187, issued August 20, 2013, which is a divisional of U.S. Patent Application 12/240,916, titled "Non-Volatile Memory Module," filed September 29, 2008, now U.S. Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007, the contents of which are incorporated by reference herein in their entirety.

[0003] This application may also be considered related to U.S. Patent Application No. 15/000,834, filed January 19, 2016 (abandoned), which is a continuation of U.S. Patent Application No. 14/489,332, filed September 17, 2014, now U.S. Patent No. 9,269,437, which is a continuation of U.S. Patent Application No. 14/173,219, filed February 5, 2014, now U.S. Patent No. 8,904,099, which is a continuation of U.S. Patent Application No. 13/905,048, filed May 29, 2013, now U.S. Patent No. 6,671,243, which is a continuation U.S. Patent Application 13/536,173 above.

[0004] This application may also be considered related to U.S. Patent Application No. 15/924,866, (abandoned), which is a continuation of U.S. Patent Application No. 14/489,281, filed September 17, 2014, now U.S. Patent No. 9,921,762, which is a continuation of U.S. Patent Application No. 13/625,563, filed September 24, 2012, now U.S. Patent No. 8,904,098, which claims the benefit of U.S. Provisional Application No. 61/583,775, filed September 23, 2011.

TECHNICAL FIELD

[0005] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0006] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0007] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high

speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0008] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0009] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a

data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MBs for the EcoRAM vs. 104MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™, which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0010] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0011] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash

DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0012] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

[0013] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0014] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0015] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

[0016] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0017] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0018] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0019] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0020] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0021] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0022] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0024] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0025] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0026] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0027] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0029] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0030] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0031] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0032] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0033] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0034] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0035] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0036] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0037] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0038] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0039] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0040] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0041] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0042] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0043] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0044] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0045] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0047] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0048] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0049] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0050] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0052] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0053] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0054] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0055] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0056] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0057] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0060] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0061] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0062] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0063] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0064] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0065] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0066] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0067] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0068] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0069] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

[0070] Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to

allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

[0071] Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

[0072] Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0074] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0075] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to

implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0076] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0077] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only

Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0078] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0079] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0080] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0081] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0082] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or

commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0083] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0084] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0085] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then

the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0086] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0087] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0088] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory

module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0089] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In

the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0090] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0091] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH

510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0092] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0093] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the

DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0094] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0095] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate

and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0096] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0097] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or

provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0098] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0099] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[00100] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[00101] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[00102] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from

the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[00103] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[00104] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[00105] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a

first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[00106] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[00107] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00108] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502

monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00109] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061

field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A_0A_1 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00110] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00111] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash

memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00112] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00113] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00114] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00115] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up

the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[00116] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[00117] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor

array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[00118] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[00119] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[00120] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[00121] FIG. 12 is a block diagram of an example memory system 1010 compatible with certain embodiments described herein. The memory system 1010 can be coupled to a host computer system and can include a volatile memory subsystem 1030, a non-volatile memory

subsystem 1040, and a controller 1062 operatively coupled to the non-volatile memory subsystem 1040. In certain embodiments, the memory system 1010 includes at least one circuit 1052 configured to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030.

[00122] In certain embodiments, the memory system 1010 comprises a memory module. The memory system 1010 may comprise a printed-circuit board (PCB) 1020. In certain embodiments, the memory system 1010 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 1010 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for

example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[00123] In certain preferred embodiments, the memory system 1010 is in electrical communication with the host system. In other embodiments, the memory system 1010 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 1010 can be in communication with a disk controller of a computer system, for example. The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.

[00124] The volatile memory subsystem 1030 comprises a plurality of volatile memory elements 1032 and the non-volatile memory subsystem 1040 comprises a plurality of non-volatile memory elements 1042. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem 1040 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 1030. In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and

synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00125] In certain embodiments, the second plurality of non-volatile memory elements 1042 comprises one or more flash memory elements. Types of flash memory elements 1042 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank 1040 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 1042. In addition, nonvolatile memory elements 1042 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 1042 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00126] FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements

1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

[00127] Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one

embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

[00128] In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

[00129] In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one

or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10

percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[00131] As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 1070 is also programmed to perform address/address translation between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 1042 because of their superior read speed and compact structure.

[00132] The memory system 1010 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 1062 and the non-volatile memory subsystem 1040 are operatively decoupled (e.g., isolated) from the volatile memory subsystem 1030 by the at least one circuit 1052 and a second state in which the

volatile memory subsystem 1030 is operatively coupled to the controller 1062 to allow data to be communicated between the volatile memory subsystem 1030 and the nonvolatile memory subsystem 1040 via the controller 1062. The memory system 1010 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 1010 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[00133] The memory system 1010 may further comprise a voltage monitor 1050. The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 1050 may transmit a signal to the controller 1062 indicative of the detected condition. The controller 1062 of certain embodiments responds to the signal from the voltage monitor circuit 1050 by transmitting a signal to the at least one circuit 1052 to operatively couple the controller to the volatile memory system 1030, such that the memory system 1010 enters the second state. For example, the voltage monitor 1050 may send a signal to the MCU 1060 which responds by accessing the data on the volatile memory system 1030 and by executing a write cycle on the nonvolatile memory subsystem 1040. During this write cycle, data is read from the volatile memory subsystem 1030 and is transferred to the non-volatile memory subsystem 1040 via the MCU 1060. In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.

[00134] The isolation or operational decoupling of the volatile memory subsystem 1030 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 1010 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 1030. For example, in one embodiment during such periods of operation, the controller 1062 and the nonvolatile memory subsystem 1040 do not add a significant capacitive load to the volatile memory system 1030 when the memory system 1010 is in the first state. In certain such embodiments, the capacitive load of the controller 1062 and the non-volatile memory subsystem 1040 do not significantly affect the signals propagating between the volatile memory subsystem 1030 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 1052 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 1052 comprises other types of isolation devices.

[00135] Power may be supplied to the volatile memory subsystem 1030 from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. In certain embodiments, the memory system 1010 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 1010 enters the second state in response to a trigger condition. In certain embodiments, the memory system 1010 has a third state in which the controller 1062 is operatively decoupled from the volatile memory subsystem 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory

subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.

[00136] In certain embodiments, the second power supply 1080 does not comprise a battery. Because a battery is not used, the second power supply 1080 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. 12-14, the second power supply 1080 comprises a step-up transformer 1082, a step-down transformer 1084, and a capacitor bank 1086 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 1080 is configured to power the volatile memory subsystem 1030 for less than thirty minutes. In certain embodiments, the second power supply 1080 may comprise a battery. For example, in certain embodiments, the second power supply 1080 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 1030 for no more than thirty minutes.

[00137] In certain embodiments, the capacitor bank 1086 of the second power supply 1080 is charged by the first power supply while the memory system 1010 is in the first state. As a result, the second power supply 1080 is fully charged when the memory system 1010 enters the second state. The memory system 1010 and the second power supply 1080 may be located on the same printed circuit board 1020. In other embodiments, the second power supply 1080 may not be on

the same printed circuit board 1020 and may be tethered to the printed circuit board 1020, for example.

[00138] When operating in the first state, in certain embodiments, the step-up transformer 1082 keeps the capacitor bank 1086 charged at a peak value. In certain embodiments, the step-down transformer 1084 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 1032 and 3.0V to the non-volatile flash memory elements 1042) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. 12-14, the memory module 1010 further comprises a switch 1090 (e.g., FET switch) that switches power provided to the controller 1062, the volatile memory subsystem 1030, and the non-volatile memory subsystem 1040, between the power from the second power supply 1080 and the power from the first power supply (e.g., system power) received via the interface 1022. For example, the switch 1090 may switch from the first power supply to the second power supply 1080 when the voltage monitor 1050 detects a low voltage condition. The switch 1090 of certain embodiments advantageously ensures that the volatile memory elements 1032 and non-volatile memory elements 1042 are powered long enough for the data to be transferred from the volatile memory elements 1032 and stored in the non-volatile memory elements 1042. In certain embodiments, after the data transfer is complete, the switch 1090 then switches back to the first power supply and the controller 1062 transmits a signal to the at least one circuit 1052 to operatively decouple the controller 1062 from the volatile memory subsystem 1030, such that the memory system 1010 reenters the first state.

[00139] When the memory system 1010 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 via the

controller 1062. The host system can then resume accessing the volatile memory subsystem 1030 of the memory module 1010. In certain embodiments, after the memory system 1010 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 1030 rather than the non-volatile memory subsystem 1040 because the volatile memory elements 1032 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 1030 to the nonvolatile memory bank 1040, or from the non-volatile memory bank 1040 to the volatile. memory bank 1030, takes less than one minute per GB.

[00140] In certain embodiments, the memory system 1010 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 1010 copies the entire contents of the volatile memory subsystem 1030 into the nonvolatile memory subsystem 1040 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are copied back into the volatile memory subsystem 1030 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 1040 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 1010. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 1010 and can improve the performance of the memory system 1010. In certain other embodiments, the entire contents of the volatile memory subsystem 1030 are not copied into the non-volatile memory subsystem 1040 on each backup operation, but only a partial copy is performed. In certain embodiments, other

management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 1040 are performed in the controller 1062.

[00141] The memory system 1010 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 1030 which then writes the data to non-volatile storage which is not part of the memory system 1010, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 1010 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 1010 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 1010 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 1010 may be operated as a write-through cache or as some other type of cache.

[00142] FIG. 16 schematically illustrates an example power module 1100 of the memory system 1010 in accordance with certain embodiments described herein. The power module 1100 provides power to the various components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. In certain embodiments, the power module 1100 comprises one or more of the components described above with respect to FIG. 12. For example, in certain embodiments, the power module 1100 includes the second power supply 1080 and the switch 1090.

[00143] The power module 1100 provides a plurality of voltages to the memory system 1010 comprising non-volatile and volatile memory subsystems 1030, 1040. The plurality of voltages comprises at least a first voltage 1102 and a second voltage 1104. The power module 1100 comprises an input 1106 providing a third voltage 1108 to the power module 1100 and a voltage conversion element 1120 configured to provide the second voltage 1104 to the memory system 1010. The power module 1100 further comprises a first power element 1130 configured to selectively provide a fourth voltage 1110 to the conversion element 1120. In certain embodiments, the first power element 1130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 1130 is configured to receive a 1.8V input system voltage as the third voltage 1108 and to output a modulated 5V output as the fourth voltage 1110.

[00144] The power module 1100 further comprises a second power element 1140 can be configured to selectively provide a fifth voltage 1112 to the conversion element 1120. The power module 1100 can be configured to selectively provide the first voltage 1102 to the memory system 1010 either from the conversion element 1120 or from the input 1106.

[00145] The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element

1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.

[00146] In certain embodiments, the power module 1100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 1100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 1102 in the second state from the first power element 1130 rather than from the input 1106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 1102 from the first power element 1130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 1120 and the first and second power elements 1130, 1140 (e.g., the sources of the pre-regulated fourth voltage 1110 in the second state and the pre-regulated fifth voltage 1112 in the third state) can smooth out potential voltage spikes.

[00147] In certain embodiments, the second power element 1140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. 16, the second power element 1140 comprises a capacitor array 1142, a buck-boost converter 1144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 1146 which limits the charge current to the capacitor array 1142 and stops charging the capacitor array 1142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 1142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For

example, in one example embodiment, the buck-boost converter 1144 receives a 1.8V system voltage (first voltage 1108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 1146. The voltage/current limiter 1146 limits the current going to the capacitor array 1142 to 1A and stops charging the array 1142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 1140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 1142.

[00148] The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate

amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.

[00149] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 1120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.

[00150] FIG. 17 is a flowchart of an example method 1200 of providing a first voltage 1102 and a second voltage 1104 to a memory system 1010 including volatile and nonvolatile memory subsystems 1030, 1040. While the method 1200 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-15, other memory systems are also compatible with embodiments of the method 1200. During a first condition, the method 1200

comprises providing the first voltage 1102 to the memory system 1010 from an input power supply 1106 and providing the second voltage 1104 to the memory system 1010 from a first power subsystem in operational block 1210. For example, in one embodiment, the first power subsystem comprises the first power element 1130 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, other first power subsystems are used.

[00151] The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.

[00152] The method 1200 further comprises charging a second power subsystem in operational block 1240. In certain embodiments, the second power subsystem comprises the second power element 1140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 1140 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, some other second power subsystem is used.

[00153] The method 1200 further comprises detecting a third condition in an operational block 1250 and during the third condition, providing the first voltage 1102 and the second

voltage 1104 to the memory system 1010 from the second power subsystem 1140 in an operational block 1260. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 1200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 1140 is charged before detecting the second condition.

[00154] In certain embodiments, the memory system 1010 comprises a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 1010 also comprises a controller 1062 operatively coupled to the volatile memory subsystem 1030 and operatively coupled to the non-volatile memory subsystem 1040. The controller 1062 can be configured to allow data to be communicated between the volatile memory subsystem 1030 and the host system when the memory system 1010 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040 when the memory system 1010 is operating in a second state.

[00155] Although the memory system 1010 having extra storage capacity of the non-volatile memory subsystem 1040 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In various embodiments, there may be at least 200, 300, or 400 percent more

storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In other embodiments, the non-volatile memory subsystem 1040 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 1030. In some embodiments, the non-volatile memory subsystem 1040 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 1030. In one embodiment, the non-volatile memory subsystem 1040 includes less than 100 percent more storage capacity than does the volatile memory subsystem 1030.

[00156] The extra storage capacity of the non-volatile memory subsystem 1040 can be used to improve the backup capability of the memory system 1010. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 1040 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem 1040 allows the volatile memory subsystem 1030 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to backup the volatile memory subsystem 1030 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 1030 is copied to a first, previously erased portion of the nonvolatile memory subsystem 1040 via the controller 1062. Since the non-volatile memory subsystem 1040 has more storage capacity than does the volatile memory subsystem 1030, there is a second portion of the non-volatile memory subsystem 1040 which does not have data from the volatile memory subsystem 1030 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 1062 of the memory system 1010 restores the data to the volatile memory subsystem 1030 by copying the backed-up data from the non-volatile memory

subsystem 40 back to the volatile memory subsystem 1030. After the data is restored, the memory system 1010 erases the non-volatile memory subsystem 1040. While the first portion of the non-volatile memory subsystem 1040 is being erased, it may be temporarily inaccessible.

[00157] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 1040 is completely erased, the volatile memory subsystem 1030 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 1040 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 1040, the memory system 1010 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 1040.

[00158] FIG. 18 is a flowchart of an example method 1300 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. In certain embodiments, the non-volatile memory subsystem 1040 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 1300 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-14, the method 1300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 1310, the method 1300 comprises communicating data between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation. The method 1300 further comprises storing a first copy of data from the

volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a first time when the memory system 1010 is in a second mode of operation in an operational block 1320.

[00159] In an operational block 1330, the method 1300 comprises restoring the first copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030. The method 1300 further comprises erasing the first copy of data from the non-volatile memory subsystem 1040 in an operational block 1340. The method further comprises storing a second copy of data from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a second time when the memory system 1010 is in the second mode of operation in an operational block 1350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 1040.

[00160] In some embodiments, the memory system 1010 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 1040. The method 1300 can also include restoring the second copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 in an operational block 1360. The operational blocks of method 1300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem 1030 at operational block 1360 before the first copy of data is completely erased in the operational block 1340.

[00161] FIG. 19 schematically illustrates an example clock distribution topology 1400 of a memory system 1010 in accordance with certain embodiments described herein. The clock

distribution topology 1400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 1010. A clock source 1402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 1402 may feed a clock generator 1404 which provides a clock signal 1406 to the controller 1062, which may be an FPGA. In one embodiment, the clock generator 1404 generates a 125 MHz clock signal 1406. The controller 1062 receives the clock signal 1406 and uses it to clock the controller 1062 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 1062.

[00162] The clock signal 1406 can also be input into a clock divider 1410 which produces a frequency-divided version of the clock signal 1406. In an example embodiment, the clock divider 1410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 1406. A non-volatile memory phase-locked loop (PLL) block 1412 can be included (e.g., in the controller 1062) which distributes a series of clock signals to the non-volatile memory subsystem 1040 and to associated control logic. For example, a series of clock signals 1414 can be sent from the controller 1062 to the non-volatile memory subsystem 1040. Another clock signal 1416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 1040. For example, the clock signal 1416 may clock the portion of the controller 1062 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 1040. A feedback clock signal 1418 is fed back into the non-volatile memory PLL block 1412. In one embodiment, the PLL block 1412 compares the feedback clock 1418 to the reference clock 1411 and varies the phase and frequency of its output until the reference 1411 and feedback 1418 clocks are phase and frequency matched.

[00163] A version of the clock signal 1406 such as the backup clock signal 1408 may be sent from the controller to the volatile memory subsystem 1030. The clock signal 1408 may be, for example, a differential version of the clock signal 1406. As described herein, the backup clock signal 1408 may be used to clock the volatile memory subsystem 1030 when the memory system 1010 is backing up the data from the volatile memory subsystem 1030 into the non-volatile memory subsystem 1040. In certain embodiments, the backup clock signal 1408 may also be used to clock the volatile memory subsystem 1030 when the memory system 1010 is copying the backed-up data back into the volatile memory subsystem 1030 from the nonvolatile memory subsystem 1040 (also referred to as restoring the volatile memory subsystem 1030). The volatile memory subsystem 1030 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the nonvolatile memory subsystem 1040 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 1030 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 1040) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 1030 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 1010.

[00164] In one embodiment, the backup clock 1408 and the volatile memory system clock signal 1420 are received by a multiplexer 1422, as schematically illustrated by FIG. 19. The multiplexer 1422 can output either the volatile memory system clock signal 1420 or the backup clock signal 1408 depending on the backup state of the memory system 1010. For example, when the memory system 1010 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 1420 may be provided by the multiplexer 422 to the volatile memory PLL block 1424. When the memory system 1010 is performing a backup (or restore) operation, the backup clock signal 1408 may be provided.

[00165] The volatile memory PLL block 1424 receives the volatile memory reference clock signal 1423 from the multiplexer 1422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 1030 and associated control logic. For example, in one embodiment, the PLL block 1424 generates a series of clock signals 1426 which clock the volatile memory elements 1032. A clock signal 1428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 1430 may be sent to the controller 1062. A feedback clock signal 1432 is fed back into the volatile memory PLL block 1424. In one embodiment, the PLL block 1424 compares the feedback clock signal 1432 to the reference clock signal 1423 and varies the phase and frequency of its output until the reference clock signal 1423 and the feedback clock signal 1432 clocks are phase and frequency matched.

[00166] The clock signal 1430 may be used by the controller 1062 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 1030. For example, control logic in the controller 1062 may be used to control the volatile memory subsystem 1030 during a backup or restore operation. The clock signal 1430 may be used as a reference clock signal for the PLL block 1434 which can generate one or more clocks 1438 used by logic in the controller 1062. For example, the PLL block 1434 may generate one or more clock signals 1438 used to drive logic circuitry associated with

controlling the volatile memory subsystem 1030. In certain embodiments, the PLL block 1434 includes a feedback clock signal 1436 and operates in a similar manner to other PLL blocks described herein.

[00167] The clock signal 1430 may be used as a reference clock signal for the PLL block 1440 which may generate one or more clock signals used by a sub-block 1442 to generate one or more other clock signals 1444. In one embodiment, for example, the volatile memory subsystem 1030 comprises DDR2 SDRAM elements and the sub-block 1442 generates one or more DDR2 compatible clock signals 1444. A feedback clock signal 1446 is fed back into the PLL block 1440. In certain embodiments, the PLL block 1440 operates in a similar manner to other PLL blocks described herein.

[00168] While described with respect to the example embodiment of FIG. 19, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 1030 operates on the volatile memory clock signal 1420 and there is no backup clock signal 1408. In some embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a restore operation and not during a backup operation.

[00169] FIG. 20 is a flowchart of an example method 1500 of controlling a memory system 1010 operatively coupled to a host system. Although described with respect to the memory

system 1010 described herein, the method 1500 is compatible with other memory systems. The memory system 1010 may include a clock distribution topology 1400 similar to the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 1040.

[00170] In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[00171] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

[00172] FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKE signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice

from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[00173] In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[00174] In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630

into self-refresh mode by sending a signal via the CKe line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKe line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

[00175] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00176] FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation in an operational block 1610. For example, the memory

system 1010 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[00177] In an operational block 1620, the method further comprises transferring data words from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 when the memory system 1010 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.

[00178] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The

40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[00179] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[00180] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1040 write each backed-up data word to the controller 1062 which writes a first portion of the data word to the volatile memory subsystem 1030 and then a second portion of the data word to the volatile memory 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00181] The method 1600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile memory subsystem 1030 comprises DRAM elements, only the slice of the DRAM which is currently being accessed

(e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 1600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[00182] In addition, the connection between the DRAM elements and the controller 1062 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 1062 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 1062 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 1030. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[00183] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module comprising:
 - a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
 - a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;
 - a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;
 - a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;
 - a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and
 - a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:
 - a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and
 - at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes.

ABSTRACT

In certain embodiments, a memory module includes a printed circuit board (PCB) having an interface that couples it to a host system for provision of power, data, address and control signals. First, second, and third buck converters receive a pre-regulated input voltage and produce first, second and third regulated voltages. A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage. Synchronous dynamic random access memory (SDRAM) devices are coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, and a voltage monitor circuit monitors an input voltage and produces a signal in response to the input voltage having a voltage amplitude that is greater than a threshold voltage.

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

**Title of
Invention**

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

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LEGAL NAME OF INVENTOR

Inventor: Chi-She Chen

Date (Optional): Oct-17-2014

Signature: *Chi-She Chen*

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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LEGAL NAME OF INVENTOR

Inventor: Jeffrey C. Solomon

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Scott H. Milton

Date (Optional): 10/17/2014

Signature: Scott H. Milton

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Jayesh BhaktaDate (Optional): 9-17-2014Signature: Jayesh Bhakta

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: TBA

SERIAL NO.: TBA

ART UNIT: TBA

FILING DATE: TBA

EXAMINER: TBA

TITLE: FLASH-DRAM HYBRID MEMORY MODULEY

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Commissioner:

Applicants respectfully request amendment of this application as directed herein.

Amendments to the Specification are discussed on page **2**.

Remarks begin on page **3**.

In the Specification

Please insert the following paragraph as the first paragraph, and renumber the subsequent paragraphs accordingly:

[0001] This application is a continuation of U.S. Patent Application No. 17/328,019, filed May 24, 2021, titled “FLASH-DRAM HYBRID MEMORY MODULE”, now US Patent No. 11,232,054, which is a continuation of U.S. Patent Application No. 17/138,766, filed December 30, 2020, titled “Flash-Dram Hybrid Memory”, now U.S. Patent No. 11,016,918, which is a continuation of U.S. Patent Application No. 15/934,416, filed March 23, 2018, titled “Flash-Dram Hybrid Memory Module,” which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled “Non-Volatile Memory Module,” now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

REMARKS

Entry of the foregoing amendment prior to substantive examination is respectfully requested. Claim 1 is pending.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Vorys, Sater, Seymour and Pease LLP
1909 K Street, N.W.
9th Floor
Washington, D.C. 20006
Phone: (202) 467-8860
Fax: (202) 467-8900

Respectfully submitted,

/Khaled Shami/
Khaled Shami
Reg. No. 38,745
Date: January 24, 2022



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/582,797	01/24/2022	Chi-She Chen	022955-000090

CONFIRMATION NO. 3546

FORMALITIES LETTER



151145
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Date Mailed: 02/02/2022

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

The application is informal since it does not comply with the regulations for the reason(s) indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- Replacement drawings in compliance with 37 CFR 1.84 and 37 CFR 1.121(d) are required. The drawings submitted are not acceptable because:
 - The drawings submitted to the Office are not electronically reproducible because portions of figures 17 - 19, & 22 are missing and/or blurry.

Applicant is cautioned that correction of the above items may cause the specification and drawings page count to exceed 100 pages. If the specification and drawings exceed 100 pages, applicant will need to submit the required application size fee.

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of

small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **320** basic filing fee.
- \$ **160** surcharge.
- \$ **700** search fee.
- \$ **800** examination fee.
- \$ **(0)** previous unapplied payment amount.
- \$ **1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

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Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at
(571) 272-4000 or **(571) 272-4200** or **1-888-786-0101**.

/afessehay/

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/582,797

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	1	minus 20 = *			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1	minus 3 = *				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
			TOTAL			TOTAL	1820

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/582,797	01/24/2022	1731	0.00	022955-000090	1	1

CONFIRMATION NO. 3546

FILING RECEIPT



0000000131509999

151145
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Date Mailed: 02/02/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 151145

Domestic Priority data as claimed by applicant

This application is a CON of 17/328,019 05/24/2021 PAT 11232054
which is a CON of 17/138,766 12/30/2020 PAT 11016918
which is a CON of 15/934,416 03/23/2018 ABN
which is a CON of 14/840,865 08/31/2015 PAT 9928186
which is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/01/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/582,797**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

501

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Chi-She Chen, et al.

Appl. No.: 17/582,797

Filing Date: January 24, 2022

Exam: Not Yet Assigned

Art Unit: 1731

Conf.: 3546

Title: **FLASH-DRAM HYBRID MEMORY MODULE**

PRELIMINARY AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Commissioner:

Please enter the following amendments:

Amendment to the Drawings begin on page **2**;

Remarks begin on page **3**.

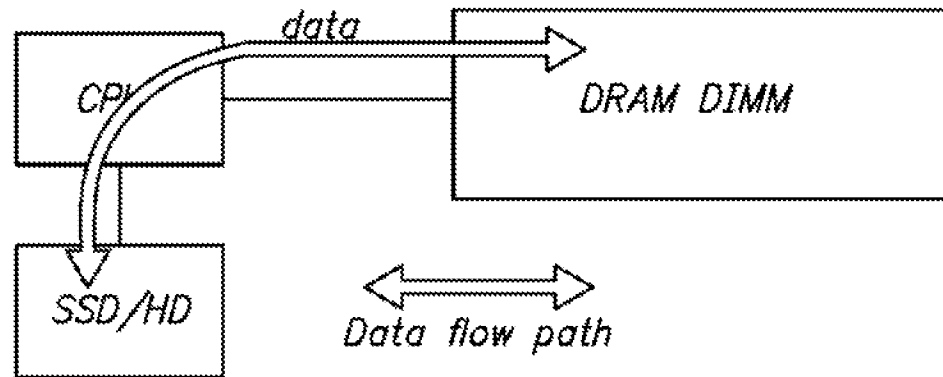


FIG. 1
(PRIOR ART)

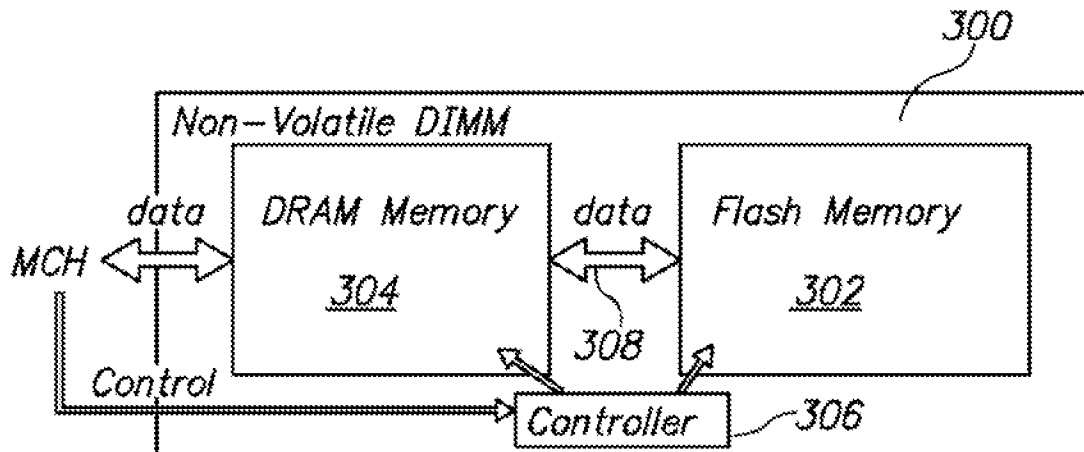
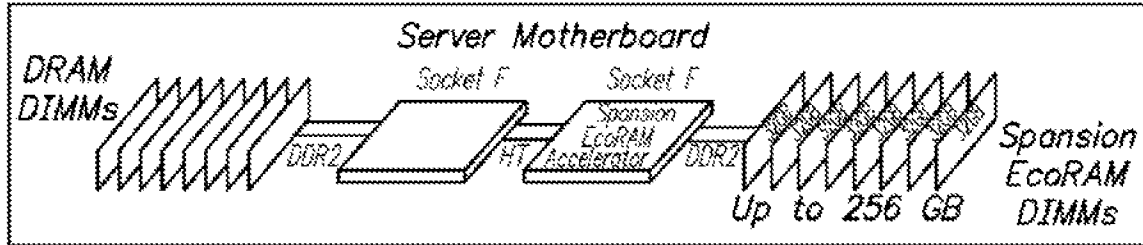


FIG. 3A

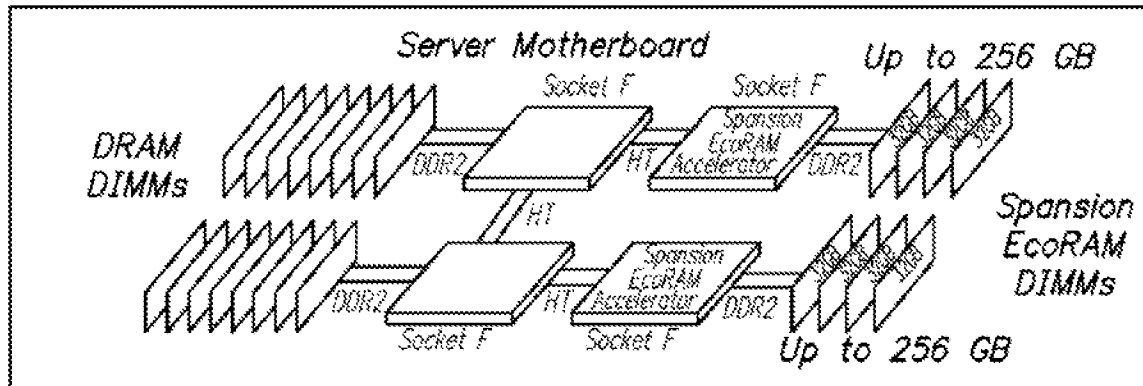
Spansion EcoRAM Configurations

256GB Spansion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spansion EcoRAM Solution

256GB Spansion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spansion EcoRAM Solution

**FIG. 2
(PRIOR ART)**

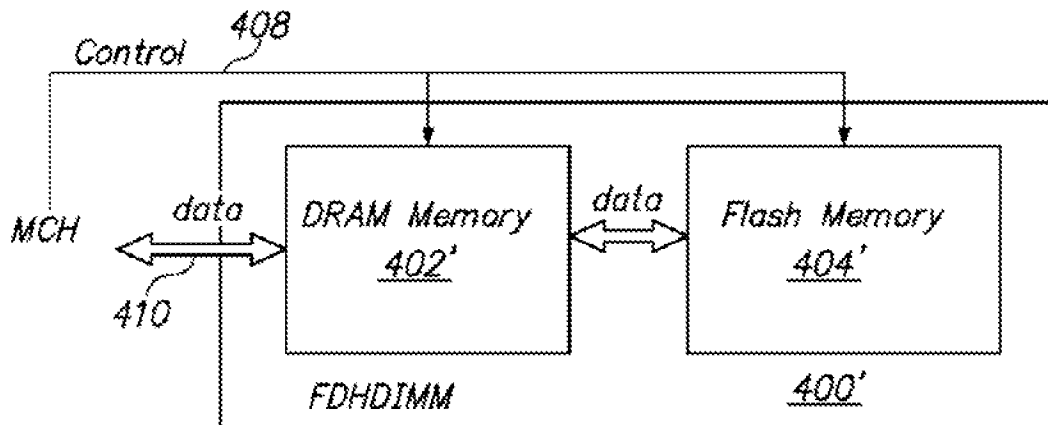


FIG. 4B

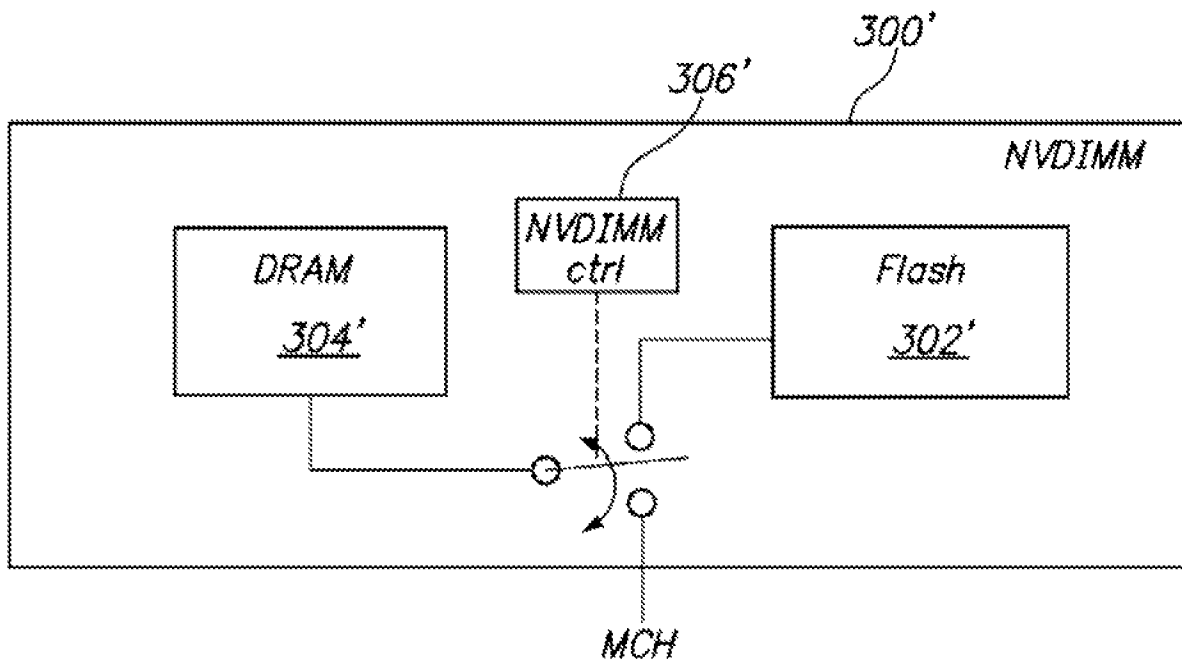


FIG. 3B

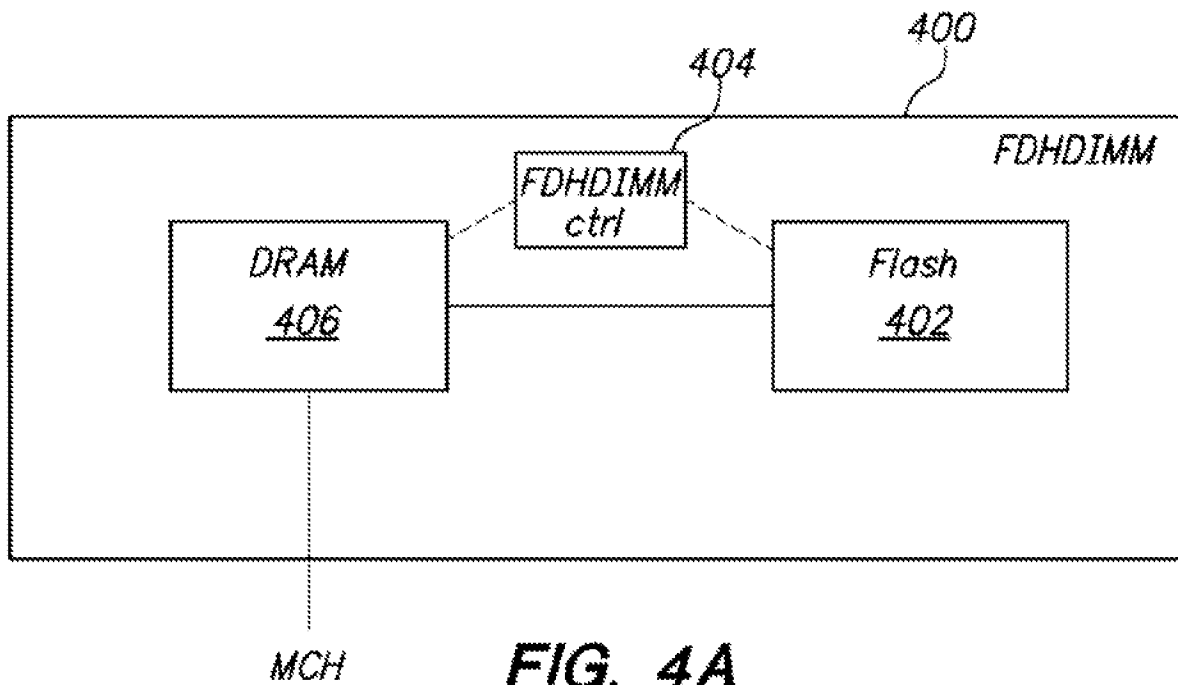
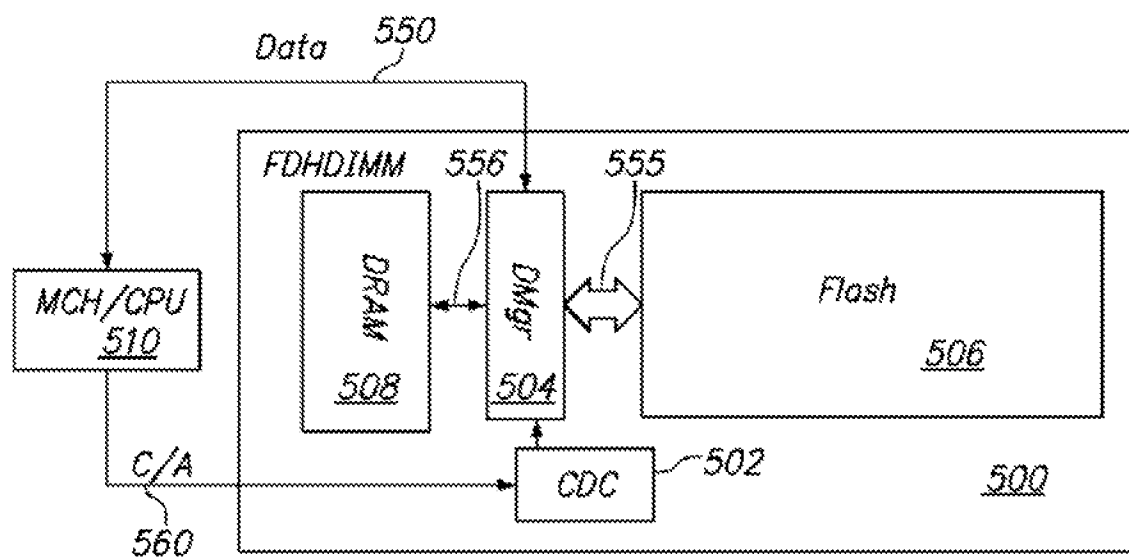
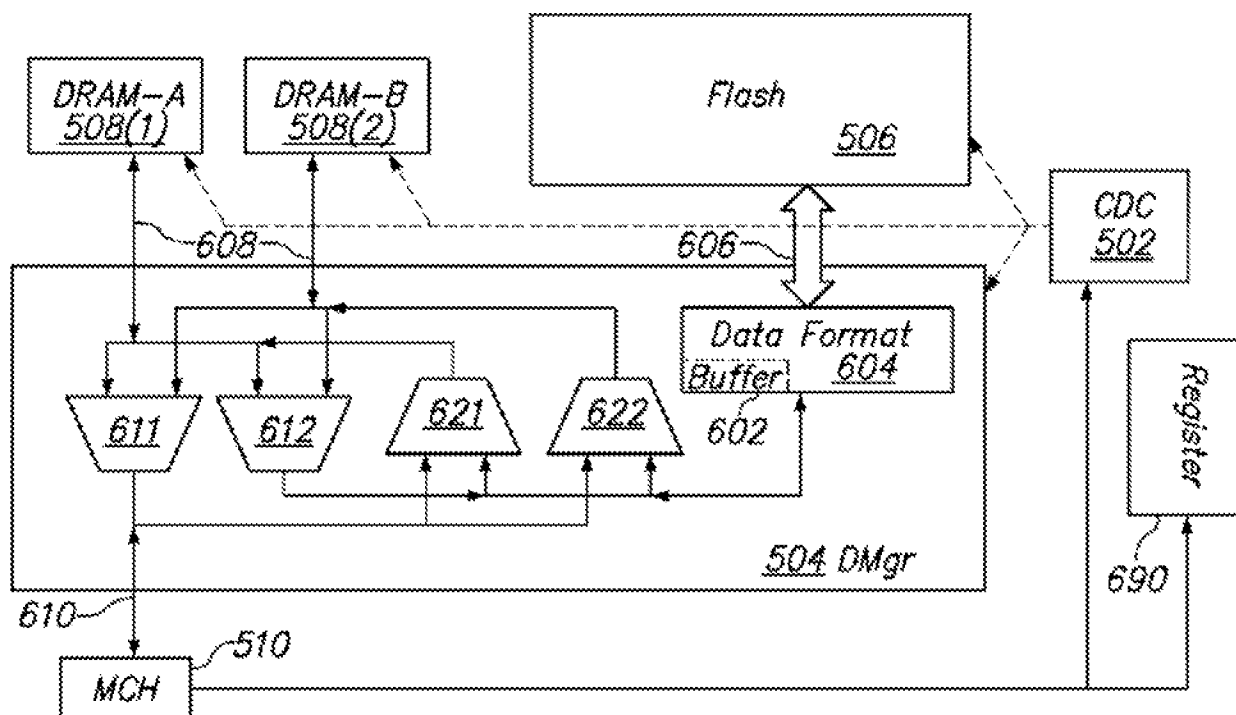


FIG. 4A

**FIG. 5A****FIG. 6**

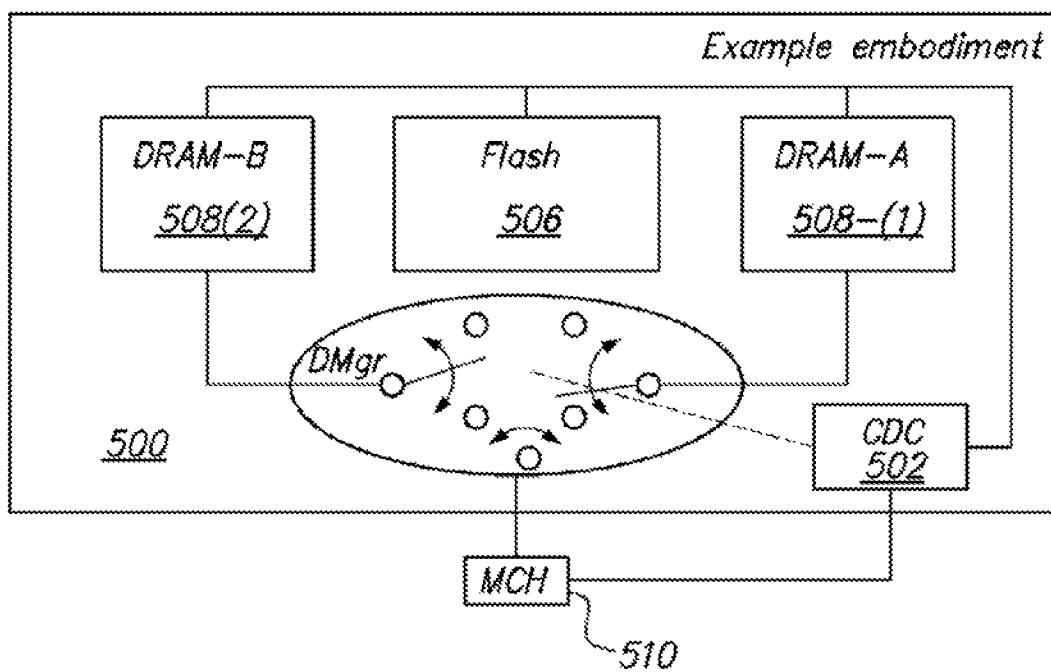
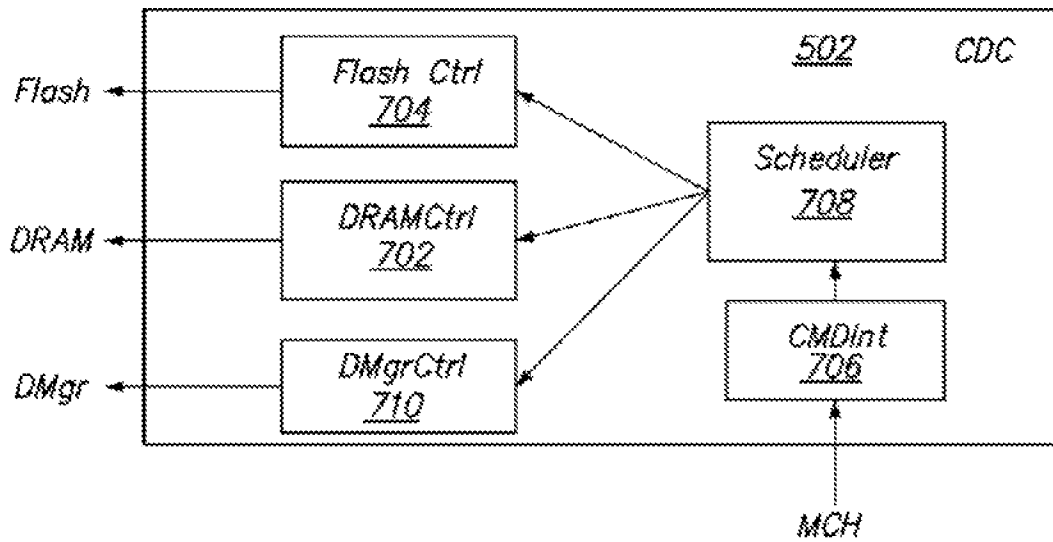
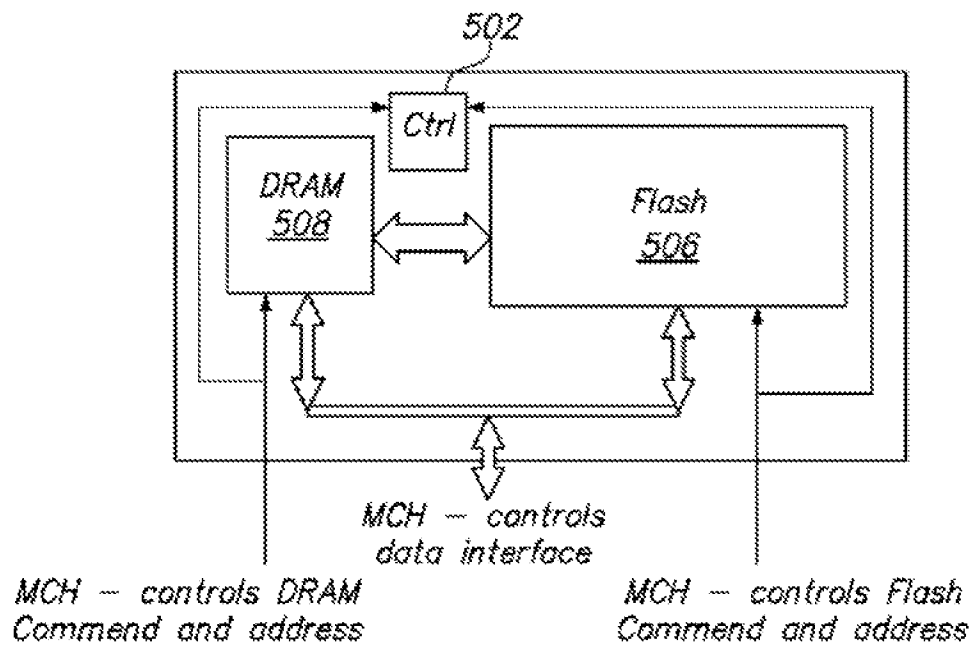
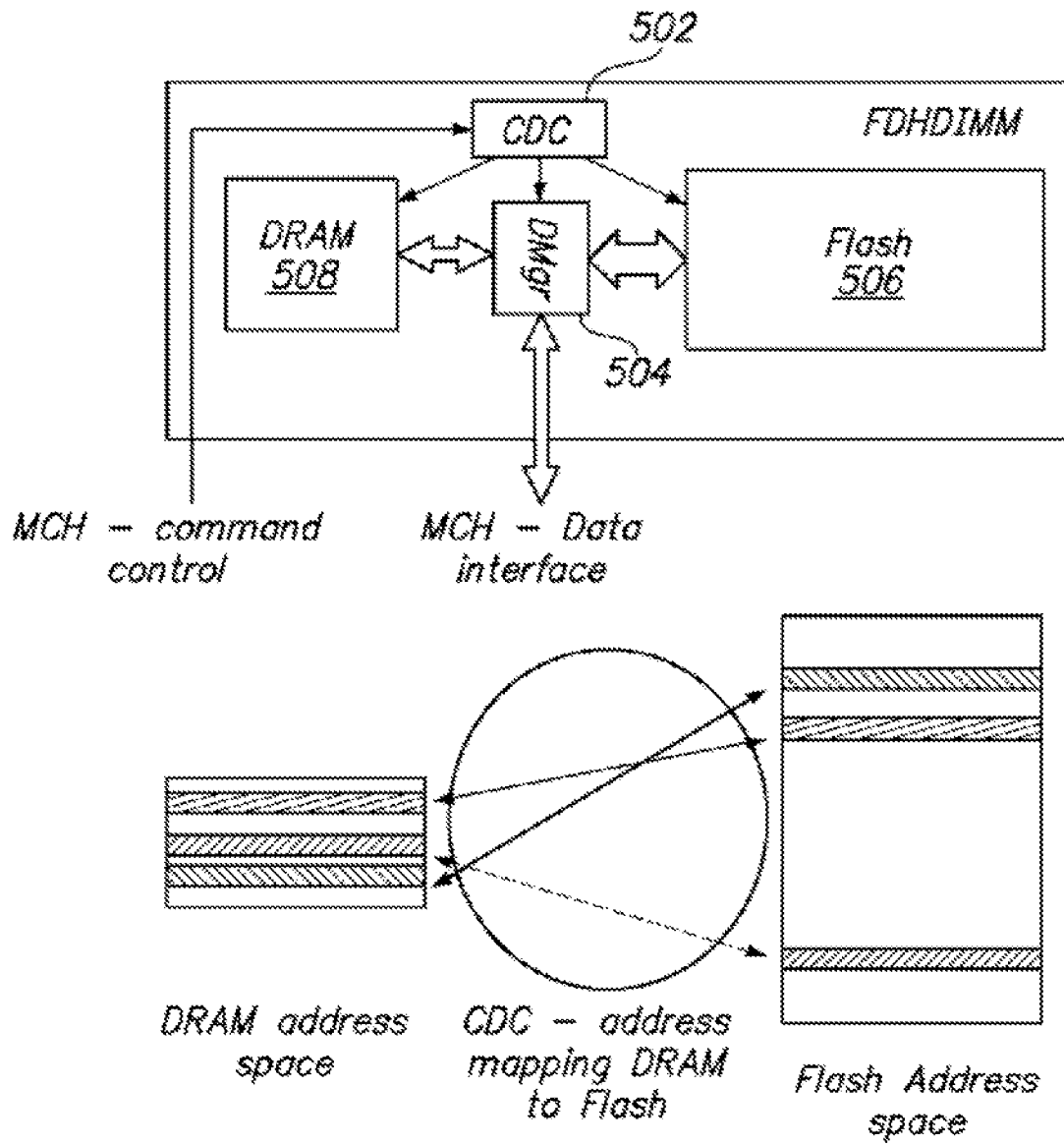
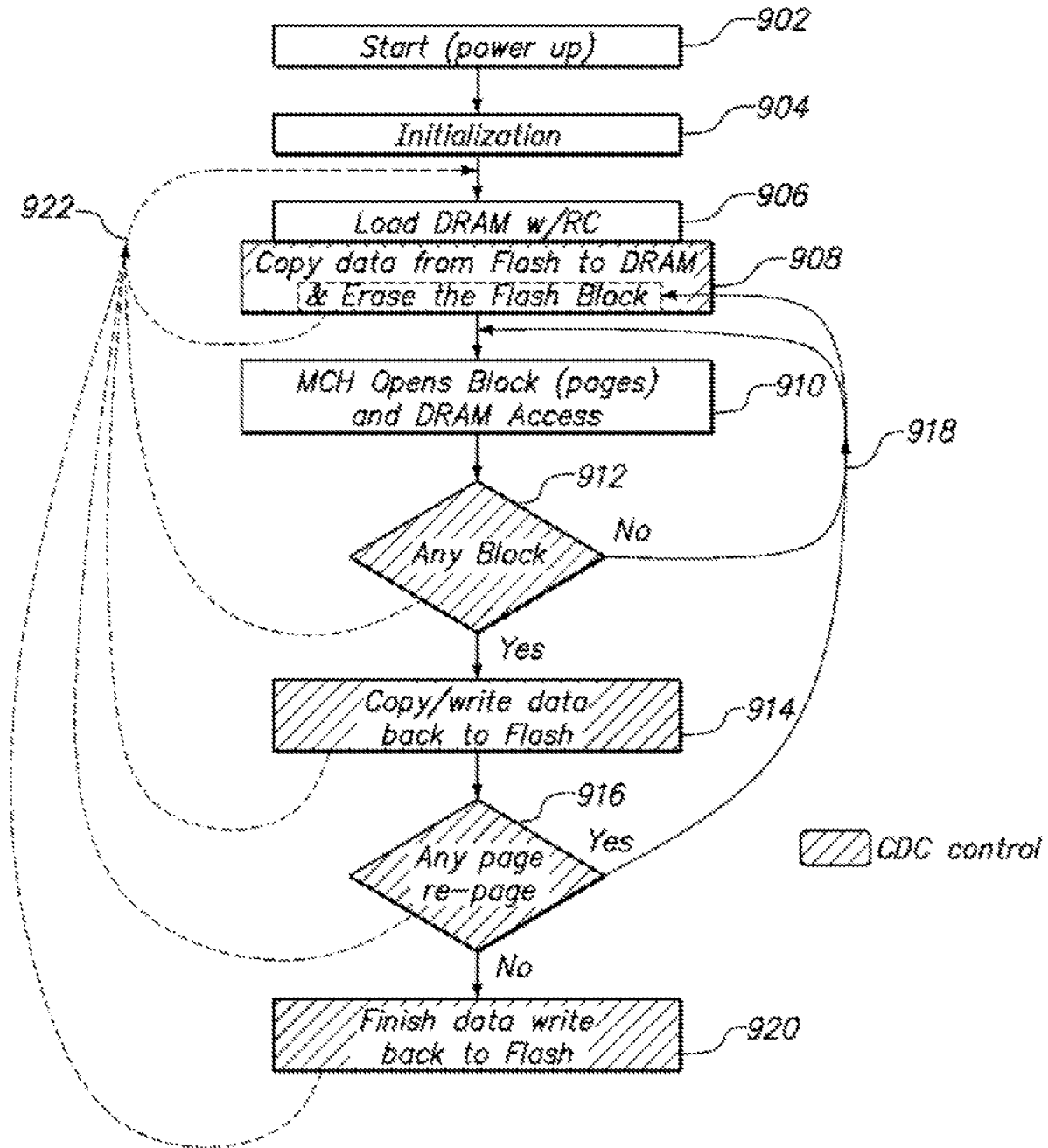
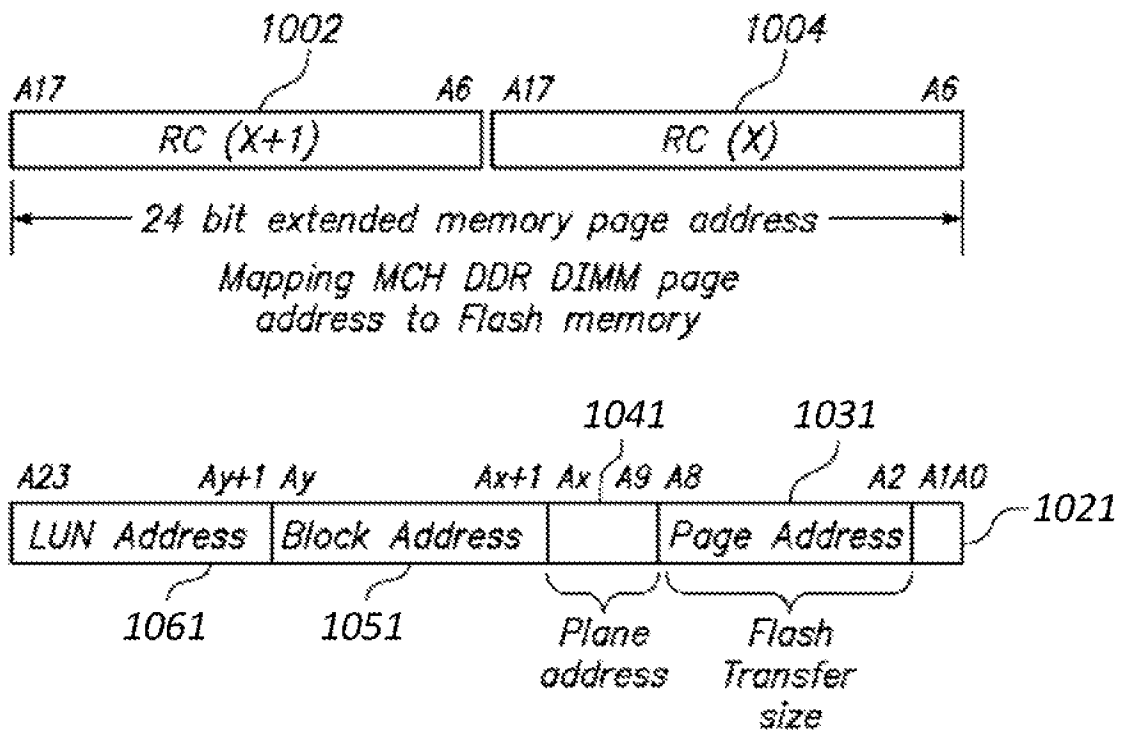


FIG. 5B

**FIG. 7****FIG. 8A**



**FIG. 9**

**FIG. 10**

REPLACEMENT SHEET

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

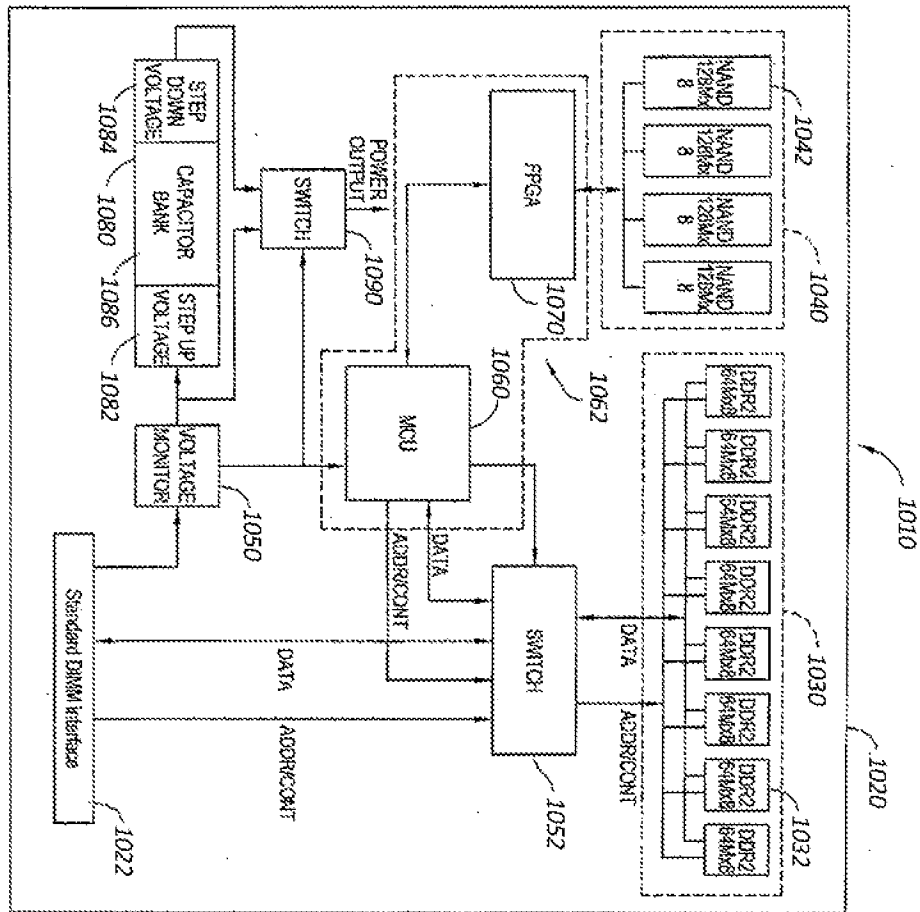


FIG. 12

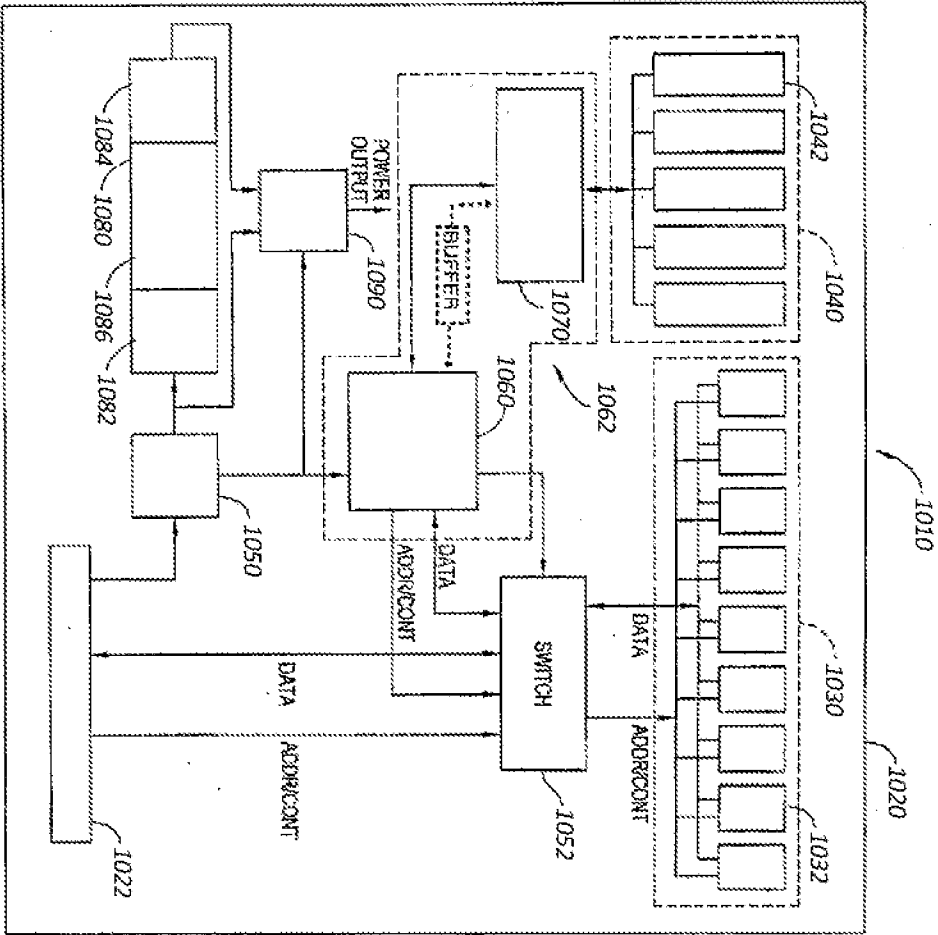


FIG. 13

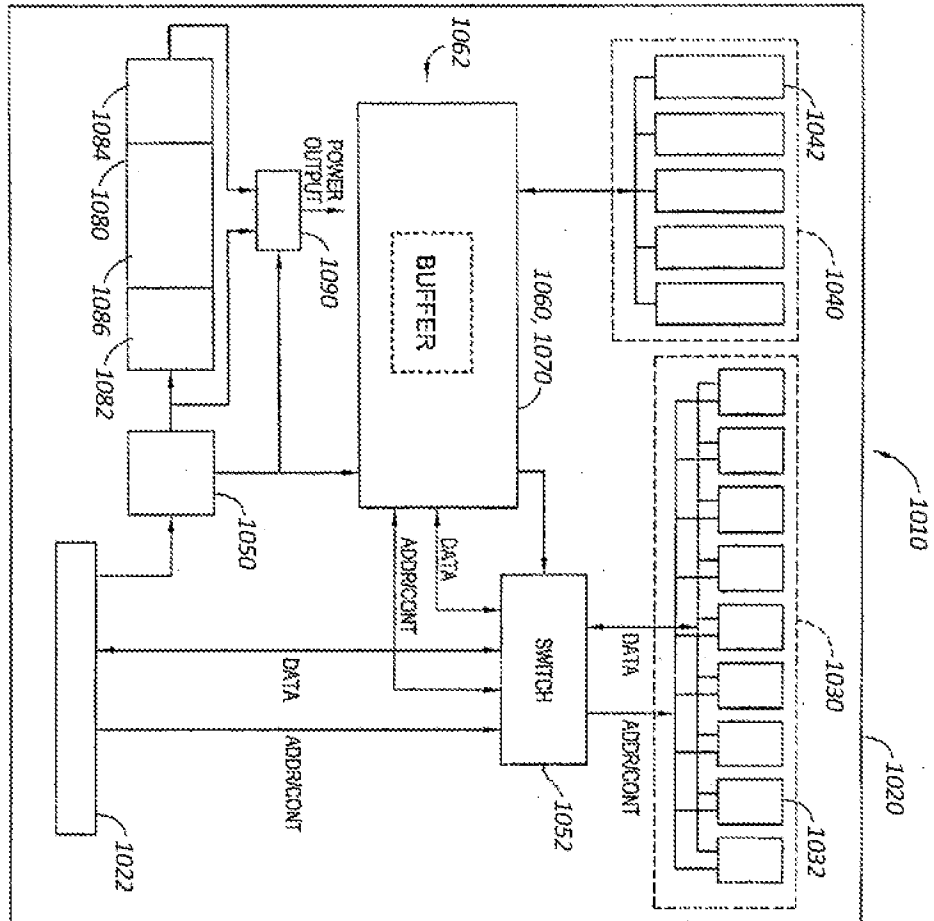
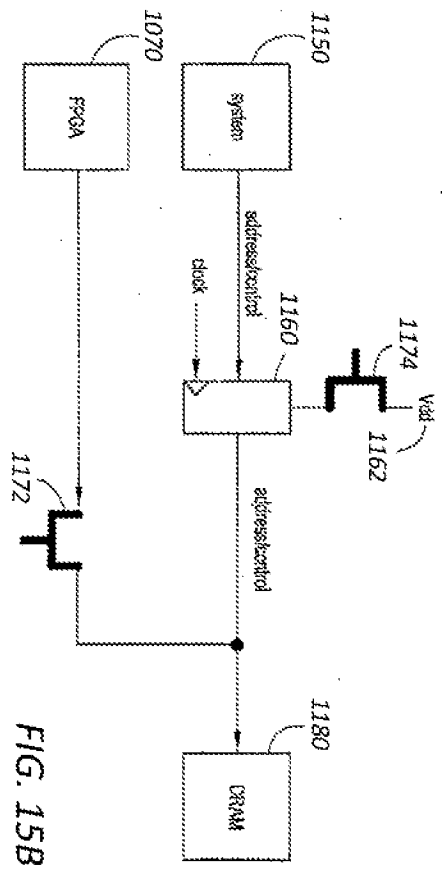
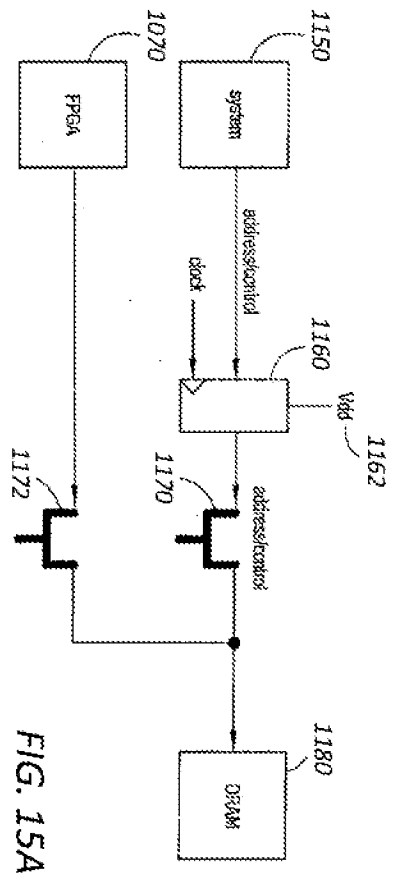


FIG. 14



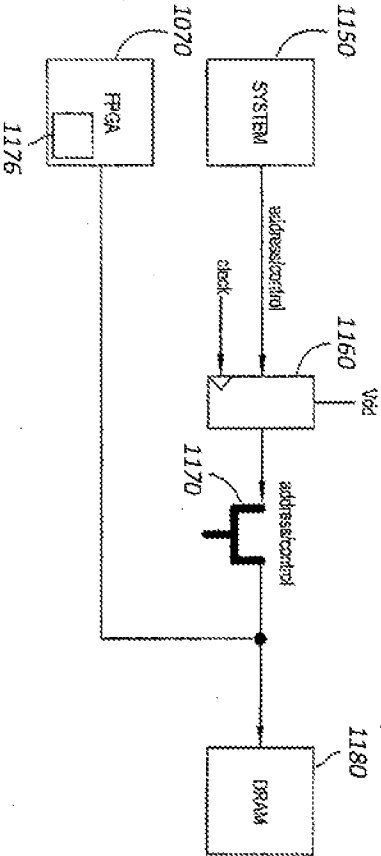


FIG. 15C

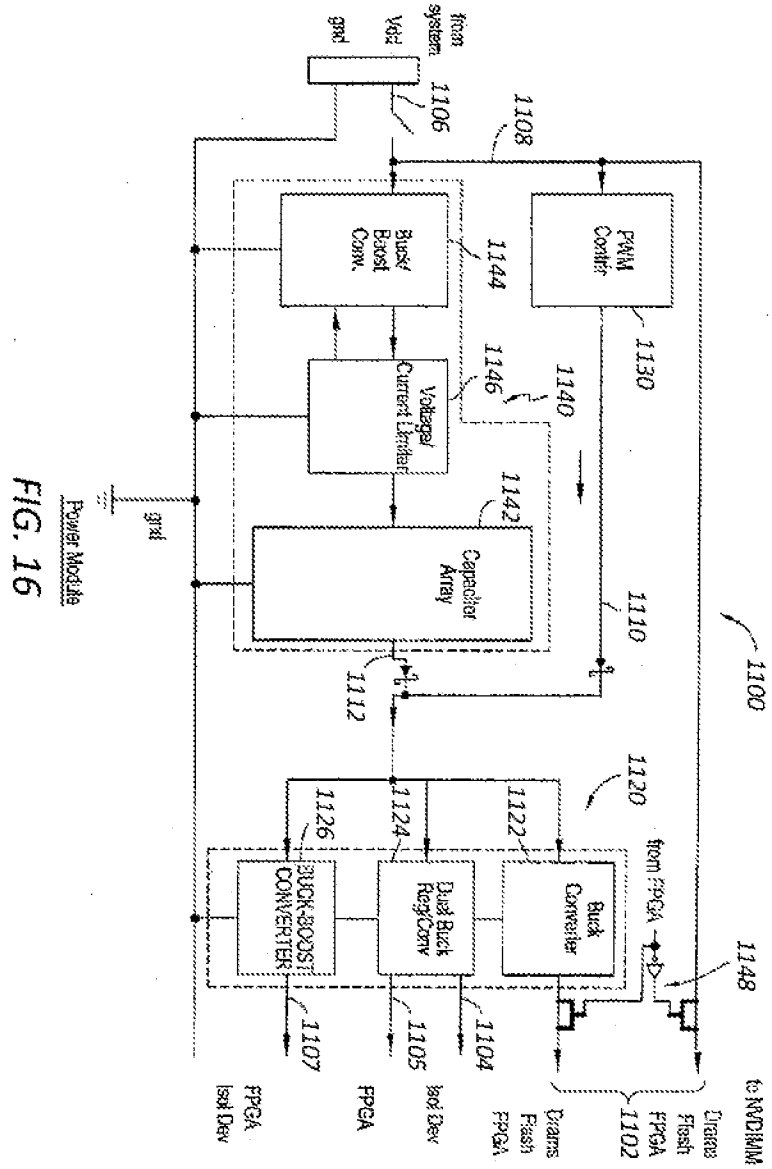


FIG. 16

Power Module

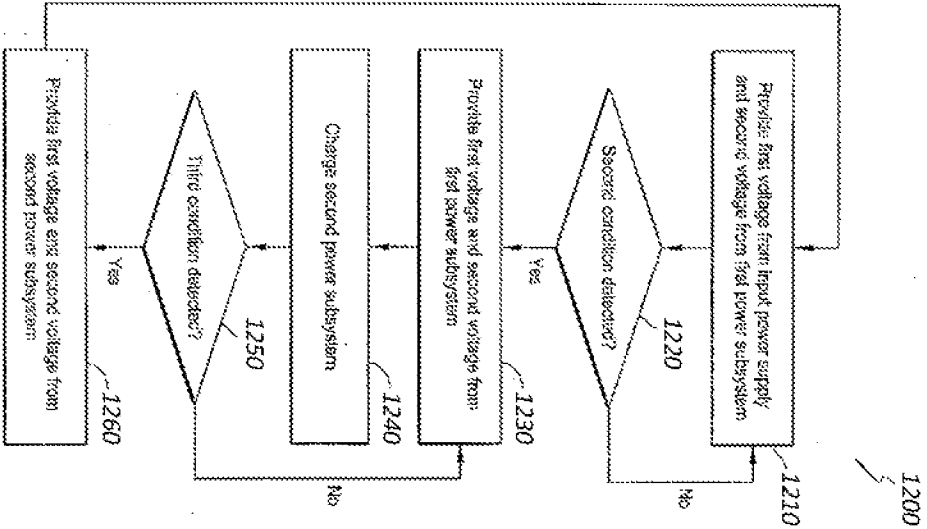


FIG. 17

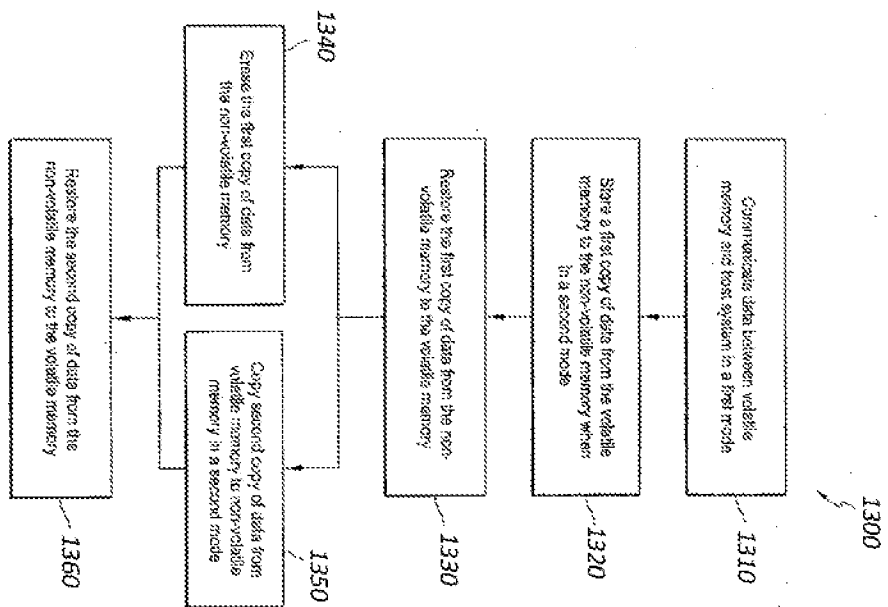


FIG. 18

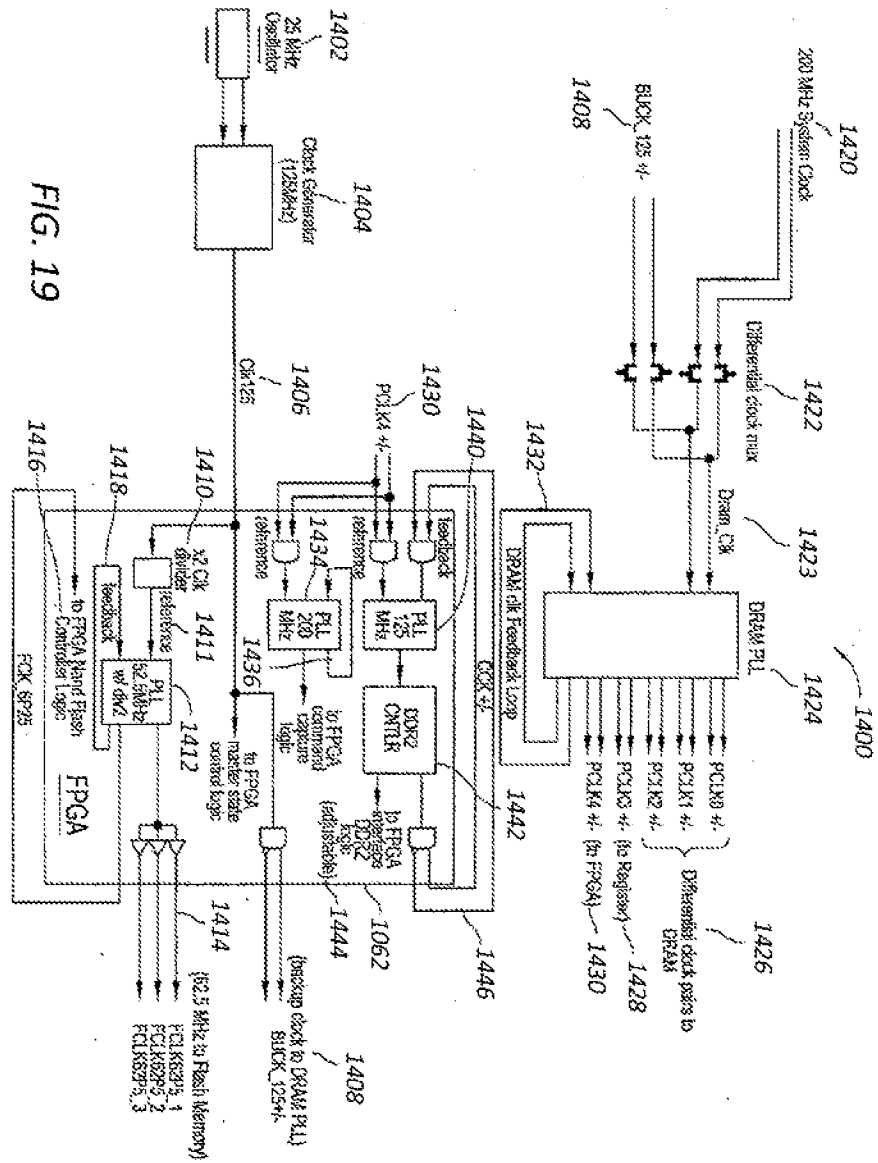


FIG. 19

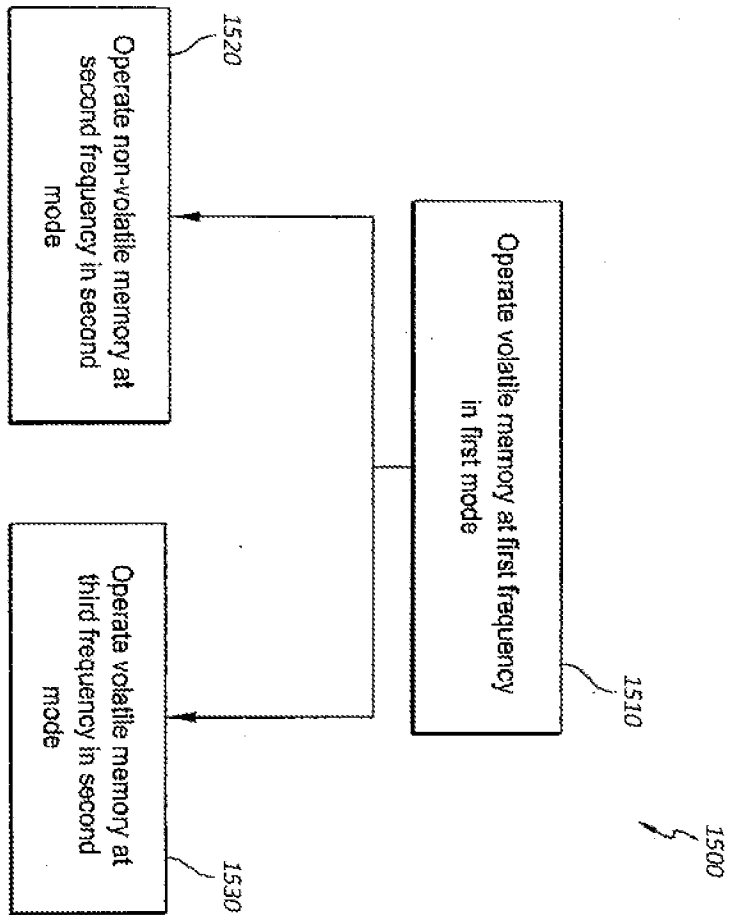


FIG. 20

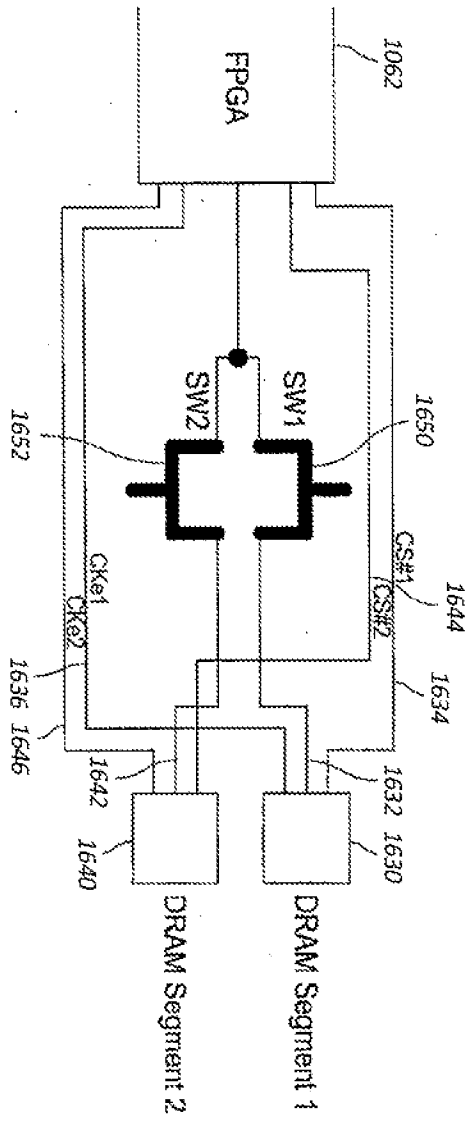


FIG. 21

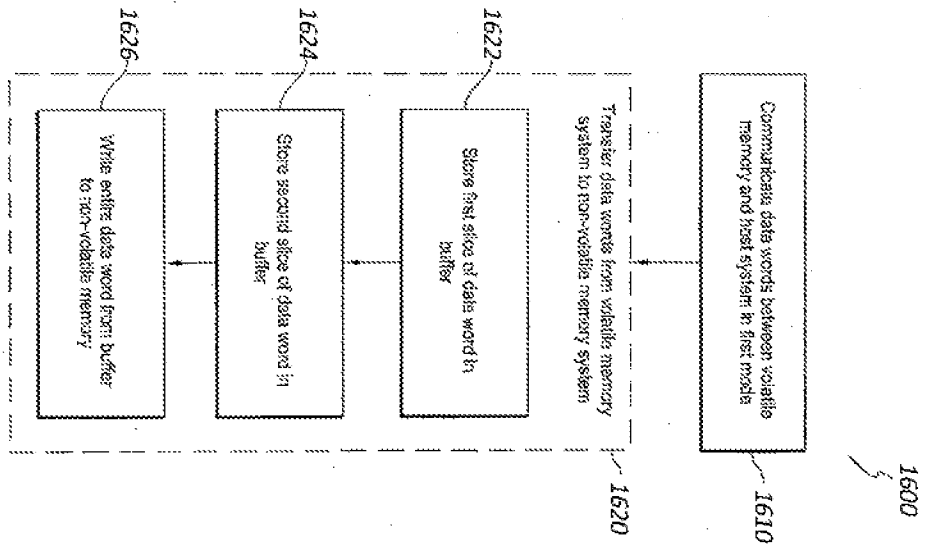


FIG. 22

Electronic Patent Application Fee Transmittal

Application Number:	17582797			
Filing Date:	24-Jan-2022			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Yayoi Bourne			
Attorney Docket Number:	022955-000090			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	320	320
UTILITY SEARCH FEE	1111	1	700	700
UTILITY EXAMINATION FEE	1311	1	800	800
Pages:				
Claims:				
Miscellaneous-Filing:				
LATE FILING FEE FOR OATH OR DECLARATION	1051	1	160	160
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1980

Electronic Acknowledgement Receipt

EFS ID:	45385028
Application Number:	17582797
International Application Number:	
Confirmation Number:	3546
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000090
Receipt Date:	04-APR-2022
Filing Date:	24-JAN-2022
Time Stamp:	11:48:53
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1980
RAM confirmation Number	E202244B49101498
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	022955-000090_RNFMP.pdf	421552	no	3
			9a89c778d707d67e4f0ec06de84875ee2fcb2be2		
Warnings:					
Information:					
2		022955-000090_RNFMP_PA.pdf	6371403	yes	25
			9afe3ed9a4daea78de44d874851f2a3a34ceeeea		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Drawings-only black and white line drawings		2	2	
	Applicant Arguments/Remarks Made in an Amendment		3	3	
	Drawings-only black and white line drawings		4	25	
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	46679	no	2
			9d1243a43dc1c7b60741c7a239f0ce56ba52aba9		
Warnings:					
Information:					
Total Files Size (in bytes):			6839634		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Chi-She Chen, et al.
Appl. No.: 17/582,797
Filing Date: January 24, 2022
Exam: Not Yet Assigned
Art Unit: 1731
Conf.: 3546
Title: **FLASH-DRAM HYBRID MEMORY MODULE**

RESPONSE TO NOTICE TO FILE MISSING PARTS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Commissioner:

This paper is submitted in response to the Notice to File Missing Parts mailed February 2, 2022, making a response due on or before April 2, 2022. Accordingly, this response is timely filed. Submitted herewith are the following:

- 1) Preliminary Amendment
- 2) Replacement Drawings
- 3) Fee payment of \$1,980, for the surcharge (\$160), basic filing fee (\$320), the search fee (\$700), and the examination fee (\$800).

It is submitted that the amendment submitted herewith overcomes the requirement for additional claim fees. The remaining required fees are filed by EFS-WEB.

In view of the foregoing, Applicant respectfully submits that the application is in condition for examination on the merits. The Examiner is invited to contact the undersigned attorney if it is believed that such contact will expedite the prosecution of the application.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

/Khaled Shami/

Khaled Shami, Registration No.: 38745
Customer No.: 151145

Date: April 4, 2022

VORYS, SATER, SEYMOUR and PEASE LLP
1909 K Street, NW, 9th Floor
Washington, DC 20006-1152
Tel: (202) 467-8800



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/582,797	01/24/2022	Chi-She Chen	022955-000090

CONFIRMATION NO. 3546

FORMALITIES LETTER



151145
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Date Mailed: 02/02/2022

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

The application is informal since it does not comply with the regulations for the reason(s) indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- Replacement drawings in compliance with 37 CFR 1.84 and 37 CFR 1.121(d) are required. The drawings submitted are not acceptable because:
 - The drawings submitted to the Office are not electronically reproducible because portions of figures 17 - 19, & 22 are missing and/or blurry.

Applicant is cautioned that correction of the above items may cause the specification and drawings page count to exceed 100 pages. If the specification and drawings exceed 100 pages, applicant will need to submit the required application size fee.

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of

small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **320** basic filing fee.
- \$ **160** surcharge.
- \$ **700** search fee.
- \$ **800** examination fee.
- \$ **(0)** previous unapplied payment amount.
- \$ **1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at
(571) 272-4000 or **(571) 272-4200** or **1-888-786-0101**.

/afessehay/

Amendments to the drawings:

Please enter the twenty two (22) sheets of Replacement Drawings attached herewith.

Remarks

Each of the required items identified in the Notice of Missing Parts has been provided.

Applicant submits herewith replacement sheets for all figures. Applicant submits that the drawing sheets are in compliance with 37 CFR 1.84 and 1.121(d). No new matter has been added.

Applicant reserves the right to reintroduce any cancelled subject matter to this application or to any child applications, or to reintroduce the full scope of any the original or previous claim amended herein, in a continuing application.

Accordingly, the Examiner is respectfully requested to enter the above amendments. Favorable consideration is requested.

Conclusion

In view of the foregoing, Applicant respectfully submits that the application is in condition for allowance. The Examiner is invited to contact the undersigned attorney if it is believed that such contact will expedite the granting of the application.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of this paper, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Vorys, Sater, Seymour and Pease LLP
1909 K Street, N.W. 9th Floor
Washington, D.C. 20006
Phone: (202) 467-8843
Fax: (202) 467-8800

Respectfully submitted,
/Khaled Shami/
Khaled Shami
Reg. No. 38745
Date: April 4, 2022



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/582,797	01/24/2022	1731	1980	022955-000090	1	1

CONFIRMATION NO. 3546

UPDATED FILING RECEIPT

151145

Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152



0000000132930211

Date Mailed: 04/06/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 151145

Domestic Priority data as claimed by applicant

This application is a CON of 17/328,019 05/24/2021 PAT 11232054
which is a CON of 17/138,766 12/30/2020 PAT 11016918
which is a CON of 15/934,416 03/23/2018 ABN
which is a CON of 14/840,865 08/31/2015 PAT 9928186
which is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/01/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/582,797**

Projected Publication Date: 07/14/2022

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

501

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/582,797

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	1 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/582,797	01/24/2022	Chi-She Chen	022955-000090

CONFIRMATION NO. 3546

PUBLICATION NOTICE



OC000000135164891

151145
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2022-0222191-A1

Publication Date:07/14/2022

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	10/12/2022	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2135	
			NOTIFICATION DATE	DELIVERY MODE
			10/12/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

Office Action Summary

Application No.

17/582,797

Applicant(s)

Chen et al.

Examiner

HASHEM FARROKH

Art Unit

2135

AIA (FITF) Status

No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 01/24/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 04/04/2022 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

1. The present application is being examined under the pre-AIA first to invent provisions.
2. The instant application having application No. 17/582,797 has a total of 20 claims pending in the application; there are 3 independent claim and 17 dependent claims, all of which are ready for examination by the examiner.

ACKNOWLEDGEMENT OF REFERENCES CITED BY APPLICANT

Information Disclosure Statement

3. As required by **M.P.E.P. 2001.06(b)** and **37 C.F.R. 1.98(d)**, since the instant application has been identified as a continuation application of an earlier filed application and is relied upon for an earlier filing date under **35 U.S.C. 120**, the examiner has reviewed the prior art cited in the earlier related application as required by **M.P.E.P. 707.05** and **904** and as stated in **M.P.E.P. 2001.06(b)**, no separate citation of the same prior art need be made by the applicants in the instant application.

INFORMATION CONCERNING CLAIMS:

Double Patenting

4. A rejection based on double patenting of the “same invention” type finds its support in the language of 35 U.S.C. 101 which states that “whoever invents or discovers any new and useful process... may obtain a patent therefor...” (Emphasis added). Thus, the term “same invention,” in this context, means an invention drawn to

identical subject matter. See *Miller v. Eagle Mfg. Co.*, 151 U.S. 186 (1894); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Ockert*, 245 F.2d 467, 114 USPQ 330 (CCPA 1957).

A statutory type (35 U.S.C. 101) double patenting rejection can be overcome by canceling or amending the claims that are directed to the same invention so they are no longer coextensive in scope. The filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. 101.

Claim 1 is rejected under 35 U.S.C. 101 as claiming the same invention as that of claim 1 of prior U.S. Patent No. 11,016,918 B2. This is a statutory double patenting rejection.

1. Claim 1 of instant application (Application No. 17/582,797) is compared to claim 1 of Patent No. 11,016,918 B2 in the following table:

Patent No. 11,016,918 B2	Application No. 17/582,797
Claim 1: 1. A memory module comprising: a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between	Claim 1: 1. A memory module comprising: a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between

the memory module and the host system; a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising: a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated	the memory module and the host system; a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising: a plurality of synchronous dynamic random access memory (SDRAM)
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voltage, and at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.	devices coupled to the first regulated voltage, and at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes.
--	--

Conclusion

The prior art made of record and not relied upon are as follows:

- 1. Chen et al. (US 20120271990 A1).*
- 2. Bates (US 20070136523 A1).*
- 3. Prete et al. (US 20080126624 A1).*

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Application/Control Number: 17/582,797
Art Unit: 2135

Page 7

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2135*

October 6, 2022

<i>Notice of References Cited</i>	Application/Control No. 17/582,797		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2135	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	10-2012	Chen; Chi-She	G06F3/065	711/E12.008
*	B	US-20070136523-A1	06-2007	Bonella; Randy M.	G06F12/0866	711/E12.019
*	C	US-20080126624-A1	05-2008	Prete; Edoardo	G11C7/106	710/53
	D					
	E					
	F					
	G					
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	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
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	P					
	Q					
	R					
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
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US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	"17582797"	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 08:56 AM
L2	1	("11232054").pn.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 08:57 AM
L3	1	("11016918").pn.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 10:06 AM
L5	732	(buck ADJ converter) WITH (regulated near2 voltage) WITH (voltage near2 (amplitude OR value))	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:02 AM
L6	5	(plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies)) WITH (coupled OR connected) WITH (regulated near2 voltage)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:16 AM
L7	5	6 AND ((printed ADJ circuit ADJ board) OR PCB)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:18 AM
L8	0	7 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:20 AM
L9	5	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies)) WITH (coupled OR connected) WITH (regulated near2 voltage)).clm.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:23 AM
L10	72	("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:33 AM
L11	129	("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:34 AM

L12	217531	(G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:41 AM
L13	5	6 AND 12	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:41 AM
L14	5	(plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2 voltage)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:44 AM
L15	5	11 AND 14	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:44 AM
L16	5	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:45 AM

		voltage)).clm.					
L17	5	10 AND 16	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 11:46 AM
L18	3	("20120271990" "20070136523" "20080126624").pn.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 12:01 PM
L19	5	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR (SDRAM OR DRAM)) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2 voltage)).clm.	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 12:07 PM
L20	5	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR (SDRAM OR DRAM)) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2 voltage))	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2022/10/06 12:07 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

1 - 50

☐ Power control unit with digitally supplied system parameters

Methods and apparatuses provide voltage regulation for a processor. Control or configuration parameters for a voltage regulator (VR) are provided digitally over a configuration bus to a VR controller. The parameters may be provided directly from a storage element, or via a processing element or...

TAHOE RSCH LTD

US7685441 | US Patents | 2010-03-23 | 4

☐ Power control unit with digitally supplied system parameters

Methods and apparatuses provide voltage regulation for a processor. Control or configuration parameters for a voltage regulator (VR) are provided digitally over a configuration bus to a VR controller. The parameters may be provided directly from a storage element, or via a processing element or...

TAHOE RSCH LTD

US8037328 | US Patents | 2011-10-11 | 4

☐ Voltage regulator with multi-level, multi-phase buck architecture

A voltage regulator having a multi-level, multi-phase architecture is disclosed. The circuit includes a two-level buck converter and an N-level buck converter each coupled to an output node, wherein N is an integer value of three or more. During operation, the two-level buck converter provides one...

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US10885851 | US Patents | 2021-01-05 | 4

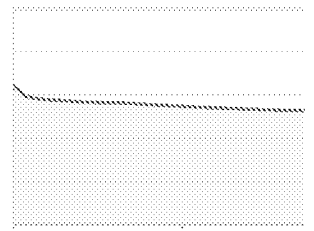
☐ Voltage regulator with multi-level multi-phase buck architecture

A voltage regulator having a multi-level, multi-phase architecture is disclosed. The circuit includes a two-level buck converter and an N-level buck converter each coupled to an output node, wherein N is an integer value of three or more. During operation, the two-level buck converter provides one...

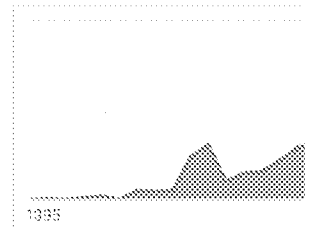
APPLE INC

WO2021050183A1 | WIPO Applications | 2021-03-18 | 4

Relevance



Publication Date



First Our Assignees by Relevance

TAHOE RSCH LTD (4)
APPLE INC (104)
QUALCOMM INC (37)
NETLIST INC (19)
HEWLETT PACKARD DEV... (8)
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Patents Assigned to Netlist, Inc. - Justia Patents Search

A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage. Synchronous dynamic random access memory (SDRAM) devices are ...

<https://patents.google.com> > patent

DC-DC converter with a dynamically adapting load-line

Nov 27, 2018 — G05F1/461 Regulating voltage or current wherein the variable actually regulated by the final control device is dc using an operational ...

<https://patents.google.com> > patent

Multilevel buck converter with a flying capacitor and charge ...

Jan 9, 2018 — The multilevel DC-DC converter 110 includes an output node VOUT coupled between the central high-side primary switch 134 and the central low- ...

<https://www.analog.com> > media > Section1

point-of-load (POL) - Practical Power Solutions

Switching regulators, such as the buck (step-down) converter shown in B, transfer energy from input to output via an inductor that has the voltage polarity ...

72 pages

<https://www.ti.com> > lit > pdf > slvt293

Understanding output voltage limitations of DC/DC buck ...

The buck converter is used to generate a lower output voltage from a higher DC input voltage. Texas Instruments Incorporated. Power Management. By John Tucker.

Missing: coupled plurality SDRAM

<https://www.ti.com> > lit > snvs559

Switching regulator fundamentals (Rev. C) - Texas Instruments

The buck converter uses a transistor as a switch that alternately connects and disconnects the input voltage to an inductor (see Figure 5).

<https://www.eng.auburn.edu> > Erickson_DC_2_DC

DC-DC Power Converters

by RW Erickson · Cited by 258 — A basic dc-dc converter circuit known as the buck converter is illustrated ... double-throw (SPDT) switch is connected to the dc input voltage V_g as shown.

19 pages

Missing: SDRAM | Must include: SDRAM

<https://www.maximintegrated.com> > technical-documents

DC to DC Buck Converter Tutorial & Diagram - Maxim Integrated

Voltage-mode PWM · Current-mode PWM · Point-of-Load (POL) Boost Regulators · nanoPower Boost Converter · Buck Regulators · DDR Memory Power Supply Buck Controller.

Missing: plurality SDRAM

<https://www.quora.com> > is-it-good-use-buck-converter1...

Is it good use buck converter to supply microcontroller? - Quora

Yes of course. It's very popular, majority of MCU's and processors are powered by buck converters or its derivatives like the multiphase buck converter on ...

5 answers · Top answer: Buck topology based S.M.P.S design are one of the simplest, smaller a...

People also ask

Does a buck converter regulate voltage?

What is a voltage buck converter?

What is the purpose of a buck converter?

What is a buck converter and how does it work?

Feedback

<https://www.renesas.com/document/whp/bc...pdf>

How to Reconfigure a Buck Converter for Multiple Outputs

Introduction. Power supply circuits come in the form of voltage step-up (boost) or the more common step-down (buck). DC/DC converter.

Missing: plurality SDRAM

Related searches

point-of-load converter

limitations of buck converter

buck converter voltage drop under load

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maximum duty cycle buck converter

buck converter dropout voltage

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	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2135	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	10949339		2021-03-16	Netlist, Inc.	
	2	10860506		2020-12-08	Netlist, Inc.	
	3	10217523		2019-02-26	Netlist, Inc.	
	4	10474595		2019-11-12	Netlist, Inc.	
	5	9858218		2018-01-02	Netlist, Inc.	
	6	7619912		2009-11-17	Netlist, Inc.	
	7	11232054		2022-01-25	Netlist, Inc.	
	8	6026465				

**INFORMATION DISCLOSURE
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First Named Inventor	CHEN
Art Unit	2135
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	1	20100110748		2010-05-06	Best, Scott C.	
	2	20060212651				
	3	20030028733				

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	1	288pin Load Reduced DIMM based on 8Gb B-die, Datasheet, Samsung, Rev. 1.6, 2017.	

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Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

2	288pin Load Reduced DIMM based on 8Gb B-die, Datasheet, Samsung, Rev. 1.4, 2017.
3	DDR Data Buffer Definition, JESD82-32A, JEDEC Standard, August 2019.
4	DDR4 SDRAM, DESD79-4C, JEDEC Standard, January 2020.
5	PMIC50x0 Power Management IC Specification, Rev. 1.0, JESD301-1, June 2020.
6	DDR5 SDRAM, JESD79-5, July 2020.
7	DDR5 Registering Clock Driver Definition (DDR5RCD01), JESD82-511, August 2021.
8	DDR Data Buffer Definition (DDR4DB01), JESD82-32, JEDEC Standard, November 2016.
9	DDR4 Registering Clock Driver - DDR4RCD01, JESD82-31, JEDEC Standard, August 2016.
10	Definition of teh SSTE32882 Registering Clock Driver with Parity and Quad Chip Selects for DDR3 RDIMM Applications, JESD82-29, JEDEC Standard, December 2009.
11	Definition of teh SSTE32882 Registering Clock Driver with Parity and Quad Chip Selects for DDR3 RDIMM 1.25V Applications, JESDxx-x Proposal, JEDEC Standard, May 2009.
12	LRDIMM DDR3 Memory Initialization Chapter Proposal, JEDEC Committee Letter Item No. 142.35, November 2009.

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Attorney Docket Number	022955-000090

13	DDR3 LRDIMM Design Specification Body, JEDEC Committee Letter Item No. 2192.44, November 2009.
14	MB Initialization sequence Item 142.35, JEDEC presentation, June 2009.
15	Memory Buffer Membist for LRDIMM, DDR3 MB TG item #142.43, JEDEC Presentation, June 2009.
16	PR2018-00362, Petition for Inter Partes Review of US Patent No. 9,606,907, filed December 22, 2017.
17	PR2020-01421, Petition for Inter Partes Review of US Patent No. 10,217,523, filed August 21, 2020.
18	PR2018-00303, Petition for Inter Partes Review of US Patent No. 9,535,623, filed December 14, 2017.
19	PR2020-01042, Petition for Inter Partes Review of US Patent No. 10,474,595, filed June 8, 2020.
20	PR2020-01044, Petition for Inter Partes Review of US Patent No. 9,858,218, filed June 8, 2020.
21	Proposed DDR4 DB BCOM Protocol, Committee Item No. 0311.14, JEDEC.
22	Proposed DDR4 DB Buffer Control Words, Committee Item No. 0311.12, JEDEC, May 2012.
23	PR2022-000063, Decision Granting Inter Partes Review of US Patent No. 10,217,523, entered May 5, 2022.

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24	288pin Load Reduced DIMM based on 16Gb B-die, Datasheet, Samsung, Rev. 0.0, September 2018.
25	288pin Load Reduced DIMM based on 16Gb B-die, Datasheet, Samsung, Rev. 1.0, February 2019.
26	DDR4 Registering Clock Driver - DDR4RCD02, JESD82-31A, JEDEC Standard, August 2019.
27	DDR4 SDRAM Load Reduced DIMM Design Specification, Rev. 1.00, JEDEC Standard No. 21C, August 2015.
28	IPR2023-00406, Petition for Inter Partes Review of US Patent No. 11,016,918, filed January 2022.
29	IPR2023-00405, Petition for Inter Partes Review of US Patent No. 11,232,054, filed January 2022.
30	IPR2022-00418, Patent Owner's Preliminary Response, filed June 9, 2022.
31	IPR2022-00418, Patent Owner's Response, filed November 25, 2022.
32	Declaration of Ron Maltiel, IPR2022-00418 Exhibit 1003, filed in IPR on January 14, 2022.
33	Netlist, Inc.'s Preliminary Proposed Claim Constructions in Civil Action Nos. 6:21-cv-00431-ADA and 6:21-cv-00430-ADA, IPR2022-00418 Exhibit 1012, filed in IPR on January 14, 2022.
34	Micron's Preliminary Identificaiton of Proposed Claim Constructions in Civil Action Nos. 6:21-cv-00431-ADA and 6:21-cv-00430-ADA, IPR2022-00418 Exhibit 1013, filed in IPR on January 14, 2022.

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Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

35	Double Data Rate (DDR) SCRAM Specification, JESD79, JEDEC Standard, (IPR2022-00418 Exhibit 1018) June 2000.
36	1.8 Volt Intel StrataFlash Wireless MEemory (L18), Datasheet, Intel, (IPR2022-00418 Exhibit 1018) April 2003.
37	DDR2SCRAM, 1 Gb: x4, x8, x16 DDR2 SD RAM, Datasheet, Micron, (IPR2022-00418 Exhibit 2003) 2007.
38	Declaration of Dr. Steven Przybylski, IPR2022-00418 Exhibit 2004, November 25, 2022.
39	Transcript of Ron Maltiel, IPR2022-00418 Exhibit 2005, November 25, 2022.
40	FSA1208 Low-Power, Eight-Port, High-Speed Isolation Switch, Fairchild, (IPR2022-00418 Exhibit 2006) December 2014.

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Application Number	17582797
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First Named Inventor	CHEN
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See attached certification statement.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-02-13
Name/Print	Khaled Shami	Registration Number	38745

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
02/13/2023 10:10:10 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

FLASH-DRAM HYBRID MEMORY MODULE

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 61588621

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE
ADDRESS -

FIRST NAMED
INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585

PAYMENT TRANSACTION ID
E20232CM11255902

PAYMENT AUTHORIZED BY
Lisa Elliott

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1806	SUBMISSION OF AN INFORMATION DISCLOSURE STATEMENT	260.00	1	260.00
TOTAL AMOUNT:				\$260.00

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National Stage of an International Application under 35 U.S.C. 371

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

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Documents

TOTAL DOCUMENTS: 58

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-02-13 IDS3.pdf	8	Information Disclosure Statement (IDS) Form (SB08)	625 KB
NPL--IDS3-01- 288pin_Samsung.pdf	42	Non Patent Literature	744 KB
NPL--IDS3-02- 288pin_Rev14_Samsung.pdf	41	Non Patent Literature	24411 KB
NPL--IDS3-03-JESD82- 32A.pdf	6	Non Patent Literature	542 KB
NPL--IDS3-04-JESD79- 4C.pdf	3	Non Patent Literature	1690 KB

NPL--IDS3-05-JESD301-1.pdf	3	Non Patent Literature	115 KB
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Digest

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NPL--IDS3-39-IPR202200418-Ex2005_Segment2of5.pdf	8D8BFC8EE1485B4041400BAB22648BAA36F0C7860D8D2A41A 0AC021F95D36F42341FAD1D2EC4B7A289D1EEEACFB349E7 017D6715E0AFB92D55047D7788E3323

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NPL--IDS3-39-IPR202200418- Ex2005__Segment4of5.pdf	56CCFC1894AABEAEC85597E4382372F6B9F8135A776E952B7 7F0AC280958B2DAC8312330202D6FEE4AA5F5C562D1AFFD1E 8E4AAEF6EF9FBC59E9AFCCCEF96ACE
NPL--IDS3-39-IPR202200418- Ex2005__Segment5of5.pdf	E121B2E615AD2ABF26A464F1E62932BAFEA2FC0746AFE1C64 90E2C7CA5927FE2E607962617268964475B1FF348DBB9D56AA 72DA2F1BD3E9C5F11EDE2A2C0826C
NPL--IDS3-40-IPR202200418- Ex2006.pdf	A14D950F370A30329273649BB4AFC21104FA00A8C5EC906D63 D175E54A6277AE30DAD4C3785151E71F947D5EE697799A29B 7A0BAF117CE5E12EC6EE6304C9AA7

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17582797
	Filing Date	2022-01-24
	First Named Inventor	CHEN
	Art Unit	2135
	Examiner Name	FARROKH
	Attorney Docket Number	022955-000090

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Attorney Docket Number	022955-000090

1	IPR2017-00692 Paper 1, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017. (Exhibit 1020 of IPR2022-00996 and IPR2022-00999).
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Attorney Docket Number	022955-000090

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-02-13
Name/Print	Khaled Shami	Registration Number	38745

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APPLICATION #
17/582,797

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Title of Invention

FLASH-DRAM HYBRID MEMORY MODULE

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 61588381

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

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NPL--IDS2-19-IPR2022-00996-Exhibit1062-AnalogDialogue-2022-05-17_Segment1of2.pdf	527353315AE7B3F066D0C12BDA2C34EF4F753D02C630C8CE7B072EBFA8FB79F9D16DC199E413BB4D8FCC10AFC625840451B528E64999F36D966476716D44B898
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NPL--IDS2-20-IPR2022-00996-Exhibit1063-LMC6953Datasheet-2022-05-17.pdf	D490D2E3C5F06A8EFE2A5E8AC8C177EAC243EE4E070C0DAF86620C92A0406843F0F384913174E952D0ADAD15012FDB36CA91B88F14AAAC9B0561780C4FAE22A3
NPL--IDS2-21-IPR2022-00996-Exhibit1067-LatticeXPDatasheet-2022-05-17_Segment1of3.pdf	171EA8517480D496A8E02694B2F19790392F9CA15CAF2F5AA5E0C027872B3012FB7C0DAAD1558C19CB6C0537EDC4313FD939BE679E72CBD0781CFE66DD7FC232
NPL--IDS2-21-IPR2022-00996-Exhibit1067-LatticeXPDatasheet-2022-05-17_Segment2of3.pdf	37BD881F86B702680658E93E1860A42C5C86C9C50291760AEF0878638D1C1BEEF5177C23D9AD398DB5B2B832CB325454D7AB7DF7C8AFF38FF9C4A9F771EBE149
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NPL--IDS2-24-IPR2022-00996-EX2003-Belloni-2022-09-09.pdf	142401550F016BBC4C1B643CC24BF2853DA9C3352BCF585B2 18FE3D970F2119DD1D44EA74F1296F0063D7F9146589E94059 7339A88450B571DAD25BF97C78823
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NPL--IDS2-27-IPR2022-00996-EX2007-LP2996N-2022-09-09.pdf	66627DADB7FDB5255E61420BD9D59C3DB609CA99C8671424 D5F975A57E07B82DC7A2ECD92D327C5B6F6D985BED3981F2 4ED5E9446313244F7B71ECA629DBCA9E
NPL--IDS2-28-IPR2022-00996-EX2008-LP2996-2022-09-09.pdf	4C9B54C7F9C8FD2210F6D9AD91582B47E7ABDEED76FA420E 1BD68004F9B8DB2BB047399B8483D47D768BBDA5205B985E9 6C048DF8E4373B279E7A93CD8C735EC
NPL--IDS2-29-IPR2022-00996-EX2009-LP2997-2022-09-09.pdf	4B147E1198A2925ED256BA6E57CABD9C7DA972FB95940D7C8 D0F5C7E2BBA93CF175FA3110CE426DC9949CF19B3F45C72D 12360D420E3128CCEFECC0C7BBE5CDAD
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NPL--IDS2-34-IPR2022-00996-EX2014-Qimonda-2022-09-09.pdf	E399C04A1BF146EB5E3BC33AFA69F6440268AFE127A13CCF5EECFF1ADA9A257DF92C944A4FDC521D627900EC789E2854F351127A4307536FD33D390A3370FAED

NPL--IDS2-35-IPR2022-00996-EX2015-PC66-222-920-2022-09-09.pdf	603E7A62D41452A3D55141A4ECD7D8461E99779FD85B4D552CFD25E721D0017E02A5EBA825B016EF9B5668C09F774915041FE3BF2CC31EDB2575380427B461FC
NPL--IDS2-36-IPR2022-00996-EX2016-PCIBusVariation-2022-09-09.pdf	FF81C948740B1987B55EDC188E7571F5BFE3B54A5725CEE1CA57DCCCAB438A2ED4E4E06B3482F1F60FAB2DFCE4C048F10A72A9C8B330C778E2134BE94FACOE4D
NPL--IDS2-37-IPR2022-00996-EX2017-QMDAS00628-1-2022-09-09.pdf	44640FB2362F7A09C3EFFC62288292A30F44CD6A6C40F5EFF0CD0771B2FF31877EF0C843DE6C20FB8053AEA2F83921F50E6D750AA366769C639A787FA6F79471
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NPL--IDS2-40-IPR2022-00996-EX2021-Hu-2022-09-09.pdf	7B7619ABDB5CE5B561BAC75B1274E09F2CE19DB31B22CFE7E6E967F80CE69DEA7FD54F00D9DD29D57C590C74E710B21511596B5912A0EDD97DEF06920E931C52
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NPL--IDS2-43-IPR2022-00996-EX2024-Schmid-2022-09-09.pdf	38C2925208831EE9F973CB21E1B2E2E3E9ED8F65AFF11E25A84197737BC97FF03EE6F81F0BF8BCC8B837D15875945C44AB389B8ACA68294D4C2632F2F902D3F4
NPL--IDS2-44-IPR2022-00996-EX2025-SDR-SDRAM-264Mb-2022-09-09_Segment1of2.pdf	C0D70C846F01BA177F33D3C5447490DDF70078FBCDFB4DD74B73436F04CAC9D5614E81777B46AEC4BA43D40250E9C0593EACA90EF5EAD373ACB5B3BADBCD1540
NPL--IDS2-44-IPR2022-00996-EX2025-SDR-SDRAM-264Mb-2022-09-09_Segment2of2.pdf	71A3721092AEA8F9AA0107293FBC76EA590B8E299DA45AFB3181A0A97AE1ABA24E47FFBA0FC890E6972FC3E7E4A582BCFB30AF50BE6A28BCB9D44BFCCCF0FD36
NPL--IDS2-45-IPR2022-00996-EX2026-SDR-SDRAM-64Mb-2022-09-09_Segment1of2.pdf	506C5F793939C087E447BF7A9F86832420C131549B6A2A29ED364351093BA568B144DAFB112FBA905FED335CAC011F7178D907EF961A46C213D188F647C24C80
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NPL--IDS2-48-IPR2022-00996-EX2029-PCIOOverview-2022-09-09.pdf	177AB36A9ADB7662BFF4B88C454F77A5CD2F017CCF5B607D CA8180128009C241EEB4E19EC7DF305868CDA31CF21F8C94C C658D875D7D743D33719C02FFEE5692
NPL--IDS2-49-IPR2022-00996-Exhibit1008-61512871FileHist-2022-05-17.pdf	CF9C0C0BDBC720DB26308BEEFA6F3BCBC34359CAAED1864 6289A7EF03BF329260FEE501955AA8C641544D3A6C5CD88B8 FD05F0C52869460F2762EE86257AF0B0

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2135	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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10	6856556		2005-02-15	Siliconsystems, Inc.
11	7023187		2006-04-04	Intersil Americas Inc.
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
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8	First Amended Complaint, Document 14, Case 1:21-cv-01453, filed January 18, 2022.
9	Exhibits to First Amended Complaint, Document 14-1, Case 1:21-cv-01453, filed January 18, 2022.
10	Exhibits to First Amended Complaint, Document 14-2, Case 1:21-cv-01453, filed January 18, 2022.
11	Exhibits to First Amended Complaint, Document 15, Case 1:21-cv-01453, filed January 18, 2022.
12	Second Amended Complaint, Document 18-1, Case 1:21-cv-01453, filed January 25, 2022.
13	Exhibits to Second Amended Complaint, Document 18-2, Case 1:21-cv-01453, filed January 25, 2022.
14	Exhibits to Second Amended Complaint, Document 18-3, Case 1:21-cv-01453, filed January 25, 2022.
15	Exhibits to Second Amended Complaint, Document 18-4, Case 1:21-cv-01453, filed January 25, 2022.
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17	Exhibits to Second Amended Complaint, Document 18-6, Case 1:21-cv-01453, filed January 25, 2022.
18	Notice of Supplemental Authority with Exhibits, Document 32, Case 1:21-cv-01453, filed May 12, 2022.

**INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2135	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

19	Defendant's Answer, Document 40, Case 1:21-cv-01453, filed August 22, 2022.
20	Defendant's First Amended Answer, Document 45, Case 1:21-cv-01453, filed September 12, 2022.
21	Defendant's Second Amended Answer, Document 58, Case 1:21-cv-01453, filed November 1, 2022.
22	Exhibits to Defendant's Second Amended Answer, Document 58-1, Case 1:21-cv-01453, filed November 1, 2022.
23	Exhibits to Defendant's Second Amended Answer, Document 58-2, Case 1:21-cv-01453, filed November 1, 2022.
24	Exhibits to Defendant's Second Amended Answer, Document 58-3, Case 1:21-cv-01453, filed November 1, 2022.
25	Exhibits to Defendant's Second Amended Answer, Document 58-4, Case 1:21-cv-01453, filed November 1, 2022.
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27	Exhibits to Defendant's Second Amended Answer, Document 58-6, Case 1:21-cv-01453, filed November 1, 2022.
28	Exhibits to Defendant's Second Amended Answer, Document 58-7, Case 1:21-cv-01453, filed November 1, 2022.
29	Exhibits to Defendant's Second Amended Answer, Document 58-8, Case 1:21-cv-01453, filed November 1, 2022.

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30	Exhibits to Defendant's Second Amended Answer, Document 58-9, Case 1:21-cv-01453, filed November 1, 2022.
31	Exhibits to Defendant's Second Amended Answer, Document 58-10, Case 1:21-cv-01453, filed November 1, 2022.
32	Answer to Second Amended Counterclaims, Document 62, Case 1:21-cv-01453, filed November 15, 2022.
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35	Petition for Inter Partes Review of US Patent No. 11,016,918, IPR2022-00996, filed May 17, 2022.
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37	Institution Decision, IPR2022-00996, dated December 7, 2022.
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Application Number	17582797
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Attorney Docket Number	022955-000090

41	Wolfe Declaration on US Patent No. 11,016,918, IPR2022-00996, filed May 17, 2022.
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43	IPR2014-00994 Paper 1, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014. (Exhibit 1014 of IPR2022-00996 and IPR2022-00999).
44	IPR2014-00994 Paper 8, Institution Decision, dated December 16, 2014. (Exhibit 1015 of IPR2022-00996 and IPR2022-00999).
45	IPR2014-01370 Paper 8, Corrected Petition for Inter Partes Review of US Patent No. 8,301,833, filed September 22, 2014. (Exhibit 1016 of IPR2022-00996 and IPR2022-00999).
46	IPR2014-01370 Paper 13, Institution Decision, dated March 13, 2015. (Exhibit 1017 of IPR2022-00996 and IPR2022-00999).
47	IPR2017-00649 Paper 1, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017. (Exhibit 1018 of IPR2022-00996 and IPR2022-00999).
48	IPR2017-00649 Paper 7, Institution Decision, dated July 24, 2017. (Exhibit 1019 of IPR2022-00996 and IPR2022-00999).
49	US60/941586 File History. (Exhibit 1005 of IPR2022-00996 and IPR2022-00999).
50	US12/131873 File History. (Exhibit 1006 of IPR2022-00996 and IPR2022-00999).

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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
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CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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See attached certification statement.

- ☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-02-13
Name/Print	Khaled Shami	Registration Number	38745

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APPLICATION #
17/582,797

RECEIPT DATE / TIME
02/13/2023 05:14:52 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

FLASH-DRAM HYBRID MEMORY MODULE

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 61587828

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 90

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
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NPL--IDS1-02- 21cv01453Document1- 20211015.pdf	CC5E769526C1BE878068F53C8876B08E1C7D17217FD2BA9C8 978AC8C0A6C869B69F6970E5020F6CB3908EA637485CCEE8F CA964B8B515422FE05405DABA00FE3
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NPL--IDS1-08- 21cv01453Document14- 20220118.pdf	24FA43D8D957120DEDE29F6807465527AE313EDBB91B9A634 E47F9E578475F7579DF3FB44FC9C2C41E4F0C41189C481A13 E2A678775F2A7A67056C8E592706ED

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APPLICATION #
17/582,797

RECEIPT DATE / TIME
02/13/2023 05:29:16 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

FLASH-DRAM HYBRID MEMORY MODULE

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 61588211

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 100

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NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment20of28.pdf	386AA0297593BED28698C14B1D57E54A663A95E37AE02435C B9F3F64E4B0A93BE5A760B990C54F8902A5F7094681F2CB5FF E11CD279A27BDA9194BDA447D555E
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment21of28.pdf	B70A89B700351210D5D8B869601FF939EBA5B4AB4FE47DB62 B7612103406B5F54765F9CFBDCD5A6FA10D8C575F86820DB1 18E26603752B04163473153DFAC69C
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NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment23of28.pdf	6A165ACBB84DAF98790B62E65F28926565F0A6DA6EA2A4C14 885A3016AF9427CAA8A5B5330C3E9CBE585DAD2F1F54166B1 F511CC05447F1FB8099A3355E8C1B3
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment24of28.pdf	4E52F9F1D590D4FE256A4D76AD8606CB525817D180A5DA81D F993E21BDA3FCABEF8784F864D6B821298B6B6DFC51A74CC 2F2C8E74209994EF559BC6480356738
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment25of28.pdf	98F3BECBCF2DF8D70F6B43A37515FC2A56B66B55CA42674D EFAE7DD5FD0DA702F74268606CB60933E38A92FF1A237B111 848DB72C96B9D7146917D5F1B641C68
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment26of28.pdf	5AB9900313B7D67496C1E3DCE3E3E4AC386DDB9FC9B032E2 FEA371E7183165A7615229B7135A49A44C2352A34D00829B86 755A7779BF5283D40A4A01E1D1CF23
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment27of28.pdf	1D2EBE985A09604A69A9906A2671639AE8BF7FC640DE8B9E3 4E29BC5ECCE4C9984BE9B0ACB47E52974D6F70CA0ED1BB30 2722C720536EEF504D09CAC045BEFB0
NPL--IDS1-16- 21cv01453Document18-5- 20220125_Segment28of28.pdf	3842B418DC90AAC15AAA1A1AC052508962FE5F34BB98C4021 8BB410EAD677873AA6F475392C17763324198AF586DE091EAF DDA5542FA3E8C057614900C321042
NPL--IDS1-17- 21cv01453Document18-6- 20220125_Segment1of3.pdf	B10D43A69E69085DF8EE505D6E4BE59EB79E604CFAEC1C0F A42BBCE05ADA67AF8552CB34861C432E7021B944569A80DE9 1D2079DB9334DA2ABE768C64E93DD32
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NPL--IDS1-17- 21cv01453Document18-6- 20220125_Segment3of3.pdf	BF91AE285F305BEC7AE2CDF6798BD1384D99A2DFB60C9502 2F138899AB5E13AE1A4965F10D17FC25EC54B70DFBD2E2C61 CF54A811B0439B6DF0B0B9E4C616DC6
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NPL--IDS1-19- 21cv01453Document40- 20220822__Segment1of2.pdf	DED346FAE8124691CEF20D12C11F64E93B91E94590358976A9 BBF908A8AEAFB45135597860CB52E1C7A594116343434C3B2 EC79F5980E369B61EC3D36078976B
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NPL--IDS1-20- 21cv01453Document45- 20220912__Segment1of2.pdf	63F39C76BD97F9FA837CC03BBEBD26FC68FAD0CFA568B760 07527003392AF0FE85E4C62926F282C24D96E0C138A846499F F24B5FE66C9430DDAFD384E767CF3F
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NPL--IDS1-21- 21cv01453Document58- 20221101__Segment1of2.pdf	CB1995900608A0C5959F9B1299AB02827ADC3B845EC79E087 B45105E9C547162F75AAC4297BEC0343E1060F25F631C75A36 3FE4CADCED0064EAB40139BB525D0
NPL--IDS1-21- 21cv01453Document58- 20221101__Segment2of2.pdf	BE591E733B1B787D4B20126E6B099DABA7F7B62A3B2C53FB7 F386EC3BCD81964A3FC1B6B0314A58EDA04FCE40CB34CDB2 50AC1B3B990B7A40E0B715EAA620CCC
NPL--IDS1-22- 21cv01453Document58-1- 20221101__Segment1of2.pdf	CE276C0149DF8639A9568AA093C81DCEB4DBE3347CB3FAF6 BE73D18273B8DC9A2B77FC98B902D9C5924C57A65E92BF37C 19D8DE9EFDA5772BC6E10CB197173ED
NPL--IDS1-22- 21cv01453Document58-1- 20221101__Segment2of2.pdf	0E44A1466F273494D500542718B135A7D4435E39FAE4FD6DBB CC9CF78BCDC8EC96725DF97BA9EDCB93EEFE8ED887692A 28A88D0F1091648CDFE7653F3D96DCE
NPL--IDS1-23- 21cv01453Document58-2- 20221101.pdf	B79C1D46CDC09020050614DE991E6D99D22C8D3BEEE1E8CC3 D31BB1B34855F856A2AA3208C8C07863EA81B6BF22A9C5763 9265A1814A3C63EACB1897689833AE8
NPL--IDS1-24- 21cv01453Document58-3- 20221101.pdf	1504EEA4CDE4D1506C87C73CA8BFD6770B61C5A67641F4AE 35F16603883F84F986A338D6AF19BFED4A38F4C5C4EC5F49 BC479D0F3CF358330A8AADFAE9071F4

NPL--IDS1-25- 21cv01453Document58-4- 20221101.pdf	988A11199F945AAE26970E7642FEFE8836EFE5F8F74F6060F4 C89F86F8C5F509C8E93CB70EF9410DE2CE9361B13D0AADFB E0752F026B2A91E782E9DEB0C7C430
NPL--IDS1-26- 21cv01453Document58-5- 20221101_Setment1of2.pdf	24B7DBECA6D93202F5EA4E2A9AE02CF09A071D8B20FB96AD E741000341745F22E62C01D8035AEF59C69ECF74A72E6E77FE FFE7BF883F078A8DF3259FE48044AA
NPL--IDS1-26- 21cv01453Document58-5- 20221101_Setment2of2.pdf	C49EC64F67F5809C50A9C98EBF59538FA63046956CAEEDE6E 685E3F2CC1F007EDA47DF2B6A14EC70EB1D13791D5BA9BF2 173F95002F4C46C0BC468E9B66DA334
NPL--IDS1-27- 21cv01453Document58-6- 20221101.pdf	A1BE488C07C67D7E2029500535A360538FE977A98AED41690D E3FB64D48F0E3011DF009C793391D56A212674C9AF5D45E727 92D70D0863B40BB6E581D1E19718
NPL--IDS1-28- 21cv01453Document58-7- 20221101.pdf	D93B3321DE919A17310EA623F7C883AB97888D4D6AD543B15 3F76D1A237FABC8C7578B51EDF967C731B583B9C01B4164F5 666A1A369D61EA6F6579F0D28C6CB8
NPL--IDS1-29- 21cv01453Document58-8- 20221101_Segment1of2.pdf	9658BCC32E13A589E5AA2183325E7B4EFFCAFE56EEFBC186 E50C627E8DDB3DDCA663B84E80B0F31BB695B438CEA2676D F001881AB8AAC3ADE1077217CCF4A130
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NPL--IDS1-30- 21cv01453Document58-9- 20221101.pdf	5F493FC18E733A570C3853E55B3D8DFAB2396C3EF63AB158A 6FF78A26FA3D8D724CB08D975DFD26DA7786893007DD33B86 8536FBCB9FB4506962C2C72AF18A73
NPL--IDS1-31- 21cv01453Document58-10- 20221101.pdf	218683DF4B679091C4670F62F4CEA2B8F65EB6788C5C75C732 43423FCEF523CE6E7622F4EF5402722427FFD4361582EB2F29 700D9914E69D89919BE316621964
NPL--IDS1-32- 21cv01453Document62- 20221115.pdf	9924183EB39984F2AF47ED3193F4322C8870D08F9F5C1915FE 22AEE11929B51BFE9986FEC5E2CCC2F86E3B35CE2A6BE8E4 93050D88A9B7B3A61A5CF65AEF70F7

NPL--IDS1-33- 21cv00463Document16- 20220412.pdf	D1BC90C1B601CF1A558238AA1A5EF6B6D73AD32C392BF8D7 12CA1AC07FD445876FE67B51E37CF695F0252331105F04A17A DF9D32DFD7E036DB61905662AD5B1C
NPL--IDS1-34- 21cv00463Document23- 20220503_Segment1of2.pdf	5FC7BDF92916812947C213F4C3C2B6E5F32C44B247364E532B 7CFC4F4310F8FF1713FA030262F51D3F4A04B6E0EDEFDF06F A76A9BC6CBDF62BC8603A997B8229
NPL--IDS1-34- 21cv00463Document23- 20220503_Segment2of2.pdf	3642D7FB47CD264A0FEA283D394522AC35ACFB5485ADF3AA FEE67136866DECB9CBB0644DB8FA360C3D892DE6D0FAE9FE 0DAF932EAE3661429E17FD05C682CE46
NPL--IDS1-35-IPR2022-00996- PetitionIPR-2022-05- 17_Segment1of3.pdf	66032F4C3E49653726F1221818ACE69552B41AA5A18B567630 C91F7E3A50255AA71412381CC55CC3AD5083A2ACE189A851F D5FAA7EC9D48B232CFE290703F61F
NPL--IDS1-35-IPR2022-00996- PetitionIPR-2022-05- 17_Segment2of3.pdf	8D071FAD5A3628EAF32E9556193508978B59B50CA23BA849A4 66CDD19613F2AB8150BBC2EB9029DE367C532CBB8B101C2D 06A699CC3C644144E7CF606F1E9C9C
NPL--IDS1-35-IPR2022-00996- PetitionIPR-2022-05- 17_Segment3of3.pdf	1C86529CFE826215E0237E0F82CE2C86FEAE3F55BE08E680F B0DAC20BA309BE3F4139F71FA91F758A5787234DF31FE1683 D0495FAAC24AFE9E9234FBE1717B7C
NPL--IDS1-36-IPR2022-00996- PatOwnPrelimResp-2022-09- 09_Segment1of2.pdf	CAA42C5AEB91870E1AF6D4C4E7F1C60A210A6C9097B5E740 E7A9B43D6CE8C1785232EFE81AF66B648399EA9A4417151E8 8982DDD247FD9A65304DFDBD8AAEE42
NPL--IDS1-36-IPR2022-00996- PatOwnPrelimResp-2022-09- 09_Segment2of2.pdf	3859679AFD7669A2E99A50C4A4C8C4CB5F89479A398A05BA5 53A96556B37204668D8A6E3CCDB391CED6DD788088056A270 599E27D32027FA7F661B53477E1C28
NPL--IDS1-37-IPR2022-00996- InstitDecision-2022-12- 07_Segment1of2.pdf	B6DE886629D482D7580861894E1EE3DFF6CC35E963A2C62D1 1BB9E37A7247FA187CC96C9D82B73A83522C9F229DAE00D64 79B7BAAFE1E2C24E0A8A430F209BFB
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NPL--IDS1-38-IPR2022-00999-PetitionIPR-2022-05-17_Segment1of3.pdf	513E79905219A8EEEC769276A7081A7DDEACB19D06DB12A3A DBC6F04802EA584E8BCAD49C446686536F4825EDD9E7AE0F3 DD7EF32502AC6C78B5A93F10E0AD22
NPL--IDS1-38-IPR2022-00999-PetitionIPR-2022-05-17_Segment2of3.pdf	1517A7ABDA68E1A15BA1E0826B2F652DC48055EDE0DB98C0 5B9EA9326F5B50F5FC67345C753282DFD2267645CB9122B419 A072584718E93DFBAD1580BAAF71E5
NPL--IDS1-38-IPR2022-00999-PetitionIPR-2022-05-17_Segment3of3.pdf	3ED4E6D1EE68A4AB1C3DCF0EDC4929C57B286EECCC2248D 15C71AC45F0CF9EA69332455F09D737AFBA856B9D02556C05 804FC5FA7CE5E83B32ED08CFC2E11B3A
NPL--IDS1-39-IPR2022-00999-PatOwnPrelimResp-2022-09-08_Segment1of2.pdf	4690906614D6846EA60453E0DAD676ACE0486AF3AD6F4CD15 045A6616D20B095BDE764F6352B0679FE6985BE74A334CAA7 AEE125D8FFE495D73DE555667FE842
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NPL--IDS1-40-IPR2022-00999-InstitDecision-2022-12-06_Segment1of2.pdf	AA62B5B60D330565AC010B8A9C0FE679B7F127BEEB78817AC 03C49AC743145DA1E96FFB1116085DA6C914217899146259ED 254AE18ED1D955DC50A4CD5AEE942
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NPL--IDS1-41-IPR2022-00996-Exhibit1003-WolfeDecl-918Pat-2022-05-17_Segment1of10.pdf	5C44783A37A559D81CCEE8B5375A8019FEE71D27ECD80B561 81B5EEF33947CF16E6D1E747988B5B826EFF3DBF8CFAC73E BE97F2BE704FD44D42CA0A17402EAD4
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NPL--IDS1-42-IPR2022-00999-Exhibit1003-WolfeDecl-054Pat-2022-05-17__Segment3of9.pdf	8CB0ACCAB4A74E8B6290FDE3C85E606C7E1F2FED8FB45D460AD0C3C42F9919ECD51AB8B462740582C143F1EA9F224C2E66B44B327E1C68CAE2DDFAB097E08091

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NPL--IDS1-42-IPR2022-00999-Exhibit1003-WolfeDecl-054Pat-2022-05-17_Segment7of9.pdf	C622FC2A00C057C4F1AA60CDC91565B80938A8673E747FF61 42A4FA155FA98BF9005210C2D8045C523F5C2FD5278EFF3356 D23D29E16268A8302037A69B7445E
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NPL--IDS1-45-IPR2022-00996-Exhibit1016-IPR2014-01370-Paper8-2022-05-	82C641239B01ED6789413A93212056FCD91402905D662644221 C25F7301F6F718EC38386E72FDE0F3320BEEAB5486FB36A44 05FD94BA99F2B70FBA99DCBA3839

17_Segment1of2.pdf	
NPL--IDS1-45-IPR2022-00996-Exhibit1016-IPR2014-01370-Paper8-2022-05-17_Segment2of2.pdf	3BB3B603250BE450CD66D02ECA6ACFD9D6FF903CBCCF4038133FE2AEA35CE15EA66706F1B01F85F22C9EC72A6FD5B16432EE5CF520BFF00820050A792BB2221A
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NPL--IDS1-47-IPR2022-00996-Exhibit1018-IPR2017-00649-Paper1-2022-05-17_Segment2of2.pdf	87F02A47F19C47CF17D2EEE7F4121395D35438782CDD724856A31E5970AD9701CDF2EBEADD89A77917F85213115F5675A50F0DACA65535F3B77E855EC71488D2
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NPL--IDS1-49-IPR2022-00996-Exhibit1005-60941586FileHist-2022-05-17.pdf	D79A7FB579CD46BF49A5755F8D885D539B7840A0AFCE8EBDCCA18DAFE4BAFF7CA71F4EC010894D0DCE59A9DC63ED14AD75F328A4387DCDEB7BFBA7EDD0E14840
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NPL--IDS1-50-IPR2022-00996-Exhibit1006-12131873FileHist-2022-05-17_Segment1of2.pdf	A62CB90FF22716350399C8286FD95E5A531532F60169BCE09D8D9E7E0B2CE36A61E6D150D3E1AB705D178DAC138EB1225E047DDC2F394C9BF0C08C655F4048C3

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional) 022955-000090
Application Number 17/582,797	Filed January 24, 2022	
For FLASH-DRAM HYBRID MEMORY MODULE		
Art Unit 2138	Examiner FARROKH	

This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application.

The requested extension and fee are as follows (check time period desired and enter the appropriate fee below):

	<u>Fee</u>	<u>Small Entity Fee</u>	<u>Micro Entity Fee</u>	
☐ One month (37 CFR 1.17(a)(1))	\$220	\$88	\$44	\$ _____
☐ Two months (37 CFR 1.17(a)(2))	\$640	\$256	\$128	\$ _____
☒ Three months (37 CFR 1.17(a)(3))	\$1,480	\$592	\$296	\$ 1480
☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$928	\$464	\$ _____
☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,264	\$632	\$ _____

☐ Applicant asserts small entity status. See 37 CFR 1.27.☐ Applicant certifies micro entity status. See 37 CFR 1.29.
Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously.☐ A check in the amount of the fee is enclosed.☐ Payment by credit card. Form PTO-2038 is attached.☐ The Director has already been authorized to charge fees in this application to a Deposit Account.☐ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to
Deposit Account Number _____.☒ Payment made via USPTO's patent electronic filing system (Patent Center or EFS-Web).**WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.**

I am the

☐ applicant.☒ attorney or agent of record. Registration number **38745**.☐ attorney or agent acting under 37 CFR 1.34. Registration number _____./Khaled Shami/

Signature

April 12, 2023

Date

Khaled Shami

Typed or printed name

202-467-8800

Telephone Number

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.☒ * Total of 1 forms are submitted.

A Federal agency may not conduct or sponsor, and a person is not required to respond to, nor shall a person be subject to a penalty for failure to comply with an information collection subject to the requirements of the Paperwork Reduction Act of 1995, unless the information collection has a currently valid OMB Control Number. The OMB Control Number for this information collection is 0651-0031. Public burden for this form is estimated to average 6 minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the information collection. Send comments regarding this burden estimate or any other aspect of this information collection, including suggestions for reducing this burden to the Chief Administrative Officer, United States Patent and Trademark Office, P.O. Box 1450, Alexandria, VA 22313-1450 or email InformationCollection@uspto.gov. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS.** If filing this completed form by mail, send to: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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Electronic Patent Application Fee Transmittal

Application Number:	17582797			
Filing Date:	24-Jan-2022			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Lisa Elliott			
Attorney Docket Number:	022955-000090			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 3 months with \$0 paid	1253	1	1480	1480
Miscellaneous:				
Total in USD (\$)				1480

Electronic Acknowledgement Receipt

EFS ID:	47832726
Application Number:	17582797
International Application Number:	
Confirmation Number:	3546
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Lisa Elliott
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000090
Receipt Date:	12-APR-2023
Filing Date:	24-JAN-2022
Time Stamp:	16:24:24
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1480
RAM confirmation Number	E20234BG25339706
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Request for Reconsideration-After Non-Final Rejection	022955-000090__Resp_10-12-2 2_NFOA.pdf	127139	no	7
			1d5b70088752cd4c46aa7ade3b44118c8fa 6cc70		
Warnings:					
Information:					
2	Extension of Time	EOTx3_large_022955-90.pdf	109762	no	2
			32356f964227423fa705ad37c7e4a021d144 4681		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	38123	no	2
			f04f35518bfa4075263dfa637c4af1979d31a fbc		
Warnings:					
Information:					
Total Files Size (in bytes):			275024		
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Commissioner:

In response to the non-final Office Action mailed October 12, 2022, Applicant submits the following Amendment and Remarks.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 6.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (New) A dual in-line memory module (DIMM) comprising:

a printed circuit board (PCB) including a set of voltage supply lines electrically connected to corresponding edge connections of a first set of edge connections, sixty four data lines electrically connected to corresponding edge connections of a second set of edge connections, and a set of address and control lines electrically connected to corresponding edge connections of a third set of edge connections;

a DIMM interface including the first, second and third set of edge connections, the DIMM interface configured to (i) fit into a memory socket connector of a host system, and (ii) provide a conduit for power, sixty four data signals, and address and control signals between the host system and the DIMM via the first, second and third set of edge connections, respectively;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) receive and monitor an input voltage from the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to other portions of the controller, wherein the controller is configured to perform one or more operations in response to the trigger signal, the one or more operations including a write operation to write data into the non-volatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a plurality of components coupled to the PCB, the plurality of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices; and

first, second, third and fourth buck converters coupled to the PCB and to the input

voltage supply line and configured to (i) receive power from the input voltage supply line, (ii) generate first, second, third and fourth regulated voltages from the power received from the input voltage supply line, and (iii) output the first, second, third and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively;

wherein each component of the plurality of components (i) is not electrically connected to the input voltage supply line, (ii) is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and (iii) receives power only from one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices (i) is not electrically connected to the input voltage supply line, and (ii) receives power only from one or more of the first, second, third and fourth regulated voltage lines.

3. (New) The DIMM of claim 2, wherein the controller is further configured to control a plurality of power transition sequences.
4. (New) The DIMM of claim 3, wherein the plurality of power transition sequences includes first and second power transition sequences, the first power transition sequence includes turning power on or off to the first regulated voltage line in response to a first trigger condition of a plurality of trigger conditions, and the plurality of trigger conditions includes a request from the host.
5. (New) The DIMM of claim 4, wherein the input voltage supply line is a first input voltage supply line, and wherein the first, second, third and fourth buck converters are coupled to a second input voltage supply line, and

wherein, during the second power transition sequence, the first, second, third and fourth buck converters are further configured to (i) receive power from the first input voltage supply line or the second input voltage supply line in response to a second trigger condition of the plurality of trigger conditions, (ii) generate the first, second, third and fourth regulated voltages from either the power received from the first input voltage supply line or the power received from the second input voltage supply line, and (iii) output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively.

6. (New) The DIMM of claim 2, wherein the controller is further configured to generate a

plurality of trigger signals in connection with a plurality of trigger conditions that include (i) a request from the host system and (ii) detecting input voltage deviation from one or more threshold voltages.

7. (New) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages specified as certain voltage values.

8. (New) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages corresponding to percentages of an operating voltage.

9. (New) the DIMM of claim 4, wherein the DIMM is configured to operate in a plurality of states including a first state and a second state, wherein the DIMM is further configured (i) to operate in the first state while power is turned on to the first regulated voltage line, and (ii) to operate in the second state while power is turned off to the first regulated voltage line.

10. (New) The DIMM of claim 9, wherein the first power transition sequence further includes turning power on to the plurality of DDR SDRAM devices, during both the first and second states, via one or more of the second, third and fourth regulated voltage lines.

11. (New) The DIMM of claim 10 further comprising:

a register coupled to the PCB, to the first regulated voltage line, and to the set of address and control lines, wherein:

during the first state, the register is configured to (i) receive and register address and control signals from the host via the set of address and control lines, and (ii) output registered address and control signals to the plurality of the DDR SDRAM devices, and

during the second state, the register does not couple address and control signals from the host to the plurality of the DDR SDRAM devices.

12. (New) The DIMM of claim 2, wherein the DIMM interface comprises a standard 240-pin DDR2 edge connector.

13. (New) The DIMM of claim 2, wherein the controller is further configured to periodically perform the one or more operations at regular intervals of time.

14. (New) The DIMM of claim 13, wherein the regular intervals of time are every millisecond.
15. (New) The DIMM of claim 2, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

REMARKS

The non-final Office Action mailed October 12, 2022, has been carefully considered.

Reconsideration in view of the following remarks is respectfully requested.

Upon entry of the foregoing amendment, claims 2-15 are pending in this application, with claim 2 being the independent claim.

Canceled Claims

Claim 1 has been canceled without prejudice or disclaimer of the subject matter contained therein.

Rejection(s) Pursuant to Judicially-Created Double Patenting

Claim 1 stands rejected pursuant to the judicially-created doctrine of obviousness-type double patenting as allegedly being unpatentable over claim 1 of U.S. pat. no. 11,016,918 B2. Claim 1 has been canceled without prejudice or disclaimer of its subject matter, and the double-patenting rejection thereof is moot.

Newly-Added Claims

Claims 2-15 have been added to further particularly point out and distinctly claim certain aspects of the invention.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: April 12, 2023
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/582,797	Filing Date 01/24/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)	
AMENDMENT	04/12/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	x \$100 =	0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 =	0	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE	0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =		
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =		
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE		
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/TAWANA E CAMPBELL/		
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This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	07/13/2023	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	DELIVERY MODE
			07/13/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

Notice of Pre-AIA or AIA Status

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This Office Action is response to the communications file on 04/12/2023. There are a total of 14 claims pending in the application. Claim 1 has been canceled and claims 2-15 have been added.

ACKNOWLEDGEMENT OF REFERENCES CITED BY APPLICANT

Information Disclosure Statement

3. As required by **M.P.E.P. 2001.06(b)** and **37 C.F.R. 1.98(d)**, since the instant application has been identified as a continuation application of an earlier filed application and is relied upon for an earlier filing date under **35 U.S.C. 120**, the examiner has reviewed the prior art cited in the earlier related application as required by **M.P.E.P. 707.05** and **904** and as stated in **M.P.E.P. 2001.06(b)**, no separate citation of the same prior art need be made by the applicants in the instant application.

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 02/13/2023 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement has been considered by the Examiner. The US patent and or the patent application with missing publication data and/or patentee or applicant show as lined-through. The

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documents listed as NPL in the IDS' are shown by lined-through because the corresponding document are not readily identifiable from the description listed in the IDS.

INFORMATION CONCERNING CLAIMS:

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL. —The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

1. *Claim 2-15 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.*

2. *The independent claim 2, in part, recites the limitation:*

*“first, second, third and **fourth** buck converters coupled to PCB and to the input voltage supply line and configured to (i) receive power from the input voltage supply line, (ii) generate first, second, third and **fourth regulated voltages** from the power received from the input voltage supply line, and (iii) output the first, second, third and fourth*

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regulated voltages via first, second, third, and fourth regulated voltage lines, respectively”

The specification does not appear to describe/support the limitation(s) as claimed above. The Abstract of instant application appears to be closest portion of the specification describing the subject matter. For convenience, the Abstract reproduced as shown below:

Abstract

“In certain embodiments, a memory module includes a printed circuit board (PCB) having an interface that couples it to a host system for provision of power, data, address and control signals. First, second, and third buck converters receive a pre-regulated input voltage and produce first, second and third regulated voltages. **A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage.** Synchronous dynamic random access memory (SDRAM) devices are coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, and a voltage monitor circuit monitors an input voltage and produces a signal in response to the input voltage having a voltage amplitude that is greater than a threshold voltage” *(emphasis added)*.

As shown above, the fourth voltage is not produced (e.g., generated) by a buck converter. The dependent claims 3-15 are rejected by virtue of their dependency from the independent claim 2.

Response to Remarks

In non-final office action mailed 10/12/2022, claim 1 was rejected under the statutory type (35 U.S.C. 101) as claiming the same invention as that of prior U.S.

Patent No. 11,016,918 B2. Applicant has canceled the claim 1. Therefore, the statutory double-patenting rejection is withdrawn.

Conclusion

*Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).*

*A shortened statutory period for reply to this final action is set to expire **THREE MONTHS** from the mailing date of this action. In the event a first reply is filed within **TWO MONTHS** of the mailing date of this final action and the advisory action is not mailed until after the end of the **THREE-MONTH** shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than **SIX MONTHS** from the date of this final action.*

*Any inquiry concerning this communication or earlier communications from the examiner should be directed to **HASHEM FARROKH** whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.*

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2138

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2138*

Notice of References Cited	Application/Control No. 17/582,797		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2138	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	10-2012	Chen; Chi-She	G06F3/065	711/E12.008
*	B	US-20070136523-A1	06-2007	Bonella; Randy M.	G06F12/0866	711/E12.019
*	C	US-20080126624-A1	05-2008	Prete; Edoardo	G11C7/106	710/53
	D					
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	N					
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF

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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135 2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

U.S.PATENTS

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	11016918		2021-05-25	Netlist, Inc.	
	2	8301833		2012-10-10	Netlist, Inc.	
	3	8874831		2014-10-28	Netlist, Inc.	
	4	9158684		2015-10-13	Netlist, Inc.	
	5	9928186		2018-03-27	Netlist, Inc.	
	6	7724604		2010-05-25	SMART Modular Technologies, Inc.	
	7	7719866		2010-05-18	Agence Spatiale Europeenne	
	8	7721130		2010-05-18	Qimonda AG	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
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Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

9	6798709	2004-09-28	Samsung Electronics Co., Ltd.
10	6856556	2005-02-15	Siliconsystems, Inc.
11	7023187	2006-04-04	Intersil Americas Inc.
12	7692938	2010-04-06	Northern Power Systems, Inc.
13	5563839	1996-10-18	Simtek Corporation
14	6693840	2004-02-17	Matsushita Electric Industrial Co., Ltd.
15	7061214	2006-06-13	Texas Instruments Incorporated
16	5630096	1997-05-13	Microunity Systems Engineering, Inc.
17	10672458	2020-06-02	Rambus Inc.
18	6067241	2000-05-23	International Power Devices, Inc.

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	20190004985		2019-01-03	Netlist, Inc.	
	2	20060174140		2006-08-03	Shaun Harris, et al.	
	3	20060080515		2006-04-13	LEFTHAND NETWORKS, INC.	
	4	20080238536		2008-10-02	ELPIDA MEMORY, INC	
	5	20100257304		2010-10-07	GOOGLE INC.	
	6	20100205470		2010-08-12	STEC, Inc.	
	7	20080101147		2008-05-01	Hossein Amidi	
	8	20070136523		2007-06-14	Randy Bonella, et al.	
	9	20090034354		2009-02-05	MICRON TECHNOLOGY, INC.	

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Complaint, Document 1, Case 2:21-cv-00463, filed December 20, 2021.	
	2	Complaint, Document 1, Case 1:21-cv-01453, filed October 15, 2021.	
	3	Exhibits of Complaint, Document 1-1, Case 1:21-cv-01453, filed October 15, 2021.	
	4	Exhibits of Complaint, Document 1-2, Case 1:21-cv-01453, filed October 15, 2021.	
	5	Exhibits of Complaint, Document 1-3, Case 1:21-cv-01453, filed October 15, 2021.	
	6	Opening Brief, Document 11, Case 1:21-cv-01453, filed December 20, 2021.	
	7	Exhibits to Opening Brief, Document 11, Case 1:21-cv-01453, filed December 20, 2021.	

**INFORMATION DISCLOSURE
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First Named Inventor	CHEN	
Art Unit	2135	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

8	First Amended Complaint, Document 14, Case 1:21-cv-01453, filed January 18, 2022.
9	Exhibits to First Amended Complaint, Document 14-1, Case 1:21-cv-01453, filed January 18, 2022.
10	Exhibits to First Amended Complaint, Document 14-2, Case 1:21-cv-01453, filed January 18, 2022.
11	Exhibits to First Amended Complaint, Document 15, Case 1:21-cv-01453, filed January 18, 2022.
12	Second Amended Complaint, Document 18-1, Case 1:21-cv-01453, filed January 25, 2022.
13	Exhibits to Second Amended Complaint, Document 18-2, Case 1:21-cv-01453, filed January 25, 2022.
14	Exhibits to Second Amended Complaint, Document 18-3, Case 1:21-cv-01453, filed January 25, 2022.
15	Exhibits to Second Amended Complaint, Document 18-4, Case 1:21-cv-01453, filed January 25, 2022.
16	Exhibits to Second Amended Complaint, Document 18-5, Case 1:21-cv-01453, filed January 25, 2022.
17	Exhibits to Second Amended Complaint, Document 18-6, Case 1:21-cv-01453, filed January 25, 2022.
18	Notice of Supplemental Authority with Exhibits, Document 32, Case 1:21-cv-01453, filed May 12, 2022.

**INFORMATION DISCLOSURE
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Examiner Name	FARROKH
Attorney Docket Number	022955-000090

10	Defendant's Answer, Document 40, Case 1:21-cv-01453, filed August 22, 2022.
20	Defendant's First Amended Answer, Document 45, Case 1:21-cv-01453, filed September 12, 2022.
21	Defendant's Second Amended Answer, Document 58, Case 1:21-cv-01453, filed November 1, 2022.
22	Exhibits to Defendant's Second Amended Answer, Document 58-1, Case 1:21-cv-01453, filed November 1, 2022.
23	Exhibits to Defendant's Second Amended Answer, Document 58-2, Case 1:21-cv-01453, filed November 1, 2022.
24	Exhibits to Defendant's Second Amended Answer, Document 58-3, Case 1:21-cv-01453, filed November 1, 2022.
25	Exhibits to Defendant's Second Amended Answer, Document 58-4, Case 1:21-cv-01453, filed November 1, 2022.
26	Exhibits to Defendant's Second Amended Answer, Document 58-5, Case 1:21-cv-01453, filed November 1, 2022.
27	Exhibits to Defendant's Second Amended Answer, Document 58-6, Case 1:21-cv-01453, filed November 1, 2022.
28	Exhibits to Defendant's Second Amended Answer, Document 58-7, Case 1:21-cv-01453, filed November 1, 2022.
29	Exhibits to Defendant's Second Amended Answer, Document 58-8, Case 1:21-cv-01453, filed November 1, 2022.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

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First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

30	Exhibits to Defendant's Second Amended Answer, Document 58-9, Case 1:21-cv-01455, filed November 1, 2022.
31	Exhibits to Defendant's Second Amended Answer, Document 58-10, Case 1:21-cv-01455, filed November 1, 2022.
32	Answer to Second Amended Counterclaims, Document 62, Case 1:21-cv-01453, filed November 15, 2022.
33	Defendant's Answer, Document 16, Case 2:21-cv-00463, filed April 12, 2022.
34	First Amended Complaint, Document 23, Case 2:21-cv-00463, filed May 3, 2022.
35	Petition for Inter Partes Review of US Patent No. 11,046,048, IPR2022-00006, filed May 17, 2022.
36	Patent Owner Preliminary Response, IPR2022-00006, filed September 8, 2022.
37	Institution Decision, IPR2022-00006, dated December 7, 2022.
38	Petition for Inter Partes Review of US Patent No. 11,232,054, IPR2022-00000, filed May 17, 2022.
39	Patent Owner Preliminary Response, IPR2022-00000, filed September 8, 2022.
40	Institution Decision, IPR2022-00000, dated December 7, 2022.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
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Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

41	Wolfe Declaration on US Patent No. 11,010,910, IPR2022-00996, filed May 17, 2022.
42	Wolfe Declaration on US Patent No. 11,232,06, IPR2022-00999, filed May 17, 2022.
43	IPR2014-00994 Paper 1, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014. (Exhibit 1014 of IPR2022-00996 and IPR2022-00999).
44	IPR2014-00994 Paper 8, Institution Decision, dated December 10, 2014. (Exhibit 1015 of IPR2022-00996 and IPR2022-00999).
45	IPR2014-01370 Paper 8, Corrected Petition for Inter Partes Review of US Patent No. 8,301,833, filed September 22, 2014. (Exhibit 1016 of IPR2022-00996 and IPR2022-00999).
46	IPR2014-01370 Paper 13, Institution Decision, dated March 13, 2015. (Exhibit 1017 of IPR2022-00996 and IPR2022-00999).
47	IPR2017-00649 Paper 1, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017. (Exhibit 1018 of IPR2022-00996 and IPR2022-00999).
48	IPR2017-00649 Paper 7, Institution Decision, dated July 24, 2017. (Exhibit 1019 of IPR2022-00996 and IPR2022-00999).
49	US60/941586 File History. (Exhibit 1005 of IPR2022-00996 and IPR2022-00999).
50	US12/131073 File History. (Exhibit 1006 of IPR2022-00996 and IPR2022-00999).

If you wish to add additional non-patent literature document citation information please click the Add button

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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Application Number	17582797
Filing Date	2022-01-24
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Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	07/07/2023
--------------------	------------------	-----------------	------------

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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Application Number	17582797
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First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-02-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L21	1	"17582797"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 04:25 PM
L22	4	("11016918").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 04:28 PM
L23	6	"15934416"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 04:35 PM
L24	0	((dual ADJ in\$1line ADJ memory ADJ module) AND ((printed ADJ circuit ADJ board) OR PCB) AND (buck ADJ converters) AND DDR).clm.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/06 04:47 PM
L25	9	((dual ADJ in\$1line ADJ memory ADJ module) AND ((printed ADJ circuit ADJ board) OR PCB) AND (buck ADJ converters) AND DDR)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/06 04:48 PM
L26	0	25 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2023/07/06 04:49 PM
L27	31	((printed ADJ circuit ADJ board) OR PCB) AND (buck ADJ converters) AND (((double ADJ data ADJ	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA,	OR	OFF	ON	2023/07/06 04:53 PM

		rate) OR DDR) adj5 SDRAM)	EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L28	0	(((printed ADJ circuit ADJ board) OR PCB) AND (buck ADJ converters) AND (((double ADJ data ADJ rate) OR DDR) adj5 SDRAM)).clm.	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/06 04:54 PM
L29	0	27 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2023/07/06 04:54 PM
L30	10	"14489269"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 04:56 PM
L31	39	("10949339" "10860506" "10217523" "10474595" "9858218" "7619912" "11232054" "6026465" "20100110748" "20060212651" "20030028733").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 05:22 PM
L32	108	("11016918" "8301833" "874831" "9158684" "9928186" "7724604" "7719866" "7721130" "6798709" "6856556" "7023187" "7692938" "5563839" "8693840" "7061214" "5630096" "10672458" "6067241" "20190004985" "20060174140" "20060080515" "20080238536" "20100257304" "20100205470"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/06 05:30 PM

		"20080101147" "20070136523" "20090034354").pn.					
L33	841	(buck near2 converter) SAME (coupl\$3 OR connected OR attached) SAME ((printed ADJ circuit ADJ board) OR PCB)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:25 AM
L34	1	1 AND 33	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:25 AM
L35	74	("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:46 AM
L36	7	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2 voltage)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:46 AM
L37	5	35 AND 36	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2023/07/07 11:47 AM

			JPO; DERWENT; IBM_TDB)				
L38	10797	(buck near5 converters).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:48 AM
L39	5	35 AND 38	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:50 AM
L40	989	(buck SAME PCB)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:52 AM
L41	51	(buck SAME PCB).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:53 AM
L42	4	35 AND 41	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 11:53 AM
L43	0	((plurality OR multiple OR several OR many OR two) near2 (DDR adj5 SDRAM)) SAME (regulated ADJ voltage)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 12:02 PM
L44	2250	((plurality OR multiple OR several OR many OR two) near2 (DDR adj5 SDRAM))	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 12:08 PM
L45	0	1 AND 44	(US-PGPUB; USPAT;	OR	ON	ON	2023/07/07

			USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				12:09 PM
L46	15687	((plurality OR multiple OR several OR many OR two) near2 (DDR OR SDRAM))	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 12:10 PM
L47	1	1 AND 46	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 12:10 PM
L48	107	(buck SAME (DDR OR SDRAM))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 12:13 PM
L49	10	48 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2023/07/07 12:14 PM
L50	4	(31 OR 32) AND 48	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY,	OR	ON	ON	2023/07/07 12:24 PM

			CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L51	3	((plurality OR multiple OR several OR many OR two) near2 (DDR OR SDRAM)) SAME ((buck OR buck\$1boost) near5 (converters OR regulators))	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 01:26 PM
L52	3	((((plurality OR multiple OR several OR many OR two) near2 (DDR OR SDRAM)) SAME ((buck OR buck\$1boost) near5 (converters OR regulators))))).clm.	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 01:31 PM
L53	20	(voltage near2 monitor) SAME monitor\$3 SAME (input near2 voltage) SAME (generat\$3 OR creat\$3 OR provid\$3) SAME (trigger\$3 near2 signal) SAME (when OR "if" OR based OR in\$1case OR in\$1event) SAME (input near2 voltage) SAME ((exceed\$3 OR greater	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH,	OR	OFF	ON	2023/07/07 01:42 PM

		OR higher OR more OR above) near5 threshold)	TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L54	4	(voltage near2 monitor) SAME monitor\$3 SAME (input near2 voltage) SAME (generat\$3 OR creat\$3 OR provid\$3) SAME (trigger\$3 near2 signal) SAME (when OR "if" OR based OR in\$1case OR in\$1event) SAME (input near2 voltage) SAME ((exceed\$3 OR greater OR higher OR more OR above) near5 threshold)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 01:42 PM
L55	4	(voltage near2 monitor) SAME monitor\$3 SAME (input near2 voltage) SAME (generat\$3 OR creat\$3 OR provid\$3 OR produc\$3) SAME (trigger\$3 near2 signal) SAME (when OR "if" OR based OR in\$1case OR in\$1event) SAME (input near2 voltage) SAME ((exceed\$3 OR greater OR higher OR more OR above) near5 threshold)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 01:51 PM
L56	154	(voltage near2 monitor) SAME monitor\$3 SAME (input near2 voltage) SAME (generat\$3 OR creat\$3 OR provid\$3 OR produc\$3) SAME (signal) SAME (when OR "if" OR based OR in\$1case OR in\$1event) SAME (input near2 voltage) SAME ((exceed\$3 OR greater OR higher OR more OR above) near5 threshold)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 01:53 PM
L57	19	56 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2023/07/07 01:56 PM
L58	39597	(trigger\$3 near5 condition) SAME signal	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 02:07 PM
L59	1	1 AND ((trigger\$3 near5 condition) SAME signal)	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2023/07/07 02:08 PM

			JPO; DERWENT; IBM_TDB)				
L60	178518	(voltage near5 monitor\$3).clm.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 02:45 PM
L61	10	35 AND 60	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 02:47 PM
L62	672603	(G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 04:56 PM
L63	0	55 AND 62	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA,	OR	ON	ON	2023/07/07 04:57 PM

			EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L64	0	53 AND 62	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 04:57 PM
L65	135	("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/07/07 04:58 PM
L66	17	65 AND (buck near5 converters)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/07/07 04:59 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17582797
	Filing Date	2022-01-24
	First Named Inventor	CHEN
	Art Unit	2135 2138
	Examiner Name	FARROKH
	Attorney Docket Number	022955-000090

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

1	IPR2017-00692 Paper 1, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017. (Exhibit 1020 of IPR2022-00996 and IPR2022-00999).
2	IPR2017-00692 Paper 25, Institution Decision, dated July 5, 2018. (Exhibit 1021 of IPR2022-00996 and IPR2022-00999).
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Application Number	17582797
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Attorney Docket Number	022955-000090

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34	Micron's Preliminary Identificaiton of Proposed Claim Constructions in Civil Action Nos. 6:21-cv-00431-ADA and 6:21-cv-00430-ADA, IPR2022-00418 Exhibit 1013, filed in IPR on January 14, 2022.

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That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-02-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
08/08/2023 12:17:41 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

FLASH-DRAM HYBRID MEMORY MODULE

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 62578850

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
022955-000090 Supplemental NFOA to amend title.pdf	3	Supplemental Response or Supplemental Amendment	146 KB
SADS.pdf	9	Application Data Sheet	263 KB

Warning: This is not a USPTO supplied ADS fillable form. Data in the form cannot be automatically loaded to other USPTO systems.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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022955-000090 Supplemental
NFOA to amend title.pdf

33B8CF85246E16AC90E0D1BBE88EC18E1516CAB1797EC8969
BB5E12AC9A7F3A1FD008001DEC5F93D3E1334F110194E8297
1DCB0C1F09698B52A6584E775B95FF

SADS.pdf

7A20BE0811F0FC121B335BDC0657D8AB63223419D9A6ECABA
BC7BB26E958EB93A4FF3F3A9758BA5B62C94FD9DB3E893ED
0705E3FEA55881AB90B5D1FE9166C61

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

SUPPLEMENTAL AMENDMENT

Commissioner:

In response to the non-final Office Action mailed October 12, 2022, Applicant submits the following Amendment and Remarks.

Amendments to the Specification, beginning on page **2**.

Remarks begin on page **3**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

AMENDMENT TO THE SPECIFICATION

Please amend the title of the application as follows:

~~FLASH-DRAM HYBRID MEMORY~~ WITH ON-MODULE POWER MANAGEMENT

REMARKS

The non-final Office Action mailed October 12, 2022, has been carefully considered.

Reconsideration in view of the remarks filed on April 12, 2023, is respectfully requested.

Herein, the Applicant has amended the title to read as “MEMORY WITH ON-MODULE POWER MANAGEMENT”. A Supplemental Application Data Sheet is also provided to show the amendment to the title.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: August 8, 2023
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Walnut	State/Province	CA	Country of Residence ⁱ US
Mailing Address of Inventor:				
Address 1		944 Crystal Water Lane		
Address 2				
City	Walnut	State/Province	CA	
Postal Code	91789	Country ⁱ	US	
Inventor 2				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jeffrey	C.	Solomon	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Irvine	State/Province	CA	Country of Residence ⁱ US
Mailing Address of Inventor:				
Address 1		6 Silver Fir		
Address 2				
City	Irvine	State/Province	CA	
Postal Code	92604	Country ⁱ	US	
Inventor 3				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Scott	H.	Milton	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	49 Statehouse Place			
Address 2				
City	Irvine	State/Province	CA	
Postal Code	92602	Country ⁱ	US	

Inventor 4

Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	12220 Rose Street			
Address 2				
City	Cerritos	State/Province	CA	
Postal Code	90703	Country ⁱ	US	

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.**Add****Correspondence Information:**

Enter either Customer Number or complete the Correspondence Information section below.	
For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	151145
Email Address	Add Email Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE MEMORY WITH ON-MODULE POWER MANAGEMENT		
Attorney Docket Number	022955-000090	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)			
Customer Number	151145					
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove	
Registration Number						
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove	
Registration Number						
Additional Representative Information blocks may be generated within this form by selecting the Add button.						

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	022955-000090
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
	Continuation of		17328019	2021-05-24	
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17328019	Continuation of	17138766	2020-12-30	11016918	2021-05-25
Prior Application Status		Abandoned		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
17138766	Continuation of		15934416	2018-03-23	
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
15934416	Continuation of	14840865	2015-08-31	9928186	2018-03-27
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14840865	Continuation of	14489269	2014-09-17	9158684	2015-10-13
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14489269	Continuation of	13559476	2012-07-26	8874831	2014-10-28
Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
13559476	Claims benefit of provisional		61512871	2011-07-28	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30

Prior Application Status	Abandoned	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		

Prior Application Status	Pending	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12131873	Claims benefit of provisional	60941586	2007-06-01		

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

- ☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.
- ☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

[Clear](#)

- | | | |
|---|--|--------------------------------------|
| ☒ Assignee | ☐ Legal Representative under 35 U.S.C. 117 | ☐ Joint Inventor |
| ☐ Person to whom the inventor is obligated to assign. | ☐ Person who shows sufficient proprietary interest | |

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information For Applicant:

Address 1	111 Academy Way		
Address 2	Suite 100		
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92617
Phone Number	US	Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1						
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.						
If the Assignee or Non-Applicant Assignee is an Organization check here. ☒						
Organization Name	Netlist, Inc.					
Mailing Address Information For Assignee including Non-Applicant Assignee:						
Address 1	111 Academy Way					
Address 2	Suite 100					
City	Irvine	State/Province	CA			
Country i	US	Postal Code	92617			
Phone Number	US	Fax Number				
Email Address						
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.						

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	022955-000090
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Khaled Shami/ /Khaled Shami/			Date (YYYY-MM-DD)	2022-01-24 2023-08-07
First Name	Khaled	Last Name	Shami	Registration Number	38745
Additional Signature may be generated within this form by selecting the Add button.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/582,797	Filing Date 01/24/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	08/08/2023		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	x \$100 = 0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 = 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/LAVINIA D JOHNSON/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	08/18/2023	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	
			DELIVERY MODE	
			08/18/2023	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

Advisory Action Before the Filing of an Appeal Brief	Application No. 17/582,797	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (FITF) Status No

--The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

THE REPLY FILED 08 August 2023 FAILS TO PLACE THIS APPLICATION IN CONDITION FOR ALLOWANCE.

NO NOTICE OF APPEAL FILED

1. ☒ The reply was filed after a final rejection. No Notice of Appeal has been filed. To avoid abandonment of this application, applicant must timely file one of the following replies: (1) an amendment, affidavit, or other evidence, which places the application in condition for allowance; (2) a Notice of Appeal (with appeal fee) in compliance with 37 CFR 41.31; or (3) a Request for Continued Examination (RCE) in compliance with 37 CFR 1.114 if this is a utility or plant application. Note that RCEs are not permitted in design applications. The reply must be filed within one of the following time periods:

a) ☐ The period for reply expires ____ months from the mailing date of the final rejection.

b) ☒ The period for reply expires on: (1) the mailing date of this Advisory Action; or (2) the date set forth in the final rejection, whichever is later. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of the final rejection.

c) ☐ A prior Advisory Action was mailed more than 3 months after the mailing date of the final rejection in response to a first after-final reply filed within 2 months of the mailing date of the final rejection. The current period for reply expires ____ months from the mailing date of the prior Advisory Action or SIX MONTHS from the mailing date of the final rejection, whichever is earlier.

Examiner Note: If box 1 is checked, check either box (a), (b) or (c). ONLY CHECK BOX (b) WHEN THIS ADVISORY ACTION IS THE FIRST RESPONSE TO APPLICANTS FIRST AFTER-FINAL REPLY WHICH WAS FILED WITHIN TWO MONTHS OF THE FINAL REJECTION. ONLY CHECK BOX (c) IN THE LIMITED SITUATION SET FORTH UNDER BOX (c). See MPEP 706.07(f).

Extensions of time may be obtained under 37 CFR 1.136(a). The date on which the petition under 37 CFR 1.136(a) and the appropriate extension fee have been filed is the date for purposes of determining the period of extension and the corresponding amount of the fee. The appropriate extension fee under 37 CFR 1.17(a) is calculated from: (1) the expiration date of the shortened statutory period for reply originally set in the final Office action; or (2) as set forth in (b) or (c) above, if checked. Any reply received by the Office later than three months after the mailing date of the final rejection, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

NOTICE OF APPEAL

2. ☐ The Notice of Appeal was filed on _____. A brief in compliance with 37 CFR 41.37 must be filed within two months of the date of filing the Notice of Appeal (37 CFR 41.37(a)), or any extension thereof (37 CFR 41.37(e)), to avoid dismissal of the appeal. Since a Notice of Appeal has been filed, any reply must be filed within the time period set forth in 37CFR 41.37(a).

AMENDMENTS

3. ☐ The proposed amendments filed after a final rejection, but prior to the date of filing a brief, will not be entered because

a) ☐ They raise new issues that would require further consideration and/or search (see NOTE below);

b) ☐ They raise the issue of new matter (see NOTE below);

c) ☐ They are not deemed to place the application in better form for appeal by materially reducing or simplifying the issues for appeal; and/or

d) ☐ They present additional claims without canceling a corresponding number of finally rejected claims.

NOTE: _____ (See 37 CFR 1.116 and 41.33(a)).

4. ☐ The amendments are not in compliance with 37 CFR 1.121. See attached Notice of Non-Compliant Amendment (PTOL-324).

5. ☐ Applicant's reply has overcome the following rejection(s): _____

6. ☐ Newly proposed or amended claim(s) _____ would be allowable if submitted in a separate, timely filed amendment canceling the non-allowable claim(s).

7. ☐ For purposes of appeal, the proposed amendment(s): (a) ☐ will not be entered, or (b) ☐ will be entered, and an explanation of how the new or amended claims would be rejected is provided below or appended.

AFFIDAVIT OR OTHER EVIDENCE

8. ☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____

9. ☐ The affidavit or other evidence filed after final action, but before or on the date of filing a Notice of Appeal will not be entered because applicant failed to provide a showing of good and sufficient reasons why the affidavit or other evidence is necessary and was not earlier presented. See 37 CFR 1.116(e).

10. ☐ The affidavit or other evidence filed after the date of filing the Notice of Appeal, but prior to the date of filing a brief, will not be entered because the affidavit or other evidence failed to overcome all rejections under appeal and/or appellant fails to provide a showing of good and sufficient reasons why it is necessary and was not earlier presented. See 37 CFR 41.33(d)(1).

11. ☐ The affidavit or other evidence is entered. An explanation of the status of the claims after entry is below or attached.

REQUEST FOR RECONSIDERATION/OTHER

12. ☒ The request for reconsideration has been considered but does NOT place the application in condition for allowance because:
See Continuation Sheet.

13. ☐ Note the attached Information *Disclosure Statement(s)*. (PTO/SB/08) Paper No(s). _____

14. ☐ Other: _____

STATUS OF CLAIMS

15. The status of the claim(s) is (or will be) as follows:

Claim(s) allowed: _____

Claim(s) objected to: _____

Claim(s) rejected: 2-15.

Claim(s) withdrawn from consideration: 1.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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Continuation of REQUEST FOR RECONSIDERATION/OTHER 12. The request for reconsideration has been considered but does NOT place the application in condition for allowance because: In the Office Action mailed on 07/13/2023, claims 2-15 were rejected. Applicant request for consideration does not appears to address the rejection.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

SUPPLEMENTAL AMENDMENT

Commissioner:

In response to the non-final Office Action mailed October 12, 2022, Applicant submits the following Amendment and Remarks.

Amendments to the Specification, beginning on page **2**.

Remarks begin on page **3**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

AMENDMENT TO THE SPECIFICATION

Please amend the title of the application as follows:

~~FLASH-DRAM HYBRID MEMORY~~ WITH ON-MODULE POWER MANAGEMENT

REMARKS

The non-final Office Action mailed October 12, 2022, has been carefully considered.

Reconsideration in view of the remarks filed on April 12, 2023, is respectfully requested.

Herein, the Applicant has amended the title to read as “MEMORY WITH ON-MODULE POWER MANAGEMENT”. A Supplemental Application Data Sheet is also provided to show the amendment to the title.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: August 8, 2023
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	09/21/2023	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	DELIVERY MODE
			09/21/2023	ELECTRONIC

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The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Telephonic
Khaled Shami	Attorney	

Date of Interview: 13 September 2023

Issues Discussed:

35 U.S.C. 112

In the final office action mailed on 07/13/2023, claims 2-13 were rejected under 36 USC 112(a), first paragraph as failing to comply to written description requirement. The Examiner rejected the independent claim 2 indicating the claimed specification does not describe/support the limitation of a fourth buck converter providing or supplying a fourth regulated voltage. During the interview the Applicant stated paragraph 0148 and Fig 16 of application describes/shows a dual buck converter, which provides two regulated voltages. Therefore there are four voltages supplied by three buck converters 1122, 1124 and 1126, wherein the dual buck converter 1124 is being or equivalent to two buck converters. The Examiner agreed.

☒ Attachment

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Applicant Initiated Interview Request Form

Application No.: 17/582,797

First Named Applicant: Chi-She Chen

Examiner: Farrokh, Hashem

Art Unit: 2138

Status of Application: Final/Advisory

Tentative Participants:

(1) Examiner Farrokh (2) Mr. Khaled Shami

(3) Mr. Marwan Fawal

(4)

Proposed Date of Interview: Sept. 13, 2023

Proposed Time: 3:00 (☐ AM ☒ PM)**Type of Interview Requested:**(1) ☒ Telephonic(2) ☐ Personal(3) ☐ Video ConferenceExhibit To Be Shown or Demonstrated: ☐ YES☒ NO

If yes, provide brief description: _____

Issues To Be Discussed

Issues (Rej., Obj., etc)	Claims/ Fig. #s	Prior Art	Discussed	Agreed	Not Agreed
(1) 112(a)	2-15		☐	☐	☐
(2)			☐	☐	☐
(3)			☐	☐	☐
(4)			☐	☐	☐

☐ Continuation Sheet Attached☐ Proposed Amendment or Arguments Attached

Brief Description of Arguments to be Presented: Mr. Shami to show support in the specification for claimed 4th buck converter and 4th regulated voltage

An interview was conducted on the above-identified application on _____

NOTE: This form should be completed and filed by applicant in advance of the interview (see MPEP § 713.01). If this form is signed by a registered practitioner not of record, the Office will accept this as an indication that he or she is authorized to conduct an interview on behalf of the principal (37 CFR 1.32(a)(3)) pursuant to 37 CFR 1.34. This is not a power of attorney to any above named practitioner. See the Instruction Sheet for this form, which is incorporated by reference. By signing this form, applicant or practitioner is certifying that he or she has read the Instruction Sheet. After the interview is conducted, applicant is advised to file a statement of the substance of this interview (37 CFR 1.133(b)) as soon as possible. This application will not be delayed from issue because of applicant's failure to submit a written record of this interview.

/Khaled Shami

Applicant/Applicant's Representative Signature

Khaled Shami

Typed/Printed Name of Applicant or Representative

38745

Registration Number, if applicable

Examiner/SPE Signature

650-745-5353

Applicant's/Applicant's Representative's Telephone Number

Instruction Sheet for:
APPLICANT INITIATED INTERVIEW REQUEST FORM
(Not to be Submitted to the USPTO)

1. If this form is signed by a registered practitioner not of record, the authority to submit the Applicant Initiated Interview Request Form is pursuant to limited authority to act in a representative capacity under 37 CFR 1.34 and further proof of authority to act in a representative capacity may be required. See 37 CFR 1.34.

The Office will accept the signed form as an indication that the registered practitioner not of record is authorized to conduct an interview on behalf of the principal in pursuant to 37 CFR 1.34.

For more information, see the "Conducting an Interview with a Registered Practitioner Acting in a Representative Capacity" notice which is available on the USPTO Web site at: <http://www.uspto.gov/patents/law/notices/2010.jsp>.

2. This is not a power of attorney to any named practitioner. Accordingly, any registered practitioner not of record named on the form does not have authority to sign a request to change the correspondence address, a request for express abandonment, a disclaimer, a power of attorney, or other document requiring the signature of the applicant, assignee of the entire interest or an attorney of record. If appropriate, a separate power of attorney to the named practitioner should be executed and filed in the US Patent and Trademark Office.
3. Any interview concerning an unpublished application under 35 U.S.C. § 122(b) with a registered practitioner not of record who obtains authorization through use of the PTOL-413A will be conducted based on the information and files supplied by the practitioner in view of the confidentiality requirements of 35 U.S.C. § 122(a).

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013). <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to: 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law; 2) a Federal, state, local, or international agency, in response to its request; 3) a contractor of the USPTO having need for the information in order to perform a contract; 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record; 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record; 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations; 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals; 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)); 9) the Office of Personnel Management (OPM) for personnel research purposes; and 9) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Additional Uses

Additional USPTO uses of the information in this record may include disclosure to: 1) the International Bureau of the World Intellectual Property Organization, if the record is related to an international application filed under the Patent Cooperation Treaty; 2) the public i) after publication of the application pursuant to 35 U.S.C. 122(b), ii) after issuance of a patent pursuant to 35 U.S.C. 151, iii) if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections, or an issued patent, or iv) without publication of the application or patent under the specific circumstances provided for by 37 CFR 1.14(a)(1)(v)-(vii); and/or 3) the National Archives and Records Administration, for inspection of records.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
10/13/2023 05:31:44 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 62996054

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 16

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NPL-- Ex_1073_Netlist_FAC_Segm ent_04of10.pdf	5	Non Patent Literature	12163 KB
NPL-- Ex_1073_Netlist_FAC_Segm ent_01of10.pdf	21	Non Patent Literature	18258 KB
NPL-- Ex_1073_Netlist_FAC_Segm ent_05of10.pdf	1	Non Patent Literature	21819 KB
NPL-- Ex_1073_Netlist_FAC_Segm ent_02of10.pdf	10	Non Patent Literature	21195 KB

NPL-- Ex_1073_Netlist_FAC_Segment_06of10.pdf	2	Non Patent Literature	23814 KB
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NPL-- Ex_1073_Netlist_FAC_Segment_09of10.pdf	3	Non Patent Literature	14195 KB
NPL-- Ex_1073_Netlist_FAC_Segment_10of10.pdf	5	Non Patent Literature	14286 KB
NPL-- Ex_2063_Netlist_Demonstratives_Segment_06of06.pdf	14	Non Patent Literature	7416 KB
NPL-- Ex_2063_Netlist_Demonstratives_Segment_01of06.pdf	32	Non Patent Literature	24114 KB
NPL-- Ex_2063_Netlist_Demonstratives_Segment_02of06.pdf	26	Non Patent Literature	23780 KB
NPL-- Ex_2063_Netlist_Demonstratives_Segment_04of06.pdf	32	Non Patent Literature	24053 KB

NPL-- Ex_2063_Netlist_Demonstrati ves_Segment_03of06.pdf	35	Non Patent Literature	24549 KB
NPL-- Ex_2063_Netlist_Demonstrati ves_Segment_05of06.pdf	41	Non Patent Literature	23385 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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t_07of10.pdf	C5ADDE7F3F1F42AD5549F1A59DB3
NPL-- Ex_1073_Netlist_FAC_Segmen t_08of10.pdf	C330F910049308499F93A07F5205A580CAC867D712352742429 1AD38C6341438D0E43C158ECC04F6547285E3B9F18DCCA39C 6783F9D0023A2C0D97D25A795D48
NPL-- Ex_1073_Netlist_FAC_Segmen t_09of10.pdf	8112FDFB472004F3D1A80D215089BAA2143115B4479655290B F8A660485EEF88377F412E1477D64B800DE8A5AB0D931310FA 66EBCA7F3D4FFA0130D3833C1747
NPL-- Ex_1073_Netlist_FAC_Segmen t_10of10.pdf	6BA735D896E1BAAF50B5A254E2ADBE8BCFD4806A63B4955B 2E776D124FB055EAF5AE729A304ED3976F040154141835AF9F 4029F6C5E4B016EF65B95B9B83F976
NPL-- Ex_2063_Netlist_Demonstrative s_Segment_06of06.pdf	C3B52EEC9B67028D43F4AB4FD9DED3C95C6BFD7F0AFC8FE C76231342F553F3C517CDF1F3C37C9C95F8C482062A0A9510 ABA74BC7B715598DE0343CCE21C256DD
NPL-- Ex_2063_Netlist_Demonstrative s_Segment_01of06.pdf	4AE60767351529E543248372771F1F67DF7C69F8542407EB499 34C4122902D09FF246DC11644B836B6CE476DE649462484413 A5C310A1DD63B9EAE543C3B8B02
NPL-- Ex_2063_Netlist_Demonstrative s_Segment_02of06.pdf	6BEB18CBC9E4FFF4744FF76D0C7D3AE72681C994E278FC2C 63B9F7A53D91DE2D795F4A7112D094249D1A9EB18AAB00DEE D1F0CFF47416CE9572D52DC6DE0E02A
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described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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	1	5563839	A	1996-10-08	HERDT et al.	
	2	6067241	A	2000-05-23	LU	
	3	6693840	B2	2004-02-17	SHIMADA et al.	
	4	6798709	B2	2004-09-28	SIM et al.	
	5	6856556	B1	2005-02-15	HAJECK	
	6	7023187	B2	2006-04-04	SHEARON et al.	
	7	7061214	B2	2006-06-13	MAYEGA et al.	
	8	7692938	B2	2010-04-06	PETTER	

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9	7719866	B2	2010-05-18	BOLDO	
10	7721130	B2	2010-05-18	PRETE et al.	
11	7724604	B2	2010-05-25	AMIDI et al.	
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	2	20060174140	A1	2006-08-03	HARRIS et al.	
	3	20070136523	A1	2007-06-14	BONELLA et al.	
	4	20080101147	A1	2008-05-01	AMIDI	
	5	20080238536	A1	2008-10-02	HAYASHI et al.	

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6	20090034354	A1	2009-02-05	RESNICK	
7	20100205470	A1	2010-08-12	MOSHAYEDI et al.	
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	1	Transcript, Day 2 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 17, 2023, 375 pages.	☐
	2	Transcript, Day 4 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 19, 2023, 338 pages.	☐
	3	Exhibit 1003, IPR 2022-00999, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,232,054, 431 pages, executed on May 16, 2022.	☐

**INFORMATION DISCLOSURE
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Attorney Docket Number	022955-000090

4	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 69 pages.	☐
5	Exhibit 1015, IPR 2014-00994, Decision - Denying Institution of Inter Partes Review, issued December 16, 2014, 16 pages.	☐
6	Exhibit 1016, Corrected Petition for Inter Partes Review of US Patent No. 8,301,833, filed September 22, 2014, 67 pages.	☐
7	Exhibit 1017, IPR 2014-01370, Decision - Denying Institution of Inter Partes Review, issued March 13, 2015, 19 pages.	☐
8	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
9	Exhibit 1019, IPR 2017-00649, Decision - Denying Institution of Inter Partes Review, issued July 24, 2017, 17 pages.	☐
10	Exhibit 1020, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017, 78 pages.	☐
11	Exhibit 1021, IPR 2017-00692, Final Written Decision, issued July 5, 2018, 42 pages.	☐
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13	Exhibit 1026, JEDEC STANDARD, DDR2 SDRAM SPECIFICATION, JESD79-2B (Revision of JESD7-2A), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, January 2005, 113 pages.	☐
14	Exhibit 1027, JEDEC STANDARD, FBDIMM: Advanced Memory Buffer (AMB), JESD82-20, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 190 pages.	☐

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15	Exhibit 1028, JEDEC STANDARD, FBDIMM Specification: DDR2 SDRAM Fully Buffered, DIMM (FBDIMM) Design Specification, JESD205, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 129 pages.	☐
16	Exhibit 1031, PCI Local Bus Specification, Revision 2.2, December 18, 1998, 322 pages.	☐
17	Exhibit 1032, Power Electronics: Converters, Applications, and Design, Second Edition, copyright 1995 by John Wiley & Sons, Inc., 37 pages.	☐
18	Exhibit 1040, Texas Instruments, TPS51020, Dual, Voltage Mode, DDR Selectable, Synchronous, Step-Down Controller for Notebook System Power, SLUS564B, July 2003 (revised December 2003), 27 pages.	☐
19	Exhibit 1041, Fairchild Semiconductor, FAN5026, Dual DDR/Dual-Output PWM Controller, October 2005, 17 pages.	☐
20	Exhibit 1042, Murata Manufacturing Co., Ltd, Power Supply Reference Guide for XILINX FPGAs, September 2006, 33 pages.	☐
21	Exhibit 1043, Murata Products, Power Supply Reference Guide for ALTERA FPGAs, Catalog No. DC-03-A, 2008, 40 pages.	☐
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23	Exhibit 1046, JEDEC STANDARD, DDR3 SDRAM Specification, JESD79-3A (Revision of JESD79-3), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, September 2007, 189 pages.	☐
24	Exhibit 1048, Murata Manufacturing, DC-DC Converter Specification(DRAFT), MPD4S014S, January 21, 2008, 11 pages.	☐
25	Exhibit 1049, Micron, NAND Flash Memory, MT29F2G08AABWP/MT29F2G16AABWP, MT29F4G08BABWP/MT29F4G16BABWP, MT29F8G08FABWP, copyright 2004 by Micron Technology, Inc., 57 pages.	☐

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27	Exhibit 1061, Analog Devices, Super Sequencer with Margining Control and Auxiliary ADC Inputs, ADM1066, copyright 2006 by Analog Devices, Inc., 36 pages.	☐
28	Exhibit 1062, Analog Devices, Analog Dialogue, Volume 40, Number 2, 2006, 24 pages.	☐
29	Exhibit 1063, National Semiconductor, LMC6953, PCI Local Bus Supervisor, October 1996, 12 pages.	☐
30	Exhibit 1067, Lattice Semiconductor Corporation, LatticeXP Family Data Sheet, Version 04.2, March 2006, 127 pages.	☐
31	Exhibit 1068, Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed October 15, 2021, 82 pages.	☐
32	Exhibit 1069, First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed January 18, 2022, 122 pages.	☐
33	Exhibit 1071, Complaint, filed December 20, 2021, 49 pages.	☐
34	Exhibit 1072, The Samsung Defendants' Answer, filed April 12, 2022, 37 pages.	☐
35	Exhibit 1073, First Amended Complaint, filed May 3, 2022, 71 pages.	☐
36	Exhibit 1075, IPR 2022-00999, Deposition of William Mangione-Smith, Ph.D. regarding US Patent No. 11,232,054, 245 pages, dated June 12, 2023.	☐

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37	Exhibit 1076, IPR 2022-00999, redline comparing Declarations - Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,232,054, 165 pages, executed on March 29, 2022.	☐
38	Exhibit 1077, Netlist, Technology Tutorial: In the Matter of Certain Memory Modules and Components Thereof and Products Containing Same, Inv No. 337-TA-1023, May 8, 2017, 11 pages.	☐
39	Exhibit 1078, Analog Devices, Step-Down DC-to-DC Controller, ADP1821, copyright 2005-2007 by Analog Devices, Inc., 24 pages.	☐
40	Exhibit 2001, IPR 2022-00999, Declaration of Dr. Sunil P. Khatri regarding US Patent No. 11,232,054, 68 pages, executed on September 8, 2022.	☐
41	Exhibit 2003, BELLONI et al., 24.6 A 4-Output Single-Inductor DC-DC Buck Converter with Self-Boosted Switch Drivers and 1.2A Total Output Current, ISSCC 2008 / Session 24 / Analog Power Techniques / 24.6, 2008 IEEE International Solid-State Circuits Conference, 3 pages.	☐
42	Exhibit 2004, MA et al., Single-Inductor Multiple-Output Switching Converters With Time-Multiplexing Control in Discontinuous Conduction Mode, IEEE Journal of Solid-State Circuits, Volume 38, Number 1, January 2003, pages 89-100 (12 pages).	☐
43	Exhibit 2006, Micron, Technical Note: Power Solutions for DDR2 Notebook PCs, TN-47-05 DDR2 Power Solutions for Notebooks, copyright 2004 by Micron Technology, Inc., 16 pages.	☐
44	Exhibit 2007, Texas Instruments, LP2996-N, LP2996A DDR Termination Regulator, SNOSA40K, November 2002 (revised December 2016), 39 pages.	☐
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48	Exhibit 2011, BERGVELD, Battery Management Systems: Design by Modelling, copyright 2001 by Royal Philips Electronics N.V., 328 pages.	☐
49	Exhibit 2012, ROMO, DDR Memories: Comparison and Overview, NXP, pages 70-73 (4 pages), September 21, 2007.	☐
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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-10-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	17582797	Filing Date	2022-01-24	Docket Number (if applicable)	022955-000090	Art Unit	2138
First Named Inventor	CHEN			Examiner Name	FARROKH		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

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☒ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☒ Other Response After Final Action filed August 8, 2023

☒ Enclosed

☒ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☒ Other Request for Continued Examination (RCE)

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
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SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Doc code: RCEX

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc. CONFIRMATION NO.: 3546
SERIAL NO.: 17/582,797 ART UNIT: 2138
FILING DATE: 2022-01-24 EXAMINER: FARROKH
TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Commissioner:

Notice of Prior and Concurrent Proceedings

Applicant hereby calls to the attention of the Patent and Trademark Office the following proceedings involving patents that are in the family of the above-identified patent application:

Proceeding	Status
<i>Micron Technology, Inc.; Micron Semiconductor Products, Inc.; and Micron Technology Texas LLC v. Netlist, Inc.</i> , Case No. IPR2022-00418 (P.T.A.B.), filed January 14, 2022	Final Written Decision - Determining All Challenged Claims Unpatentable entered August 28, 2023
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00615 (P.T.A.B.), filed February 17, 2022	Institution Granted October 19, 2022
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00996 (P.T.A.B.), filed May 17, 2022	Institution Granted December 7, 2022
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00999 (P.T.A.B.), filed May 17, 2022	Institution Granted December 6, 2022
2:21-cv-00463-JRG (EDTX Case asserting '918 and '054 against Samsung)	Closed - Termination Date August 11, 2023
2:22-cv-00203-JRG (EDTX Case asserting '918 and '054 against Micron)	Open - Filing Date June 10, 2022
1:22-cv-00134-DII (WDTX Case asserting the '833 against Micron)	Open - Filing Date April 28, 2021

Information Disclosure Statement

Listed on accompanying two IDS Forms PTO/SB/08a are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). Due to the Request for Continued Examination (RCE) filed concurrently, this Information Disclosure Statement is being filed before the mailing date of a first Office Action on the merits. No statement or fee is required.

Copies of documents **NPL 1-50 and NPL 1-36** (as listed on the two PTO/SB/08a forms) are submitted. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications are submitted.

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 17/328,019, filed 05-24-2021 (now U.S. Pat. No. 11,232,054), 17/138,766, filed 12-30-2020 (now U.S. Pat. No. 11,016,918), 15/934,416, filed 03-23-2018 (now abandoned), 14/840,865, filed 08-31-2015 (now U.S. Pat. No. 9,928,186); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next

communication from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

It is respectfully requested that the Examiner initial and return a copy of the enclosed two IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 22-0585.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

Date: October 13, 2023

/Khaled Shami/

Khaled Shami, Registration No.: 38745

Customer No.: 151145

VORYS, SATER, SEYMOUR and PEASE LLP

1909 K Street, NW, 9th Floor

Washington, DC 20006-1152

Tel: (202) 467-8800

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797	
	Filing Date		2022-01-24	
	First Named Inventor	CHEN		
	Art Unit	2138		
	Examiner Name	FARROKH		
	Attorney Docket Number	022955-000090		

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

1	Exhibit 2014, Qimonda, Internet Data Sheet, Rev. 1.25, June 2007, 24 pages.	☐
2	Exhibit 2015, Siemens, 3.3V 2M x 64/72-Bit 1 BANK SDRAM Module, HYS64/72V2200GU-8/-10, 3.3V 4M x 64/72-Bit 2 BANK SDRAM Module, HYS64/72V4200GU-8/-10, 168 pin unbuffered DIMM Modules, downloaded from www.datasheetcatalog.com; 1998, 18 pages	☐
3	Exhibit 2016, PCI Bus Variation - Technology Note, copyright 2006 by Euresys, 9 pages.	☐
4	Exhibit 2017, Qimonda, Internet Data Sheet, Rev. 1.42, September 2007, 27 pages.	☐
5	Exhibit 2018, Transcend, Product Support, What is the difference between SDRAM, DDR1, DDR2, DDR3 and DDR4?, available online at https://www.transcend-info.com/support/faq-296#:~:text=DDR3's prefetch buffer width is,according to the temperature variation , downloaded August 24, 2022, 2 pages.	☐
6	Exhibit 2021, Texas Instruments: Application Report, Low Dropout Operation in a Buck Converter, SLUA928A - December 2018 - revised March 2019, 18 pages.	☐
7	Exhibit 2022, Electronic Design, Simple Soft-Start Circuitry Provides Long Startup Times, available online at http://electronicdesign.com/power/simple-soft-start-circuitry-provides-long-startup-times , published June 22, 1998, 3 pages.	☐
8	Exhibit 2024, SCHMID, Understanding Hard Drive Performance, available online at https://www.tomshardware.com/reviews/understanding-hard-drive-performance,1557-5.html , downloaded August 15, 2022, published March 5, 2007, 17 pages.	☐
9	Exhibit 2025, Micron, SDR SDRAM, MT48LC64M4A2 - 16 Meg x 4 x 4 Banks, MT48LC32M8A2 - 8 Meg x 8 x 4 Banks, MT48LC16M16A2 - 4 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 94 pages.	☐
10	Exhibit 2026, Micron, SDR SDRAM, MT48LC16M4A2 - 4 Meg x 4 x 4 Banks, MT48LC8M8A2 - 2 Meg x 8 x 4 Banks, MT48LC4M16A2 - 1 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 84 pages.	☐
11	Exhibit 2027, Transcend, DDR2 SO-DIMM, copyright 2012 by Transcend, 3 pages.	☐

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Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

12	Exhibit 2028, Micron Technical Note: DDR3 Point-to-Point Design Support, copyright 2013 by Micron Technology, Inc., 27 pages.	☐
13	Exhibit 2030, IPR 2022-00996 and IPR 2022-00999, Videoconference Deposition of Andrew Wolfe, Ph.D. regarding US Patent Nos. 11,016,918 and 11,232,054, 311 pages, dated March 16, 2023.	☐
14	Exhibit 2031, IPR 2022-00996, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,016,918, 452 pages, executed on May 16, 2022.	☐
15	Exhibit 2031, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
16	Exhibit 2032, Claim Construction Order, filed December 14, 2022, 35 pages.	☐
17	Exhibit 2033, Defendants' Objections to Claim Construction Memorandum Order, filed December 29, 2022, 11 pages.	☐
18	Exhibit 2043, Samsung Unveils New Power Management Solutions for DDR5 Modules, May 18, 2021, 6 pages.	☐
19	Exhibit 2044, Micron, DDR2 SDRAM RDIMM, MT9HTF3272Y - 256MB, MT9HTF6472PY - 512MB, MT9HTF12872PY - 1GB, copyright 2003 by Micron Technology, Inc., 23 pages.	☐
20	Exhibit 2046, JEDEC Standard No. 21C, Page 4.20.11-1, PC2-6400/PC2-5300/PC2-4200/PC2-3200, DDR2 Unbuffered SO-DIMM, Reference Design Specification, Revision 2.5, July 2008, 60 pages.	☐
21	Exhibit 2048, Micron, DDR2 SDRAM FBDIMM, MT18HTF12872FD - 1GB, MT18HTF25672FD - 2GB, copyright 2005 by Micron Technology, Inc., 39 pages.	☐
22	Exhibit 2056, IPR 2022-00615, Remote Videotaped Deposition by Virtual Zoom of Andrew Wolfe, Ph.D. regarding US Patent No. 7,619,912, 297 pages, dated January 4, 2023.	☐

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23	Exhibit 2057, Micron Technical Brief: Micron DDR5: Key Module Features, DDR4 vs. DDR5 DIMM, copyright 2020 by Micron Technology, Inc., 7 pages.	☐
24	Exhibit 2063, Netlist, Demonstrative Exhibit for IPR 2022-00996 and IPR 2022-00999, filed September 6, 2023, 180 pages.	☐
25	Exhibit 2102, LEE et al., PCI Express Power and Mini Card Solutions - Design Note 346, Linear Technology - Design Notes, copyright 2004 by Linear Technology Corporation, 2 pages.	☐
26	Paper 1, IPR 2022-00996, Petition for Inter Partes Review of US Patent No. 11,016,918, filed May 17, 2022, 151 pages.	☐
27	Paper 1, IPR 2022-00999, Petition for Inter Partes Review of US Patent No. 11,232,054, filed May 17, 2022, 149 pages.	☐
28	Paper 7, IPR 2022-00996, Patent Owner Preliminary Response, filed September 9, 2022, 85 pages.	☐
29	Paper 7, IPR 2022-00999, Patent Owner Preliminary Response, filed September 8, 2022, 82 pages.	☐
30	Paper 10, IPR 2022-00996, Decision - Granting Institution of Inter Partes Review, dated December 7, 2022, 56 pages.	☐
31	Paper 21, IPR 2022-00996, Patent Owner Response, filed March 28, 2023, 91 pages.	☐
32	Paper 22, IPR 2022-00999, Patent Owner Response, filed March 28, 2023, 90 pages.	☐
33	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐

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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

34	Paper 26, IPR 2022-00999, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
35	Paper 31, IPR 2022-00996, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐
36	Paper 33, IPR 2022-00999, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-10-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
10/13/2023 05:28:08 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 62993059

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585

PAYMENT TRANSACTION ID
E20230CH28438201

PAYMENT AUTHORIZED BY
Lisa Elliott

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1801	REQUEST FOR CONTINUED EXAMINATION (RCE) - 1ST REQUEST (SEE 37 CFR 1.114)	1360.00	1	1360.00
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
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PATENT CENTER # 62993059

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 96

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-10-13 - IDS Pleading - 022955-000090.pdf	3	Miscellaneous Incoming Letter	146 KB
2023-10-13 - IDS1 - 022955- 000090.pdf	10	Information Disclosure Statement (IDS) Form (SB08)	1444 KB
2023-10-13 - IDS2 - 022955- 000090.pdf	7	Information Disclosure Statement (IDS) Form (SB08)	1406 KB
2023-10-13 - RCE - 022955- 000090.pdf	3	Request for Continued Examination (RCE)	1368 KB
2023-10-13 - Resp to 7-13-23 FOA - 022955-000090.pdf	8	Amendment Submitted/Entered with Filing of Continued	152 KB

		Prosecution Application (CPA)/Request for Continued Examination(RCE)	
NPL--Day 4 redacted.pdf	338	Non Patent Literature	733 KB
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NPL-- Ex_1071_Complaint_EDTX.p df	49	Non Patent Literature	4082 KB
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage

submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

RESPONSE AFTER FINAL REJECTION

Commissioner:

In response to the Final Office Action mailed July 13, 2023, and the Advisory Action mailed August 18, 2023, Applicant submits the following Amendment and Remarks.

Listing of Claims begins on page **2**.

Remarks begin on page **6**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) comprising:

a printed circuit board (PCB) including a set of voltage supply lines electrically connected to corresponding edge connections of a first set of edge connections, sixty four data lines electrically connected to corresponding edge connections of a second set of edge connections, and a set of address and control lines electrically connected to corresponding edge connections of a third set of edge connections;

a DIMM interface including the first, second and third set of edge connections, the DIMM interface configured to (i) fit into a memory socket connector of a host system, and (ii) provide a conduit for power, sixty four data signals, and address and control signals between the host system and the DIMM via the first, second and third set of edge connections, respectively;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) receive and monitor an input voltage from the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to other portions of the controller, wherein the controller is configured to perform one or more operations in response to the trigger signal, the one or more operations including a write operation to write data into the non-volatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a plurality of components coupled to the PCB, the plurality of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices; and

first, second, third and fourth buck converters coupled to the PCB and to the input

voltage supply line and configured to (i) receive power from the input voltage supply line, (ii) generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and (iii) output the first, second, third and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively;

wherein each component of the plurality of components (i) is not electrically connected to the input voltage supply line, (ii) is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and (iii) receives power only from one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices (i) is not electrically connected to the input voltage supply line, and (ii) receives power only from one or more of the first, second, third and fourth regulated voltage lines.

3. (Previously presented) The DIMM of claim 2, wherein the controller is further configured to control a plurality of power transition sequences.

4. (Currently amended) The DIMM of claim 3, wherein the plurality of power transition sequences includes first and second power transition sequences, the first power transition sequence includes turning power on or off to the first regulated voltage line in response to a first trigger condition of a plurality of trigger conditions, ~~and the plurality of trigger conditions includes a request from the host.~~

5. (Currently amended) The DIMM of claim 4, wherein the input voltage supply line is a first input voltage supply line, and wherein the first, second, third and fourth buck converters are coupled to a second input voltage supply line, and

wherein, during the second power transition sequence, the first, second, third and fourth buck converters are further configured to (i) receive power from the first input voltage supply line or the second input voltage supply line in response to a second trigger condition of the plurality of trigger conditions, (ii) generate, respectively, the first, second, third and fourth regulated voltages from either the power received from the first input voltage supply line or the power received from the second input voltage supply line, and (iii) ~~output~~ output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively.

6. (Currently amended) The DIMM of claim 24, wherein the controller is further configured to generate a respective trigger signal of a plurality of trigger signals in connection response to with a respective trigger condition of the plurality of trigger conditions, wherein the plurality of trigger conditions ~~that~~ include (i) a request from the host system ~~and or~~ (ii) the voltage monitor circuit detecting input voltage level deviation from one or more threshold voltages.

7. (Previously presented) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages specified as certain voltage values.

8. (Previously presented) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages corresponding to percentages of an operating voltage.

9. (Previously presented) the DIMM of claim 4, wherein the DIMM is configured to operate in a plurality of states including a first state and a second state, wherein the DIMM is further configured (i) to operate in the first state while power is turned on to the first regulated voltage line, and (ii) to operate in the second state while power is turned off to the first regulated voltage line.

10. (Previously presented) The DIMM of claim 9, wherein the first power transition sequence further includes turning power on to the plurality of DDR SDRAM devices, during both the first and second states, via one or more of the second, third and fourth regulated voltage lines.

11. (Currently amended) The DIMM of claim 10 further comprising:

a register coupled to the PCB, ~~to~~ the first regulated voltage line, and ~~to~~ the set of address and control lines, wherein:

during the first state, the register is configured to (i) receive and register address and control signals from the host via the set of address and control lines, and (ii) output registered address and control signals ~~to the plurality of the DDR SDRAM devices, and~~

during the second state, the register does not ~~couple output registered~~ address and control signals ~~from the host to the plurality of the DDR SDRAM devices.~~

12. (Previously presented) The DIMM of claim 2, wherein the DIMM interface comprises a standard 240-pin DDR2 edge connector.
13. (Previously presented) The DIMM of claim 2, wherein the controller is further configured to periodically perform the one or more operations at regular intervals of time.
14. (Previously presented) The DIMM of claim 13, wherein the regular intervals of time are every millisecond.
15. (Previously presented) The DIMM of claim 2, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

REMARKS

The Final Office Action mailed July 13, 2023, has been carefully considered.

Interview Record

Applicants gratefully acknowledge the courtesy extended to the undersigned representative during the Examiner interview conducted on September 13, 2023 between Mr. Khaled Shami, Mr. Marwan Fawal, and Examiner Hashem Farrokh. The substance of the interview related to the 35 USC 112(a) rejection as discussed below.

Rejection(s) Under 35 U.S.C. § 112, First Paragraph

Claims 2-15 stand rejected under 35 U.S.C. § 112, first paragraph, as allegedly failing to comply with the written description requirement. The claims allegedly contain subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-A/A 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Applicants respectfully disagree. The subject matter at issue is a fourth buck converter and a fourth regulated voltage, and as pointed out during the Examiner interview, paragraphs [00148] – [00149] of the originally-filed description, as well as FIG. 16 of the drawings, are example locations where support for these claimed components can be found.

Specifically, voltage conversion element 1120 is shown in FIG. 16 as comprising a buck converter 1122 (*i.e.* 1st buck converter); dual buck converter 1124 (*i.e.* 2nd and 3rd buck converters); and buck-boost converter 1126 (*i.e.* 4th buck converter). In addition, paragraph [00148] states that “[t]he conversion element 1120 may comprise a plurality of subblocks 1122, 1124, 1126 as schematically illustrated by FIG. 16,” that “[t]he sub-blocks [1122, 1124, 1126] may comprise various converter circuits such as buck-converters, boost converters, and buck-

boost converter circuits,” and that “sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16.” (ll. 4-9). Hence there are four buck converters described in [0148] that are referenced to schematic illustration of FIG. 16.

FIG. 16 schematically illustrates and labels the sub-blocks 1122, 1124, 1126 of the conversion element 1120 matching the description of paragraph [0148]. Specifically, the sub-block 1122 is labeled “Buck Converter”, the sub-block 1124 is labeled “Dual Buck Reg/Conv”, and the sub-block 1126 is labeled “Buck-Boost Converter”. Hence there are four buck converters illustrated in FIG. 16, which further shows the buck converters 1122, 1124, 1126 outputting four regulated voltages: 1102, 1104, 1105 and 1107.

Further support for a fourth buck converter and fourth regulated voltage can be found in paragraph [00149], which states that “[f]or example, there may be more or less sub-blocks [in conversion element 1120] which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values.” (ll. 3-5).

For these reasons at least, Applicants respectfully maintain that the claimed fourth buck converter and fourth regulated voltage are sufficiently described in the specification to meet the written description requirement of 35 U.S.C. § 112, and the rejection thereunder should be withdrawn.

Amendment to Claims

Claims 2, 4-6 and 11 have been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references. Any remarks made herein with respect to a given claim or amendment is intended

only in the context of that specific claim or amendment, and should not be applied to other claims, amendments or aspects of the claimed arrangement.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

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/DRAFT/
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/582,797	Filing Date 01/24/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)	
AMENDMENT	10/13/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	x \$100 =	0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 =	0	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE	0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =		
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =		
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						HSLIE		
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/DEREK A SAUNDERS/		
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".								
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.								

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
10/17/2023 02:49:06 PM ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lisa Elliott

PATENT CENTER # 63014851

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
022955-000090 Supplemental Resp to 7-13- 23 FOA.pdf	8	Supplemental Response or Supplemental Amendment	152 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
022955-000090 Supplemental Resp to 7-13-23 FOA.pdf	9232DB88AFA16DFC24CF93DD50D9B19CC95244C7843EDD0D B7300698B356CA0AA6EE113BFE2723B30FA2A30FE04C4ED03 3965B9950B306382E179D1B20DE98E8

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

SUPPLEMENTAL RESPONSE AFTER FINAL REJECTION

Commissioner:

This supplemental response is being filed in order to provide the signature of Applicant's representative. The previous response, filed on October 13, 2023, was marked "DRAFT" and was not signed or dated.

In response to the Final Office Action mailed July 13, 2023, and the Advisory Action mailed August 18, 2023, Applicant submits the following Amendment and Remarks.

Listing of Claims begins on page **2**.

Remarks begin on page **6**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) comprising:

a printed circuit board (PCB) including a set of voltage supply lines electrically connected to corresponding edge connections of a first set of edge connections, sixty four data lines electrically connected to corresponding edge connections of a second set of edge connections, and a set of address and control lines electrically connected to corresponding edge connections of a third set of edge connections;

a DIMM interface including the first, second and third set of edge connections, the DIMM interface configured to (i) fit into a memory socket connector of a host system, and (ii) provide a conduit for power, sixty four data signals, and address and control signals between the host system and the DIMM via the first, second and third set of edge connections, respectively;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) receive and monitor an input voltage from the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to other portions of the controller, wherein the controller is configured to perform one or more operations in response to the trigger signal, the one or more operations including a write operation to write data into the non-volatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a plurality of components coupled to the PCB, the plurality of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices; and

first, second, third and fourth buck converters coupled to the PCB and to the input

voltage supply line and configured to (i) receive power from the input voltage supply line, (ii) generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and (iii) output the first, second, third and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively;

wherein each component of the plurality of components (i) is not electrically connected to the input voltage supply line, (ii) is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and (iii) receives power only from one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices (i) is not electrically connected to the input voltage supply line, and (ii) receives power only from one or more of the first, second, third and fourth regulated voltage lines.

3. (Previously presented) The DIMM of claim 2, wherein the controller is further configured to control a plurality of power transition sequences.

4. (Currently amended) The DIMM of claim 3, wherein the plurality of power transition sequences includes first and second power transition sequences, the first power transition sequence includes turning power on or off to the first regulated voltage line in response to a first trigger condition of a plurality of trigger conditions, ~~and the plurality of trigger conditions includes a request from the host.~~

5. (Currently amended) The DIMM of claim 4, wherein the input voltage supply line is a first input voltage supply line, and wherein the first, second, third and fourth buck converters are coupled to a second input voltage supply line, and

wherein, during the second power transition sequence, the first, second, third and fourth buck converters are further configured to (i) receive power from the first input voltage supply line or the second input voltage supply line in response to a second trigger condition of the plurality of trigger conditions, (ii) generate, respectively, the first, second, third and fourth regulated voltages from either the power received from the first input voltage supply line or the power received from the second input voltage supply line, and (iii) ~~output~~ output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively.

6. (Currently amended) The DIMM of claim 24, wherein the controller is further configured to generate a respective trigger signal of a plurality of trigger signals in connection response to with a respective trigger condition of the plurality of trigger conditions, wherein the plurality of trigger conditions ~~that~~ include (i) a request from the host system ~~and or~~ (ii) the voltage monitor circuit detecting input voltage level deviation from one or more threshold voltages.

7. (Previously presented) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages specified as certain voltage values.

8. (Previously presented) The DIMM of claim 6, wherein the one or more threshold voltages include threshold voltages corresponding to percentages of an operating voltage.

9. (Previously presented) the DIMM of claim 4, wherein the DIMM is configured to operate in a plurality of states including a first state and a second state, wherein the DIMM is further configured (i) to operate in the first state while power is turned on to the first regulated voltage line, and (ii) to operate in the second state while power is turned off to the first regulated voltage line.

10. (Previously presented) The DIMM of claim 9, wherein the first power transition sequence further includes turning power on to the plurality of DDR SDRAM devices, during both the first and second states, via one or more of the second, third and fourth regulated voltage lines.

11. (Currently amended) The DIMM of claim 10 further comprising:

a register coupled to the PCB, ~~to~~ the first regulated voltage line, and ~~to~~ the set of address and control lines, wherein:

during the first state, the register is configured to (i) receive and register address and control signals from the host via the set of address and control lines, and (ii) output registered address and control signals ~~to the plurality of the DDR SDRAM devices, and~~

during the second state, the register does not ~~couple output registered~~ address and control signals ~~from the host to the plurality of the DDR SDRAM devices.~~

12. (Previously presented) The DIMM of claim 2, wherein the DIMM interface comprises a standard 240-pin DDR2 edge connector.
13. (Previously presented) The DIMM of claim 2, wherein the controller is further configured to periodically perform the one or more operations at regular intervals of time.
14. (Previously presented) The DIMM of claim 13, wherein the regular intervals of time are every millisecond.
15. (Previously presented) The DIMM of claim 2, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

REMARKS

The Final Office Action mailed July 13, 2023, has been carefully considered.

Interview Record

Applicants gratefully acknowledge the courtesy extended to the undersigned representative during the Examiner interview conducted on September 13, 2023 between Mr. Khaled Shami, Mr. Marwan Fawal, and Examiner Hashem Farrokh. The substance of the interview related to the 35 USC 112(a) rejection as discussed below.

Rejection(s) Under 35 U.S.C. § 112, First Paragraph

Claims 2-15 stand rejected under 35 U.S.C. § 112, first paragraph, as allegedly failing to comply with the written description requirement. The claims allegedly contain subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-A/A 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Applicants respectfully disagree. The subject matter at issue is a fourth buck converter and a fourth regulated voltage, and as pointed out during the Examiner interview, paragraphs [00148] – [00149] of the originally-filed description, as well as FIG. 16 of the drawings, are example locations where support for these claimed components can be found.

Specifically, voltage conversion element 1120 is shown in FIG. 16 as comprising a buck converter 1122 (*i.e.* 1st buck converter); dual buck converter 1124 (*i.e.* 2nd and 3rd buck converters); and buck-boost converter 1126 (*i.e.* 4th buck converter). In addition, paragraph [00148] states that “[t]he conversion element 1120 may comprise a plurality of subblocks 1122, 1124, 1126 as schematically illustrated by FIG. 16,” that “[t]he sub-blocks [1122, 1124, 1126] may comprise various converter circuits such as buck-converters, boost converters, and buck-

boost converter circuits,” and that “sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16.” (ll. 4-9). Hence there are four buck converters described in [0148] that are referenced to schematic illustration of FIG. 16.

FIG. 16 schematically illustrates and labels the sub-blocks 1122, 1124, 1126 of the conversion element 1120 matching the description of paragraph [0148]. Specifically, the sub-block 1122 is labeled “Buck Converter”, the sub-block 1124 is labeled “Dual Buck Reg/Conv”, and the sub-block 1126 is labeled “Buck-Boost Converter”. Hence there are four buck converters illustrated in FIG. 16, which further shows the buck converters 1122, 1124, 1126 outputting four regulated voltages: 1102, 1104, 1105 and 1107.

Further support for a fourth buck converter and fourth regulated voltage can be found in paragraph [00149], which states that “[f]or example, there may be more or less sub-blocks [in conversion element 1120] which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values.” (ll. 3-5).

For these reasons at least, Applicants respectfully maintain that the claimed fourth buck converter and fourth regulated voltage are sufficiently described in the specification to meet the written description requirement of 35 U.S.C. § 112, and the rejection thereunder should be withdrawn.

Amendment to Claims

Claims 2, 4-6 and 11 have been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references. Any remarks made herein with respect to a given claim or amendment is intended

only in the context of that specific claim or amendment, and should not be applied to other claims, amendments or aspects of the claimed arrangement.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: October 17, 2023
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/582,797		Filing Date 01/24/2022		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
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APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	10/17/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0			x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0			x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
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151145 7590 11/08/2023
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

EXAMINER	
FARROKH, HASHEM	
ART UNIT	PAPER NUMBER
2138	

DATE MAILED: 11/08/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/08/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

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If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

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151145 7590 11/08/2023
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

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(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

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nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/08/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2138	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via Patent Center or EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	11/08/2023	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
DATE MAILED: 11/08/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/582,797	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (FITF) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 10/17/2023.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-15. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>10/13/2023</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION:

2. This office action is in response to the communication(s) filed on 10/17/2023. There are a total of 14 claims pending in the application. Claims 2, 4-6 and 11 have been amended; claim 1 has been canceled; and no new claims have been added.

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 10/13/2023 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the Examiner. However, the documents listed as non-patent literature documents and shown by lined-through have not be considered by the Examiner. The reason for not considering is that title of actual documents are not readily identifiable and/or are different from IDS list. For example, in IDS listing many documents include an Exhibit number, which are not included in the actual documents. Furthermore, many of documents listed appear to have large number pages with small portion of pages may be relevant to the claims of current application. It is being suggested that applicant(s) include a title for each of one-to-one correspond to the title that has been listed, point to the portion(s) of the listed non-literature documents, which is/are directly relevant to the claims of the instant application. See MPEP 609.04(a) for more information.

RESPONSE TO REMARKS:

4. Applicant's arguments have been fully considered and the have been persuasive. In the final office action mailed 07/13/2023, the Examiner rejected claims 2-15 under 35 USC § 112(a), first paragraph, indication there is no description/support for a fourth buck converter as claimed in the independent claim 2. However, The Applicant states that each of the sub-blocks 1122, 1124, and 1126 in the conversion element 1120 in the Fig.16 matching description of paragraph [0148] describe or support buck converters,

including the fourth buck converter. Specifically, sub-block 1122 is buck converter, sub-block 1126 is buck-boost converter, and sub-block 1124 is a dual buck converter is generating two regulated output voltages. Thus, the three buck converters generate or supply 4 regulated voltages (e.g., page 6 and 7 of the Remarks, see also the interview summary mailed 09/21/2023).

REASON FOR ALLOWANCE:

5. *Claims 2-14 are considered to be allowable.*

The reason for allowance of the independent claim 2 are the following: in interpreting the pending claim(s), in view of the specification, the Examiner finds the claimed invention to be patentably distinct from the prior art of record. The prior art of record neither anticipates , nor renders obvious the recited combination as a whole including the limitation(s):

“first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line, generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and output the first, second, third and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively; wherein each component of the plurality of components is not electrically connected to the input voltage supply line, is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only from one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices is not electrically connected to the input voltage supply line, and receives power only from one or more of the first, second, third and fourth regulated voltage lines.”

6. *Dependent claims 3-15 are allowable at least for the reasons indicated above including all the limitations of the allowable base claim 2 upon which they depend.*

7. *Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably*

accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

DIRECTION OF FUTURE CORRESPONDENCES:

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

9. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

10. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF
711	110	11/01/2023	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF
Inventor/Assignee Search Updated	10/31/2023	HF
EAST Text, CPC, FACET Search Updated	10/31/2023	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	11/01/2023	HF
G06F13/28	G06F13/00	11/01/2023	HF
G06F2212/205	G06F2212/00	11/01/2023	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	14	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	01 November 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	2	6

<i>Issue Classification</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	3	/ 06
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	110

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	14	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	01 November 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	2	6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1	9	10												
1	2	10	11												
2	3	11	12												
3	4	12	13												
4	5	13	14												
5	6	14	15												
6	7														
7	8														
8	9														

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 14	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner) _____ (Date) _____		01 November 2023 O.G. Print Claim(s) 2	O.G. Print Figure 6

Bibliographic Data

Application No: 17/582,797

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /HASHEM FARROKH/

Examiner's Signature

Initials

Title:

MEMORY WITH ON-MODULE POWER MANAGEMENT

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2022	711	2138	
RULE			

APPLICANTS

Netlist, Inc., Irvine, CA, UNITED STATES

INVENTORS

Chi-She Chen, Walnut, CA, UNITED STATES

Jeffrey C. Solomon, Irvine, CA, UNITED STATES

Scott H. Milton, Irvine, CA, UNITED STATES

Jayesh Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17328019 05/24/2021 PAT 11232054

17328019 is a CON of 17138766 12/30/2020 PAT 11016918

17138766 is a CON of 15934416 03/23/2018ABN

15934416 is a CON of 14840865 08/31/2015 PAT 9928186

14840865 is a CON of 14489269 09/17/2014 PAT 9158684

14489269 is a CON of 13559476 07/26/2012 PAT 8874831

13559476 has PRO of 61512871 07/28/2011

13559476 is a CIP of 12240916 09/29/2008 PAT 8301833

12240916 is a CON of 12131873 06/02/2008ABN

12131873 has PRO of 60941586 06/01/2007

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

02/01/2022

STATE OR COUNTRY

UNITED STATES

ADDRESS

FILING FEE RECEIVED

\$1,980

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797	
	Filing Date		2022-01-24	
	First Named Inventor	CHEN		
	Art Unit	2138		
	Examiner Name	FARROKH		
	Attorney Docket Number	022955-000090		

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

1	Exhibit 2014, Qimonda, Internet Data Sheet, Rev. 1.25, June 2007, 24 pages.	☐
2	Exhibit 2015, Siemens, 3.3V 2M x 64/72 Bit 1 BANK SDRAM Module, HYS64/72V2200GU-8/-10, 3.3V 4M x 64/72 Bit 2 BANK SDRAM Module, HYS64/72V4200GU-8/-10, 168 pin unbuffered DIMM Modules, downloaded from www.datasheetcatalog.com; 1998, 18 pages	☐
3	Exhibit 2016, PCI Bus Variation Technology Note; copyright 2006 by Euresys, 9 pages.	☐
4	Exhibit 2017, Qimonda, Internet Data Sheet, Rev. 1.42, September 2007, 27 pages.	☐
5	Exhibit 2018, Transcend, Product Support, What is the difference between SDRAM, DDR1, DDR2, DDR3 and DDR4?, available online at https://www.transcend-info.com/support/faq-296#:~:text=DDR3's prefetch buffer width is,according to the temperature variation , downloaded August 24, 2022, 2 pages.	☐
6	Exhibit 2021, Texas Instruments: Application Report, Low Dropout Operation in a Buck Converter, SLUA928A, December 2018 - revised March 2019, 18 pages.	☐
7	Exhibit 2022, Electronic Design, Simple Soft-Start Circuitry Provides Long Startup Times, available online at http://electronicdesign.com/power/simple-soft-start-circuitry-provides-long-startup-times , published June 22, 1998, 3 pages.	☐
8	Exhibit 2024, SCHMID, Understanding Hard Drive Performance, available online at https://www.tomshardware.com/reviews/understanding-hard-drive-performance,1557-5.html , downloaded August 15, 2022, published March 5, 2007, 17 pages.	☐
9	Exhibit 2025, Micron, SDR SDRAM, MT48LC64M4A2 - 16 Meg x 4 x 4 Banks, MT48LC32M8A2 - 8 Meg x 8 x 4 Banks, MT48LC16M16A2 - 4 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 94 pages.	☐
10	Exhibit 2026, Micron, SDR SDRAM, MT48LC16M4A2 - 4 Meg x 4 x 4 Banks, MT48LC8M8A2 - 2 Meg x 8 x 4 Banks, MT48LC4M16A2 - 1 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 84 pages.	☐
11	Exhibit 2027, Transcend, DDR2 SO-DIMM, copyright 2012 by Transcend, 3 pages.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit		2138
Examiner Name	FARROKH	
Attorney Docket Number		022955-000090

12	Exhibit 2028, Micron Technical Note: DDR3 Point to Point Design Support, copyright 2013 by Micron Technology, Inc. 27 pages.	☐
13	Exhibit 2030, IPR 2022-00996 and IPR 2022-00999, Videoconference Deposition of Andrew Wolfe, Ph.D. regarding US Patent Nos. 11,016,918 and 11,232,054, 311 pages, dated March 16, 2023.	☐
14	Exhibit 2031, IPR 2022-00996, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,016,918, 452 pages, executed on May 16, 2022.	☐
15	Exhibit 2034, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
16	Exhibit 2032, Claim Construction Order, filed December 14, 2022, 35 pages.	☐
17	Exhibit 2033, Defendants' Objections to Claim Construction Memorandum Order, filed December 29, 2022, 11 pages.	☐
18	Exhibit 2043, Samsung Unveils New Power Management Solutions for DDR5 Modules, May 18, 2021, 6 pages.	☐
19	Exhibit 2044, Micron, DDR2 SDRAM RDIMM, MT0HTF3272Y, 256MB, MT0HTF6472PY, 512MB, MT0HTF12872PY, 1GB, copyright 2003 by Micron Technology, Inc., 23 pages.	☐
20	Exhibit 2046, JEDEC Standard No. 21G, Page 4-20-11-1, PG2-6400/PG2-5300/PG2-4200/PG2-3200, DDR2 Unbuffered SO-DIMM, Reference Design Specification, Revision 2.5, July 2008, 60 pages.	☐
21	Exhibit 2048, Micron, DDR2 SDRAM FBDIMM, MT18HTF12872FD, 1GB, MT18HTF25672FD, 2GB, copyright 2005 by Micron Technology, Inc., 39 pages.	☐
22	Exhibit 2056, IPR 2022-00645, Remote Videotaped Deposition by Virtual Zoom of Andrew Wolfe, Ph.D. regarding US Patent No. 7,619,912, 297 pages, dated January 4, 2023.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

23	Exhibit 2057, Micron Technical Brief: Micron DDR5: Key Module Features, DDR4 vs. DDR5 DIMM, copyright 2020 by Micron Technology, Inc., 7 pages.	☐
24	Exhibit 2069, Netlist, Demonstrative Exhibit for IPR 2022-00990 and IPR 2022-00999, filed September 8, 2023, 180 pages.	☐
25	Exhibit 2102, LEE et al., PCI Express Power and Mini Card Solutions - Design Note 346, Linear Technology - Design Notes, copyright 2004 by Linear Technology Corporation, 2 pages.	☐
26	Paper 1, IPR 2022-00996, Petition for Inter Partes Review of US Patent No. 11,016,918, filed May 17, 2022, 151 pages.	☐
27	Paper 1, IPR 2022-00999, Petition for Inter Partes Review of US Patent No. 11,232,054, filed May 17, 2022, 149 pages.	☐
28	Paper 7, IPR 2022-00996, Patent Owner Preliminary Response, filed September 8, 2022, 85 pages.	☐
29	Paper 7, IPR 2022-00999, Patent Owner Preliminary Response, filed September 8, 2022, 82 pages.	☐
30	Paper 10, IPR 2022-00996, Decision - Granting Institution of Inter Partes Review, dated December 7, 2022, 56 pages.	☐
31	Paper 21, IPR 2022-00996, Patent Owner Response, filed March 28, 2023, 91 pages.	☐
32	Paper 22, IPR 2022-00999, Patent Owner Response, filed March 28, 2023, 90 pages.	☐
33	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐

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34	Paper 26, IPR 2022-00000, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
35	Paper 31, IPR 2022-00006, Patent Owner's Sur Reply, filed August 4, 2023, 48 pages.	☐
36	Paper 32, IPR 2022-00000, Patent Owner's Sur Reply, filed August 4, 2023, 48 pages.	☐

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Examiner Signature	/HASHEM FARROKH/	Date Considered	10/30/2023
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-10-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc. CONFIRMATION NO.: 3546
SERIAL NO.: 17/582,797 ART UNIT: 2138
FILING DATE: 2022-01-24 EXAMINER: FARROKH
TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Commissioner:

Notice of Prior and Concurrent Proceedings

Applicant hereby calls to the attention of the Patent and Trademark Office the following proceedings involving patents that are in the family of the above-identified patent application:

Proceeding	Status
<i>Micron Technology, Inc.; Micron Semiconductor Products, Inc.; and Micron Technology Texas LLC v. Netlist, Inc.</i> , Case No. IPR2022-00418 (P.T.A.B.), filed January 14, 2022	Final Written Decision - Determining All Challenged Claims Unpatentable entered August 28, 2023
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00615 (P.T.A.B.), filed February 17, 2022	Institution Granted October 19, 2022
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00996 (P.T.A.B.), filed May 17, 2022	Institution Granted December 7, 2022
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00999 (P.T.A.B.), filed May 17, 2022	Institution Granted December 6, 2022
2:21-cv-00463-JRG (EDTX Case asserting '918 and '054 against Samsung)	Closed - Termination Date August 11, 2023
2:22-cv-00203-JRG (EDTX Case asserting '918 and '054 against Micron)	Open - Filing Date June 10, 2022
1:22-cv-00134-DII (WDTX Case asserting the '833 against Micron)	Open - Filing Date April 28, 2021

/HASHEM FARROKH/

10/30/2023

Information Disclosure Statement

Listed on accompanying two IDS Forms PTO/SB/08a are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). Due to the Request for Continued Examination (RCE) filed concurrently, this Information Disclosure Statement is being filed before the mailing date of a first Office Action on the merits. No statement or fee is required.

Copies of documents **NPL 1-50 and NPL 1-36** (as listed on the two PTO/SB/08a forms) are submitted. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications are submitted.

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 17/328,019, filed 05-24-2021 (now U.S. Pat. No. 11,232,054), 17/138,766, filed 12-30-2020 (now U.S. Pat. No. 11,016,918), 15/934,416, filed 03-23-2018 (now abandoned), 14/840,865, filed 08-31-2015 (now U.S. Pat. No. 9,928,186); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next

communication from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

It is respectfully requested that the Examiner initial and return a copy of the enclosed two IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 22-0585.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

Date: October 13, 2023

/Khaled Shami/

Khaled Shami, Registration No.: 38745

Customer No.: 151145

VORYS, SATER, SEYMOUR and PEASE LLP

1909 K Street, NW, 9th Floor

Washington, DC 20006-1152

Tel: (202) 467-8800

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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	1	5563839	A	1996-10-08	HERDT et al.	
	2	6067241	A	2000-05-23	LU	
	3	6693840	B2	2004-02-17	SHIMADA et al.	
	4	6798709	B2	2004-09-28	SIM et al.	
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	6	7023187	B2	2006-04-04	SHEARON et al.	
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Application Number	17582797
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Art Unit	2138
Examiner Name	FARROKH
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9	7719866	B2	2010-05-18	BOLDO	
10	7721130	B2	2010-05-18	PRETE et al.	
11	7724604	B2	2010-05-25	AMIDI et al.	
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	1	20060080515	A1	2006-04-13	SPIERS et al.	
	2	20060174140	A1	2006-08-03	HARRIS et al.	
	3	20070136523	A1	2007-06-14	BONELLA et al.	
	4	20080101147	A1	2008-05-01	AMIDI	
	5	20080238536	A1	2008-10-02	HAYASHI et al.	

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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

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	1	Transcript, Day 2 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 17, 2023, 375 pages.	☐
	2	Transcript, Day 4 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 19, 2023, 338 pages.	☐
	3	Exhibit 1003, IPR 2022-00999, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,232,054, 431 pages, executed on May 16, 2022.	☐

**INFORMATION DISCLOSURE
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Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

4	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 69 pages.	☐
5	Exhibit 1015, IPR 2014-00994, Decision - Denying Institution of Inter Partes Review, issued December 16, 2014, 16 pages.	☐
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7	Exhibit 1017, IPR 2014-01370, Decision - Denying Institution of Inter Partes Review, issued March 13, 2015, 19 pages.	☐
8	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
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10	Exhibit 1020, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017, 78 pages.	☐
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13	Exhibit 1026, JEDEC STANDARD, DDR2 SDRAM SPECIFICATION, JESD79-2B (Revision of JESD7-2A), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, January 2005, 113 pages.	☐
14	Exhibit 1027, JEDEC STANDARD, FBDIMM: Advanced Memory Buffer (AMB), JESD82-20, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 190 pages.	☐

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15	Exhibit 1028, JEDEC STANDARD, FBDIMM Specification: DDR2 SDRAM Fully Buffered, DIMM (FBDIMM) Design Specification, JESD205, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 129 pages.	☐
16	Exhibit 1031, PCI Local Bus Specification, Revision 2.2, December 18, 1998, 322 pages.	☐
17	Exhibit 1032, Power Electronics: Converters, Applications, and Design, Second Edition, copyright 1995 by John Wiley & Sons, Inc., 37 pages.	☐
18	Exhibit 1040, Texas Instruments, TPS51020, Dual, Voltage Mode, DDR Selectable, Synchronous, Step-Down Controller for Notebook System Power, SLUS564B, July 2003 (revised December 2003), 27 pages.	☐
19	Exhibit 1041, Fairchild Semiconductor, FAN5026, Dual DDR/Dual-Output PWM Controller, October 2005, 17 pages.	☐
20	Exhibit 1042, Murata Manufacturing Co., Ltd, Power Supply Reference Guide for XILINX FPGAs, September 2006, 33 pages.	☐
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22	Exhibit 1045, JEDEC STANDARD, Double Data Rate (DDR) SDRAM Specification, JESD79, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, June 2000, 77 pages.	☐
23	Exhibit 1046, JEDEC STANDARD, DDR3 SDRAM Specification, JESD79-3A (Revision of JESD79-3), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, September 2007, 189 pages.	☐
24	Exhibit 1048, Murata Manufacturing, DC-DC Converter Specification(DRAFT), MPD4S014S, January 21, 2008, 11 pages.	☐
25	Exhibit 1049, Micron, NAND Flash Memory, MT29F2G08AABWP/MT29F2G16AABWP, MT29F4G08BABWP/MT29F4G16BABWP, MT29F8G08EABWP, copyright 2004 by Micron Technology, Inc., 57 pages.	☐

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Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

26	Exhibit 1058, LENK, Simplified Design of Switching Power Supplies, Linear Technology, EDN Series for Design Engineers, copyright 1995 by Butterworth-Heinemann, 9 pages.	☐
27	Exhibit 1061, Analog Devices, Super Sequencer with Margining Control and Auxiliary ADC Inputs, ADM1066, copyright 2006 by Analog Devices, Inc., 36 pages.	☐
28	Exhibit 1062, Analog Devices, Analog Dialogue, Volume 40, Number 2, 2006, 24 pages.	☐
29	Exhibit 1063, National Semiconductor, LMC6953, PCI Local Bus Supervisor, October 1996, 12 pages.	☐
30	Exhibit 1067, Lattice Semiconductor Corporation, LatticeXP Family Data Sheet, Version 04.2, March 2006, 127 pages.	☐
31	Exhibit 1068, Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed October 15, 2021, 82 pages.	☐
32	Exhibit 1069, First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed January 18, 2022, 122 pages.	☐
33	Exhibit 1071, Complaint, filed December 20, 2021, 49 pages.	☐
34	Exhibit 1072, The Samsung Defendants' Answer, filed April 12, 2022, 37 pages.	☐
35	Exhibit 1073, First Amended Complaint, filed May 3, 2022, 71 pages.	☐
36	Exhibit 1075, IPR 2022-00999, Deposition of William Mangione-Smith, Ph.D. regarding US Patent No. 11,232,054, 245 pages, dated June 12, 2023.	☐

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Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

37	Exhibit 1076, IPR 2022-00999, redline comparing Declarations - Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,232,054, 105 pages, executed on March 29, 2022.	☐
38	Exhibit 1077, Netlist, Technology Tutorial: In the Matter of Certain Memory Modules and Components Thereof and Products Containing Same, Inv No. 337-TA-1023, May 8, 2017, 11 pages.	☐
39	Exhibit 1078, Analog Devices, Step-Down DC-to-DC Controller, ADP1821, copyright 2005-2007 by Analog Devices, Inc., 24 pages.	☐
40	Exhibit 2001, IPR 2022-00999, Declaration of Dr. Sunil P. Khatri regarding US Patent No. 11,232,054, 68 pages, executed on September 8, 2022.	☐
41	Exhibit 2003, BELLONI et al., 24.6 A 4-Output Single-Inductor DC-DC Buck Converter with Self-Boosted Switch Drivers and 1.2A Total Output Current, ISSCC 2008 / Session 24 / Analog Power Techniques / 24.6, 2008 IEEE International Solid-State Circuits Conference, 3 pages.	☐
42	Exhibit 2004, MA et al., Single-Inductor Multiple-Output Switching Converters With Time-Multiplexing Control in Discontinuous Conduction Mode, IEEE Journal of Solid-State Circuits, Volume 38, Number 1, January 2003, pages 89-100 (12 pages).	☐
43	Exhibit 2006, Micron, Technical Note: Power Solutions for DDR2 Notebook PCs, TN-47-05 DDR2 Power Solutions for Notebooks, copyright 2004 by Micron Technology, Inc., 16 pages.	☐
44	Exhibit 2007, Texas Instruments, LP2996-N, LP2996A DDR Termination Regulator, SNOSA40K, November 2002 (revised December 2016), 39 pages.	☐
45	Exhibit 2008, JAMECO ELECTRONICS, Jameco Part Number 843868, National Semiconductor: LP2996: DDR Termination Regulator, June 2006, 19 pages.	☐
46	Exhibit 2009, JAMECO ELECTRONICS, Jameco Part Number 843930, National Semiconductor: LP2997: DDR-II Termination Regulator, June 2006, 13 pages.	☐
47	Exhibit 2010, National Semiconductor: LP2998: DDR-II and DDR-I Termination Regulator, December 12, 2007, 18 pages.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

48	Exhibit 2011, BERGVELD, Battery Management Systems: Design by Modelling, copyright 2001 by Royal Philips Electronics N.V., 328 pages.	☐
49	Exhibit 2012, ROMO, DDR Memories: Comparison and Overview, NXP, pages 70-73 (4 pages), September 21, 2007.	☐
50	Exhibit 2013, JEDEC Standard No. 21-C, Page 4.20.3-1; PC133 SDRAM Unbuffered SO-DIMM, Reference Design Specification, Revision 1.02, 20 pages, downloaded August 16, 2022.	☐

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	10/30/2023
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2023-10-13
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L67	92	("5563839" "6067241" "6693840" "6798709" "6856556" "1023187" "7061214" "7692938" "7719866" "7721130" "1724604" "10672458" "20060080515" "20060174140" "20070136523" "20080101147" "20080238536" "20090034354" "20100205470" "20100257304").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/23 04:41 PM
L68	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 11:04 AM
L69	7	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies OR modules)) WITH (coupled OR connected) SAME (regulated near2 voltage)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/10/31 12:03 PM
L70	7	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies OR modules)) WITH (coupled OR connected) SAME (regulated near2 voltage)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 12:03 PM
L71	5	70 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 12:04 PM

L72	7	((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM) near2 (devices OR chips OR dies OR modules)) WITH (coupled OR connected) SAME (regulated near2 voltage))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2023/10/31 12:07 PM
L73	5	72 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 12:07 PM
L74	5	72 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 12:09 PM
L75	7	72 AND ((G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/10/31 12:09 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

AMENDMENT

Commissioner:

In response to the Notice of Allowance mailed November 8, 2023, Applicant submits the following Amendment and Remarks, accompanied by a Request for Continued Examination (RCE) and Information Disclosure Statement (IDS).

Listing of Claims begins on page 2.

Remarks begin on page 10.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)
2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to
~~corresponding edge connections of a~~ the first set of edge connections, a set of~~sixty four~~ data lines electrically connected to ~~the corresponding edge connections of a~~ a second set of edge connections, and a set of address and control lines electrically connected to the
~~corresponding edge connections of a third set of edge connections;~~

~~a DIMM interface including the first, second and third set of edge connections, the DIMM interface configured to (i) fit into a memory socket connector of a host system, and (ii) provide a conduit for power, sixty four data signals, and address and control signals between the host system and the DIMM via the first, second and third set of edge connections, respectively;~~

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) ~~receive and monitor an input voltage from~~ supplied via

the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, ~~one or more operations~~ in response to the trigger signal, ~~the one or more operations including~~ a write operation to write data into the ~~non-volatile~~ nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a ~~plurality set~~ of components coupled to the PCB, the ~~plurality set~~ of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to ~~(i)~~ receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to ~~(ii)~~ generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and ~~(iii)~~ to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line;

wherein each component of the ~~plurality set~~ of components ~~(i) is not electrically connected to the input voltage supply line, (ii) is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and (iii) receives power only from~~ via the one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices (i) is not electrically connected to the input voltage supply line, and (ii) receives power only from one or more of the first, second, third and fourth regulated

voltage lines~~[[.]], and~~

wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.

3 – 5. (Canceled)

6. (Currently amended) The DIMM of claim 2~~[[4]]~~, wherein the controller is further configured to periodically perform one or more write operations at regular intervals of time in response to an additional trigger signal that is associated with an additional trigger condition, the additional trigger condition including a request from the host system or detection of a generate a respective trigger signal of a plurality of trigger signals in response to a respective trigger condition of the plurality of trigger conditions, wherein the plurality of trigger conditions include (i) a request from the host system ~~[[or]]~~ (ii) the voltage monitor circuit detecting input voltage level deviation from one or more threshold voltages.

7. (Currently amended) The DIMM of claim 6, wherein the first preset threshold voltage or the one or more threshold voltages include threshold voltages specified as certain voltage values.

8. (Currently amended) The DIMM of claim 6, wherein the first preset threshold voltage or the one or more threshold voltages include threshold voltages specified as a percentage corresponding to percentages of an operating voltage of the DIMM.

9 – 10. (Canceled)

11. (Currently amended) The DIMM of claim 2~~[[10]]~~ further comprising:

a register coupled to the PCB, the first regulated voltage line, and the set of address and control lines, wherein:

during the first state, the register is configured to receive power via the first regulated voltage line, [(i)] receive and register address and control signals from the host system via the set of address and control lines, and [(ii)] output registered address and control signals, and

during the second state, the register does not output registered address and control signals.

12 -13. (Canceled)

14. (Currently amended) The DIMM of claim 6[[13]], wherein the regular intervals of time are every millisecond.

15. (Previously presented) The DIMM of claim 2, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

16. (New) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to monitor an input voltage supplied via the input voltage

supply line, generate a trigger signal upon detecting a first trigger condition, and transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line of the set of voltage supply lines and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines,

wherein, in response to a second trigger condition, the controller is configured to turn power via the first regulated voltage line off or on, and

wherein, in response to a third trigger condition, the controller is configured to turn power via the second regulated voltage line off or on.

17. (New) The DIMM of claim 16, wherein the second or third trigger condition includes a request from the host system or detection of a voltage level deviation from one or more threshold voltages.
18. (New) The DIMM of claim 17, wherein the controller is further configured to periodically perform write operations in response to a fourth trigger condition, each of the write operations being performed in a respective time period of a plurality of time periods.
19. (New) The DIMM of claim 18, wherein each time period is about a millisecond.
20. (New) The DIMM of claim 17, wherein the first preset threshold voltage and the one or more threshold voltages include a threshold voltage specified as a voltage value and/or a threshold voltage specified as a percentage of an operating voltage of the DIMM.
21. (New) The DIMM of claim 16, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.
22. (New) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:
 - a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;
 - a controller including a voltage monitor circuit and nonvolatile memory, the controller

coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to monitor an input voltage supplied via the input voltage supply line, to generate a trigger signal upon detecting a first trigger condition, and to transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines, and

wherein, in response to a second trigger condition, the controller is further configured to turn off power via the first regulated voltage line while power via the second, third and fourth regulated voltage lines is turned on.

23. (New) The DIMM of claim 22, wherein the second trigger condition includes a request from the host system or detection of a voltage level deviation from one or more threshold voltages.
24. (New) The DIMM of claim 23, wherein the controller is further configured to periodically perform write operations in response to a third trigger condition, each of the write operations being performed in a respective time period of a plurality of time periods.
25. (New) The DIMM of claim 24, wherein each time period is about a millisecond.
26. (New) The DIMM of claim 23, wherein the first preset threshold voltage and the one or more threshold voltages include a threshold voltage specified as a voltage value and/or a threshold voltage specified as a percentage of an operating voltage of the DIMM.
27. (New) The DIMM of claim 22, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

REMARKS

By this Amendment, Applicant has **amended claims 2, 6-8, 11 and 14; canceled claims 3-5, 9-10, and 12-13; and added new claims 16-27.**

Claims 2, 6-8, 11, and 14-27 are now pending.

Information Disclosure Statement

Acknowledgement of the Information Disclosure Statement filed concurrently herewith is respectfully requested.

Amendment to Claims

Claims 2, 6-8, 11 and 14 have been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references.

For the examiner's convenience, claim 2 is reproduced hereinbelow in clean version for ease of reading:

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) monitor an input voltage supplied via the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines, and

wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.

Canceled Claims

Claims 3-5, 9-10, and 12 have been canceled without prejudice or disclaimer of the subject matter contained therein.

Newly-Added Claims

Claims 16-27 have been added to further particularly point out and distinctly claim certain aspects of the invention.

Conclusion

Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: February 8, 2024
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Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745



UNITED STATES
PATENT AND TRADEMARK OFFICE

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Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

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02/08/2024 04:28:49 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Yayoi Bourne

PATENT CENTER # 64262523

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Michael Messinger

Documents

TOTAL DOCUMENTS: 11

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
RCE - 022955-90.pdf	3	Request for Continued Examination (RCE)	1322 KB
NPL--3--Decision Granting Institution and Joinder_11016918__IPR202 300406.pdf	7	Non Patent Literature	36 KB
NPL--4--Decision Granting Institution and Joinder_11232054__IPR202 300405.pdf	7	Non Patent Literature	36 KB
NPL--2--IPR Petition_11232054__IPR202 300405.pdf	147	Non Patent Literature	8935 KB
NPL--1--IPR	151	Non Patent Literature	10266 KB

Petition_11016918__IPR202
300406.pdf

NPL --5--Decision Final

Written Decision Determining_11016918__IP R202200996.pdf	91	Non Patent Literature	3097 KB
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NPL--6--Decision Final

Written Decision Determining_11232054__IP R202200999.pdf	80	Non Patent Literature	19492 KB
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IDS - 022955-90.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1241 KB
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022955-000090 2-8-24 Amendment.pdf	13	-	147 KB
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022955-000090 2-8-24 Amendment-AMSB.pdf	(1-1)	1	Amendment Submitted/Entered with Filing of Continued Prosecution Application (CPA)/Request for Continued Examination(RCE)	109 KB
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022955-000090 2-8-24 Amendment-CLM.pdf	(2-9)	8	Claims	132 KB
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022955-000090 2-8-24 Amendment-REM.pdf	(10-13)	4	Applicant Arguments/Remarks Made in an Amendment	117 KB
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Digest

DOCUMENT

MESSAGE DIGEST(SHA-512)

RCE - 022955-90.pdf

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NPL--3--Decision Granting
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0406.pdf

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NPL--4--Decision Granting
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NPL--2--IPR
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NPL--6--Decision Final Written Decision Determining_11232054__IPR2 02200999.pdf	44B52A818BBEDC12A6F43CDA745E0EF3821BA909367F91972 689BD71C23EBF00C9B7DFCC58D47A05927AE3D4555C37639 9395E5CD0B91E1DA1BA5D0AC2D0C40D
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022955-000090 2-8-24 Amendment-REM.pdf	64720DEB8227E73E63CD14871B0CFAEB9784FACA2BD76F88 B1A1724EDBFED1A6A44C8AF9719FF44DD346EDC35D3A0B4 AF01607C396D2F9BFF19EC7A4DC21B449

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)
2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to
~~corresponding edge connections of a~~ the first set of edge connections, a set of~~sixty four~~ data lines electrically connected to ~~the corresponding edge connections of a~~ a second set of edge connections,
and a set of address and control lines electrically connected to ~~the corresponding edge connections of a~~ a third set of edge connections;

~~a DIMM interface including the first, second and third set of edge connections, the DIMM interface configured to (i) fit into a memory socket connector of a host system, and (ii) provide a conduit for power, sixty four data signals, and address and control signals between the host system and the DIMM via the first, second and third set of edge connections, respectively;~~

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) ~~receive and~~ monitor an input voltage ~~from~~ supplied via

the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, ~~one or more operations~~ in response to the trigger signal, ~~the one or more operations including~~ a write operation to write data into the ~~non-volatile~~ nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a ~~plurality set~~ of components coupled to the PCB, the ~~plurality set~~ of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to ~~(i)~~ receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, ~~the first, second, third and fourth buck converters being further configured to~~ (ii) generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and ~~(iii)~~ to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line of the set of components;

wherein each component of the ~~plurality set~~ of components ~~(i) is not electrically connected to the input voltage supply line,~~ (ii) is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and ~~(iii) receives power only from~~ via the one or more of the first, second, third and fourth regulated voltage lines, ~~and wherein the plurality of DDR SDRAM devices (i) is not electrically connected to the input voltage supply line, and (ii) receives power only from one or more of the first, second, third and fourth regulated~~

voltage lines[[.]], and

wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.

3 – 5. (Canceled)

6. (Currently amended) The DIMM of claim 2[[4]], wherein the controller is further configured to periodically perform one or more write operations at regular intervals of time in response to an additional trigger signal that is associated with an additional trigger condition, the additional trigger condition including a request from the host system or detection of a generate a respective trigger signal of a plurality of trigger signals in response to a respective trigger condition of the plurality of trigger conditions, wherein the plurality of trigger conditions include (i) a request from the host system [[or]] (ii) the voltage monitor circuit detecting input voltage level deviation from one or more threshold voltages.

7. (Currently amended) The DIMM of claim 6, wherein the first preset threshold voltage or the one or more threshold voltages include threshold voltages specified as certain voltage values.

8. (Currently amended) The DIMM of claim 6, wherein the first preset threshold voltage or the one or more threshold voltages include threshold voltages specified as a percentage corresponding to percentages of an operating voltage of the DIMM.

9 – 10. (Canceled)

11. (Currently amended) The DIMM of claim 2[[10]] further comprising:

a register coupled to the PCB, the first regulated voltage line, and the set of address and control lines, wherein:

during the first state, the register is configured to receive power via the first regulated voltage line, [(i)] receive and register address and control signals from the host system via the set of address and control lines, and [(ii)] output registered address and control signals, and

during the second state, the register does not output registered address and control signals.

12 -13. (Canceled)

14. (Currently amended) The DIMM of claim 6[[13]], wherein the regular intervals of time are every millisecond.

15. (Previously presented) The DIMM of claim 2, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

16. (New) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to monitor an input voltage supplied via the input voltage

supply line, generate a trigger signal upon detecting a first trigger condition, and transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line of the set of voltage supply lines and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines,

wherein, in response to a second trigger condition, the controller is configured to turn power via the first regulated voltage line off or on, and

wherein, in response to a third trigger condition, the controller is configured to turn power via the second regulated voltage line off or on.

17. (New) The DIMM of claim 16, wherein the second or third trigger condition includes a request from the host system or detection of a voltage level deviation from one or more threshold voltages.
18. (New) The DIMM of claim 17, wherein the controller is further configured to periodically perform write operations in response to a fourth trigger condition, each of the write operations being performed in a respective time period of a plurality of time periods.
19. (New) The DIMM of claim 18, wherein each time period is about a millisecond.
20. (New) The DIMM of claim 17, wherein the first preset threshold voltage and the one or more threshold voltages include a threshold voltage specified as a voltage value and/or a threshold voltage specified as a percentage of an operating voltage of the DIMM.
21. (New) The DIMM of claim 16, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.
22. (New) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:
 - a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;
 - a controller including a voltage monitor circuit and nonvolatile memory, the controller

coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to monitor an input voltage supplied via the input voltage supply line, to generate a trigger signal upon detecting a first trigger condition, and to transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines, and

wherein, in response to a second trigger condition, the controller is further configured to turn off power via the first regulated voltage line while power via the second, third and fourth regulated voltage lines is turned on.

23. (New) The DIMM of claim 22, wherein the second trigger condition includes a request from the host system or detection of a voltage level deviation from one or more threshold voltages.
24. (New) The DIMM of claim 23, wherein the controller is further configured to periodically perform write operations in response to a third trigger condition, each of the write operations being performed in a respective time period of a plurality of time periods.
25. (New) The DIMM of claim 24, wherein each time period is about a millisecond.
26. (New) The DIMM of claim 23, wherein the first preset threshold voltage and the one or more threshold voltages include a threshold voltage specified as a voltage value and/or a threshold voltage specified as a percentage of an operating voltage of the DIMM.
27. (New) The DIMM of claim 22, wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

AMENDMENT

Commissioner:

In response to the Notice of Allowance mailed November 8, 2023, Applicant submits the following Amendment and Remarks, accompanied by a Request for Continued Examination (RCE) and Information Disclosure Statement (IDS).

Listing of Claims begins on page **2**.

Remarks begin on page **10**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

REMARKS

By this Amendment, Applicant has **amended claims 2, 6-8, 11 and 14; canceled claims 3-5, 9-10, and 12-13; and added new claims 16-27.**

Claims 2, 6-8, 11, and 14-27 are now pending.

Information Disclosure Statement

Acknowledgement of the Information Disclosure Statement filed concurrently herewith is respectfully requested.

Amendment to Claims

Claims 2, 6-8, 11 and 14 have been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references.

For the examiner's convenience, claim 2 is reproduced hereinbelow in clean version for ease of reading:

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) monitor an input voltage supplied via the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and

first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line,

wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines, and

wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.

Canceled Claims

Claims 3-5, 9-10, and 12 have been canceled without prejudice or disclaimer of the subject matter contained therein.

Newly-Added Claims

Claims 16-27 have been added to further particularly point out and distinctly claim certain aspects of the invention.

Conclusion

Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: February 8, 2024
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797RECEIPT DATE / TIME
02/08/2024 04:28:49 PM Z ETATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Yayoi Bourne

PATENT CENTER # 64262523

AUTHORIZED BY Michael Messinger

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE
ADDRESS -FIRST NAMED
INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585PAYMENT TRANSACTION ID
E202428G29348203PAYMENT AUTHORIZED BY
Yayoi BournePRE-AUTHORIZED ACCOUNT
220585PRE-AUTHORIZED CATEGORY
37 CFR 1.16 (National application filing, search, and examination fees); 37
CFR 1.17 (Patent application and reexamination processing fees); 37 CFR
1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21
(Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1820	REQUEST FOR CONTINUED EXAMINATION (RCE)- 2ND AND SUBSEQUENT REQUEST (SEE 37 CFR 1.114)	2000.00	1	2000.00
TOTAL AMOUNT:				\$2,000.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

1	Copy of Petition for Inter Partes Review of US Patent No.: 11,016,918, IPR No.: IPR2023-00406, 151 pages, filed January 6, 2023.	☐
2	Copy of Petition for Inter Partes Review of US Patent No.: 11,232,054, IPR No.: IPR2023-00405, 147 pages, filed January 6, 2023.	☐
3	Copy of Decision Granting Institution of Inter Partes Review and Granting Motion for Joinder for US Patent No.: 11,016,918, IPR No.: IPR2023-00406, 7 pages, entered June 23, 2023.	☐
4	Copy of Decision Granting Institution of Inter Partes Review and Granting Motion for Joinder for US Patent No.: 11,232,054, IPR No.: IPR2023-00405, 7 pages, entered June 23, 2023.	☐
5	Copy of Decision of Final Written Decision Determining All Challenged Claims Unpatentable Dismissing Petitioner's Motion to Exclude for US Patent No.: 11,016,918, IPR No.: IPR2022-00996, 91 pages, entered December 6, 2023.	☐
6	Copy of Decision of Final Written Decision Determining All Challenged Claims Unpatentable Dismissing Petitioner's Motion to Exclude for US Patent No.: 11,232,054, IPR No.: IPR2022-00999, 80 pages, entered December 5, 2023.	☐

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-02-08
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	17582797	Filing Date	2022-01-24	Docket Number (if applicable)	022955-000090	Art Unit	2138
First Named Inventor	CHEN			Examiner Name	FARROKH		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 220585

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-02-08
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/582,797	Filing Date 01/24/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	02/08/2024	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 19	Minus	** 20	= 0	x \$100 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$480 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/BRENDA L TURNER/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

151145 7590 02/28/2024
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

EXAMINER	
FARROKH, HASHEM	
ART UNIT	PAPER NUMBER
2138	

DATE MAILED: 02/28/2024

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	05/28/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 02/28/2024
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	05/28/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2138	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	02/28/2024	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW				
9th Floor				
Washington, DC 20006-1152			ART UNIT	PAPER NUMBER
			2138	
DATE MAILED: 02/28/2024				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 17/582,797	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (FITF) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 02/08/2024.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2,6-8,11 and 14-27. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>02/08/2024</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION:

2. This office action is in response to the communication(s) filed on 02/08/2024. There are a total of 19 claims pending in the application. Claims 2, 6-8, 11 and 14 have been amended; claims 1, 3-5, 9-10, and 12-13 have been canceled; and new claims 16-27 have been added.

INFORMATION CONCERNING IDS:

The information disclosure statement (IDS) submitted on 02/08/2024 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the Examiner.

RESPONSE TO REMARKS:

3. Applicant's arguments have been fully considered and they have been persuasive. Applicant has amended the independent claim 2 and added the new independent claims 16 and 22. The amended independent claim 2 includes the allowable subject matters that previously made the claim allowable. The new independent claims 16 and 22 recite substantially similar features as independent claim 2.

REASON FOR ALLOWANCE:

5. Claims 2, 6-8, 11, and 14-27 are considered to be allowable. The reason for allowance of the independent claim 2 are the following: in interpreting the pending claim(s), in view of the specification, the Examiner finds the claimed invention to be patentably distinct from the prior art of record. The prior art of record neither anticipates, nor renders obvious the recited combination as a whole including the limitation(s):

“a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the

Art Unit: 2138

set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines; and first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line[;]], wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage line”

Independent claims 16 and 22 include similar limitations and are allowed for the reasons indicated above with respect to the independent claim 1.

6. *Dependent claims 2, 6-8, 11, 14-15, 17-21, and 23-27 are allowable at least for the reasons indicated above including all the limitations of the allowable base claims upon which they depend.*

7. *Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”*

DIRECTION OF FUTURE CORRESPONDENCES:

Art Unit: 2138

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

9. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

10. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

Notice of References Cited	Application/Control No. 17/582,797		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2138	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	10-2012	Chen; Chi-She	G06F3/065	711/E12.008
*	B	US-20070136523-A1	06-2007	Bonella; Randy M.	G06F12/0866	711/E12.019
*	C	US-20080126624-A1	05-2008	Prete; Edoardo	G11C7/106	710/53
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF
711	110	11/01/2023	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF
Inventor/Assignee Search Updated	10/31/2023	HF
EAST Text, CPC, FACET Search Updated	10/31/2023	HF
Inventor/Assignee Search Updated	02/22/2024	HF
EAST Text, CPC, FACET Search Updated	02/22/2024	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	11/01/2023	HF
G06F13/28	G06F13/00	11/01/2023	HF
G06F2212/205	G06F2212/00	11/01/2023	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 19	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner) _____ (Date) 22 February 2024		O.G. Print Claim(s) 1	O.G. Print Figure 6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	13	/ 28
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	110

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	19	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner)	22 February 2024 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10	11	19										
1	2	6	11	12	20										
	3		12	13	21										
	4		13	14	22										
	5	6	14	15	23										
2	6	7	15	16	24										
3	7	8	16	17	25										
4	8	9	17	18	26										
	9	10	18	19	27										

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	19
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	22 February 2024	O.G. Print Claim(s)
(Primary Examiner)	(Date)	1
		O.G. Print Figure
		6

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L79	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 01:54 PM
L80	4	("11016918").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 01:58 PM
L81	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 02:04 PM
L82	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	ON	2024/02/22 02:04 PM

			IBM_TDB)				
L83	0	((first near2 second near2 third near2 fourth) OR four OR "4" OR plurality) near5 (buck ADJ converters)) SAME (coupled OR connected OR attached) SAME ((printed ADJ circuit ADJ board) OR PCB) SAME (input near2 voltage near2 supply near2 line) SAME (receiv\$3 near2 power) SAME (supply near2 line)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 03:01 PM
L84	0	((double ADJ data ADJ rate) OR DDR) near2 SDRAM) SAME (devices OR chips OR dies OR ICs) SAME (operable OR configured) SAME ((receiv\$3 OR output\$4) near2 data) SAME (data ADJ lines) SAME based SAME (control near5 address signals)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 03:18 PM
L85	0	(buck ADJ converters) SAME (coupled OR connected OR attached) SAME ((printed ADJ circuit ADJ board) OR PCB) SAME (input near2 voltage near2 supply near2 line) SAME (receiv\$3 near2 power) SAME (supply near2 line)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/02/22 03:24 PM
L86	51	(buck ADJ converters) SAME (coupled OR connected OR attached) SAME ((printed ADJ circuit ADJ board) OR PCB)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/02/22 03:25 PM
L87	8	((buck ADJ converters) SAME (coupled OR connected OR attached) SAME ((printed ADJ circuit ADJ board) OR PCB)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/02/22 03:25 PM
L88	2	86 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/02/22 03:26 PM
L89	3	87 AND ("BHAKTA; JAYESH".ine. OR	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/02/22 03:27 PM

		"CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)					
L90	5	86 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/02/22 03:29 PM
L91	7	86 AND ((G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/02/22 03:31 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

1	Copy of Petition for Inter Partes Review of US Patent No.: 11,016,918, IPR No.: IPR2023-00406, 151 pages, filed January 6, 2023.	☐
2	Copy of Petition for Inter Partes Review of US Patent No.: 11,232,054, IPR No.: IPR2023-00405, 147 pages, filed January 6, 2023.	☐
3	Copy of Decision Granting Institution of Inter Partes Review and Granting Motion for Joinder for US Patent No.: 11,016,918, IPR No.: IPR2023-00406, 7 pages, entered June 23, 2023.	☐
4	Copy of Decision Granting Institution of Inter Partes Review and Granting Motion for Joinder for US Patent No.: 11,232,054, IPR No.: IPR2023-00405, 7 pages, entered June 23, 2023.	☐
5	Copy of Decision of Final Written Decision Determining All Challenged Claims Unpatentable Dismissing Petitioner's Motion to Exclude for US Patent No.: 11,016,918, IPR No.: IPR2022-00996, 91 pages, entered December 6, 2023.	☐
6	Copy of Decision of Final Written Decision Determining All Challenged Claims Unpatentable Dismissing Petitioner's Motion to Exclude for US Patent No.: 11,232,054, IPR No.: IPR2022-00999, 80 pages, entered December 5, 2023.	☐

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EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	02/22/2024
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***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-02-08
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Bibliographic Data

Application No: 17/582,797

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /HASHEM FARROKH/

Examiner's Signature

Initials

Title: MEMORY WITH ON-MODULE POWER MANAGEMENT

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2022	711	2138	
RULE			

APPLICANTS

Netlist, Inc., Irvine, CA, UNITED STATES

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Jeffrey C. Solomon, Irvine, CA, UNITED STATES

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Jayesh Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17328019 05/24/2021 PAT 11232054

17328019 is a CON of 17138766 12/30/2020 PAT 11016918

17138766 is a CON of 15934416 03/23/2018ABN

15934416 is a CON of 14840865 08/31/2015 PAT 9928186

14840865 is a CON of 14489269 09/17/2014 PAT 9158684

14489269 is a CON of 13559476 07/26/2012 PAT 8874831

13559476 has PRO of 61512871 07/28/2011

13559476 is a CIP of 12240916 09/29/2008 PAT 8301833

12240916 is a CON of 12131873 06/02/2008ABN

12131873 has PRO of 60941586 06/01/2007

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UNITED STATES PATENT AND TRADEMARK OFFICE

USPTO Automated Interview Request (AIR)

May 10 2024

This paper requesting to schedule and/or conduct an interview is appropriate because:

This submission is requested to be accepted as an authorization for this interview to communicate via the internet. Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned concerning scheduling of the interview via video conference, instant messaging, or electronic mail, and to conduct the interview in accordance with office practice including video conferencing.

Name(s) :

Michael Milan Stojsavljevic

S-signature:

/Michael Milan Stojsavljevic/

Registration Number:

80857

U.S. Application Number:

17582797

Confirmation Number:

3546

E-mail Address:

mmstojsavljevic@vorys.com

Phone Number:

+1 2163733437

Proposed Time of Interview:

5-16-2024 1:30 PM ET

Preferred Interview Type:

Video Conference

I am the applicant or applicant's representative for this application.

Topic for Discussion:

Presenting example IDS documents and listings to confirm that each document in the IDS is considered by the Examiner.



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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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	1	5563839	A	1996-10-08	HERDT et al.	
	2	6067241	A	2000-05-23	LU	
	3	6693840	B2	2004-02-17	SHIMADA et al.	
	4	6798709	B2	2004-09-28	SIM et al.	
	5	6856556	B1	2005-02-15	HAJECK	
	6	7023187	B2	2006-04-04	SHEARON et al.	
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	8	7692938	B2	2010-04-06	PETTER	

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9	7719866	B2	2010-05-18	BOLDO	
10	7721130	B2	2010-05-18	PRETE et al.	
11	7724604	B2	2010-05-25	AMIDI et al.	
12	10672458	B1	2020-06-02	SHAEFFER et al.	

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	1	20060080515	A1	2006-04-13	SPIERS et al.	
	2	20060174140	A1	2006-08-03	HARRIS et al.	
	3	20070136523	A1	2007-06-14	BONELLA et al.	
	4	20080101147	A1	2008-05-01	AMIDI	
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Attorney Docket Number	022955-000090

6	20090034354	A1	2009-02-05	RESNICK	
7	20100205470	A1	2010-08-12	MOSHAYEDI et al.	
8	20100257304	A1	2010-10-07	RAJAN et al.	

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	1	Transcript, Day 2 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 17, 2023, 375 pages.	☐
	2	Transcript, Day 4 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 19, 2023, 338 pages.	☐
	3	Exhibit 1003, IPR 2022-00999, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,232,054, 431 pages, executed on May 16, 2022.	☐

**INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

4	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 69 pages.	☐
5	Exhibit 1015, IPR 2014-00994, Decision - Denying Institution of Inter Partes Review, issued December 16, 2014, 16 pages.	☐
6	Exhibit 1016, Corrected Petition for Inter Partes Review of US Patent No. 8,301,833, filed September 22, 2014, 67 pages.	☐
7	Exhibit 1017, IPR 2014-01370, Decision - Denying Institution of Inter Partes Review, issued March 13, 2015, 19 pages.	☐
8	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
9	Exhibit 1019, IPR 2017-00649, Decision - Denying Institution of Inter Partes Review, issued July 24, 2017, 17 pages.	☐
10	Exhibit 1020, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017, 78 pages.	☐
11	Exhibit 1021, IPR 2017-00692, Final Written Decision, issued July 5, 2018, 42 pages.	☐
12	Exhibit 1022, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 14, 2022, 76 pages.	☐
13	Exhibit 1026, JEDEC STANDARD, DDR2 SDRAM SPECIFICATION, JESD79-2B (Revision of JESD7-2A), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, January 2005, 113 pages.	☐
14	Exhibit 1027, JEDEC STANDARD, FBDIMM: Advanced Memory Buffer (AMB), JESD82-20, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 190 pages.	☐

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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

15	Exhibit 1028, JEDEC STANDARD, FBDIMM Specification: DDR2 SDRAM Fully Buffered, DIMM (FBDIMM) Design Specification, JESD205, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 129 pages.	☐
16	Exhibit 1031, PCI Local Bus Specification, Revision 2.2, December 18, 1998, 322 pages.	☐
17	Exhibit 1032, Power Electronics: Converters, Applications, and Design, Second Edition, copyright 1995 by John Wiley & Sons, Inc., 37 pages.	☐
18	Exhibit 1040, Texas Instruments, TPS51020, Dual, Voltage Mode, DDR Selectable, Synchronous, Step-Down Controller for Notebook System Power, SLUS564B, July 2003 (revised December 2003), 27 pages.	☐
19	Exhibit 1041, Fairchild Semiconductor, FAN5026, Dual DDR/Dual-Output PWM Controller, October 2005, 17 pages.	☐
20	Exhibit 1042, Murata Manufacturing Co., Ltd, Power Supply Reference Guide for XILINX FPGAs, September 2006, 33 pages.	☐
21	Exhibit 1043, Murata Products, Power Supply Reference Guide for ALTERA FPGAs, Catalog No. DC-03-A, 2008, 40 pages.	☐
22	Exhibit 1045, JEDEC STANDARD, Double Data Rate (DDR) SDRAM Specification, JESD79, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, June 2000, 77 pages.	☐
23	Exhibit 1046, JEDEC STANDARD, DDR3 SDRAM Specification, JESD79-3A (Revision of JESD79-3), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, September 2007, 189 pages.	☐
24	Exhibit 1048, Murata Manufacturing, DC-DC Converter Specification(DRAFT), MPD4S014S, January 21, 2008, 11 pages.	☐
25	Exhibit 1049, Micron, NAND Flash Memory, MT29F2G08AABWP/MT29F2G16AABWP, MT29F4G08BABWP/MT29F4G16BABWP, MT29F8G08FABWP, copyright 2004 by Micron Technology, Inc., 57 pages.	☐

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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

26	Exhibit 1058, LENK, Simplified Design of Switching Power Supplies, Linear Technology, EDN Series for Design Engineers, copyright 1995 by Butterworth-Heinemann, 9 pages.	☐
27	Exhibit 1061, Analog Devices, Super Sequencer with Margining Control and Auxiliary ADC Inputs, ADM1066, copyright 2006 by Analog Devices, Inc., 36 pages.	☐
28	Exhibit 1062, Analog Devices, Analog Dialogue, Volume 40, Number 2, 2006, 24 pages.	☐
29	Exhibit 1063, National Semiconductor, LMC6953, PCI Local Bus Supervisor, October 1996, 12 pages.	☐
30	Exhibit 1067, Lattice Semiconductor Corporation, LatticeXP Family Data Sheet, Version 04.2, March 2006, 127 pages.	☐
31	Exhibit 1068, Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed October 15, 2021, 82 pages.	☐
32	Exhibit 1069, First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed January 18, 2022, 122 pages.	☐
33	Exhibit 1071, Complaint, filed December 20, 2021, 49 pages.	☐
34	Exhibit 1072, The Samsung Defendants' Answer, filed April 12, 2022, 37 pages.	☐
35	Exhibit 1073, First Amended Complaint, filed May 3, 2022, 71 pages.	☐
36	Exhibit 1075, IPR 2022-00999, Deposition of William Mangione-Smith, Ph.D. regarding US Patent No. 11,232,054, 245 pages, dated June 12, 2023.	☐

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Filing Date	2022-01-24
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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

37	Exhibit 1076, IPR 2022-00999, redline comparing Declarations - Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,232,054, 165 pages, executed on March 29, 2022.	☐
38	Exhibit 1077, Netlist, Technology Tutorial: In the Matter of Certain Memory Modules and Components Thereof and Products Containing Same, Inv No. 337-TA-1023, May 8, 2017, 11 pages.	☐
39	Exhibit 1078, Analog Devices, Step-Down DC-to-DC Controller, ADP1821, copyright 2005-2007 by Analog Devices, Inc., 24 pages.	☐
40	Exhibit 2001, IPR 2022-00999, Declaration of Dr. Sunil P. Khatri regarding US Patent No. 11,232,054, 68 pages, executed on September 8, 2022.	☐
41	Exhibit 2003, BELLONI et al., 24.6 A 4-Output Single-Inductor DC-DC Buck Converter with Self-Boosted Switch Drivers and 1.2A Total Output Current, ISSCC 2008 / Session 24 / Analog Power Techniques / 24.6, 2008 IEEE International Solid-State Circuits Conference, 3 pages.	☐
42	Exhibit 2004, MA et al., Single-Inductor Multiple-Output Switching Converters With Time-Multiplexing Control in Discontinuous Conduction Mode, IEEE Journal of Solid-State Circuits, Volume 38, Number 1, January 2003, pages 89-100 (12 pages).	☐
43	Exhibit 2006, Micron, Technical Note: Power Solutions for DDR2 Notebook PCs, TN-47-05 DDR2 Power Solutions for Notebooks, copyright 2004 by Micron Technology, Inc., 16 pages.	☐
44	Exhibit 2007, Texas Instruments, LP2996-N, LP2996A DDR Termination Regulator, SNOSA40K, November 2002 (revised December 2016), 39 pages.	☐
45	Exhibit 2008, JAMECO ELECTRONICS, Jameco Part Number 843868, National Semiconductor: LP2996: DDR Termination Regulator, June 2006, 19 pages.	☐
46	Exhibit 2009, JAMECO ELECTRONICS, Jameco Part Number 843930, National Semiconductor: LP2997: DDR-II Termination Regulator, June 2006, 13 pages.	☐
47	Exhibit 2010, National Semiconductor: LP2998: DDR-II and DDR-I Termination Regulator, December 12, 2007, 18 pages.	☐

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48	Exhibit 2011, BERGVELD, Battery Management Systems: Design by Modelling, copyright 2001 by Royal Philips Electronics N.V., 328 pages.	☐
49	Exhibit 2012, ROMO, DDR Memories: Comparison and Overview, NXP, pages 70-73 (4 pages), September 21, 2007.	☐
50	Exhibit 2013, JEDEC Standard No. 21-C, Page 4.20.3-1; PC133 SDRAM Unbuffered SO-DIMM, Reference Design Specification, Revision 1.02, 26 pages. downloaded August 16, 2022.	☐

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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

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A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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	Filing Date		2022-01-24
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	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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1	Exhibit 2014, Qimonda, Internet Data Sheet, Rev. 1.25, June 2007, 24 pages.	☐
2	Exhibit 2015, Siemens, 3.3V 2M x 64/72-Bit 1 BANK SDRAM Module, HYS64/72V2200GU-8/-10, 3.3V 4M x 64/72-Bit 2 BANK SDRAM Module, HYS64/72V4200GU-8/-10, 168 pin unbuffered DIMM Modules, downloaded from www.datasheetcatalog.com; 1998, 18 pages	☐
3	Exhibit 2016, PCI Bus Variation - Technology Note, copyright 2006 by Euresys, 9 pages.	☐
4	Exhibit 2017, Qimonda, Internet Data Sheet, Rev. 1.42, September 2007, 27 pages.	☐
5	Exhibit 2018, Transcend, Product Support, What is the difference between SDRAM, DDR1, DDR2, DDR3 and DDR4?, available online at https://www.transcend-info.com/support/faq-296#:~:text=DDR3's prefetch buffer width is,according to the temperature variation , downloaded August 24, 2022, 2 pages.	☐
6	Exhibit 2021, Texas Instruments: Application Report, Low Dropout Operation in a Buck Converter, SLUA928A - December 2018 - revised March 2019, 18 pages.	☐
7	Exhibit 2022, Electronic Design, Simple Soft-Start Circuitry Provides Long Startup Times, available online at http://electronicdesign.com/power/simple-soft-start-circuitry-provides-long-startup-times , published June 22, 1998, 3 pages.	☐
8	Exhibit 2024, SCHMID, Understanding Hard Drive Performance, available online at https://www.tomshardware.com/reviews/understanding-hard-drive-performance,1557-5.html , downloaded August 15, 2022, published March 5, 2007, 17 pages.	☐
9	Exhibit 2025, Micron, SDR SDRAM, MT48LC64M4A2 - 16 Meg x 4 x 4 Banks, MT48LC32M8A2 - 8 Meg x 8 x 4 Banks, MT48LC16M16A2 - 4 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 94 pages.	☐
10	Exhibit 2026, Micron, SDR SDRAM, MT48LC16M4A2 - 4 Meg x 4 x 4 Banks, MT48LC8M8A2 - 2 Meg x 8 x 4 Banks, MT48LC4M16A2 - 1 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 84 pages.	☐
11	Exhibit 2027, Transcend, DDR2 SO-DIMM, copyright 2012 by Transcend, 3 pages.	☐

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12	Exhibit 2028, Micron Technical Note: DDR3 Point-to-Point Design Support, copyright 2013 by Micron Technology, Inc., 27 pages.	☐
13	Exhibit 2030, IPR 2022-00996 and IPR 2022-00999, Videoconference Deposition of Andrew Wolfe, Ph.D. regarding US Patent Nos. 11,016,918 and 11,232,054, 311 pages, dated March 16, 2023.	☐
14	Exhibit 2031, IPR 2022-00996, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,016,918, 452 pages, executed on May 16, 2022.	☐
15	Exhibit 2031, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
16	Exhibit 2032, Claim Construction Order, filed December 14, 2022, 35 pages.	☐
17	Exhibit 2033, Defendants' Objections to Claim Construction Memorandum Order, filed December 29, 2022, 11 pages.	☐
18	Exhibit 2043, Samsung Unveils New Power Management Solutions for DDR5 Modules, May 18, 2021, 6 pages.	☐
19	Exhibit 2044, Micron, DDR2 SDRAM RDIMM, MT9HTF3272Y - 256MB, MT9HTF6472PY - 512MB, MT9HTF12872PY - 1GB, copyright 2003 by Micron Technology, Inc., 23 pages.	☐
20	Exhibit 2046, JEDEC Standard No. 21C, Page 4.20.11-1, PC2-6400/PC2-5300/PC2-4200/PC2-3200, DDR2 Unbuffered SO-DIMM, Reference Design Specification, Revision 2.5, July 2008, 60 pages.	☐
21	Exhibit 2048, Micron, DDR2 SDRAM FBDIMM, MT18HTF12872FD - 1GB, MT18HTF25672FD - 2GB, copyright 2005 by Micron Technology, Inc., 39 pages.	☐
22	Exhibit 2056, IPR 2022-00615, Remote Videotaped Deposition by Virtual Zoom of Andrew Wolfe, Ph.D. regarding US Patent No. 7,619,912, 297 pages, dated January 4, 2023.	☐

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23	Exhibit 2057, Micron Technical Brief: Micron DDR5: Key Module Features, DDR4 vs. DDR5 DIMM, copyright 2020 by Micron Technology, Inc., 7 pages.	☐
24	Exhibit 2063, Netlist, Demonstrative Exhibit for IPR 2022-00996 and IPR 2022-00999, filed September 6, 2023, 180 pages.	☐
25	Exhibit 2102, LEE et al., PCI Express Power and Mini Card Solutions - Design Note 346, Linear Technology - Design Notes, copyright 2004 by Linear Technology Corporation, 2 pages.	☐
26	Paper 1, IPR 2022-00996, Petition for Inter Partes Review of US Patent No. 11,016,918, filed May 17, 2022, 151 pages.	☐
27	Paper 1, IPR 2022-00999, Petition for Inter Partes Review of US Patent No. 11,232,054, filed May 17, 2022, 149 pages.	☐
28	Paper 7, IPR 2022-00996, Patent Owner Preliminary Response, filed September 9, 2022, 85 pages.	☐
29	Paper 7, IPR 2022-00999, Patent Owner Preliminary Response, filed September 8, 2022, 82 pages.	☐
30	Paper 10, IPR 2022-00996, Decision - Granting Institution of Inter Partes Review, dated December 7, 2022, 56 pages.	☐
31	Paper 21, IPR 2022-00996, Patent Owner Response, filed March 28, 2023, 91 pages.	☐
32	Paper 22, IPR 2022-00999, Patent Owner Response, filed March 28, 2023, 90 pages.	☐
33	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐

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34	Paper 26, IPR 2022-00999, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
35	Paper 31, IPR 2022-00996, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐
36	Paper 33, IPR 2022-00999, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

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APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/28/2024 04:31:47 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 65669799

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
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Documents

TOTAL DOCUMENTS: 100

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
4-NPL-- Ex_2017_Qimonda_256- MBit_Synchronous_DRAM_S D.pdf	27	Non Patent Literature	583 KB
6-NPL-- Ex_2021_Low_Dropout_Oper ation_Buck_Converter.pdf	18	Non Patent Literature	700 KB
2-NPL-- Ex_2015_Siemens_168_Pin_ Unbuffered_DIMM_Module.p df	18	Non Patent Literature	785 KB
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NPL--3- Ex_1003_Decl_Andrew_Wolfe. pdf	CCB9024FA690A3A7A4B43B35E180269F7A5F28995B5E853DE 8B456BB72227E6A6801F1275B0A4F46730D4EC5F65B799DED B67A8024DD6614EA357F41DD46C644
NPL--19- Ex_1041_Fairchild_Semiconduc tor_FAM2056_Datasheet.pdf	5FC5357F311A95AE99036D72FBAED153F1C307603C63039E61 83B364A4A1D7C98BD97ED579C1AD2B23CD49B96980F3B21B B61B153E5A0E83D7526C80DB777A85
NPL--18- Ex_1040_Tex_Instruments_TP S51020_Datasheet.pdf	FC8C6A16E54318CEC400C0F047ECE454ECC946528A0AC5F0 F2C51210E718E21B442688210CB0605962C5EEBE098394FC35 F716D1606682CD563D59C14D2EE5C3
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NPL--12- Ex_1022_Petition_IPR_833_Mi	6A24F4B7FEEE3C0C9B8E33BD34977099C025CCF0E88862999 80327D934828EB7949CBC6DEE85E3A3FE6560221F8FC2A457

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NPL--29- Ex_1063_PCI_Local_Bus_Pow er_Supervisor.pdf	F8AE55E50D0982D53AD4D20B7B53E6AD91AE6E861D319745F 4985F7F6093AC49C672C831ACB61A2432A8760F9BEBFEC555 93BA2B5FFB9039134F802B504CCC88
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NPL--48- Ex_2011_Battery_Management _Systems.pdf	2C103A77346A35C61C580F89BF7BA818A7450682E62409E09D 27BCA20956147C8A3132135B674061D8A8377E3E29D19F53DE C49DDBB6A5AAA904D1B8C0FD1E65
NPL--40- Ex_2001_Decl_Sunil_Khatri.pdf	01B66CABE30B13CD6A047D14CF02A3BE7CD1D97A7B1D9884 7ED57544213764402A3099B9EC258D03AD9CC507B48CA7E83 1312DFCD94F92160F2AF89BC484C492

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	Filing Date		2022-01-24
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	3	6418065	B2	2002-07-09	SATO et al.	
	4	6434044	B1	2002-08-13	GONGWER et al.	
	5	7034569	B1	2006-04-25	BALASUBRAMANIAN et al.	
	6	7161820	B2	2007-01-09	FUNABA et al.	
	7	7166934	B2	2007-01-23	DEWEY	
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9	7426649	B2	2008-09-16	BRITTAIN et al.	
10	7613880	B2	2009-11-03	MIURA	
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	2	RENESAS, ISL6537 ACPI Regulator/Controller for Dual Channel DDR Memory Systems Data Sheet, Rev 6.00, 16 pages, July 18, 2007.	☐
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	6	RENESAS, ISL6534 Dual PWM with Linear Data Sheet, Rev 2.00, 29 pages, November 18, 2005.	☐
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Attorney Docket Number	022955-000090

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SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/28/2024 04:28:40 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 65697055

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 11

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NPL--1-MachTrans_KR20050073902 A.pdf	7	Non Patent Literature	76 KB
NPL--5-TKDM.pdf	8	Non Patent Literature	402 KB
NPL--7-MAXIM.pdf	14	Non Patent Literature	779 KB
NPL--6-ISL6534.pdf	29	Non Patent Literature	928 KB
NPL--2-ISL6537.pdf	16	Non Patent Literature	2034 KB
NPL--3-ISL6532B.pdf	16	Non Patent Literature	2254 KB

FOR--1-KR20050073902A.pdf	10	Foreign Reference	4164 KB
NPL--4-NXP.pdf	48	Non Patent Literature	4730 KB
022955-000090 Resp to 2-28-24 RCE Response to Notice of Allowance.pdf	4	Amendment Submitted/Entered with Filing of Continued Prosecution Application (CPA)/Request for Continued Examination(RCE)	158 KB
022955-000090 RCE form.pdf	3	Request for Continued Examination (RCE)	1334 KB
IDS3 - 022955-000090.pdf	5	Information Disclosure Statement (IDS) Form (SB08)	1242 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
NPL--1-MachTrans__KR20050073902A.pdf	AD310DA693A1EC9DE0859389404E4187F2927144AAF3A95FC6BD4BAB043051A72093CEBFB3E7B003E2B127A92945F8510B5E2ADD0DD8B356FA3DDD57EF81C8764
NPL--5-TKDM.pdf	B235489F9D5A1B531859DB22B9672BE8BC04E9FC3148A1EEBEC364779CFE1F20F99031A6282871F8E7BDE5C2C08EE6827FA7C701365F1E9482DAFC8C9D5A2ED8
NPL--7-MAXIM.pdf	53D804854A96F5F71676E2B1DD1487225A9158DDF8A1038185510AD6766473B10545985AD2B8083649C1058F4ECE5116861D2A0FBFD7AF4C85CAD45E321B6ACD
NPL--6-ISL6534.pdf	EF58A5C0B922434265FCCE1DBD0ED91DA91ECBE6C420CCE2A1D07B3720A71E3B77EF0C314491082D327A2AAD6785461A8F6C846FE32FE8C8C0A3B2E5FF72AE8D
NPL--2-ISL6537.pdf	C5FA232152FEBBC9E4DD32401268E661E47E9D89BD711225FC

52A6DB75DB6B1B04C2BAE7C7638A4B8F282D89D7ACDA5B0C
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NPL--3-ISL6532B.pdf CF8063AAE5ADAEC818F41A528F53F0A03949E17BE29015DC0
C09961515CFA464322515962267743521FA74A20C95174A5233
9EA8999A21A6C0AA765991B9C41F

FOR--1-KR20050073902A.pdf 4595FC5184CB055FA8E26F8620555DEAFFC00CACE9A4C1477
C9AB5891108E03B64866C641D7E828975E1B65B2E2C89B9E8
DFFF13DA90F5F48C9E9C0822A37938

NPL--4-NXP.pdf 3F0AE4C63649DE0EB71006DF85F57A37330D3990E2EAC1EA1
A0150C4863DD522EC7AC39CAA7CF807CF23FE41F4BD3E58D
B3185B95DF423C9E4605DC0E65DBEDB

022955-000090 Resp to 2-28-
24 RCE Response to Notice of
Allowance.pdf A6AABCC0D7E2095224749D7D85121EC0774E1524E12DFF903
5F11CA8642D5CAAEEC1AA53AB9D55B447B75D71AC91CF5D
A2CF71BB1FE9A7D4C9EC673DE925F6A9

022955-000090 RCE form.pdf 39D5F896E9CB2A57BDAA73606678254660F9B433BF935A6964
3B70D5F39BDE38E04028620309A4F50E44594EB1E3CB3DA3B
E6805C819D91900C0CC4CFB24D782

IDS3 - 022955-000090.pdf E7C30BD96605E7FD9C4AD28A330307F840EAA0043FAAC990C
11AE82AAEA1BC1BA93276A1A17701826C9FFD69E5397253BF
B1FC5F337CA0298BD22A537ED5648A

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the

International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/28/2024 04:28:40 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 65697055

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE
ADDRESS -

FIRST NAMED
INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220565

PAYMENT TRANSACTION ID
E20245RG30167823

PAYMENT AUTHORIZED BY
LISA MARGARET ELLIOTT

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1820	REQUEST FOR CONTINUED EXAMINATION (RCE)- 2ND AND SUBSEQUENT REQUEST (SEE 37 CFR 1.114)	2000.00	1	2000.00
TOTAL AMOUNT:				\$2,000.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

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REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	17/582,797	Filing Date	2022-01-24	Docket Number (if applicable)	022955-000090	Art Unit	2138
First Named Inventor	Chi-She Chen			Examiner Name	Hashem Farrokh		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No 220585

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop RCE

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

RESPONSE AFTER NOTICE OF ALLOWANCE

Commissioner:

In response to the Notice of Allowance issued February 28, 2024, Applicant submits the following:

Remarks begin on page 3.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

REMARKS

The Notice of Allowance mailed February 28, 2024, has been carefully considered. The following remarks and accompanying Information Disclosure Statements are being submitted with a Request for Continued Examination.

Interview Record

Applicants gratefully acknowledge the courtesy extended to the undersigned representative during the Examiner interview conducted on May 22, 2024 between Michael Stojavljevic and Examiner Hashem Farrokh. The substance of the interview related to previously submitted Information Disclosure Statements (IDS) and listing that were not considered, as discussed below. Specifically, two IDS listings and accompanying documents were proposed, the documents having the IDS listings inserted as headers. Accordingly, it was agreed that the header (e.g., titles) provided to the documents corresponding to IDS listings would enable identification and ensure consideration.

Notice of Prior and Concurrent Proceedings

Applicant hereby calls to the attention of the Patent and Trademark Office the following proceedings involving patents that are in the family of the above-identified patent application:

Proceeding	Status
<i>Micron Technology, Inc.; Micron Semiconductor Products, Inc.; and Micron Technology Texas LLC v. Netlist, Inc.</i> , Case No. IPR2022-00418 (P.T.A.B.), filed January 14, 2022	Final Written Decision - Determining All Challenged Claims Unpatentable entered August 28, 2023
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00615 (P.T.A.B.), filed February 17, 2022	Pending Director Review Responsive to Request Email entered May 20, 2023, following Judgment Final Written Decision – Determining All Challenged Claims Unpatentable entered April 17, 2024
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00996 (P.T.A.B.), filed May 17, 2022	Final Written Decision – Determining All Challenged Claims Unpatentable entered December 6, 2023

Proceeding	Status
<i>Samsung Electronics Co., Ltd. v. Netlist, Inc.</i> , Case No. IPR2022-00999 (P.T.A.B.), filed May 17, 2022	Final Written Decision – Determining All Challenged Claims Unpatentable entered December 5, 2023
2:21-cv-00463-JRG (EDTX Case asserting ‘918 and ‘054 against Samsung)	Closed - Termination Date August 11, 2023
2:22-cv-00203-JRG (EDTX Case asserting ‘918 and ‘054 against Micron)	Open - Filing Date June 10, 2022
1:22-cv-00134-DII (WDTX Case asserting the ‘833 against Micron)	Open - Filing Date April 28, 2021

Information Disclosure Statement

Listed on accompanying three IDS Forms PTO/SB/08a are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). Due to the Request for Continued Examination (RCE) filed concurrently, this Information Disclosure Statement is being filed before the mailing date of a first Office Action on the merits. No statement or fee is required.

Copies of documents **NPL 1-50** and **NPL 1-36** (as listed on two PTO/SB/08a forms) are submitted. NPL 1-50 and NPL 1-36 were previously submitted, but not considered. The Notice of Allowance dated November 8, 2023 at page 2 states that these IDS listings and the above Notice of Prior and Concurrent Proceedings were not considered because the “title of actual documents were not readily identifiable and/or are different from IDS list. For example, in the IDS listing

many documents include an Exhibit number, which are not included in the actual documents.” Accordingly, the documents are being submitted again with proper headings that reflect the listing and exhibit numbers provided in the corresponding IDS listing. Moreover, a third IDS (as listed on a third PTO/SB/08a form) is being submitted in a similar manner with documents not previously submitted. Specifically, the third IDS lists six NPL documents and 16 patent publications. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications are submitted.

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 17/328,019, filed 05-24-2021 (now U.S. Pat. No. 11,232,054), 17/138,766, filed 12-30-2020 (now U.S. Pat. No. 11,016,918), 15/934,416, filed 03-23-2018 (now abandoned), 14/840,865, filed 08-31-2015 (now U.S. Pat. No. 9,928,186); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next communication from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

It is respectfully requested that the Examiner initial and return a copy of the enclosed two IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 22-0585.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

Date: May 28, 2024
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	05/28/2024	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	
			DELIVERY MODE	
			05/28/2024	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Video Conference
MileTyson Ovich	Attorney	

Date of Interview: 22 May 2024

Issues Discussed:

Other

An applicant initiated interview with the Applicant, Mr. Mike Tyson Ovich (Reg. # 80,857), was conducted on 05/22/2024 (a copy of the interview agenda is attached). In the Notice of Allowance (NOA) mailed 11/08/2023, the Examiner stated the IDS' filed 10/13/2023 list/include a number of NPL documents not readily identifiable and some of documents contain many number pages. The Examiner suggested the Applicant provide the documents with a unique title or sequence number that can be identified from the corresponding document title or sequence number listed in the IDS'. The Examiner also suggested, since some of documents have large number pages and some portions of each document may include subject matters not related to the claims of instant application, the Applicant identify the portion(s) of NPL documents, which are more relevant to claims of instant application. The interview was video conference in order to see and communicate the best way for the Applicant to provide the IDS'

☒ Attachment

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Interview Agenda for Application No.: 17/582,797; Attn. Docket No. 022955-000090

Attendees:

Primary Examiner Hashem Farrokh, Art Unit 2138

Mike Stojcavljevic, USPTO Registration No. 80,857

Summary:

Discuss the IDS list below and appended documents corresponding to the IDS list to ensure consideration by the Examiner. The non-patent literature listed in the IDS of the Notice of Allowance dated 8 November 2023 (NOA) were shown lined-through and were not considered by the Examiner. The reasoning provided on page 2 of the NOA stated that the:

“title of actual documents are not readily identifiable and are different from the IDS list. For example, in IDS listing many documents include an Exhibit number, which are not included in the actual documents... It is being suggested that applicant(s) include a title for each of one-to-one corresponding to the title that has been listed, point to the portions(s) of the listed non-literature documents, which is/are directly relevant to the claims of the instant application.”

Accordingly, the documents appended correspond to the brief example IDS list below and are provided as an example of documents that include a title that correspond to the title that has been listed. Additionally, most of the documents were cited as exhibits in related proceedings, such that the documents are wholly relevant, but listings will point to portions of the listed non-literature documents where appropriate.

	1	Exhibit 2014, Qimonda, Internet Data Sheet, Rev. 1.25, June 2007, 24 pages.	
	...		
	3	Exhibit 2016, PCI Bus Variation -- Technology Note, copyright 2006 by Euresys, 9 pages.	
	...		

台湾HYB39SC256[80/16]E-6供应商

德多邦，专业PCB打样工厂，24小时加急出货

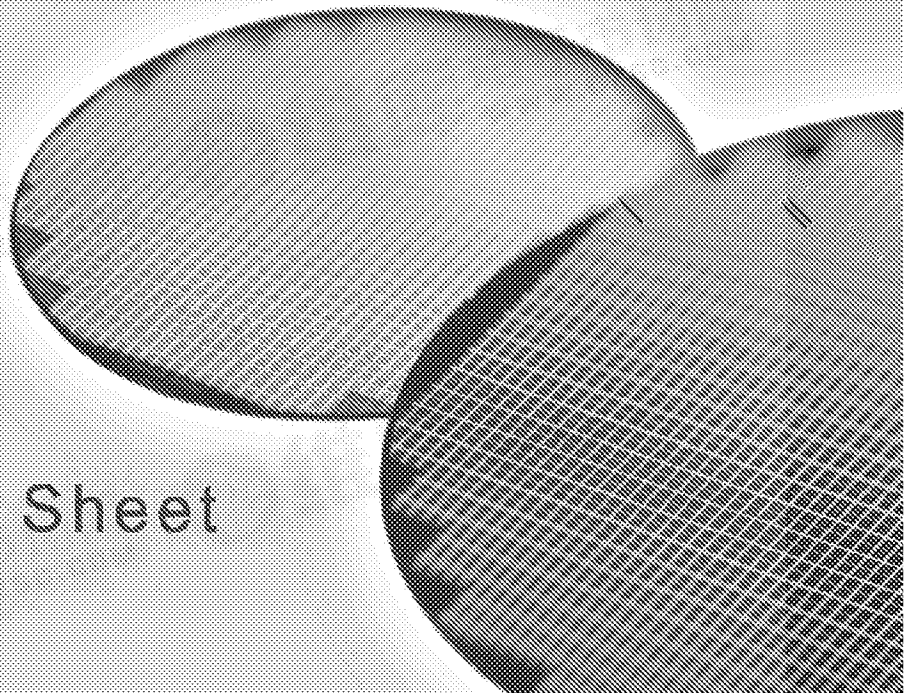
June 2007

HYB39SC256[80/16]0FE HYI39SC256[80/16]0FF

256-MBit Synchronous DRAM
Green Product
SDRAM

Internet Data Sheet

Rev. 1.25



Netlist Ex 2014
Samsung v Netlist
#P2022-00999



HY[B/I]39SC256[80/16]0F[E/F]
256-MBit Synchronous DRAM

HYB39SC256[80/16]0FE, HYI39SC256[80/16]0FF	
Revision History: 2007-06, Rev. 1.26	
Page	Subjects (major changes since last revision)
All	Adapted internet edition
41	Corrected in figure 28 (auto refresh) tRC to tRFC
Previous Revision: 2007-06, Rev. 1.24	
8	Corrected figure 1
11,12	Corrected block diagram
Previous Revision: 2007-06, Rev. 1.23	
18	Added text for Auto Refresh Command (CBR)
Previous Revision: 2007-06, Rev. 1.22	
6	Corrected ball DQ0 to 2,8A for data signals x16 organization
6	Corrected data Data Signal Bus [7:0] for data signals x8 organization
Previous Revision: 2007-06, Rev. 1.21	
13	Corrected operation command "Power Down / Clock suspend ..." in truth table
Previous Revision: 2007-05, Rev. 1.2	
13	Corrected operation command "Power Down Exit" to X (WE#)
15	Corrected text to "After the mode register is set a NOP command is required", chapter 3.2
19	Corrected text to "One clock delay is required for mode entry and exit", chapter 3.4
21	Corrected the line "Input Capacitances: CK" in table 10, chapter 4
21	Corrected to A0-A12 in table 10, chapter 4
22	Corrected tCK MIN in table 13
22	Corrected CLE setup time in table 13
Previous Revision: 2007-05, Rev. 1.11	
6	Corrected A6 position from H to 3H in table 3
Previous Revision: 2006-09, Rev. 1.1	

We Listen to Your Comments

Any information within this document that you feel is wrong, unclear or missing at all?

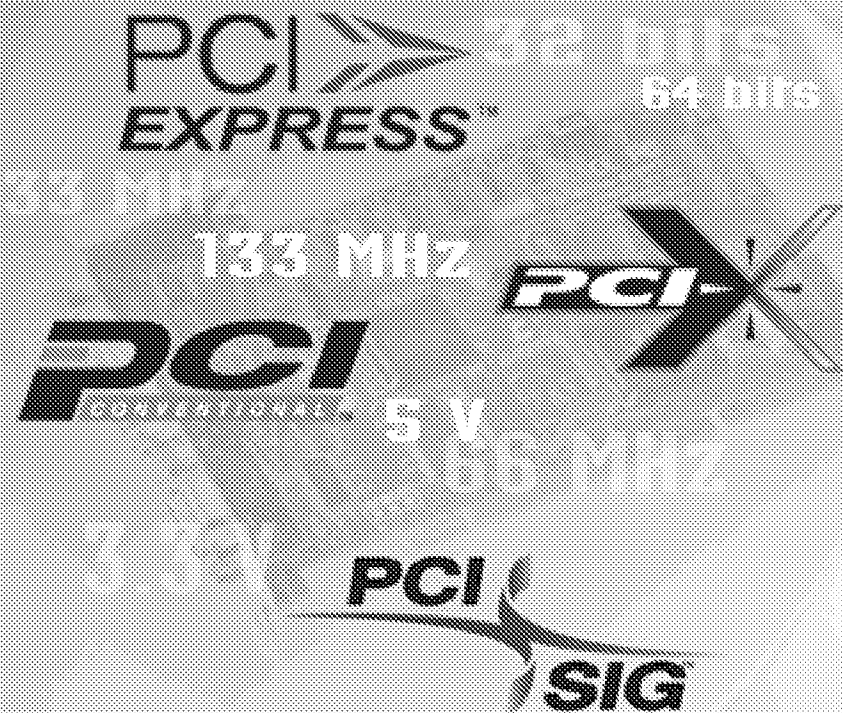
Your feedback will help us to continuously improve the quality of this document.

Please send your proposal (including a reference to this document) to:

techdoc@qimonda.com

Technology Note

PCI Bus Variation



www.euresys.com

info@euresys.com

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EURESYS
Excellence in vision

PCI Bus Variation

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References

PCI

The PCI bus specification is handled by the PCI-SIG (PCI Special Interest Group), which is an association of numerous industrial actors in the field.

Euresys s.a. is a member of the PCI-SIG (vendor ID h1805).

The web site address of PCI-SIG is <http://www.pcisig.com>

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Other product and company names listed are trademarks or trade names of their respective manufacturers.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REQUEST FOR SUSPENSION OF ACTION

Commissioner:

In connection with the Request for Continuation of Examination (RCE) filed May 28, 2024, Applicant hereby requests a two (2) month suspension of action.

Applicant hereby authorizes to be charged the processing fee set forth in 37 CFR 1.17(i) to our Deposit Account No. 22-0585.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

Date: May 29, 2024
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745



UNITED STATES
PATENT AND TRADEMARK OFFICE

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Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/29/2024 09:02:35 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 65725464

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE
ADDRESS -

FIRST NAMED
INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585

PAYMENT TRANSACTION ID
E20245SL03577505

PAYMENT AUTHORIZED BY
LISA MARGARET ELLIOTT

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1830	PROCESSING FEE, EXCEPT IN PROVISIONAL APPLICATIONS	140.00	1	140.00
TOTAL AMOUNT:				\$140.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/29/2024 09:02:35 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 65725464

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
022955-000090 Request for Suspension of Action.pdf	1	Letter Requesting Suspension of Action	103 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
022955-000090 Request for Suspension of Action.pdf	1A627646B89079663584265C3D10CCA55CB2B8DE1A32223F6 D9AA5507E7ECA997E86943A7BCD6FC68DB72B304CBE6E4E6 9BA1430071AEE0C2AA75488FE170484

by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	08/08/2024	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	DELIVERY MODE
			08/08/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Continued Examination Under 37 CFR 1.114

2. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 05/28/2024 has been entered.

Applicant's request for Suspension of Action

3. Applicant's request filed 05/29/2024, for suspension of action in this application under 37 CFR 1.103(b) or (c) is denied as being improper. The request was not filed at the time of filing a CPA or RCE, and/or (2) not accompanied by the requisite fee as set forth in 37 CFR 1.17(i). See MPEP § 709.

The RCE was filed on 05/28/2024 (e.g., different from time of suspension request).

DETAILED ACTION:

4. This office action is in response to the communication(s) filed on 05/28/2024. There are a total of 19 claims pending in the application. Claims 2, 6-8, 11 and 14 have been amended; claims 1, 3-5, 9-10, and 12-13 have been canceled; and new claims 16-27 have been added.

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 05/28/2024 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the Examiner.

RESPONSE TO REMARKS:

5. Applicant's arguments have been fully considered and based on Trial Termination (Trial-Term) or Final Written Decision, by United State Patent and Trademark. The Trial-Term appears, for most part, to be based IDS's that may or may not filed by the Applicant. The Trial-Terms Final Written Decision considered that Applications 17/138,766 (now US 11,016,918 B2); and 17/328,019 (now US 11,232,054 B2 not patentable. Therefore, the Examiner considers the claims of Application 17/582,797, that substantially include limitations similar to claims of Applications 17/138,766 and 17/328,019, not to be patentable. Therefore, the Examiner as shown below rejected the claims.

INFORMATION CONCERNING CLAIMS:

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 2, 6-8, 11, 15-18, and 20-24 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Harris et al. "Harris" (US 2006/0174140 A1) in view of FBDIMM Standards (Exhibitions 1027, 2028 provided by Applicant) and Amidi et al. "Amidi" (US 7,724,604 B2). The rejection is based on information provided in Trial Termination or Final Written Decision (hereinafter Trial-Trim, dated 12/06/2023) by the USPTO Trial and Appeal Board on Application 17/138,766 (now US patent 11,016,918 B2), which appears, in most part, to be based on Exhibition 1001, IPR 2022-00996. The instant Application, which is a continuation of US Patent 11,016,918 B2 includes substantially the same or similar claim limitations.

6. *Regarding claim 2, Harris teaches or suggests:*

"A dual in-line memory module (DIMM) (e.g., Fig. 1A, ¶ 0009, memory module 100A) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:" (e.g., Harris: Fig. 1A, ¶ 0009, a fully buffered DIMM wherein a buffer/logic component 112 is provided for buffering command/address; ¶ 0010, a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels may be implemented as the on-board VRM 102 for purposes of the present patent disclosure ¶ 0013, It is contemplated that local supply voltage conversion for double-rank DIMMs can be accommodated with a form factor design of approximately about one square inch (both sides of the printed circuit board), and at a component height compatible with applicable Joint Electron Device Engineering Council (JEDEC) standards; Fig 1B, ¶ 0014, ¶ 0010, Reference numerals 120-1 through 120-K refer to K supply voltage paths

which may be coupled to various voltage sources provided within the electronic system; Trial-Trim: pages 42-56, power, data, address and control signal; Trial-Trim: pages 18-84). *Fig. 1B Harris shows a block diagram of memory module 100B comprising DRAM devices 110, inputs a plurality voltages 120 from external source and generates a plurality of regulated voltages 106 and 108. The memory module 100B also comprises Buffer 112 that receives command, address, and data from the external source via interface path 114.*

“a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected the first set of edge connections, a set of data lines electrically connected to second set of edge connections, and a set of address and control lines electrically connected to third set of edge connections;” **(e.g., Harris: Fig. 3, ¶ 0017, a block diagram of an exemplary memory assembly 300 according to one embodiment. A memory controller 302 is operable to drive a bidirectional memory link 304 to which a plurality of memory boards 306-1 through 306-M are coupled in a daisy-chain fashion at their respective buffers; Fig. 4, ¶ 0018, an exemplary computer system 400 according to one embodiment... A number of memory modules 410 are coupled to the memory control block 408 via a memory link 411; Trial-Term: pages 18-84).** *Figs 3 and 4 of Harris shows a plurality of Boards 306 or modules connected to external device (e.g., memory controller). Fig. 3 of Harris that the plurality of boards or modules 306 comprising edge connections to connect to the external memory controller or device.*

“a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of

data lines based on control and address signals received from the host system via the set of address and control lines;” (e.g., **Harris: Fig. 1A, ¶ 0009, a Dual In-line Memory Module (DIMM) wherein each of the memory devices 110-1 through 110-N comprises a Double Data Rate (DDR) dynamic random access memory (DRAM) device having a particular density, e.g., 256 Mb, 512 Mb, 1 Gb or 2 Gb, etc. Also, the memory devices can be of any known or heretofore unknown DDR type; Trial-Term: pages 18-84**). *Harris teaches supplying voltages to Fully-Buffered DIMM (FBDIMM). However, Harris does not appear to expressly teach while:*

*FBDIMM Standards discloses: “and first, second, third and **fourth buck converters** coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, **third and fourth buck** converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line, wherein each component of the set of is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines,” (e.g., **Trial-Trim: Pages 20-40 and 56-59; Trial-Trim: pages 18-84**). *Trial-Trim discloses that combining Harris with FBDIMM Standard teaches or render obvious all limitation recited above comprising regulated voltage converters (e.g., including buck converters) recited in the claim.**

“wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control

power delivery to one or more **components** of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is **turned off** and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.” (e.g., **Trial-Trim: Pages 16-41 and 64-65, The combination of Harris and the FBDIMM Standards discloses an S3 sleep mode switch that selectively switches off the second voltage to turn off the power to one or more registers while the DRAMS remains powered on to refresh data in DRAMS; Trial-Trim: pages 18-84**). *Trial-Trim discloses the combination of Harris provides or maps regulated voltages as specified in the claim and DIMMs configured to operate in a plurality mode at least the normal mode or state and sleep mode or state.*

Amidi disclose: “a controller including a voltage **monitor** circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage **monitor** circuit configured to (i) **monitor** an input voltage supplied via the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;” (e.g., **Amidi: Fig. 5, col. 4, lines 14-22, Power management block 500 includes the actual power management module 510, an incoming system supply 520, an incoming battery supply 530, and an outgoing memory power supply 540. Power management block 500 may thus be used to attempt to ensure a stable power supply even in the face of disruptions in system supply 520, for example, Figs 5-6 of Amidi and the corresponding text descriptions; TRIAL-TRM: pages 70-64; pages 18-84**).

Disclosures by Harris, FBDIMM Standards, and Amidi are analogous because they are in the same field of endeavor and/or solving a similar or common problem.

It would have been obvious to a person of having ordinary skill in the art before the effective filing date of the claimed invention to modify the Voltage Distribution System and Method for a Memory Assembly taught by Harris to include the voltage (power), data, and address/control signals disclosed by FBDIMM Standards; furthermore, to include Clock and Power Fault Detection disclosed by Amidi.

The motivation for including the voltages, data, address, and control signal as taught by page 41 of TRIAL-TRM is to look/obtain information concerning voltages for the FBDIMM; furthermore, the motivation to include Clock and Power Fault Detection is to attempt to ensure a stable power supply even in the face of disruptions in system supply 520, for example.

Therefore, it would have been obvious to combine teaching of FBDIMM Standards and Amidi with Harris to obtain the invention as specified in the claim.

7. *Regarding claim 16, Harris teaches or suggests:*

“A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:” (e.g., Harris: Fig. 1A, ¶ 0009, a fully buffered DIMM wherein a buffer/logic component 112 is provided for buffering command/address; ¶ 0010, a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels may be implemented as the on-board VRM 102 for purposes of the present patent disclosure ¶ 0013, It is contemplated that local supply voltage conversion for double-rank DIMMs can be accommodated with a form factor design of approximately about one square inch (both sides of the printed circuit board), and at a component height compatible with applicable Joint Electron Device Engineering Council (JEDEC) standards;

Fig 1B, ¶ 0014, ¶ 0010, Reference numerals 120-1 through 120-K refer to K supply voltage paths which may be coupled to various voltage sources provided within the electronic system; Trial-Term: pages 42-56, power, data, address and control signal; Trial-Term: pages 18-84). *Fig. 1B Harris shows a block diagram of memory module 100B comprising DRAM devices 110, inputs a plurality voltages 120 from external source and generates a plurality of regulated voltages 106 and 108. The memory module 100B also comprises Buffer 112 that receives command, address, and data from the external source via interface path 114.*

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;”
(e.g., Harris: Fig. 3, ¶ 0017, a block diagram of an exemplary memory assembly 300 according to one embodiment. A memory controller 302 is operable to drive a bidirectional memory link 304 to which a plurality of memory boards 306-1 through 306-M are coupled in a daisy-chain fashion at their respective buffers; Fig. 4, ¶ 0018, an exemplary computer system 400 according to one embodiment... A number of memory modules 410 are coupled to the memory control block 408 via a memory link 411; Trial-Term: pages 18-84). *Figs 3 and 4 of Harris shows a plurality of Boards 306 or modules connected to external device (e.g., memory controller). Fig. 3 of Harris that the plurality of boards or modules 306 comprising edge connections to connect to the external memory controller or device.*

“a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM)

devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines;” (e.g., **Harris: Fig. 1A, ¶ 0009, a Dual In-line Memory Module (DIMM) wherein each of the memory devices 110-1 through 110-N comprises a Double Data Rate (DDR) dynamic random access memory (DRAM) device having a particular density, e.g., 256 Mb, 512 Mb, 1 Gb or 2 Gb, etc. Also, the memory devices can be of any known or heretofore unknown DDR type; Trial-Term: pages 18-84**). *Harris teaches supplying voltages to Fully-Buffered DIMM (FBDIMM). However, Harris does not appear to expressly teach while:*

*FBDIMMM Standards discloses: “and first, second, **third and fourth buck converters** coupled to the PCB and to the input voltage supply line of the set of voltage supply lines and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and **fourth buck converters** being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line, wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines,” (e.g., **Trial-Trim: Pages 20-40 and 56-59; Trial-Trim: pages 18-84**).* *Trial-Trim discloses that combining Harris with FBDIMM Standard teaches or render obvious all limitation recited above comprising regulated voltage converters (e.g., including buck converters) recited in the claim.*

“wherein, in response to a second **trigger** condition, the controller is configured to turn power via the first regulated voltage line off or on, and wherein, in response to a third trigger condition, the controller is configured to turn power via the second regulated voltage line off or on.” (e.g., **Trial-Trim: Pages 16-41 and 64-65, The combination of Harris and the FBDIMM Standards discloses an S3 sleep mode switch that selectively switches off the second voltage to turn off the power to one or more registers while the DRAMS remains powered on to refresh data in DRAMS; Trial-Trim: pages 18-84**). *Trial-Trim discloses that FBDIMM Standards teaches/provides S3 mode switch that turns components off/on.*

Amidi discloses: “a controller including a voltage **monitor** circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage **monitor** circuit configured to **monitor** an input voltage supplied via the input voltage supply line, generate a **trigger** signal upon detecting a first trigger condition, and transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;” (e.g., **Amidi: Fig. 5, col. 4, lines 14-22, Power management block 500 includes the actual power management module 510, an incoming system supply 520, an incoming battery supply 530, and an outgoing memory power supply 540. Power management block 500 may thus be used to attempt to ensure a stable power supply even in the face of disruptions in system supply 520, for example, Figs 5-6 of Amidi and the corresponding text descriptions; TRIAL-TRM: pages 70-64; pages 18-84**). *The motivation for combining is based on the same rational presented for rejection of the independent claim 1.*

8. *Regarding claim 22, Harris teaches or suggests:*

“A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host

system is provided to the DIMM via the set of power conduits, data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits, the DIMM comprising:” (e.g., Harris: Fig. 1A, ¶ 0009, a fully buffered DIMM wherein a buffer/logic component 112 is provided for buffering command/address; ¶ 0010, a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels may be implemented as the on-board VRM 102 for purposes of the present patent disclosure ¶ 0013, It is contemplated that local supply voltage conversion for double-rank DIMMs can be accommodated with a form factor design of approximately about one square inch (both sides of the printed circuit board), and at a component height compatible with applicable Joint Electron Device Engineering Council (JEDEC) standards; Fig 1B, ¶ 0014, ¶ 0010, Reference numerals 120-1 through 120-K refer to K supply voltage paths which may be coupled to various voltage sources provided within the electronic system; Trial-Trim: pages 42-56, power, data, address and control signal; Trial-Trim: pages 18-84). *Fig. 1B Harris shows a block diagram of memory module 100B comprising DRAM devices 110, inputs a plurality voltages 120 from external source and generates a plurality of regulated voltages 106 and 108. The memory module 100B also comprises Buffer 112 that receives command, address, and data from the external source via interface path 114.*

“a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits, the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, a set of data lines electrically connected to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections;” (e.g., Harris: Fig. 3, ¶ 0017, a block diagram of an exemplary memory assembly 300 according to one embodiment. A memory controller 302 is operable to drive a

bidirectional memory link 304 to which a plurality of memory boards 306-1 through 306-M are coupled in a daisy-chain fashion at their respective buffers; Fig. 4, ¶ 0018, an exemplary computer system 400 according to one embodiment... A number of memory modules 410 are coupled to the memory control block 408 via a memory link 411; Terial-Term: pages 18-84). *Figs 3 and 4 of Harris shows a plurality of Boards 306 or modules connected to external device (e.g., memory controller). Fig. 3 of Harris that the plurality of boards or modules 306 comprising edge connections to connect to the external memory controller or device.*

“a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on control and address signals received from the host system via the set of address and control lines;” **(e.g., Harris: Fig. 1A, ¶ 0009, a Dual In-line Memory Module (DIMM) wherein each of the memory devices 110-1 through 110-N comprises a Double Data Rate (DDR) dynamic random access memory (DRAM) device having a particular density, e.g., 256 Mb, 512 Mb, 1 Gb or 2 Gb, etc. Also, the memory devices can be of any known or heretofore unknown DDR type; Trial-Term: pages 18-84). Harris teaches supplying voltages to Fully-Buffered DIMM (FBDIMM). However, Harris does not appear to teach while:**

FBDIMMM Standards discloses: “first, second, third and fourth **buck converters** coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, **third and fourth buck converters** being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including a first component configured to receive power via at least the first regulated voltage line, a second component configured to receive power via at least the second regulated voltage line, a third component configured to receive power via at least the

third regulated voltage line, and a fourth component configured to receive power via at least the fourth regulated voltage line, wherein each component of the set of components is electrically connected to one or more of the first, second, third and fourth regulated voltages lines, and receives power only via the one or more of the first, second, third and fourth regulated voltage lines,” (e.g., **Trial-Trim: Pages 20-40 and 56-59; Trial-Trim: pages 18-84**). *Trial-Trim discloses that combining Harris with FBDIMM Standard teaches or render obvious all limitation recited above comprising regulated voltage converters (e.g., including buck converters) recited in the claim.*

“wherein, in response to a second **trigger** condition, the controller is further configured to turn off power via the first regulated voltage line while power via the second, third and fourth regulated voltage lines is turned on.” (e.g., **Trial-Trim: Pages 16-41 and 64-65, The combination of Harris and the FBDIMM Standards discloses an S3 sleep mode switch that selectively switches off the second voltage to turn off the power to one or more registers while the DRAMS remains powered on to refresh data in DRAMS; Trial-Trim: pages 18-84**). *Trial-Trim discloses that FBDIMM Standards teaches/provides S3 mode switch that turns components off/on.*

Amidi discloses: “a controller including a voltage **monitor** circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage **monitor** circuit configured to **monitor** an input voltage supplied via the input voltage supply line, to generate a **trigger** signal upon detecting a first trigger condition, and to transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the first trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM;” (e.g., **Amidi: Fig. 5, col. 4, lines 14-22, Power management block 500 includes the actual power management module 510, an incoming system supply 520, an incoming battery supply 530, and an outgoing memory power supply 540. Power management block 500 may thus be used to attempt to ensure a stable power supply even in the face of disruptions in system supply 520, for example, Figs 5-6 of Amidi and**

the corresponding text descriptions; TRIAL-TRM: pages 70-64; pages 18-84). *The motivation for combining is based on the same rational presented for rejection of the independent claim 1.*

9. *Regarding claims 6, Trial-Term teaches:*

“wherein the controller is further configured to periodically perform one or more write operations at regular intervals of time in response to an additional trigger signal that is associated with an additional trigger condition, the additional trigger condition including a request from the host system **or** detection of a voltage level deviation from one or more threshold voltages.” **(e.g., Trial-Term: pages 78-80).** *Generating trigger signals for overvoltage detection and undervoltage detection for different DIMM voltage values, requires setting of threshold voltage of different values.*

10. *Claims 7-8, 17, 20, 23, and 26 are rejected based on the same ground of rejection as claim 6.*

11. *Regarding claims 11, Trial-Term teaches:*

“a register coupled to the PCB, the first regulated voltage line, and the set of address and control lines, wherein: during the first state, the register is configured to receive power via the first regulated voltage line, receive and register address and control signals from the host system via the set of address and control lines, and output registered address and control signals, and during the second state, the register does not output registered address and control signals.” **(e.g., Trial-Trim: Pages 16-41 and 64-65, The combination of Harris and the FBDIMM Standards discloses an S3 sleep mode switch that selectively switches off the second voltage to turn off the power to one or more registers while the DRAMS remains powered on to refresh data in DRAMS; Trial-Trim: pages 18-84).** *Trial-Trim discloses the combination of Harris provides or maps regulated voltages as specified in the claim and DIMMs configured to operate in a plurality mode at least the normal mode or state and sleep mode or state.*

12. *Regarding claims 15, 21, and 27, Trial-Term teaches:*

“wherein two of the first, second, third and fourth buck converters are configured to operate as a dual-buck converter.” (e.g., **Trial-Trim: Page 63**). *It would have been obvious to implement two or more regulate voltages using dual-buck converter.*

Claims 14, 18, 19, and 25 rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Harris, FBDIMM Standards, and Amidi as applied to claims 6, 18, and 24 above, and further in view of Levkovitz et al. “Levkovitz” (US 2007/0088838 A1).

13. Regarding claim 14, Harris, FBDIMM Standards, and Amidi teach all limitations recited in claim 6 but do not expressly teach while Levkovitz discloses:

“wherein the regular intervals of time are every millisecond.” (e.g., ¶ 0075) *periodically, at pre-defined intervals dump or copy data into a non-volatile memory. Specifying a certain periodic time-interval is a design choice.*

Disclosures by Harris, FBDIMM Standards, Amidi and Levkovitz are analogous because they are in the same field of endeavor and/or solving a similar or common problem.

It would have been obvious to a person of having ordinary skill in the art before the effective filing date of the claimed invention to modify the Voltage Distribution System and Method for a Memory Assembly taught by Harris to include the voltage (power), data, and address/control signals disclosed by FBDIMM Standards; furthermore, to include Clock and Power Fault Detection disclosed by Amidi; furthermore, to include the periodically copying data to non-volatile memory disclosed by Levkovitz.

The motivation for including the voltages, data, address, and control signal as taught by page 41 of TRIAL-TRM is to look/obtain information concerning voltages for the FBDIMM; furthermore, the motivation to include Clock and Power Fault Detection is to attempt to ensure a stable power supply even in the face of disruptions in system supply 520, for example; furthermore, the motivation for copying the data to non-volatile memory as taught by paragraph [0275] of Levkovitz is to limit the data loss in case of a power outage which may erase the volatile memory unit.

Therefore, it would have been obvious to combine teaching of Levkovitz, FBDIMM Standards, and Amidi with Harris to obtain the invention as specified in the claim.

Claims 18, 19 and 25 include similar limitations.

DIRECTION OF FUTURE CORRESPONDENCES:

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

15. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

16. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2138*

<i>Notice of References Cited</i>	Application/Control No. 17/582,797		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2138	Page 1 of 1

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*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
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*	B	US-20070136523-A1	2007-06-14	Bonella; Randy M.	G06F12/0866	711/E12.019
*	C	US-20080126624-A1	2008-05-29	Prete; Edoardo	G11C7/1006	710/53
*	D	US-20060174140-A1	2006-08-03	Harris; Shaun L.	G11C5/04	713/300
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NON-PATENT DOCUMENTS		
*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	JEDEC STANDARD, FBDIMM, Advanced Memory Buffer (AMB), JES82-20 JEDEC SOLID STATE TECHNOLOGY ASSOCIATION , March 2007, 190 pages
	V	JEDEC STANDARD, FBDIMM Specification, DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification,, JESD205, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 129 pages
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

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US Classification - Searched*			
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711	110	10/06/2022	HF
711	110	11/01/2023	HF

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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
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Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
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G06F3/0685	G06F3/00	11/01/2023	HF
G06F13/28	G06F13/00	11/01/2023	HF
G06F2212/205	G06F2212/00	11/01/2023	HF

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
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	Attorney Docket Number	022955-000090	

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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Examiner Name	FARROKH
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	1	20090020608	A1	2009-01-22	BENNETT et al.	
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	1	10-2005-0073902	KR	A	2005-07-18	SAMSUNG ELECTRONICS CO LTD	Machine Translation included as NPL 1	☒

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	Art Unit	2138	
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	Attorney Docket Number	022955-000090	

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	1	Machine Translation of KR 10-2005-0073902 A, published July 18, 2005, 7 pages.	☐
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(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

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☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

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A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L92	51	("5563838" "6351827" "6418065" "6434044" "7034569" "7161820" "7166934" "7269764" "7426649" "7613880" "7644216" "7656735" "20090020608" "20090027844").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 01:19 PM
L93	2	92 AND (buck near5 (conver\$3 OR regulat\$3))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 01:25 PM
L94	16	("8301833").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 03:47 PM
L95	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 05:00 PM
L96	2	("8874831").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 09:00 AM
L97	2	("8874831").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2024/07/02 09:00 AM

			JPO; DERWENT; IBM_TDB)				
L99	2	("20060174140").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 02:16 PM
L100	10	("7724604" "20060080515" "6856556").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 02:24 PM
L101	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/03 10:03 AM
L102	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/30 10:26 AM
L103	2	("20070136523").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 01:59 PM
L104	532	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:30 PM
L105	2281	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:31 PM
L106	97	105 AND (trigger\$3 near5 signal)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:31 PM
L107	0	97 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:32 PM
L108	410	105 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:32 PM
L109	13	106 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:33 PM
L110	55	(backup OR replicat\$3 OR cop\$4 OR writ\$4	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2024/08/02 04:48 PM

		OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH milli\$1second	JPO; DERWENT; IBM_TDB)				
L111	24	110 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:48 PM
L112	530	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) near5 (milli\$1second OR ms)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:51 PM
L113	143	112 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:52 PM
L114	71	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) ADJ (milli\$1second OR ms)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:53 PM
L115	14	114 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:54 PM
L116	6	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) ADJ (milli\$1second)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:55 PM
L117	0	116 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:55 PM
L118	0	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM

		WITH millisecond					
L119	145	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH second	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM
L120	26	119 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM
L121	0	120 AND dimm	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:19 PM
L122	19	"backup may occur every millisecond"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:39 PM
L123	0	122 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:39 PM
L124	19	(backup near5 occur near5 every near5 millisecond)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:40 PM
L125	19	(backup near5 occur near5 (every OR each OR one) near5 millisecond)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:41 PM
L126	31	(buck ADJ converters) SAME (regulat\$3 near2 voltage) SAME ((printed ADJ circuit ADJ board) OR PCB)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/08/05 08:32 AM
L127	3	((buck ADJ converters) SAME (regulat\$3 near2 voltage) SAME ((printed ADJ circuit ADJ board) OR PCB)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/08/05 08:32 AM
L128	3	127 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/08/05 08:33 AM
L129	5	126 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/08/05 08:34 AM
L130	6	126 AND ((G06F1/185	(US-PGPUB; USPAT;	OR	ON	ON	2024/08/05

		OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)	USOCR; IBM_TDB)				08:35 AM
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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Attorney Docket Number	022955-000090

1	Exhibit 2014, Qimonda, Internet Data Sheet, Rev. 1.25, June 2007, 24 pages.	☐
2	Exhibit 2015, Siemens, 3.3V 2M x 64/72-Bit 1 BANK SDRAM Module, HYS64/72V2200GU-8/-10, 3.3V 4M x 64/72-Bit 2 BANK SDRAM Module, HYS64/72V4200GU-8/-10, 168 pin unbuffered DIMM Modules, downloaded from www.datasheetcatalog.com; 1998, 18 pages	☐
3	Exhibit 2016, PCI Bus Variation - Technology Note, copyright 2006 by Euresys, 9 pages.	☐
4	Exhibit 2017, Qimonda, Internet Data Sheet, Rev. 1.42, September 2007, 27 pages.	☐
5	Exhibit 2018, Transcend, Product Support, What is the difference between SDRAM, DDR1, DDR2, DDR3 and DDR4?, available online at https://www.transcend-info.com/support/faq-296#:~:text=DDR3's prefetch buffer width is,according to the temperature variation , downloaded August 24, 2022, 2 pages.	☐
6	Exhibit 2021, Texas Instruments: Application Report, Low Dropout Operation in a Buck Converter, SLUA928A - December 2018 - revised March 2019, 18 pages.	☐
7	Exhibit 2022, Electronic Design, Simple Soft-Start Circuitry Provides Long Startup Times, available online at http://electronicdesign.com/power/simple-soft-start-circuitry-provides-long-startup-times , published June 22, 1998, 3 pages.	☐
8	Exhibit 2024, SCHMID, Understanding Hard Drive Performance, available online at https://www.tomshardware.com/reviews/understanding-hard-drive-performance,1557-5.html , downloaded August 15, 2022, published March 5, 2007, 17 pages.	☐
9	Exhibit 2025, Micron, SDR SDRAM, MT48LC64M4A2 - 16 Meg x 4 x 4 Banks, MT48LC32M8A2 - 8 Meg x 8 x 4 Banks, MT48LC16M16A2 - 4 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 94 pages.	☐
10	Exhibit 2026, Micron, SDR SDRAM, MT48LC16M4A2 - 4 Meg x 4 x 4 Banks, MT48LC8M8A2 - 2 Meg x 8 x 4 Banks, MT48LC4M16A2 - 1 Meg x 16 x 4 Banks, copyright 1999 by Micron Technology, Inc., 84 pages.	☐
11	Exhibit 2027, Transcend, DDR2 SO-DIMM, copyright 2012 by Transcend, 3 pages.	☐

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12	Exhibit 2028, Micron Technical Note: DDR3 Point-to-Point Design Support, copyright 2013 by Micron Technology, Inc., 27 pages.	☐
13	Exhibit 2030, IPR 2022-00996 and IPR 2022-00999, Videoconference Deposition of Andrew Wolfe, Ph.D. regarding US Patent Nos. 11,016,918 and 11,232,054, 311 pages, dated March 16, 2023.	☐
14	Exhibit 2031, IPR 2022-00996, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,016,918, 452 pages, executed on May 16, 2022.	☐
15	Exhibit 2031, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
16	Exhibit 2032, Claim Construction Order, filed December 14, 2022, 35 pages.	☐
17	Exhibit 2033, Defendants' Objections to Claim Construction Memorandum Order, filed December 29, 2022, 11 pages.	☐
18	Exhibit 2043, Samsung Unveils New Power Management Solutions for DDR5 Modules, May 18, 2021, 6 pages.	☐
19	Exhibit 2044, Micron, DDR2 SDRAM RDIMM, MT9HTF3272Y - 256MB, MT9HTF6472PY - 512MB, MT9HTF12872PY - 1GB, copyright 2003 by Micron Technology, Inc., 23 pages.	☐
20	Exhibit 2046, JEDEC Standard No. 21C, Page 4.20.11-1, PC2-6400/PC2-5300/PC2-4200/PC2-3200, DDR2 Unbuffered SO-DIMM, Reference Design Specification, Revision 2.5, July 2008, 60 pages.	☐
21	Exhibit 2048, Micron, DDR2 SDRAM FBDIMM, MT18HTF12872FD - 1GB, MT18HTF25672FD - 2GB, copyright 2005 by Micron Technology, Inc., 39 pages.	☐
22	Exhibit 2056, IPR 2022-00615, Remote Videotaped Deposition by Virtual Zoom of Andrew Wolfe, Ph.D. regarding US Patent No. 7,619,912, 297 pages, dated January 4, 2023.	☐

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23	Exhibit 2057, Micron Technical Brief: Micron DDR5: Key Module Features, DDR4 vs. DDR5 DIMM, copyright 2020 by Micron Technology, Inc., 7 pages.	☐
24	Exhibit 2063, Netlist, Demonstrative Exhibit for IPR 2022-00996 and IPR 2022-00999, filed September 6, 2023, 180 pages.	☐
25	Exhibit 2102, LEE et al., PCI Express Power and Mini Card Solutions - Design Note 346, Linear Technology - Design Notes, copyright 2004 by Linear Technology Corporation, 2 pages.	☐
26	Paper 1, IPR 2022-00996, Petition for Inter Partes Review of US Patent No. 11,016,918, filed May 17, 2022, 151 pages.	☐
27	Paper 1, IPR 2022-00999, Petition for Inter Partes Review of US Patent No. 11,232,054, filed May 17, 2022, 149 pages.	☐
28	Paper 7, IPR 2022-00996, Patent Owner Preliminary Response, filed September 9, 2022, 85 pages.	☐
29	Paper 7, IPR 2022-00999, Patent Owner Preliminary Response, filed September 8, 2022, 82 pages.	☐
30	Paper 10, IPR 2022-00996, Decision - Granting Institution of Inter Partes Review, dated December 7, 2022, 56 pages.	☐
31	Paper 21, IPR 2022-00996, Patent Owner Response, filed March 28, 2023, 91 pages.	☐
32	Paper 22, IPR 2022-00999, Patent Owner Response, filed March 28, 2023, 90 pages.	☐
33	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐

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34	Paper 26, IPR 2022-00999, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
35	Paper 31, IPR 2022-00996, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐
36	Paper 33, IPR 2022-00999, Patent Owner's Sur-Reply, filed August 4, 2023, 48 pages.	☐

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	07/02/2024
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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	1	5563839	A	1996-10-08	HERDT et al.	
	2	6067241	A	2000-05-23	LU	
	3	6693840	B2	2004-02-17	SHIMADA et al.	
	4	6798709	B2	2004-09-28	SIM et al.	
	5	6856556	B1	2005-02-15	HAJECK	
	6	7023187	B2	2006-04-04	SHEARON et al.	
	7	7061214	B2	2006-06-13	MAYEGA et al.	
	8	7692938	B2	2010-04-06	PETTER	

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9	7719866	B2	2010-05-18	BOLDO	
10	7721130	B2	2010-05-18	PRETE et al.	
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	3	20070136523	A1	2007-06-14	BONELLA et al.	
	4	20080101147	A1	2008-05-01	AMIDI	
	5	20080238536	A1	2008-10-02	HAYASHI et al.	

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	6	20090034354	A1	2009-02-05	RESNICK	
	7	20100205470	A1	2010-08-12	MOSHAYEDI et al.	
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	1	Transcript, Day 2 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 17, 2023, 375 pages.	☐
	2	Transcript, Day 4 (redacted), Trial on the Merits, Netlist, Inc. (plaintiff) vs. Samsung Electronics Co., Ltd., et al. (defendants), April 19, 2023, 338 pages.	☐
	3	Exhibit 1003, IPR 2022-00999, Declaration of Dr. Andrew Wolfe regarding US Patent No. 11,232,054, 431 pages, executed on May 16, 2022.	☐

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4	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 60 pages.	☐
5	Exhibit 1015, IPR 2014-00994, Decision - Denying Institution of Inter Partes Review, issued December 16, 2014, 16 pages.	☐
6	Exhibit 1016, Corrected Petition for Inter Partes Review of US Patent No. 8,301,833, filed September 22, 2014, 67 pages.	☐
7	Exhibit 1017, IPR 2014-01370, Decision - Denying Institution of Inter Partes Review, issued March 13, 2015, 19 pages.	☐
8	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
9	Exhibit 1019, IPR 2017-00649, Decision - Denying Institution of Inter Partes Review, issued July 24, 2017, 17 pages.	☐
10	Exhibit 1020, Petition for Inter Partes Review of US Patent No. 8,874,831, filed January 17, 2017, 78 pages.	☐
11	Exhibit 1021, IPR 2017-00692, Final Written Decision, issued July 5, 2018, 42 pages.	☐
12	Exhibit 1022, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 14, 2022, 76 pages.	☐
13	Exhibit 1026, JEDEC STANDARD, DDR2 SDRAM SPECIFICATION, JESD79-2B (Revision of JESD7-2A), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, January 2005, 113 pages.	☐
14	Exhibit 1027, JEDEC STANDARD, FBDIMM: Advanced Memory Buffer (AMB), JESD82-20, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 190 pages.	☐

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15	Exhibit 1028, JEDEC STANDARD, FBDIMM Specification: DDR2 SDRAM Fully Buffered, DIMM (FBDIMM) Design Specification, JESD205, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, March 2007, 129 pages.	☐
16	Exhibit 1031, PCI Local Bus Specification, Revision 2.2, December 18, 1998, 322 pages.	☐
17	Exhibit 1032, Power Electronics: Converters, Applications, and Design, Second Edition, copyright 1995 by John Wiley & Sons, Inc., 37 pages.	☐
18	Exhibit 1040, Texas Instruments, TPS51020, Dual, Voltage Mode, DDR Selectable, Synchronous, Step-Down Controller for Notebook System Power, SLUS564B, July 2003 (revised December 2003), 27 pages.	☐
19	Exhibit 1041, Fairchild Semiconductor, FAN5026, Dual DDR/Dual-Output PWM Controller, October 2005, 17 pages.	☐
20	Exhibit 1042, Murata Manufacturing Co., Ltd, Power Supply Reference Guide for XILINX FPGAs, September 2006, 33 pages.	☐
21	Exhibit 1043, Murata Products, Power Supply Reference Guide for ALTERA FPGAs, Catalog No. DC-03-A, 2008, 40 pages.	☐
22	Exhibit 1045, JEDEC STANDARD, Double Data Rate (DDR) SDRAM Specification, JESD79, JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, June 2000, 77 pages.	☐
23	Exhibit 1046, JEDEC STANDARD, DDR3 SDRAM Specification, JESD79-3A (Revision of JESD79-3), JEDEC SOLID STATE TECHNOLOGY ASSOCIATION, September 2007, 189 pages.	☐
24	Exhibit 1048, Murata Manufacturing, DC-DC Converter Specification(DRAFT), MPD4S014S, January 21, 2008, 11 pages.	☐
25	Exhibit 1049, Micron, NAND Flash Memory, MT29F2G08AABWP/MT29F2G16AABWP, MT29F4G08BABWP/MT29F4G16BABWP, MT29F8G08FABWP, copyright 2004 by Micron Technology, Inc., 57 pages.	☐

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26	Exhibit 1058, LENK, Simplified Design of Switching Power Supplies, Linear Technology, EDN Series for Design Engineers, copyright 1995 by Butterworth-Heinemann, 9 pages.	☐
27	Exhibit 1061, Analog Devices, Super Sequencer with Margining Control and Auxiliary ADC Inputs, ADM1066, copyright 2006 by Analog Devices, Inc., 36 pages.	☐
28	Exhibit 1062, Analog Devices, Analog Dialogue, Volume 40, Number 2, 2006, 24 pages.	☐
29	Exhibit 1063, National Semiconductor, LMC6953, PCI Local Bus Supervisor, October 1996, 12 pages.	☐
30	Exhibit 1067, Lattice Semiconductor Corporation, LatticeXP Family Data Sheet, Version 04.2, March 2006, 127 pages.	☐
31	Exhibit 1068, Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed October 15, 2021, 82 pages.	☐
32	Exhibit 1069, First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, filed January 18, 2022, 122 pages.	☐
33	Exhibit 1071, Complaint, filed December 20, 2021, 49 pages.	☐
34	Exhibit 1072, The Samsung Defendants' Answer, filed April 12, 2022, 37 pages.	☐
35	Exhibit 1073, First Amended Complaint, filed May 3, 2022, 71 pages.	☐
36	Exhibit 1075, IPR 2022-00999, Deposition of William Mangione-Smith, Ph.D. regarding US Patent No. 11,232,054, 245 pages, dated June 12, 2023.	☐

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37	Exhibit 1076, IPR 2022-00999, redline comparing Declarations - Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,232,054, 165 pages, executed on March 29, 2022.	☐
38	Exhibit 1077, Netlist, Technology Tutorial: In the Matter of Certain Memory Modules and Components Thereof and Products Containing Same, Inv No. 337-TA-1023, May 8, 2017, 11 pages.	☐
39	Exhibit 1078, Analog Devices, Step-Down DC-to-DC Controller, ADP1821, copyright 2005-2007 by Analog Devices, Inc., 24 pages.	☐
40	Exhibit 2001, IPR 2022-00999, Declaration of Dr. Sunil P. Khatri regarding US Patent No. 11,232,054, 68 pages, executed on September 8, 2022.	☐
41	Exhibit 2003, BELLONI et al., 24.6 A 4-Output Single-Inductor DC-DC Buck Converter with Self-Boosted Switch Drivers and 1.2A Total Output Current, ISSCC 2008 / Session 24 / Analog Power Techniques / 24.6, 2008 IEEE International Solid-State Circuits Conference, 3 pages.	☐
42	Exhibit 2004, MA et al., Single-Inductor Multiple-Output Switching Converters With Time-Multiplexing Control in Discontinuous Conduction Mode, IEEE Journal of Solid-State Circuits, Volume 38, Number 1, January 2003, pages 89-100 (12 pages).	☐
43	Exhibit 2006, Micron, Technical Note: Power Solutions for DDR2 Notebook PCs, TN-47-05 DDR2 Power Solutions for Notebooks, copyright 2004 by Micron Technology, Inc., 16 pages.	☐
44	Exhibit 2007, Texas Instruments, LP2996-N, LP2996A DDR Termination Regulator, SNOSA40K, November 2002 (revised December 2016), 39 pages.	☐
45	Exhibit 2008, JAMECO ELECTRONICS, Jameco Part Number 843868, National Semiconductor: LP2996: DDR Termination Regulator, June 2006, 19 pages.	☐
46	Exhibit 2009, JAMECO ELECTRONICS, Jameco Part Number 843930, National Semiconductor: LP2997: DDR-II Termination Regulator, June 2006, 13 pages.	☐
47	Exhibit 2010, National Semiconductor: LP2998: DDR-II and DDR-I Termination Regulator, December 12, 2007, 18 pages.	☐

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Attorney Docket Number	022955-000090

48	Exhibit 2011, BERGVELD, Battery Management Systems: Design by Modelling, copyright 2001 by Royal Philips Electronics N.V., 328 pages.	☐
49	Exhibit 2012, ROMO, DDR Memories: Comparison and Overview, NXP, pages 70-73 (4 pages), September 21, 2007.	☐
50	Exhibit 2013, JEDEC Standard No. 21-C, Page 4.20.3-1; PC133 SDRAM Unbuffered SO-DIMM, Reference Design Specification, Revision 1.02, 26 pages. downloaded August 16, 2022.	☐

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Examiner Signature	/HASHEM FARROKH/	Date Considered	07/02/2024
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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-05-28
Name/Print	Khaled Shami	Registration Number	38745

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	09/13/2024	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
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<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 2

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Telephonic
Khaled Shami	Attorney	

Date of Interview: 06 September 2024

Issues Discussed:

35 U.S.C. 103

An applicant initiated interview with the Applicant, Mr. Khaled Shami (reg. # 38,745), conducted on 09/06/2024. The interview was in response to the non-final office action mailed 08/08/2024. Applicant presented a proposed amendment of claims in which claim 2 was amended, claims 1 and 3-27 canceled and claims 28-23 added. Applicant summarize the differences between the amended newly amended claims and the prior art of record in Items I through VI shown below. However, amendments fails to point the portion(s) of the specification that describe amendment (or arguments).

I. The limitation recite "a input voltage line from the memory socket...", then I.a recites power from system board provided by the set of power conduits. Specification does not describe/support the set of power conduits.

II. The specification does not describe/support a set of edge connection configure to be connected to a set of data conduits (or address and control conduits).

III. Applicant appears to state the claim limitation(s) that DIMM operates in two states no longer is being claimed. In discussion related to item number III, Applicant appears to discuss limitations or subject matters that are not described/supported by the specification, as an example the specification does not describe a plurality of packages. Furthermore, In the non-final rejection, mailed on 08/08/2024, the Trial Termination document also used as additional reference information. The Applicant arguments, for most part, appear to based on Harris and FBDIMM. The Examiner suggests that Applicant also refer to the relevant portion(s) of Trial-Term document and describe why the amendment overcome issues related to the board final decision.

IV. Regarding voltage regulation and mapping, the board has decided that combination of Harris and FBDIMM standards renders obvious limitations regarding voltage regulators (including buck converters). Applicant needs to describe what is new in claim limitations that overcome the discussion in Trial-Term document and the board decision.

V. Amidi teaches that memory (comprising backup power) may operate in backup mode, in response, for example, in response to change to voltage level (e.g., a triggering condition). The backup comprises a plurality of write operations.

VI. Paragraph [0075] teaches periodically dump or copy data to non-volatile memory. The reason for periodic of copying is to limit data loss in the event of sudden power loss. The copying or backup performed in a pre-defined time intervals. A timer may be programmed to trigger the copying at pre-defined time interval

<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 2 of 2

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

PROPOSED AMENDMENT AND INTERVIEW AGENDA

FOR DISCUSSION PURPOSES ONLY

LISTING OF PROPOSED CLAIMS

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~slot connector socket~~ of a system board of a ~~host computer~~ system, the memory ~~slot connector socket~~ including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the ~~host system board system~~ is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data ~~signal~~ signal is electrically conducted between the ~~host system board~~ and the DIMM are transferred via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the ~~host system board~~ are transferred to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB), the PCB including:

an interface having edge connections disposed along the width of the PCB and configured to fit into the memory ~~slot connector socket~~ of the ~~host system~~, the edge connections including a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits, a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits,

the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board,

a set of data lines electrically ~~connected~~ coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is

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configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines, and

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a set of address and control lines electrically ~~connected~~ coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller including a voltage monitor circuit, a data storage element configured to store a plurality of data bits, a logic element, and nonvolatile memory, the controller coupled to the PCB and electrically connected to the ~~to an~~ input voltage supply line ~~of the set of voltage supply lines~~, the controller configured to receive the system power when the input voltage is electrically conducted by the input voltage supply line, the voltage monitor circuit configured to (i) monitor an input voltage by comparing the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage ~~supplied via the input voltage supply line~~, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to the logic element ~~at least one other portion of the controller~~, wherein the controller logic element is configured to perform, in response to the trigger signal, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to write a second set of data bits into a second portion of the data storage element, and then a third write operation to write the first and second sets of data bits from the data storage element into the nonvolatile memory, wherein the DIMM is operable when the input voltage is within a range of operating voltage values of the DIMM, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, wherein the lower operating voltage value is below a specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, wherein the input voltage corresponds to the specified operating voltage value, wherein the first threshold voltage matches the lower operating voltage value, and the second threshold voltage matches the upper operating voltage value, and wherein the trigger condition occurs when the input voltage exceeds the second ~~a first preset~~ threshold voltage ~~that is above an operating voltage of the DIMM~~;

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a set of components coupled to the PCB, the set of components including a plurality of double-data-rate (DDR) synchronous dynamic random access memory (DDR SDRAM) devices;

packages coupled to the set of address and control lines and to the set of data lines, the each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and devices being operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on address and control signals received from the system via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third and fourth converter circuits being configured coupled to the PCB and to the controller, each of the first, second, third and fourth converter circuits input voltage supply lines configured to receive power from the controller when the system power is received by the controller, the first, second, third and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power received from the controller and (ii) output the first, second, third, and fourth regulated voltages via first, second, third and fourth regulated voltage lines, wherein the first regulated voltage is generated by the first converter circuit, a second component electrically connected to the second regulated voltage line and powered on when the second regulated voltage is turned on by the second converter circuit, a third component electrically connected to the third regulated voltage line and powered on when the third regulated voltage is turned on by the third converter circuit, and a fourth component electrically connected to the fourth regulated voltage line and powered on when the fourth regulated voltage is turned on by the fourth converter circuit, and the first converter circuit is a buck converter, and the second, third and fourth converter circuits are voltage reducing circuits.

wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

wherein each component of the set of components (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, and (ii) is powered, when the system power is received by the controller, (a) receives power only by one or more of the first, second, third, and fourth converter circuits, and (b) only via the one or more of the first, second, third and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM packages is configured to receive power using at least the first regulated voltage supplied from the first converter circuit via the first regulated voltage line;

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wherein the plurality of DDR SDRAM packages includes a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal, and a second group of at least three DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal.

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wherein the first group of DDR SDRAM packages is operable to be (i) accessible and not in self-refresh mode, and (ii) enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines.

wherein the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals, and

wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data lines of the set of data lines, and the total number of data lines is at least 72 data lines.

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wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.

3 – 5. (Canceled)

6 – 27 (Canceled)

28. (New) The DIMM of claim 2, wherein the second component (i) is electrically connected to both the first and second regulated voltage lines, and (ii) is powered on when the first and second

regulated voltages are turned on by the first and second converter circuits, respectively.

29. (New) The DIMM of claim 2, wherein the data storage element is a buffer configured to store the plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

30. (New) The DIMM of claim 2, wherein the PCB is configured to have a form factor with a width of 133.35 millimeters.

31. (New) The DIMM of claim 30, wherein the form factor of the PCB is selected from one of a plurality of profiles ranging in height from 18.3 millimeters to 50 millimeters.

32. (New) The DIMM of claim 2, wherein the third write operation to write the first and second sets of data bits from the data storage element into the nonvolatile memory includes the logic element combining the first set of data bits and the second set of data bits and writing the combined first and second sets of data bits into the nonvolatile memory.

33. (New) The DIMM of claim 2, wherein the third write operation to write the first and second sets of data bits from the data storage element into the nonvolatile memory includes the logic element writing the plurality of data bits from the data storage element into the nonvolatile memory.

34. (New) The DIMM of claim 2, wherein, when the system power is received by the controller, the nonvolatile memory is not powered using the first regulated voltage.

35. (New) A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB), the PCB including:

an interface having edge connections disposed along the width of the PCB and configured to fit into the memory socket, the edge connections including a first set of edge connections configured to be electrically connected to respective power conduits of the set of

power conduits, a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;

a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller including a voltage monitor circuit, a data storage element configured to store a plurality of data bits, a logic element, and nonvolatile memory, the controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power when the input voltage is electrically conducted by the input voltage supply line, the voltage monitor circuit configured to monitor the input voltage by comparing the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage, wherein the logic element is configured to periodically perform, at regular intervals of time and in response to a trigger condition, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to write a second set of data bits into a second portion of the data storage element, wherein the DIMM is operable when the input voltage is within a range of operating voltage values of the DIMM, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, wherein the lower operating voltage value is below a specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, wherein the input voltage corresponds to the specified operating voltage value, wherein the first threshold voltage matches the lower operating voltage value, and the second threshold voltage matches the upper operating voltage value, and wherein the trigger condition occurs when the input voltage exceeds the second threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converter circuits coupled to the PCB and to the controller, each of the first, second, third, and fourth converter circuits configured to receive power from the controller when the system power is received by the controller, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power received from the controller, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, the set of components including a first component electrically connected to the first regulated voltage line and powered on when the first regulated voltage is turned on by the first converter circuit, a second component electrically connected to the second regulated voltage line and powered on when the second regulated voltage is turned on by the second converter circuit, a third component electrically connected to the third regulated voltage line and powered on when the third regulated voltage is turned on by the third converter circuit, and a fourth component electrically connected to the fourth regulated voltage line and powered on when the fourth regulated voltage is turned on by the fourth converter circuit, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit;

wherein each component of the set of components (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, and (ii) is powered, when the system power is received by the controller, (a) only by one or more of the first, second, third, and fourth converter circuits, and (b) only via the one or more of the first, second, third, and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM packages is configured to receive power using at least the first regulated voltage supplied from the first converter circuit via the first regulated voltage line;

wherein the plurality of DDR SDRAM packages includes a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal, and a second group of at least three DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein the first group of DDR SDRAM packages is operable to be (i) accessible and not in

self-refresh mode, and (ii) enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;

wherein the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data lines of the set of data lines, and the total number of data lines is at least 72 data lines.

36. (New) The DIMM of claim 35, wherein each of the regular intervals of time is one millisecond duration.

37. (New) The DIMM of claim 35, wherein the logic element is further configured to read and combine first and second data bits from the first and second portions of the data storage element, respectively.

38. (New) The DIMM of claim 37, wherein the plurality of write operations includes a third write operation, and wherein the logic element is further configured to perform the first and second write operations and subsequently to perform the third write operation to write the combined data bits from the first and second portions of the data storage element into the nonvolatile memory.

39. (New) The DIMM of claim 35, wherein the data storage element is a buffer configured to store the plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

40. (New) The DIMM of claim 35, wherein the PCB is configured to have a form factor with a width of 133.35 millimeters.

41. (New) The DIMM of claim 41, wherein the form factor of the PCB is selected from one of a plurality of profiles ranging in height from 18.3 millimeters to 50 millimeters.

42. (New) The DIMM of claim 35, wherein, when the system power is received by the controller, the nonvolatile memory is not powered using the first regulated voltage.

43. (New) A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB), the PCB including:

an interface having edge connections disposed along the width of the PCB and configured to fit into the memory socket, the edge connections including a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits, a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;

a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller including a voltage monitor circuit, a data storage element configured to store a plurality of data bits, a logic element, and nonvolatile memory, the controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power when the input voltage is electrically conducted by the input voltage supply line, the voltage

monitor circuit configured to monitor the input voltage by comparing the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage, wherein the logic element is configured to periodically perform every millisecond, in response to a trigger condition, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to write a second set of data bits into a second portion of the data storage element, wherein the DIMM is operable when the input voltage is within a range of operating voltage values of the DIMM, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, wherein the lower operating voltage value is below a specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, wherein the input voltage corresponds to the specified operating voltage value, wherein the first threshold voltage matches the lower operating voltage value, and the second threshold voltage matches the upper operating voltage value, and wherein the trigger condition occurs when the input voltage exceeds the first threshold voltage and is below the second threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converter circuits coupled to the PCB and to the controller, each of the first, second, third, and fourth converter circuits configured to receive power from the controller when the system power is received by the controller, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power received from the controller, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, the set of components including a first component electrically connected to the first regulated voltage line and powered on when the first regulated voltage is turned on by the first converter circuit, a second component electrically connected to the second regulated voltage line and powered on when the second regulated voltage is turned on by the second converter circuit, a third component electrically connected to the third regulated voltage line and powered on when the third regulated voltage is turned on by the third converter circuit, and a fourth component electrically connected to the fourth regulated voltage line and powered on when the fourth regulated voltage is turned on by the fourth converter circuit, wherein each of the first, second, and third

converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit;

wherein each component of the set of components (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, and (ii) is powered, when the system power is received by the controller, (a) only by one or more of the first, second, third, and fourth converter circuits, and (b) only via the one or more of the first, second, third, and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM packages is configured to receive power using at least the first regulated voltage supplied from the first converter circuit via the first regulated voltage line;

wherein the plurality of DDR SDRAM packages includes a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal, and a second group of at least three DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein the first group of DDR SDRAM packages is operable to be (i) accessible and not in self-refresh mode, and (ii) enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;

wherein the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data lines of the set of data lines, and the total number of data lines is at least 72 data lines.

REMARKS

By this Proposed Amendment, submitted for purposes of discussion only, Applicant has **amended claim 2; canceled claims 6-8 and 11 and 14-27; and added new claims 28-43.** (Claims 3-5, 9-10, and 12-13 were previously canceled). **Claims 2 and 28-43 are now pending.**

Agenda for Examiner Interview

Points of Distinction to Discuss

- I. **The system power used by four converters is electrically connected to an input voltage line from the memory socket of a system board of a computer system in a particular form and on particular first set of edge connections of the PCB having edge connections configured to fit into the memory slot connector.**
 - a. “wherein power from the system board is provided to the DIMM via the memory socket by the set of power conduits” and
 - b. “a printed circuit board (PCB) having edge connections configured to fit into the memory socket, the edge connections including a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits”

Neither Harris nor FBDIMM, singularly or in combination, disclose this feature. Harris discloses “since the Vdd and Vcc power supply rails are rendered independent from external voltage sources, the overall power supply interface to the memory module 100A is improved to better support large memory capacity scaling, hot-pluggability, DRAM technology independence (e.g., DDR2, DDR3, DDR4, and other DRAM types) . . .” Para. [0011]. This passage makes clear that the DIMM of Harris draws power from a source external to the system, i.e. “external voltage sources,” as does a reference to “hot-pluggability” especially in light of DRAM technology independence. Each DDR generation standardizes a different set of voltages for memory modules, and the power management system on the motherboard of DDR4 and previous generations exclusively provides power via the socket to the DIMM. Thus, the ONLY way to have hot pluggability and DRAM technology independence is to have power from a source external to the host system.

This source external to the system is confirmed in Para. [0012] of Harris, which further states “a voltage-independent FBD design incorporating the embodiment of FIG. 1A is capable of replacing these power supply interface pins with as few as six +12V pins (from an external voltage source) with local conversion . . .” Paragraph [0019] is even more conclusive.

“Based on the foregoing Detailed Description, it should be appreciated that an implementation of the embodiments described herein thus provides a technology-independent voltage distribution scheme for memory devices where system board power supply and associated voltage plane(s) are eliminated.”

Later in the same paragraph, Harris discloses “reduction in the pin count on DIMM connector to system board or memory riser card (whereby a smaller connector or wider pin spacing may be accommodated); . . . reservation of extra pins for implementing addition standard or proprietary functions.” Thus, it is conclusive that power does not come from the system board or through the socket because those edge connections normally used for power are either eliminated or repurposed.

Moreover, the FBDIMM Standards and Amidi provide no cure for this defect, as they use the same FBDIMM edge connections as Harris.

II. Separate address and control signals, and data signals from system board to the claimed module in a particular form and on particular second and third sets of edge connections of the PCB having edge connections configured to fit into the memory slot connector:

- a. “data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals from the host system are transferred to the DIMM via the set of address and control conduits” and
- b. “a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits”

The JEDEC standard modules which the applied prior art use a packetized signaling protocol. Data and command/control/address signals do not travel on separate lines, because they are sent

in packetized forms called “frames” – “After initialization the host communicates with the AMBs on the channel using southbound frames of information containing commands and data.” JESD206.01, pg. 55

- c. Thus, the data signals are not on “the set of data conduits” and the address and control signals are not on “the set of address and control conduits” as recited in the claim, rather they are on the same set of conduits, referred to as “lanes” by JESD206.01.
- d. Moreover, the FBDIMM Standards and Amidi provide no cure for this defect, as they use the same FBDIMM protocols as Harris.
- e. Thus, the current claims are allowable for at least this additional reason.

III. Applicant’s additional amendments to further distinguish over the teachings of Harris and FBDIMM in the combination such that the claimed DIMM is no longer (i) operable in two states, or (ii) turning off power to any component of the set of components on the claimed DIMM, specifically:

- a. The wherein clause for the DIMM operating in two states and power is turned off to one component is completely deleted, and thus at least for this reason the rejection is no longer applicable.
- b. The claim as currently amended always provide power to all components via all regulated voltage when system power is turned on, and
- c. Additional wherein clauses are added to specifically claim that the DDR SDRAM are accessible and not in self-refresh mode.
- d. At least for the reasons above the claimed DIMM is no longer (i) operable in two states, or (ii) turning off power to any component of the set of components on the claimed DIMM.
- e. Thus, the rejection over Harris and FBDIMM in the combination over the S3 mode is not applicable to the currently amended Claim 2.

IV. Each component of the set of components

- a. is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines, and
- b. is powered, when the system power is received by the controller,

- i. only by one or more of the first, second, third, and fourth converter circuits, and
- ii. only via the one or more of the first, second, third, and fourth regulated voltage lines.

The above listed additional limitations further distinguishes over claim 2 as amended on 2/8/2024 and that none of Harris, FBDIMM, and Amidi, individually or in the combination, teach any of these additional limitations, at least because:

- iii. none of Harris, FBDIMM, and Amidi, individually or in the combination, teach Harris teach a controller configured to receive system power and provide power to each of the four converter circuits, as summarized in limitation 4.a above
- iv. none of Harris, FBDIMM, and Amidi, individually or in the combination, teach Harris teach that four converter circuits receive power from the controller and generates regulated voltages, as summarized in limitations 4.b above.
- v. none of Harris, FBDIMM, and Amidi, individually or in the combination, teach Harris teach that that each component of the set of components including the plurality of DDR SDRAM packages are powered ONLY from the converter circuits regardless of whether a trigger condition is detected by the voltage monitor circuit, and that all four converters are required to receive power from the controller when the system power is received by the controller, as summarized in limitations 4.c and 4.d above.
- c. Thus, none of Harris, FBDIMM, and Amidi, individually or in the combination, teach any of the additional limitations currently presented in sections 4a-4.d above.
- d.

V. Performing a plurality of write operations in response to a trigger condition:

- a. Currently presented limitations further distinguish over Claim 2 as amended on 2/8/2024 and further require that the logic element of the controller, in response to a trigger signal generated by a voltage monitor circuit, performs

- i. first and second write operations to write first and second sets of data bits to data storage element, and then
- ii. write operation to write the first and second set of data bits from the data storage element into the NVM
- iii. the trigger signal generated by the voltage monitor circuit that monitors the input voltage line that supplies the claimed system power to the controller, and compares the input voltage against a range of operating voltages bounded by first and second threshold voltages,
- iv. the trigger condition occurs when the input voltage exceeds the second threshold voltage.

Amidi does not teach writing to NVM in response to detection of any power faults, as required by claim 2. Amidi only teaches a “backup system” and describes ways to “initiate backup system”. However, the “backup system” that Amidi teaches is only about (i) switches the power to the DRAM to a “backup” voltage source such as a battery, and (ii) maintain the DRAM in a self-refresh state using the “backup system” power supply. Please note that Amidi expressly teaches DRAM are not accessible (because they are in self-refresh mode) until main system power is restored. (See at least in Fig. 15 and Col. 9:13-39)

Amidi does not teach that all components of a set components are powered only from the buck converters, as required by claim 2. Amidi teaches that DRAM are powered from system power, and upon detection of power failure switches the DRAM components to a backup system using a battery supplied voltage. (Fig. 5, Fig. 6, Fig. 7, and Fig. 14, and at least in Col. 8:23-61). Thus, Amidi teaches away from powering any component from the system power upon detecting a power fault.

Amidi does not teach the detection of an input voltage that exceed a threshold voltage which is above an operating voltage of the claimed DIMM, as required by claim 2.

Amidi only teaches the detection of power failure and provide many examples of power failure including when input voltage is below a reference voltage such as a 3.0V for a 3.3V rail (Fig. 14 and Col. 8:27-29)

VI. Additional currently presented limitations regarding periodically performing a plurality of write operations in response to a trigger condition, as claimed in independent claim 35:

Levkovitz teaches the periodic dump or copy from volatile to nonvolatile is **always performed every 60 seconds, every 5 minutes, every hour**, and not in response to a trigger condition, as required by the claims.

Furthermore, Levkovitz teaches that memory unit 118 ad server 110 is a volatile memory unit, and thus cannot include nonvolatile memory. Thus, Levkovitz does not teach that the nonvolatile memory is on a memory module that includes the volatile memory (plurality of DDR SDRAM), as required by the claims.

Moreover, **the periodic dump or copy**, as taught by Levkovitz, is to “limit the data loss in case of a power outage which erase the volatile memory unit 118.” Thus, Levkovitz teachings eliminates the needs for Amidi’s teachings.

Respectfully submitted,

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Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

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PATENT # -

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PATENT CENTER # 67294168

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 39

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
IDS - 022955-90.pdf	10	Information Disclosure Statement (IDS) Form (SB08)	1366 KB
FOR--EP0744748B1.pdf	30	Foreign Reference	461 KB
FOR--JP2001357669A.pdf	16	Foreign Reference	2378 KB
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Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 67294168

AUTHORIZED BY Khaled Shami

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CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Chi-She Chen

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	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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Art Unit	2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

31	7702928	B2	2010-04-20	SCHUMACHER et al.	
32	7779285	B2	2010-08-17	DOBLAR et al.	

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	2	20060294437	A1	2006-12-28	WASHBURN et al.	
	3	20080307240	A1	2008-12-11	DAHAN et al.	
	4	20090020608	A1	2009-01-22	BENNETT et al.	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		17582797
Filing Date		2022-01-24
First Named Inventor	CHEN	
Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

	1	10-2005-0073902	KR	A	2005-07-18	SAMSUNG ELECTRONICS CO LTD	Machine Translation attached	☒
	2	0 744 748	EP	B1	2000-10-18	SILICON GRAPHICS, INC.		☐
	3	0 908 892	EP	B1	2005-07-27	NEC CORPORATION; NEC Electronics Corporation		☐
	4	3707888	JP	B2	2005-10-19	HITACHI LTD	Machine Translation attached	☒
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	6	2001-318741	JP	A	2001-11-16	IDEC IZUMI CORP	Machine Translation attached	☒
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	10	2005/066968	WO	A1	2005-07-21	INTEL CORPORATION		☐

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First Named Inventor	CHEN
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Attorney Docket Number	022955-000090

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	ISL6537 "ACPI Regulator/Controller for Dual Channel DDR Memory Systems", Renesas, 16 pages, 07/18/2007, taken from internet on 16 April 2024 from: https://www.renesas.com/us/en/document/dst/isl6537-datasheet	☐
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	4	TKDM "TKDM - A Reconfigurable Co-processor in a PC's Memory Slot", Plessi, 8 pages, 12/17/2003, taken from internet on 17 April 2024 from: https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=615235db2bed5bb4823f9c3a9d383a92e1f602c7	☐
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Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
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Examiner Name	FARROKH
Attorney Docket Number	022955-000090

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Art Unit	2138	
Examiner Name	FARROKH	
Attorney Docket Number	022955-000090	

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	Filing Date		2022-01-24
	First Named Inventor	CHEN	
	Art Unit	2138	
	Examiner Name	FARROKH	
	Attorney Docket Number	022955-000090	

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☐ See attached certification statement.

☒ A certification statement is not submitted herewith.

FEE

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

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☐ An identification of an earlier application pursuant to 37 CFR 1.98 (d) (1) is attached.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-09-24
Name/Print	Khaled Shami	Registration Number	38745

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APPLICATION #
17/582,797

RECEIPT DATE / TIME
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ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 67397793

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

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Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
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Attachment showing support for all Claim elements.pdf	664A60D5E02B93DE404ECD392B24A269EF0CFF62BA367F93 CE8F0734294DFBF3B276DF753BFEAB6E3E64D7BAFB0B7913 BCA3B5B6BA5013FF49DFB57D6E55F47C

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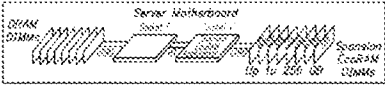
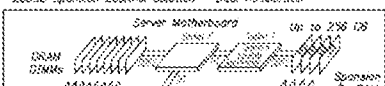
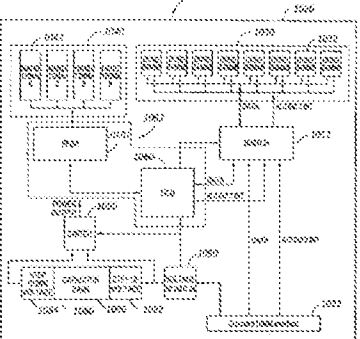
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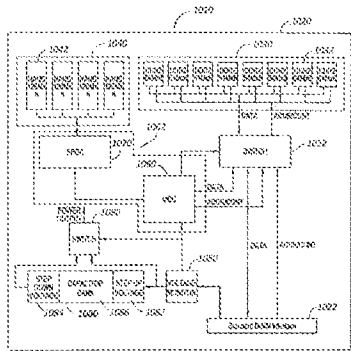
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Support for Claim Amendments

Claim Element	Limitation	Support
2.1	A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board,	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figure 2 and Fig. 12. Spanion EcceRAM Configurations 256GB Spanion EcceRAM Solution - Single Accelerator  256GB Single Accelerator Spanion EcceRAM Solution 256GB Spanion EcceRAM Solution - Dual Accelerator  256GB Single Accelerator Spanion EcceRAM Solution FIG. 2 (PRIOR ART)  FIG. 12

	The memory socket of a computer system includes pins soldered onto the system board that provide electrical conduits for system power, data, and address and control signals from the system board to be electrically connected to corresponding edge connections of the edge connector 1022 when it is inserted into the memory socket. The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector. The set of data conduits includes at least 72 data conduits. Para. [00122] “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)”. The embodiment of FIG. 13 shows nine x8 DRAM devices for a data bus bit width of 72: FIG. 13
--	---

2.2	a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” This is showing a 64 bit wide DDR2 data bus, where each bit is sent in parallel. From the DDR2 Standard a POSITA would be aware that each data line is one bit wide as illustrated in FIG. 12: FIG. 12 
2.3	the DIMM comprising: a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:	Para. [00122] “In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters.”

2.4	an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including: a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits, a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB. See also limitation 2.1.
2.5	a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically	Note respective power conduits (Vdd and gnd) of the set of power conduits of the Standard DIMM Interface 1022 are electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16, for example, in Fig. 12 the input voltage supply lines is shown electrically connected to the Voltage Monitor 1050.

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

FIG. 12

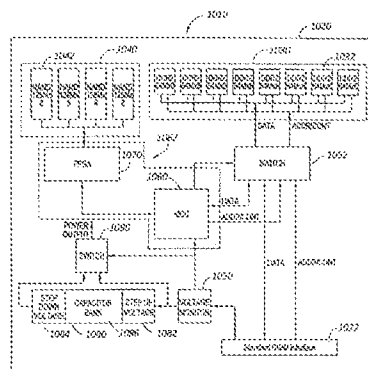
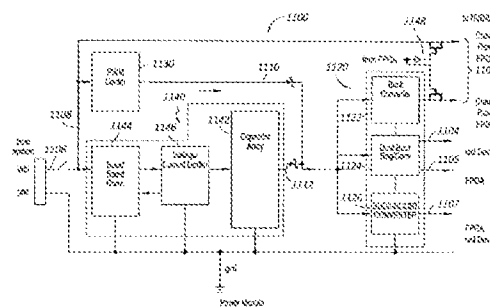


FIG. 15



Para. [00124]
 “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises **eight 64Mx8 DDR2 SDRAM** elements 1032.”

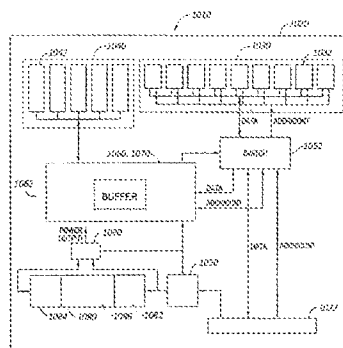
And

“In one embodiment, for example, the volatile memory subsystem 1030 comprises **a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example,**

a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to:

“FIG. 14 is a block diagram of an example memory module 1010 having **a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.**”

FIG. 14

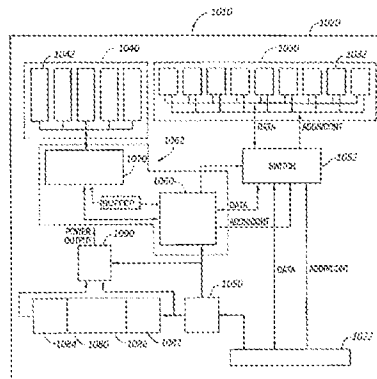


Para. [00172]

Para. [00178]

For example, FIG. 13 shows controller 1062 comprising a buffer:

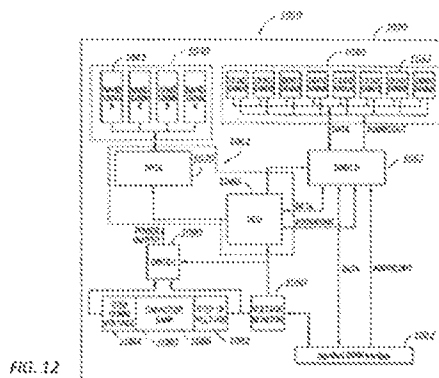
FIG. 13



Para. [00127]

“Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the **logic element 1070 comprises** an FPGA available from Lattice Semiconductor Corporation which includes **an internal flash**. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. **The internal flash can improve the speed of the memory system 1010 and save physical space**. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, **the logic element 1070 is a custom device**. In certain embodiments, **the logic element 1070 comprises various discrete electrical elements**, while in certain other embodiments, **the logic element 1070 comprises one or more integrated circuits**. FIG. 14 is a block diagram of an example memory module 1010 having **a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062** in accordance with certain embodiments described herein. **In certain embodiments, the controller 1062 includes one or more other components**. For example, in one embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.”

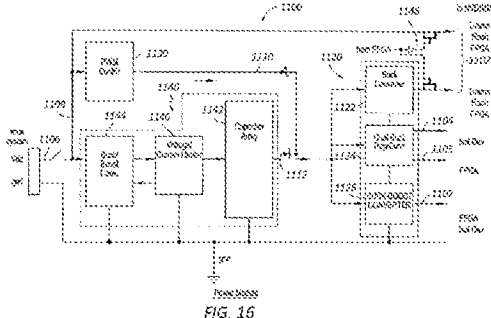
FIG. 12



		Para. [00133] “The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.” “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060)” Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070
2.7	compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and detect a trigger condition;	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
2.8	wherein, in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations, including: a first write operation to write a first set of data bits	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”

	into a first portion of the data storage element; a second write operation to write a second set of data bits into a second portion of the data storage element, and subsequently; and subsequently a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;	Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
2.9	wherein, when the system power is provided to the DIMM from the system board: the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and wherein: a specified operating voltage value is within a range of operating voltage values of the DIMM, the range of operating voltage values is bounded by an upper operating voltage value and a	For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above. Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.

	lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value;	
2.10	a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and	FIG. 12 FIG. 12 Para. [00124] “Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data

		slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” And Para. [00122] “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.” And Para. [00124] “In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”
2.11	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger signal condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, the first, second, third,	FIG. 16  FIG. 16 Para. [00145] “The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first

and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively,

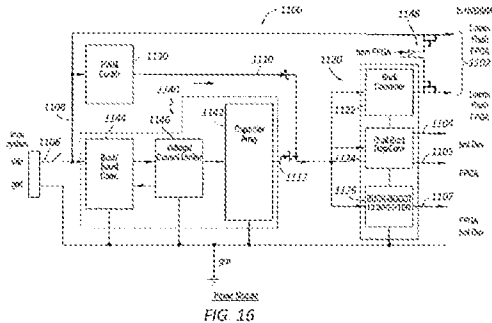
voltage 1102 is provided to the memory system 1010 from the input 1106 and **the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.**”

The conversion element 1120 is always powered from the system power through the first power element 1130 in either the first or second state, and the trigger condition here is the same as 2.8, above, disclosed in:

Para. [00130]

“In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond **in response to a trigger condition.**”

FIG. 16



		1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck- converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
2.12	the set of components including: a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;	FIG. 16

a second component electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;

a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board;

and

a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;

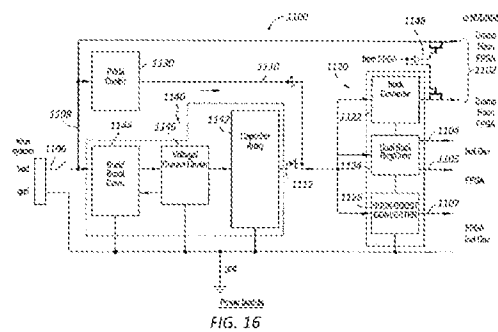
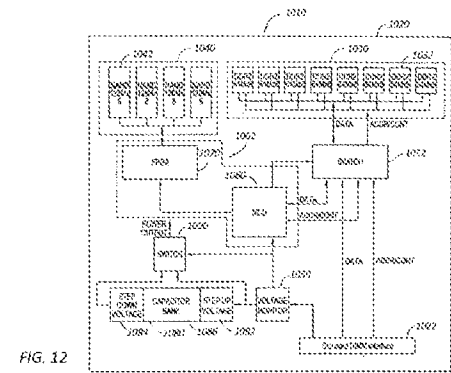


FIG. 12



On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively.

On the right, each of the DRAM are shown as being connected to the first regulated voltage, for example.

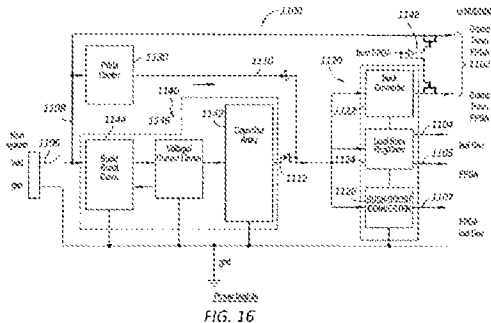
On the right, the isolation device is shown as being connected to the second regulated voltage, for example.

On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example.

On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure:

Para. [00148]

“the other reduced **voltage 1105** is 1.2V and is used to **power the controller 1062** (e.g., FPGA). The subblock 1126 can provide yet

		another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
2.13	wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit;	Para. [00148] “The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system.”
2.14		
2.15	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) when the system power is provided to the DIMM from the system board, is powered: only by one or more of the first, second,	FIG. 16  On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the

	third, and fourth converter circuits; and • only via the one or more of the first, second, third, and fourth regulated voltage lines;	system power is received from couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
2.16	wherein the plurality of DDR SDRAM packages includes: - a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal; - a second group of at least four DDR	FIG. 21

	SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and	“FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.
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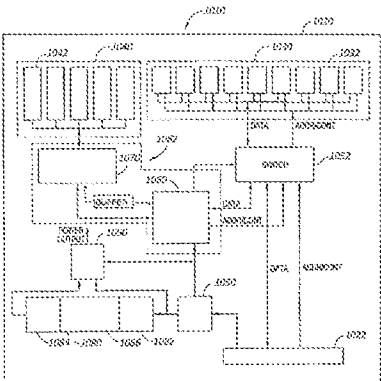
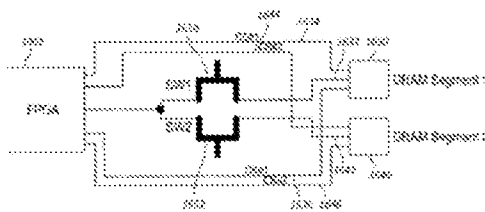
		Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
2.17	when the system power is provided to the DIMM from the system board: - the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode; - the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.

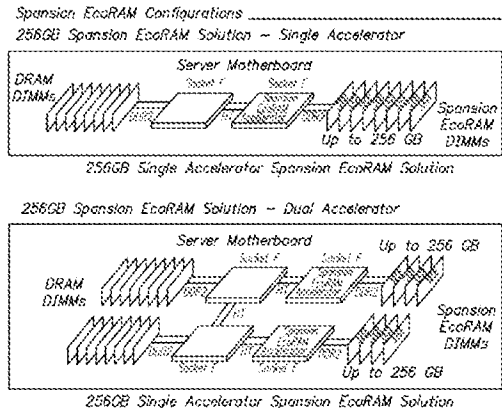
	DDR data signals in parallel via the integer number of data lines of the set of data lines; and the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
2.18	wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
28	wherein the second component is further connected to the first regulated voltage line and is powered by the first	FIG. 16

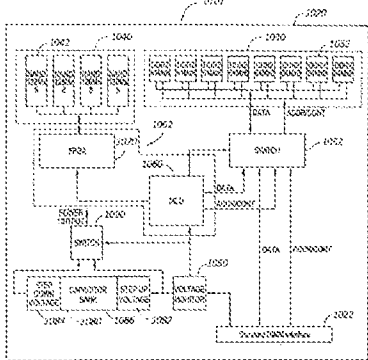
		has a very low profile (VLP) form factor with a height of 18.3 millimeters.”
32	wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
33	wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.	Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer.”
34	wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”
35	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In

	generate the third set of data bits based on the read first and second sets of data bits.	certain other embodiments, the data word may comprise more than three portions.”
36	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Difference from Claim 35 is the read from the buffer is a plurality of steps in Claim 35, and one step here. Same sentence in Para. [00178] supports both scenarios.
37	wherein, prior to performing the third write operation, the logic element is further configured to perform (i) a first read operation of the plurality of read operations to read the first set of data bits from the first portion of the data storage element, and (ii) a second read operation of the plurality of read operations to read the second set of data bits from the second portion of the data storage element, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Difference from Claim 35 is the read from the buffer is a plurality of steps in Claim 35, and specifically two steps here. Same sentence in Para. [00178] supports both scenarios. Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040.” Reading a data word or a block of words to be written into the NVM.
38	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to	Para. [00178] “In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then

	the detected trigger condition, one or more read operations.	brought out of self-refresh and a 40-bit block from the slice is read into a buffer . Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.”
39	wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.	Para. [00178] “In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer , and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer . Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.”
40	wherein the logic element is further configured to perform (i) first and second read operations of the one or more read operations, and (ii) subsequently, the first and second write operations.	Para. [00178] “In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash). ” This makes clear that the read of the slices must happen before writing the slices into the buffer.

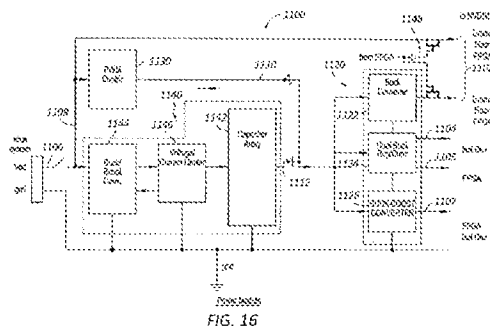
41	wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.	FIG. 13  FIG. 13 FIG. 21  FIG. 21 Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.”
42	wherein, in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.”
43	wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and wherein, in response to the additional trigger	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In

	condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.”
44.1	A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board,	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Per the JEDEC DDR2 Standard, power is supplied from the socket to designated edge connections of the standard 240-pin DDR2 edge connector. A POSITA would be aware of this standard and the configuration of both the edge connector and the socket. For example, as shown in FIG. 2 of the disclosure: Figure 2  FIG. 2 (PRIOR ART)
44.2	a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . .

	electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits	For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” This is showing a 64 bit wide DDR2 data bus, where each bit is sent in parallel. From the DDR2 Standard a POSITA would be aware that each data line is one bit wide as illustrated in FIG. 12: FIG. 12  FIG. 12
44.3	the DIMM comprising: a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:	Para. [00122] In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein.
44.4	an interface having edge connections disposed along the width dimension and	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

FIG. 16



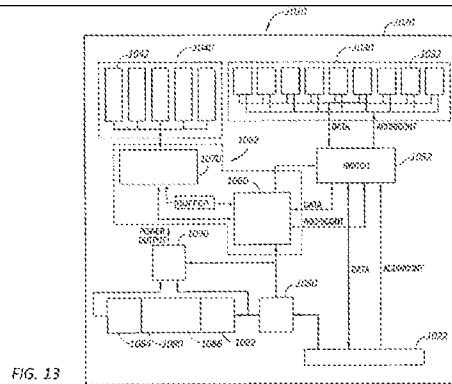
Note the data lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.

And

Para. [00172]

“In one embodiment, for example, the volatile memory subsystem 1030 comprises **a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)**, the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

The embodiment of FIG. 13 shows nine x8 DRAM devices for a data bus bit width of 72:



And

Para. [00122]

“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

Note the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.

44.6

a controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power and the input voltage via the input voltage supply line, the controller including:

- nonvolatile memory;
- a data storage element configured to store a plurality of data bits; and

Para. [00172]

“In certain embodiments, **the controller 1062 comprises a buffer** (not shown) **which is configured to store data** from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).”

The buffer provides support for the data storage element.

The controller 1062 is shown in FIG. 13:

	a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and a voltage monitor circuit electrically connected to the input voltage supply and coupled to the logic element, the voltage monitor circuit configured to:
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Para. [00127]
“Referring to FIG. 12, in certain embodiments, **the logic element 1070** comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation **which includes an internal flash**. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. **The internal flash can improve the speed of the memory system 1010 and save physical space.** Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, **the logic element 1070 comprises one or more integrated circuits.** FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. **In certain embodiments, the controller 1062 includes one or more other components.** For example, in one embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.”

FIG. 13

	second threshold voltage; and detect a trigger condition;	portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
44.8	wherein, in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations, including: - a first write operation to write a first set of data bits into a first portion of the data storage element; - a second write operation to write a second set of data bits into a second portion of the data storage element;	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”

44.9	wherein, when the system power is provided to the DIMM from the system board: the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and wherein: a specified operating voltage value is within a range of operating voltage values of the DIMM, the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value;	For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above. Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
44.10	a set of components coupled to the PCB, the set of components including a plurality of	FIG. 12

	DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and
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“Volatile memory elements 1032 compatible with certain

Para. [00172]

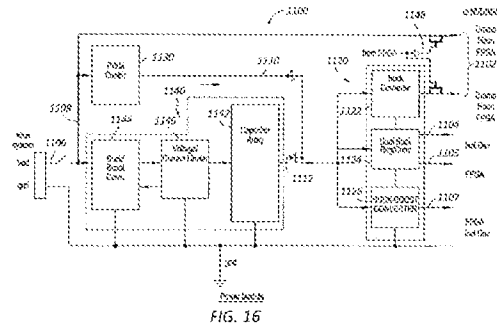
And

“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

Para. [00124]

“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. **Types of DRAM elements 1032 compatible with certain embodiments described herein include,**

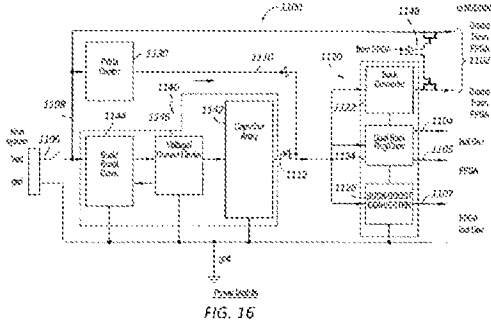
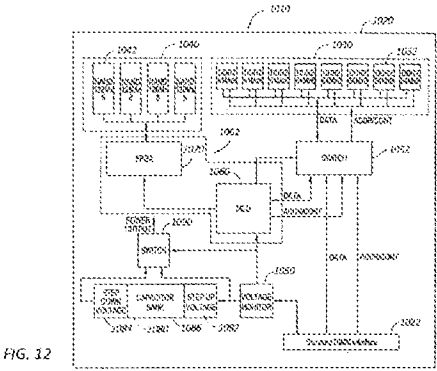
		but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”
44.11	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively,	FIG. 16 FIG. 16 Para. [00145] “The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” And The trigger condition here is the same as 2.8, above, disclosed in: Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” FIG. 16



On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively.

Para. [00148]

“The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck- converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g.,

		flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
44.12	the set of components including: a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board; a second component electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board; a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the	FIG. 16  FIG. 12  On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. On the right, each of the DRAM are shown as being connected to the first regulated voltage, for example.

	DIMM from the system board;	On the right, the isolation device is shown as being connected to the second regulated voltage, for example. On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example. On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
44.13	wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit;	Para. [00148] “The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system.”
44.14		

44.15	wherein each component of the set of components:
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(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

- only by one or more of the first, second, third, and fourth converter circuits; and
- only via the one or more of the first, second, third, and fourth regulated voltage lines;

FIG. 16

Para. [00145]

Para. [00151]

On the right side of the figure, the components are listed. The power provided to the PWM controller 1130 and from there to the converter circuits is the only power to the module. There is no alternative power source claimed.

44.16	wherein the plurality of DDR SDRAM packages includes:
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a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

FIG. 22

Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above.

	a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and	Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out
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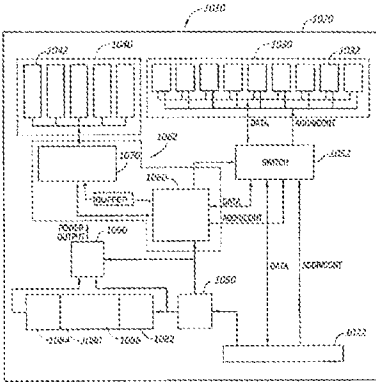
		of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
44.17	when the system power is provided to the DIMM from the system board: • the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out

	integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
44.18	wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
45	wherein the plurality of write operations further includes a third	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system

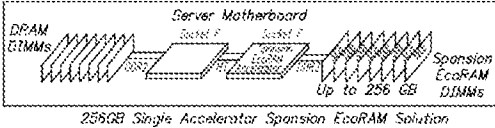
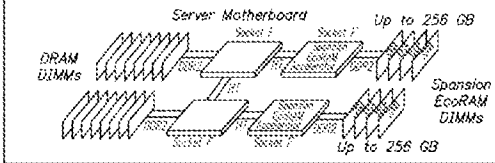
	write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform at regular intervals of time all of the first and second write operations followed by the third write operation.	1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”
46	wherein, the logic element is further configured to perform the third write operation by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
47	wherein, prior to performing the third write operation, the logic element is further configured to generate	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”

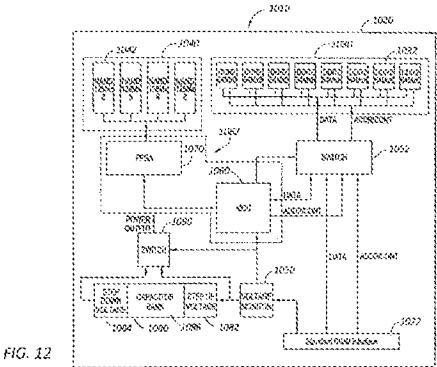
	the third set of data bits based on groups of data bits stored in the data storage element.	
48	wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102. ”
50	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00178] “ The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
51	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00178] “The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a

		32-bit block from the slice into the buffer , and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. ”
52	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time a plurality of read operations.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
53	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read data bits from the data storage element and to generate the third set of data bits based on the read data bits.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
54	wherein the width dimension is 133.35 millimeters	Para. [00122] “For example, the PCB I 020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of

		133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB I 020 has a very low profile (VLP) form factor with a height of 18.3 millimeters.”
55	wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.	Para. [00122] “For example, the PCB I 020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB I 020 has a very low profile (VLP) form factor with a height of 18.3 millimeters.”
56	wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.	Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” FIG. 13  FIG. 13 As can be seen in FIG. 13, the transfer from the DRAM occurs on a connection on the right side of the buffer, and the transfer to NVM will occur on a connection on the left side of the buffer.

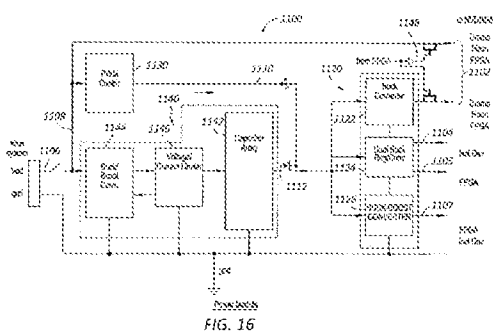
57	wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.”
58.1	A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board,	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Per the JEDEC DDR2 Standard, power is supplied from the socket to designated edge connections of the standard 240-pin DDR2 edge connector. A POSITA would be aware of this standard and the configuration of both the edge connector and the socket. For example, as shown in FIG. 2 of the disclosure: Figure 2

		<i>Sansion EcoRAM Configurations</i> <i>256GB Sansion EcoRAM Solution – Single Accelerator</i>  <i>256GB Single Accelerator Sansion EcoRAM Solution</i> <i>256GB Sansion EcoRAM Solution – Dual Accelerator</i>  <i>256GB Single Accelerator Sansion EcoRAM Solution</i> FIG. 2 (PRIOR ART)
58.2	a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” This is showing a 64 bit wide DDR2 data bus, were each bit is sent in parallel. From the DDR2 Standard a POSITA would be aware that each data line is one bit wide as illustrated in FIG. 12: FIG. 12

		 FIG. 12
58.3	the DIMM comprising: a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:	Para. [00122] In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein.
58.4	an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including: - a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits, - a second set	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

set of data lines includes at least 72 data lines; and

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;



Please note the input voltage supply line from the system on left is electrically connected to the PWM Controller 1130.

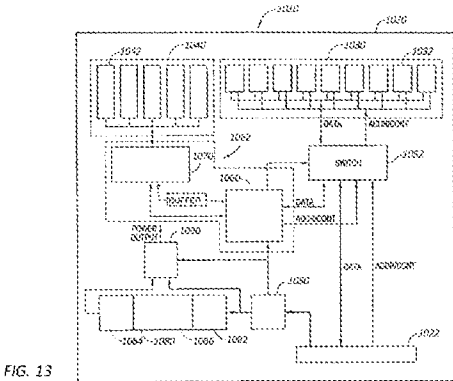
Note the data lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.

And

Para. [00172]

“In one embodiment, for example, the volatile memory subsystem 1030 comprises a **72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)**, the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

The embodiment of FIG. 13 shows nine x8 DRAM devices for a data bus bit width of 72:



And

a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to:

compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, **the logic element 1070 comprises one or more integrated circuits.** FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. **In certain embodiments, the controller 1062 includes one or more other components.** For example, in one embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.”

FIG. 13

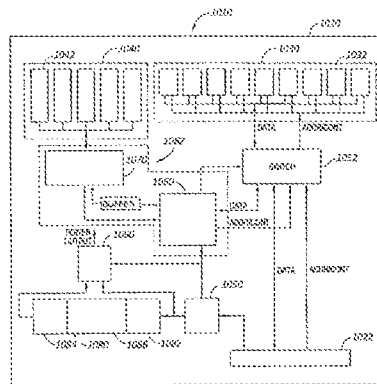


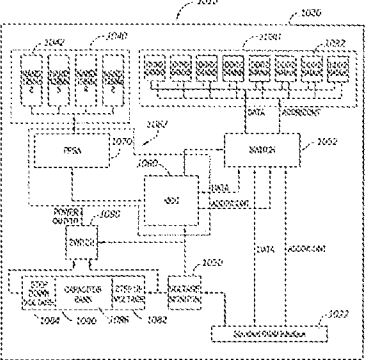
FIG. 13

Para. [00127]

“FIG. 14 is a block diagram of an example memory module 1010 having **a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.**”

FIG. 14

58.8	wherein, in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations, including: - a first write operation to write a first set of data bits into a first portion of the data storage element; - a second write operation to write a second set of data bits into a second portion of the data storage element;	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
58.9	wherein, when the system power is provided to the DIMM from the system board: - the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and wherein: - a specified operating voltage value is within a range of operating voltage values of the DIMM, - the range of operating voltage values is bounded by an upper operating voltage value and a	For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above. Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.

	lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value;	
58.10	a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and	FIG. 12  FIG. 12 Para. [00124] “Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data

slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

And

Para. [00122]

“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

And

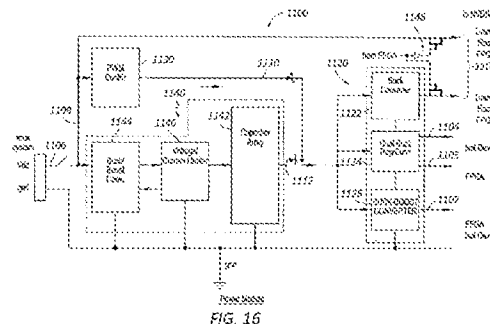
Para. [00124]

“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. **Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM).** For example, in the block diagram of FIG. 12, **the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.** The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, **volatile memory elements 1032 having bit widths of 4, 8, 16, 32,** as well as other bit widths, are compatible with certain embodiments described herein.”

58.11

first, second, third, and fourth voltage reducing circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth voltage reducing circuits configured to, in response to the detected trigger condition, receive power from the controller when the system power is provided to the DIMM from the system board, the first, second, third,

FIG. 16



Para. [00145]

“The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first

and fourth voltage reducing circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the system power supplied by the input voltage supply line , and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively,

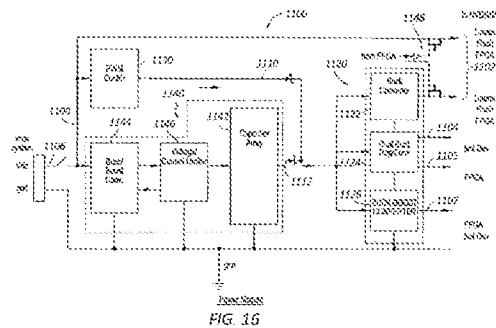
voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. **In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.**”

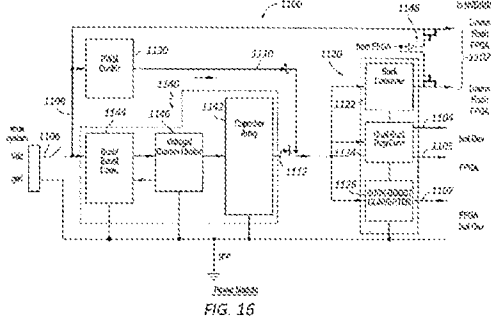
And

The trigger condition here is the same as 2.8, above, disclosed in: Para. [00130]

“In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond **in response to a trigger condition.**”

FIG. 16



		as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck- converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
58.12	the set of components including: a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board; a second component electrically connected to the second regulated voltage line and powered	FIG. 16  FIG. 12

	by the second regulated voltage when the system power is provided to the DIMM from the system board; a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;
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On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively.

On the right, each of the DRAM are shown as being connected to the first regulated voltage, for example.

On the right, the isolation device is shown as being connected to the second regulated voltage, for example.

On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example.

On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure:

Para. [00148]

“the other reduced **voltage 1105** is 1.2V and is used to **power the controller 1062** (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the **voltage 1107** may be 3.3V and may be used to **power** both the **controller 1062** and the at least one circuit 1052.”

As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.

configured to electrically conduct a first chip select signal;
a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

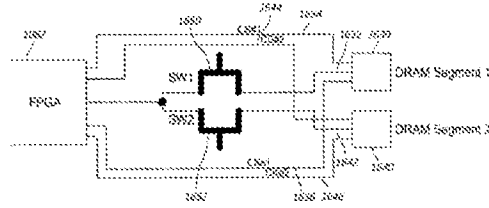


FIG. 21

Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above.

Para. [00172]

“FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from **two DRAM segments 1630, 1640** of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology **including two DRAM segments 1630, 1640** for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from **the first DRAM segment 1630 and the second DRAM segment 1640** of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. **The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKE signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.**”

Para. [00177]

“In certain embodiments, transferring each data word comprises storing **a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624**, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”

Para. [00178]

“**In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.** In certain embodiments, the buffer is included

		in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash)." This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
58.16	when the system power is received by the controller: • the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-	Para. [00173] "In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode." Para. [00178] "In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included

	bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment." Para. [00182] "In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode."
58.17	wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.	Para. [00123] "The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector." Para. [00172] "In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes." Para. [00178]

		“In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
59	wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”
60	wherein the logic element is further configured to perform the third write operation by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”

61	wherein, in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition. ”
62	wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond a plurality of read operations prior to performing the first and second write operations.	Para. [00113] “In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.”
63	wherein the logic element is further configured to perform a first and second read operations of the plurality of read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.	[00172] <u>FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.</u> <u>The controller read first and second data slices (sets of data bits) from first and second groups of DDR SDRAM packages, and then write the first and second sets of data bits into the controller 1062 (data storage element).</u>
64	wherein the width dimension is 133.35 millimeters.	Para. [00122] “For example, the PCB I 020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters . In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB I 020 has a very low profile (VLP) form factor with a height of 18.3 millimeters.”
65	wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.	Para. [00122] “ For example, the PCB I 020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50

		millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters.”
66	wherein each of the first, second, and third voltage reducing circuits is a buck converter.	Para. [00148] “The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub- blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters , boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.”
67	wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”
68	wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value , such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.”

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
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10/01/2024 02:15:28 PM Z ETATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 67393003

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She ChenCORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 9

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Supplemental IDS Letter.pdf	1	Miscellaneous Incoming Letter	142 KB
Supplemental IDS_IDS5_022955- 000090.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1355 KB
NPL-- EX1018_Petition_IPR833_S Khynix.pdf	70	Non Patent Literature	428 KB
NPL-- EX1014_Petition_IPR833_Sa nDisk.pdf	69	Non Patent Literature	1046 KB
NPL-- EX2013_JEDEC_Standard_ SDRAM_Unbuffered_SO- DIMM.pdf	26	Non Patent Literature	1891 KB

NPL--Paper25 - Sams Reply to POR - IPR2022-00996.pdf	57	Non Patent Literature	6338 KB
NPL--EX2031_Decl_MangioneSmith.pdf	144	Non Patent Literature	14332 KB
NPL--FBDIMM_Architecture-and-Protocol_JEDEC-Standard_JESD206-01_Jan2007.pdf	128	Non Patent Literature	15933 KB
Resp to 8-8-24 NFOA.pdf	45	Amendment/Request for Reconsideration-After Non-Final Rejection	972 KB

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Supplemental IDS Letter.pdf	86745618B5D3DBC3AF380EE5F02437D0DBD2C369B3A3A5D19849B8931CE5321C9AC0DC330B315667AE8288161B3CF72C1B513DCB1BB4926616373B1C69B22403
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NPL--Paper25 - Sams Reply to POR - IPR2022-00996.pdf	A8A5C782F5322FD90F7BD7AB2C8C1FD2CF8F09573D32611508E1D0678F532AC9CCFA51B22DBEE03BBF0D859F69EFA0163E196E6ECD13545E08B56B471486CD4E
NPL--EX2031_Decl_MangioneSmith.pdf	CFAE365C9A72938E273056177FB9F5BD715B930043EDAE0A0403EC7CB650B0C9A3F499F5A0995354AB61FFE5FB924D4BEFCBEEA2DE29D7B9AFF006C8EBDD4B49
NPL--FBDIMM_Architecture-and-Protocol_JEDEC-	0F3DFC3BCDC9CA0CE980D86A424F8025BA132C9E09FE115FA900C5C58C82E48EBBE945972EB81610DC81B177F82F13850

Standard_JESD206-
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797RECEIPT DATE / TIME
10/01/2024 02:15:28 PM Z ETATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 67393003

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE
ADDRESS -FIRST NAMED
INVENTOR Chi-She Chen

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FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
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New International Application Filed with the USPTO as a Receiving Office

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: 3546

SERIAL NO.: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: HASHEM FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Supplemental Information Disclosure Statement

Commissioner:

Applicant submits this Supplemental Information Disclosure Statement. The non-final Office Action issued on August 8, 2024, states that the Information Disclosure Statements submitted on May 28, 2024, have been considered by the examiner. However, some references in the Information Disclosure Statement filed on May 28, 2024, were crossed out in the List of References cited by applicant and considered by examiner issued on August 8, 2024. Therefore, the applicant re-submits those references in the Supplemental Information Disclosure Statement.

The appropriate fee for submitting an Information Disclosure Statement is also being submitted. The Commissioner is hereby authorized to charge any fees which may be required of this application, or credit any overpayment, to Deposit Account No. 22-0585.

Respectfully submitted,

Dated: October 1, 2024

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2138	
	Examiner Name	HASHEM FARROKH	
	Attorney Docket Number	022955-000090	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	Chi-She Chen
Art Unit	2138
Examiner Name	HASHEM FARROKH
Attorney Docket Number	022955-000090

1	Exhibit 2031, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
2	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
3	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 69 pages.	☐
4	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
5	Exhibit 2013, JEDEC Standard No. 21-C, Page 4.20.3-1; PC133 SDRAM Unbuffered SO-DIMM, Reference Design Specification, Revision 1.02, 26 pages. downloaded August 16, 2022.	☐
6	FBDIMM: Architecture and Protocol, JEDEC Standard JESD206.01, January 2007, 128 pages.	☐

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***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2138	
	Examiner Name	HASHEM FARROKH	
	Attorney Docket Number	022955-000090	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☒ A certification statement is not submitted herewith.

FEE

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

COPIES

☐ An identification of an earlier application pursuant to 37 CFR 1.98 (d) (1) is attached.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2024-10-01
Name/Print	Khaled Shami	Registration Number	38745

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

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EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

AMENDMENT

Commissioner:

In response to the non-final Office Action mailed August 8, 2024, Applicant submits the following Amendment and Remarks.

Listing of Claims begins on page **2**.

Remarks begin on page **22**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~slot connector~~socket of a system board of a host computer system, the memory ~~slot connector~~socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the host system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal signals is electrically conducted between the host system board system and the DIMM are transferred via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the host system board are transferred to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:

an interface having edge connections disposed along the width dimension and configured to fit into the memory-slot connector of the host system socket, the edge connections including:

a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,

a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and

a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;
~~the PCB further including~~ a set of voltage supply lines electrically connected to

the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

a set of data lines electrically ~~connected~~ coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

a set of address and control lines electrically ~~connected~~ coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller ~~including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and electrically connected to the to an input voltage supply line of the set of voltage supply lines, the controller configured to receive the input voltage via the input voltage supply line, the controller including:~~

nonvolatile memory;

a data storage element configured to store a plurality of data bits;

a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and

a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to: (i) monitor an

compare the input voltage supplied via the input voltage supply line to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and

(ii) generate a trigger signal upon detecting detect a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller;

wherein the controller is configured to perform, in response to the detected trigger conditionsignal, the logic element is configured to perform a plurality of write operations, including;

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;

wherein, when the system power is provided to the DIMM from the system board:

and wherein the trigger condition occurs when the input voltage goes

above both exceeds the first and second a first preset threshold

voltages voltage that is above an operating voltage of the DIMM; and

wherein:

a specified operating voltage value is within a range of operating voltage values of the DIMM,

the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value,

the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value,

the first threshold voltage corresponds to the lower operating voltage value, and

the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of ~~double data rate (DDR)~~ synchronous dynamic random access memory (DDR SDRAM) ~~devices~~packages coupled to the set of address and control lines and to the set of data lines, ~~the~~ each DDR SDRAM package of the plurality of DDR SDRAM packages~~devices~~ having a bit width of m bits and configured being operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on control and in response to

address and control signals received from the host system via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converter circuits ~~buck converters~~ coupled to the PCB and to the input voltage supply line, and each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, from the input voltage supply line and the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, to deliver from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages to the set of components via first, second, third, and fourth regulated voltage lines, respectively, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines, respectively, the set of components including;

a first component electrically connected to the first regulated voltage line and powered by configured to receive power via at least the first regulated voltage line when the system power is provided to the DIMM from the system board;;

a second component electrically connected to the second regulated voltage line and powered by configured to receive power via at least the second regulated voltage line when the system power is provided to the DIMM from the system board;;

a third component electrically connected to the third regulated voltage line and powered by configured to receive power via at least the third regulated voltage line when the system power is provided to the DIMM from the system board; and

a fourth component electrically connected to the fourth regulated voltage line and powered by configured to receive power via at least the fourth regulated voltage line when the system power is provided to the DIMM from the system board;;

wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and

wherein each component of the set of components;

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) receives power when the system power is provided to the DIMM from the system board, is powered:

- only by one or more of the first, second, third, and fourth converter circuits; and
- only via the one or more of the first, second, third, and fourth regulated voltage lines; and

~~wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state.~~

wherein the plurality of DDR SDRAM packages includes:

a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

when the system power is provided to the DIMM from the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR

SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and

- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

3 – 27. (Canceled)

28. (New) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when the system power is provided to the DIMM from the system board.

29. (New) The DIMM of claim 2, wherein the data storage element is a buffer configured to store the plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

30. (New) The DIMM of claim 2, wherein the width dimension is 133.35 millimeters.

31. (New) The DIMM of claim 30, wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

32. (New) The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (New) The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of

data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (New) The DIMM of claim 2, wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.

35. (New) The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (New) The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (New) The DIMM of claim 35, wherein, prior to performing the third write operation, the logic element is further configured to perform (i) a first read operation of the plurality of read operations to read the first set of data bits from the first portion of the data storage element, and (ii) a second read operation of the plurality of read operations to read the second set of data bits from the second portion of the data storage element, and to generate the third set of data bits based on the read first and second sets of data bits.

38. (New) The DIMM of claim 2, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, one or more read operations.

39. (New) The DIMM of claim 38, wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.

40. (New) The DIMM of claim 38, wherein the logic element is further configured to perform (i) first and second read operations of the one or more read operations, and (ii) subsequently, the first and second write operations.

41. (New) The DIMM of claim 2, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

42. (New) The DIMM of claim 2, wherein, in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

43. (New) The DIMM of claim 2, wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

44. (New) A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:

an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:

a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,

a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and

a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;

a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power and the input voltage via the input voltage supply line, the controller including:

nonvolatile memory;

a data storage element configured to store a plurality of data bits; and

a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and

a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to:

compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and

detect a trigger condition;

wherein, in response to the detected trigger condition, the logic element is configured to periodically perform at regular intervals of time a plurality of write operations, including:

a first write operation to write a first set of data bits into a first portion of the data storage element; and

a second write operation to write a second set of data bits into a second portion of the data storage element;

wherein, when the system power is provided to the DIMM from the system board:

- the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein:

- a specified operating voltage value is within a range of operating voltage values of the DIMM,
- the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value,
- the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value,
- the first threshold voltage corresponds to the lower operating voltage value, and
- the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, the set of components including:

a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;

a second component electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;

a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and

a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;

wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) when the system power is provided to the DIMM from the system board, is powered:

- only by one or more of the first, second, third, and fourth converter circuits; and
- only via the one or more of the first, second, third, and fourth

regulated voltage lines;

wherein the plurality of DDR SDRAM packages includes:

a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

when the system power is provided to the DIMM from the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

45. (New) The DIMM of claim 44, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform at regular intervals of time all of the first and second write operations followed by the third write operation.

46. (New) The DIMM of claim 45, wherein, the logic element is further configured to perform the third write operation by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

47. (New) The DIMM of claim 45, wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element.

48. (New) The DIMM of claim 44, wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (New) The DIMM of claim 44, wherein, in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

50. (New) The DIMM of claim 46, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (New) The DIMM of claim 44, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.

52. (New) The DIMM of claim 44, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time a plurality of read operations.

53. (New) The DIMM of claim 46, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition,

periodically at regular intervals of time one or more read operations to read data bits from the data storage element and to generate the third set of data bits based on the read data bits.

54. (New) The DIMM of claim 44, wherein the width dimension is 133.35 millimeters.

55. (New) The DIMM of claim 54, wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

56. (New) The DIMM of claim 45, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (New) The DIMM of claim 44, wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

58. (New) A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:

an interface having edge connections disposed along the width dimension and

configured to fit into the memory socket, the edge connections including:

- a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,

- a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and

- a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;

- a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

- a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

- a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the input voltage via the input voltage supply line, the controller including:

- nonvolatile memory;

- a data storage element configured to store a plurality of data bits; and

- a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element;

- a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to:

compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and

detect a trigger condition;

wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond a plurality of write operations, including:

a first write operation to write a first set of data bits into a first portion of the data storage element; and

a second write operation to write a second set of data bits into a second portion of the data storage element;

wherein, when the system power is provided to the DIMM from the system board:

- the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein:

- a specified operating voltage value is within a range of operating voltage values of the DIMM,
- the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value,
- the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value,
- the first threshold voltage corresponds to the lower operating voltage value, and
- the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in

response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth voltage reducing circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth voltage reducing circuits configured to, in response to the detected trigger condition, receive power from the controller when the system power is provided to the DIMM from the system board, the first, second, third, and fourth voltage reducing circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the system power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, the set of components including:

a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;

a second component electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;

a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and

a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;

and

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) when the system power is provided to the DIMM from the system board, is powered:

- only by one or more of the first, second, third, and fourth voltage reducing circuits; and

- only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM packages includes:

a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

when the system power is provided to the DIMM from the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

59. (New) The DIMM of claim 56, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by

the third write operation.

60. (New) The DIMM of claim 57, wherein the logic element is further configured to perform the third write operation by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (New) The DIMM of claim 56, wherein, in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

62. (New) The DIMM of claim 56, wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond a plurality of read operations prior to performing the first and second write operations.

63. (New) The DIMM of claim 60, wherein the logic element is further configured to perform a first and second read operations of the plurality of read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively.

64. (New) The DIMM of claim 56, wherein the width dimension is 133.35 millimeters.

65. (New) The DIMM of claim 62, wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

66. (New) The DIMM of claim 56, wherein each of the first, second, and third voltage reducing circuits is a buck converter.

67. (New) The DIMM of claim 56, wherein, when the system power is provided to the DIMM from the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (New) The DIMM of claim 58, wherein, when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

REMARKS

By this Proposed Amendment, submitted for purposes of discussion only, Applicant has **amended claim 2; canceled claims 6-8 and 11 and 14-27; and added new claims 28-68.** (Claims 3-5, 9-10, and 12-13 were previously canceled).

Claims 2 and 28-68 are now pending.

Amendment to Claims

Claim 2, has been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references.

For the examiner's convenience, claim 2 is reproduced hereinbelow in clean version for ease of reading:

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits, wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:

- an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:

- a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,

- a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and

- a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;

- a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;

- a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

- a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the input voltage via the input voltage supply line, the controller including:

- nonvolatile memory;

- a data storage element configured to store a plurality of data bits;

- a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and

a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to:

- compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and

- detect a trigger condition;

wherein, in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations, including:

- a first write operation to write a first set of data bits into a first portion of the data storage element;

- a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

- a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;

wherein, when the system power is provided to the DIMM from the system board:

- the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein:

- a specified operating voltage value is within a range of operating voltage values of the DIMM,

- the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value,

- the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value,

- the first threshold voltage corresponds to the lower operating voltage value, and

- the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of DDR synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines in response to address and control signals received via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively, the set of components including:

- a first component electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;

- a second component electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;

- a third component electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and

- a fourth component electrically connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;

wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) when the system power is provided to the DIMM from the system board, is powered:

- only by one or more of the first, second, third, and fourth converter circuits; and
- only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM packages includes:

a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

when the system power is provided to the DIMM from the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory

socket, and the total number of data conduits is at least 72 data conduits.

Canceled Claims

Claims 6-8 and 11 and 14-27 have been canceled without prejudice or disclaimer of the subject matter contained therein.

Newly-Added Claims

Claims 28-68 have been added to further particularly point out and distinctly claim certain aspects of the invention.

Examiner Interview

Applicants gratefully acknowledge Examiner Farrokh's courtesy and consideration during the telephonic Examiner Interview of September 6, 2024 with the Applicants' undersigned representative and Mr. Marwan Fawal, technology expert. The arguments below were raised with the Examiner but no agreement was reached.

Rejection(s) Under 35 U.S.C. § 103(a)

Claims 2, 6-8, 11, 15-18, and 20-24 stand rejected under pre-AIA 35 U.S.C. 103(a) as allegedly being unpatentable over Harris et al. "Harris" (US 2006/0174140 A1) in view of FBDIMM Standards (Exhibitions 1027, 2028 provided by Applicant) and Amidi et al. "Amidi" (US 7,724,604 B2).

Claims 14, 18, 19, and 25 rejected under pre-AIA 35 U.S.C. 103(a) as allegedly being unpatentable over Harris, FBDIMM Standards, and Amidi as applied to claims 6, 18, and 24, and further in view of Levkovitz et al. "Levkovitz" (US 2007/0088838 A1).

The Examiner asserts that the rejection is based on information provided in Trial Termination or Final Written Decision (hereinafter Trial-T[er]m (*sic*), dated 12/06/2023) by the USPTO Trial and Appeal Board on Application 17/138,766 (now US patent 11,016,918 B2), which appears, in most part, to be based on Exhibition 1001, IPR 2022-00996. The Examiner alleges that the “instant Application, which is a continuation of US Patent 11,016,918 B2 includes substantially the same or similar claim limitations.” (Office Action, p. 4, ll. 4-9).

In the Office Action, the Examiner also states that he “considers the claims of Application 17/582, 797, that substantially include limitations similar to claims of Applications 17/138, 766 and 17/328, 019, not to be patentable. Therefore, the Examiner as shown below rejected the claims.” (*Id.*, p. 3, ll. 6-9).

Support for all of the below amendments has been provided in the attachment accompanying this paper. The support is provided by claim number in a table within the attachment.

To advance the prosecution of this application, the Applicant has canceled claims 6-8, 11, 15-18, and 20-24 without prejudice or disclaimer; amended claim 2; and added new claims 28-68. Applicant acknowledges the Trial-Term cited by the Examiner in the Office Action mailed August 8, 2024, but maintains that the rejected claims 2, 6-8, 11, 15-18, and 20-24 (that is, prior to the instant amendment) contain limitations that are novel and unobvious over the references cited in Trial-Term and by the Examiner.

In fact, the Applicant will establish below that specific claim limitations included in the rejected claim 2, both before and after the instant amendment, describe a module that is structurally and operationally patentably distinguishable over the applied references. For instance, the limitations before the instant amendment (pre-amendment limitations) require that the four buck converters generate regulated voltages from the system power received through the input voltage supply line that is connected to a first set of edge connections that are connected to the set of power conduits of the memory socket, and that the power from the host system is provided to the DIMM via the set of power conduits of the memory socket. Harris expressly

teaches against providing power from the system board or from the main board voltage sources to any memory module, as will be discussed further below.

In relevant part, pre-amendment claim 2 recites:

“A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein power from the host system is provided to the DIMM via the set of power conduits,” ... “a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including a first set of edge connections configured to be electrically connected to the set of power conduits,” ... “the PCB further including a set of voltage supply lines electrically connected to the first set of edge connections, ” ... “the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines” ... “first, second, third and fourth buck converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power to the set of components via first, second, third and fourth regulated voltages lines, the first, second, third and fourth buck converters being further configured to generate first, second, third and fourth regulated voltages, respectively, from the power received from the input voltage supply line, and to output the first, second, third and fourth regulated voltages via the first, second, third, and fourth regulated voltage lines,... (Emphasis added).

As instantly amended, claim 2 recites:

A dual in-line memory module (DIMM) configured to fit into a memory socket of a system board of a computer system, the memory socket including a set of power conduits” ... , wherein system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board,” ... “PCB including... a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,” ... “a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an

input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board” ... “first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, (Emphasis added).

It can be seen that claim 2 as instantly amended still require that the system power is received by the DIMM from the memory socket of a system board of a computer system in a particular form (i.e., via the input voltage supply) and on particular pins of the memory socket (i.e. first set of edge connection electrically connected to respective power conduits of the set of power conduits of the memory socket), and that the converter circuits generates the regulated voltages from the power supplied to the DIMM from the system board via the set of power conduits of the memory socket.

As the Office Action acknowledges, the memory module 100A of Harris receives its power through an external source. (“Fig. 1B Harris shows a block diagram of memory module 100B comprising DRAM devices 110, inputs a plurality voltages 120 from e|x|ternal (sic) source and generates a plurality of regulated voltages 106 and 108,” Office Action, p. 5, ll. 3-5, emphasis added).

Applicant respectfully maintains that Harris expressly teaches away from providing power to any memory module assembly via the memory socket from the system board of the computer system.

First, Harris recognized and described, at ¶0002, significant power supply concerns particularly in advanced memory designs, implemented at the time, that **“industry standard memory modules, e.g., Dual In-line Memory Modules (DIMMS) populated with ... (DRAM) devices, are provided with power supply rails (on a relatively large number of pins) that are**

powered from system board or main board voltage sources, and are specific to the memory technology.” (Emphasis added).

Second, Harris recognized and described, at ¶0002, that “[a]s the performance of the DRAM technology goes up, and timing margin shrink, **it is becoming increasingly more difficult for the system board sources to provide tightly regulated power for the DRAM** cores as well as input/output (I/O) interface buffers.” (Emphasis added).

Third, Harris concludes, at ¶0002, and states that “[o]ne skilled in the art will recognize ... that these issues are particularly significant for [computer] systems that are deployed in infrastructures” **because “it is difficult to mix memory technology on a system board or provide upgrades to next generation DRAM technology in a cost-effective manner.”**” (Emphasis added).

In summary, Harris has recognized and teaches that (i) particularly significant power supply concerns exist for computer systems using industry standard DIMMS that are powered from the system board or the main board voltage sources, (ii) it is more difficult to provide tightly regulated power for the DRAM from the system board power sources via large number of pins of standard DIMM memory socket, and (iii) any one skilled in the art will recognize these particularly significant issues because when power is provided to the DIMMs from the system board, then it is difficult to mix DIMMs with different memory technology on such system board or being able to provide upgrades to next generation DRAM in a cost-effective manner.

In ¶0003, Harris described his invention as “**a voltage distribution system for a memory board assembly, comprising means for supplying a voltage to the memory board assembly from an external source**”.

In the context of ¶¶ 0002-0003, Harris has identified particularly significant problems associated with advanced memory designs using standard DIMMs that are powered from the system board and teaches that his innovative solution to these significant problems is by supplying power to the DIMM from an external voltage source and locally converting the external voltage by an on-board voltage regulator.

Thus, Harris’ external voltage source would not be understood by a skilled person in the art to be supplied to the standard interface DIMM from the system board, and Harris’ description

offers further detailed evidence that underscores this understanding.

Harris teaches several recognized advantages, in ¶0011, specifically for large memory capacity scaling, hot-pluggability, and DRAM technology independence:

[0011] ... Since the Vdd and Vcc power supply rails are rendered independent from external voltage sources, the overall power supply interface to the memory module 100A is improved to better support **large memory capacity scaling, hot-pluggability, DRAM technology independence (e.g., DDR2, DDR3, DDR4, and other DRAM types)**, ... as well as provide improved system-level cost sharing. Additionally, memory performance is also improved because of tighter voltage regulation.

This passage ¶ 0011, makes clear that the DIMM of Harris draws power from a source external to the system board, i.e. “external voltage sources,” as does a reference to “hot-pluggability” especially in light of DRAM technology independence. Each DDR generation standardizes a different set of voltages for memory modules, and all DDR4, DDR3, DDR2, and DDR system boards provide system power exclusively to the DIMM via the socket. Thus, the ONLY way to have hot pluggability and DRAM technology independence as sought by Harris is to have power from a source external to the system board.

This power source external to the system board is further confirmed to be received by the module on external source path 104, in ¶ 0010 of Harris, and illustrated in Fig. 1A and Fig. 1B:

[0010] In accordance with the teachings of the present patent disclosure, at least one on-board voltage regulator module (VRM) is provided as part of the memory board assembly module 100A for converting **an externally supplied voltage level available on external source path 104** into appropriate local Voltage levels that power the first and second Voltage paths, i.e., the Vdd and V paths 108, 106, respectively. ...

Harris further teaches, by way of example in ¶ 0012, that the 72 pins required by standard 2-ranks FBD module (36 DRAM devices) to receive power from the system board would be replaced by 6 pins when power is provided from an external (+12V) voltage source. Harris expressly teaches a clear advantage of providing power to DIMM from external sources is by being able to supply power to support large scale capacity DIMMs, such as a 144 DRAM devices of an 8-ranks FBD, using 24 pins from the external (+12V) voltage source.

[0012] By way of example, **a standard FBD module requires 28V pins** (for DRAM devices) and 8V pins (for buffer and logic) and associated Ground

returns, resulting in a total of 72 pins that provide a power supply interface for up to two X4 DRAM ranks (36 devices) and buffer logic. A voltage-independent FBD design incorporating the embodiment of FIG. 1A is capable of replacing these power supply interface pins with as few as six +12V pins (from an external voltage source), Alternatively, using additional power supply pins would provide the capability to support even more devices. For instance, with 12 supply pins (at +12V), the embodiment of FIG. 1A can provide enough power to Supply 4 ranks (i.e., 72 devices). Likewise, a memory assembly of 8 ranks (144 devices) may be powered with 24 pins at +12V.

Even more conclusive evidence of the advantages sought by Harris are disclosed in ¶ 0019 which eliminates any doubts that Harris does not provide power to the DIMM from the system board by expressly stating the elimination of the system board power supply and associated voltage planes. Furthermore, Harris expressly describes the cost reduction associated with the elimination of system-board-specific power supply or regulator, associated components, and heavy etch power planes on the system board. Harris further describes, and as a direct result of the elimination of the system board power supply, that additional advantages can be realized such as a smaller connector, improved signal integrity, or reservation of extra pins for implementing addition standard or proprietary functions.

[0019] Based on the foregoing Detailed Description, it should be appreciated that an implementation of the embodiments described herein thus provides a technology-independent voltage distribution scheme for memory devices wherein system board power supply and associated voltage plane(s) are eliminated. Accordingly, power Supply designs complexity as well as the cost of memory-specific power distribution are reduced. Cost savings may include, for example, elimination of system-board-specific power supply or regulator output, associated bypass capacitor arrangements, heavy etch or power planes. Also, under the technology-independent Voltage distribution scheme disclosed herein, cost of memory power regulation scales with populated memory devices, rather than being pre-provisioned for maximum memory capacity as is typically the case in existing systems. Additionally, one or more of the following advantages may be realized in an exemplary Voltage distribution embodiment: provision of tighter, lower inductance regulation which decreases Voltage variability and increases timing margins; reduction in the pin count on DIMM connector to system board or memory riser card (whereby a smaller connector or wider pin spacing may be accommodated); improved signal integrity due to reduced crosstalk on the same size connector; and reservation of extra pins for implementing additional standard or proprietary functions.

This clarifies that Harris cannot read on Claim 2, either before or after the instant amended, as the entirety of the disclosure of Harris conveys to one of ordinary skill in the art that

Harris does not use the DIMM connector to obtain power from the system board. Rather, as shown immediately above, Harris looks to remove the system board power supply and/or repurpose the memory socket pins that would have connected to the power source of the system board.

It is conclusive that Harris teaches (i) the elimination of system-board-specific power supply components and, importantly, the heavy etch power planes used to distribute the voltage supply on the system board to the DIMM connector, and (ii) that the power provided to Harris' memory module does not come from the system board or through the DIMM connector.

Thus, Harris teaches away from providing power from the system board or from the main board voltage sources to the DIMM via the DIMM connector as required by claim 2.

Thus, Claim 2 is allowable for at least this reason. New Claims 44 and 58 also recite the same language and are therefore allowable for at least this reason. The Claims which depend from Claims 2, 44, and 58 are also allowable for at least this reason.

Claim 2 has been further amended to recite "a set of data conduits including at least 72 data conduits" and "a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits." Similar to the power discussed above, the data signals are received by the DIMM from the memory socket of a system board of a computer system in a particular form (i.e., the 1-bit wide DDR per data conduit) and on particular pins of the memory socket (i.e. each of the second set of edge connections electrically connected to a respective data conduit of the set of data conduits of the memory socket). Harris makes no disclosure of signaling protocol other than disclosing that the DIMM which Harris is modifying is an FBDIMM. The FBDIMM standards outline the signaling protocol to be used on FBDIMM systems. In that protocol, address and control information as well as data are packetized in what the standard refers to as "frames." These frames have a header and footer in addition to the multiplexed address and control and data information. Thus, the frames include far more than a single bit. Moreover, they are not DDR data signals as defined in JEDEC standards. Again, these are defined in the FBDIMM standard as something different – namely "frames." The terminology in the FBDIMM

standards is purposely used by JEDEC to avoid any confusion with data signals as defined in other JEDEC standards, namely all non-FBDIMM DDR standards.

Moreover, the FBDIMM standards make clear that there are not 72 data conduits. Even if the conduits which do exist in FBDIMM, which are called “lanes” (which unmistakably convey frames including BOTH address and control and data signals) are counted as data conduits, there are only 24. The southbound link has 10 lanes, while the northbound link has 14 lanes. Thus, the FBDIMM standards, which must be relied upon as Harris lacks disclosure beyond referencing the FBDIMM standards, fail to disclose the above elements of Claim 2.

Claim 2 is allowable for at least this additional reason. New Claims 44 and 58 also recite the same language and are therefore allowable for at least this additional reason. The Claims which depend from Claims 2, 44, and 58 are also allowable for at least this additional reason.

Claim 2 has been further amended to recite:

- a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:
 - an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:
 - a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits,
 - a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and
 - a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits;
 - a set of voltage supply lines electrically connected to the first set of edge connections, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;
 - a set of data lines electrically coupled to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and
 - a set of address and control lines electrically coupled to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

The applied prior art does not disclose the claimed dedicated set of edge connections and data lines for data; rather, data in Harris and FBDIMM shares the address and control edge connections and lines because a serial protocol is utilized. Both references are JEDEC standard modules, at least insofar as they use the packetized signaling protocol laid out in the AMB/FBDIMM standards (obviously, Harris is non-standard regarding its power architecture). In that packetized signaling protocol, data and command/control/address signals do not travel on separate lines, because they are sent in packetized forms called “frames” – “After initialization the host communicates with the AMBs on the channel using southbound frames of information containing commands and data.” JESD206.01, pg. 55

Thus, the data signals are not received or outputted on “the set of data conduits” and the address and control signals are not received on “the set of address and control conduits” as recited in the claim, rather they are transferred in a packetized format on the same set of conduits, referred to as “lanes” by JESD206.01.

Moreover, the FBDIMM Standards provide no cure for this defect, as they use the same FBDIMM protocols as Harris.

Thus, Claim 2 is allowable for at least this additional reason. New Claims 44 and 58 also recite the same language and are therefore allowable for at least this additional reason. The Claims which depend from Claims 2, 44, and 58 are also allowable for at least this additional reason.

Claim 2 has further been amended to state that:

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board ...

and

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) when the system power is provided to the DIMM from the system board, is powered:

- only by one or more of the first, second, third, and fourth converter circuits; and
- only via the one or more of the first, second, third, and fourth regulated voltage lines;

These features are distinguishable from the applied references, at least because the plurality of DDR SDRAM packages is required to be powered “only by one or more of the first, second, third, and fourth converter circuits, and only via the one or more of the first, second, third, and fourth regulated voltage lines”. In contrast, Amidi, for example, is operable to switch DRAM memory supply 660 to battery power depending on a comparison to a reference voltage. Amidi Col 4, Lines 44-55 expressly state:

Signal 670 is controlled by voltage supervisory block 665, which receives system supply 605 and a reference voltage 675, and compares the two. If system supply 605 has a greater magnitude than reference voltage 675, signal 670 causes power switch multiplexer 655 to choose system supply 605 as a source for memory supply 660. If system supply 605 has a magnitude lower than reference voltage 675, then signal 670 causes power switch multiplexer 655 to choose battery power 650 for memory supply 660. Reference voltage 675 is generated in one embodiment by a resistive Voltage divider using resistors 677 and 673 in series between a voltage rail and ground.

This defect is not cured by Harris or FBDIMM at least because they do not recite a response to a trigger condition and Claim 2 is allowable for at least this additional reason. New Claims 44 and 58 also recite the same language and are therefore allowable for at least this additional reason. The Claims which depend from Claims 2, 44, and 58 are also allowable for at least this additional reason.

Moreover, the Office Action alleges that Amidi discloses "a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to (i) monitor an input voltage supplied via the input voltage supply line, (ii) generate a trigger signal upon detecting a trigger condition, and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first preset threshold voltage that is above an operating voltage of the DIMM" (Office Action, p. 7, ll. 14-28) as previously recited, prior to the present amendment.

However, Amidi merely discloses transferring control of the volatile memory to the memory module so that the memory module can provide necessary self-refresh signaling to preserve the volatile memory contents upon power supply failure.¹ (Amidi, col. 8, ll. 38-41) Moreover, in Amidi, this is performed using back up battery power. (*Id.*, l. 37) By comparison, claim 2 previously claimed "wherein the controller is configured to perform in response to the trigger signal, a write operation to write data into the nonvolatile memory," and has now been clarifyingly amended to claim a controller including a logic element that, in response to the detected trigger condition, is configured to perform a plurality of write operations. Similarly claim 44 recites a controller including a logic element that, in response to the detected trigger condition, is configured to periodically perform at regular intervals of time a plurality of write operations; and claim 58 recites a controller including a logic element that, in response to the detected trigger condition, is configured to periodically perform every millisecond a plurality of write operations. Levkovitz, cited in the Office Action for allegedly showing these features

¹ "Self-refresh," as explained to the Examiner during the Examiner Interview, is a term of art that defines a state of the DIMM, or individual DDR SDRAM device, in which it receives sufficient power to preserve its information content, but is not accessible for reading or writing. ("The Self Refresh command can be used to retain data in the DDR2 SDRAM, even if the rest of the system is powered down. When in the Self Refresh mode, the DDR2 SDRAM retains data without external clocking. The DDR2 SDRAM device has a built-in timer to accommodate Self Refresh operation." Ex 1026 JEDEC Standard JESD79-2B Jan 2005)

(Office Action, p. 6, ll. 14-18) is not properly combinable with Amidi because Amidi uses battery backup power upon detection of power loss to transfer management of volatile memory so that its contents are preserved, whereas Levkovitz performs regular write operations (backups) as a precaution under normal operating conditions and writes volatile memory information into nonvolatile memory such as hard disc, without the urgency of power loss. The contention in the Office Action that the teachings of Levkovitz are combinable with those of Amidi to arrive at the claimed performance of a plurality of write operations by a logic element of a controller in response to a trigger condition is impermissibly grounded in hindsight gleaned from familiarity with Applicant's disclosure. There can be no doubt that such a combination would either change the principle of operation of one of either of the references. A change in the principle of operation is an indicia of non-obviousness. MPEP 2143.01(VI)

Claim 2 has further been amended to state that "the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode." As explained in the note above, "self-refresh mode" is a term of art meaning that the memory device is not accessible for read or write. This is not the case in Amidi, which was cited in the Final Written Decision of the '918 and '054 patents, and relied upon in the Office Action for current limitations, and which at least fails to disclose "the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode," of the current amendment. Amidi discloses that all the DRAM are forced into self-refresh mode. Specifically, Amidi states that a "signal from the trp timer advances the state machine 1100 to self-refresh entry state 1170, with self-refresh initiated for the memory module. The state machine then advances automatically to self refresh state 1180, and stays there automatically self-refreshing the memory module. A reset signal or an exit 5 signal will move the state machine 1100 back to idle state 1110, ending the self-refresh process, such as when a system sufficiently recovers power." (Col. 7, Lines 1-8)

In the Applicant-Initiated Interview Summary filed by the Examiner on 9/13/24, Section V states "Amidi teaches that memory (comprising backup power) may operate in backup mode, in response, for example, in response [*sic*] to change to voltage level (e.g., a triggering

condition). The backup comprises a plurality of write operations.” Applicant has addressed Amidi having the DRAM enter self-refresh mode when switching to backup power supply. It should be noted that self-refresh cannot include performing ANY write operations. Rather, to avoid data loss, the DRAM cells are themselves refreshed to maintain the data as it was when power was lost. Thus, Amidi’s battery is used for backup power, but the data is not “backed up” by writing the data to another location. Rather, the data stays right where it was prior to power loss, specifically, in the DRAM.

Additionally, there is only one action which may be termed a “write operation” after the trigger condition is detected in Amidi. Amidi, Col. 8, Lines 30-36 state “If one or both of the clock collapsing or the power Supply diminishing occurs, then at module 1440, the process moves the memory module into a clock and power fault detection state. This may include setting a power fail interrupt bit in a register, indicating that a power failure was detected (or a clock failure) and the memory module assumed control.” The writing/setting of the power fail interrupt bit in a register is the only action involving a change in the state of a single bit taken post trigger condition in Amidi. Thus, Amidi does not disclose a plurality of write operations after the detection of a trigger condition.

The Office Action also cites U.S. Patent Publication No. 2007/0088838 to Levkovitz (“Levkovitz”), and specifically to paragraph [0075]. Paragraph [0075] of Levkovitz states:

In some embodiments, for example, utilizing volatile memory unit 118, a pre-defined memory “dumping” process may be used. For example, content items ready for serving may be received by ad server 110 from control server 120, may be stored in volatile memory unit 118, and optionally may not be stored in a non-volatile memory of ad server 110. Ad server 110 may serve content items directly from its volatile memory unit 118, and may further store in memory unit 118 a log tracking the activity or servings done by ad server 110. Periodically, for example, at pre-defined time intervals (e.g., every 60 seconds, every 5 minutes, every hour, etc.) or upon serving a certain number of servings of content items (e.g., every 100 impressions, every 500 impressions, etc.), ad server 110 may “dump” or copy the activity log data into a non-volatile memory unit, e.g., included in ad server 110 or database 140. This may allow ad server 110, for example, to rapidly serve content items from its rapidly-available volatile memory unit 118, instead of from a slower non-volatile memory (e.g., database 140 or other database); and the periodical

“dumping” process of activity logs may limit the data loss in case of a power outage which may erase the volatile memory unit 118.

From this passage, a POSITA would conclude that the activity log is copied to NVM as part of a regular course of operation. Thus, there would be no motivation to combine Levkovitz with Amidi, as the setting of the bit which even if it were construed as a write operation it must be for a single bit and is done only Amidi detecting power failure. There would have to be a teaching or suggestion to perform periodic writes after the trigger condition in Amidi, but after the trigger condition, Amidi very distinctly wants to conserve power as much as possible, not perform write after write. Thus, Amidi chooses to supply power to the DRAMs from backup power source (battery) and put the DIMM (DRAMs) in self-refresh. In contrast, Claims 2, 44, and 58 recite “wherein, in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations,” or “wherein, in response to the detected trigger condition, the logic element is configured to perform at regular intervals of time a plurality of write operations,” or “wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond a plurality of write operations.”

Harris and FBDIMM Standards cannot cure this deficiency because they do not disclose any kind of backup write operation. The S3 mode discussed in the FBDIMM standards is also self-refresh, so it suffers from the same deficiencies as Amidi.

Claims 2, 44, and 58 and the claims which depend therefrom are allowable for at least this additional reason.

Claim 2, 44, and 58, as currently amended, recite:

wherein the plurality of DDR SDRAM packages includes:

- a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

- a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

and

- when the system power is provided to the DIMM from the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

While Harris, the FBDIMM Standards, and Amidi all disclose ranks generally, they do not disclose the specific limitations recited above. Harris specifically discloses that its DIMM “may be configured as having one or more ranks (e.g., 2, 4, 8, or more).” Amidi does not use the term rank in the disclosure; rather, Amidi only very generally discloses “Mounted on memory card 310 are memory chips 320 (320A-I as illustrated) and resistors 330 (330A-I as illustrated).” The FBDIMM Standards, specifically, JESD205 disclose several raw card configurations with ranks. The ranks have eight devices and are not sub-grouped below the rank level. This is confirmed by the Pin Description on page 29 of that standard:

CS[1:0]A, CS[1:0]B	Chip Select (one per rank)	4
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The Pin Description only describes a single chip select for each rank.

This is further confirmed by the Pin Description of JESD 82-20. On page 82 of that standard, the table with the pin description discloses:

CS0A-CS1A, CS0B-CS1B	○	Chip Select: Used with $\overline{\text{CAS}}$, $\overline{\text{RAS}}$, and $\overline{\text{WE}}$ to specify the SDRAM command. These signals are used for selecting one of two SDRAM ranks. CS0 is used to select the first rank and CS1 is used to select the second rank.
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Thus, the chip select disclosed is used to select a rank and not some subgrouping below the rank or ranks.

As described above, the FBDIMM standards make clear that there are not at least 72 data conduits to communicate at least 72 1-bit wide DDR signals between the DIMM and the computer system. Even if conduits which do exist in FBDIMM, are used as serial communication “lanes” (which unmistakably convey frames including BOTH address and control and data) are counted as data conduits, there are only 24 lane conduits in FBDIMM. The southbound link has 10 lanes, while the northbound link has 14 lanes. Thus, the FBDIMM standards, which must be relied upon as Harris lacks disclosure beyond referencing the FBDIMM standards, fail to disclose the above elements of Claim 2.

Thus, individually or in combination, Harris, FBDIMM Standards, and Amidi cannot disclose:

wherein the plurality of DDR SDRAM packages includes:

a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal;
and

when the system power is provided to the DIMM by the system board:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

as recited by Claims 2, 44, and 58. Claims 2, 44, and 58 and the claims which depend

therefrom are allowable for at least this additional reason.

Claims 2, 44, and 58 have been further amended to recite:

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board

and have been separately amended to recite that:

- the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;

Amidi discloses switching to a backup power system upon power fault and placing the DRAM in self-refresh mode, as is discussed in detail above. FBDIMM standards disclose an “S3 mode,” discussed in part above. However, in S3 mode, the DRAM are in self refresh just like Amidi, as is disclosed on Page 21 of JESD 82-20:

6. Command/Address output control in S3 mode. When the Advanced Memory Buffer core logic is in S3 power mode, all command/address outputs, including CKE, ODT, CLK, and all other command/address pins, will be driven low. In S3 mode, 1.8V supply is on. 1.5V, Vtt, and 3.3V are off. DRAMs are in self refresh, and the CKE signals must be driven low.

Harris discussed no trigger condition and therefore no operation post trigger condition. Thus, Harris, FBDIMM standards, and Amidi, alone or in combination cannot disclose:

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board

and

- the first and second groups of DDR SDRAM packages are configured to be

accessible and not in self-refresh mode;

as recited by Claims 2, 44, and 58. Thus, Claims 2, 44, and 58 and the Claims which depend therefrom are allowable for at least this additional reason.

Conclusion

Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: October 1, 2024

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/582,797		Filing Date 01/24/2022		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	10/01/2024	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 42	Minus	** 20	= 22		x \$100 =		2200		
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0		x \$480 =		0		
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		2200	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=		x \$0 =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=		x \$0 =				
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/LAVINIA D JOHNSON/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	01/24/2025	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2138	
			NOTIFICATION DATE	DELIVERY MODE
			01/24/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

DETAIL ACTION

Notice of Pre-AIA or AIA Status

1. *The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.*
2. *This office action is in response to the communication filed on 10/01/2024. There are a total of 42 claims pending in the application. Claim 2 has been amended; claims 1 and 3-27 have been canceled; and claims 28-68 have been added.*

INFORMATION CONCERNING IDS:

3. *The information disclosure statements (IDS') submitted on 09/24/2024 and 10/01/2024 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the Examiner. The IDS listed documents marked by lined-through have not been considered by the Examiner, because a copy documents have not been provided.*

INFORMATION CONCERNING CLAIMS:

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

4. *Claims 2 and 28-68 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.*

The independent claim 2 recites, in part, the limitation(s):

- “the memory-socket including a set of power conduits, a set of data conduits including at least 72 data conduits, and a set of address and control conduits”.

- “a 1-bit wide double data rate (DDR) data signal is electrically conducted between the system board”

- “wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines”

- “in response to the detected trigger condition, the logic element is configured to perform a plurality of write operations, including: a first write operation to write a first set of data bits into a first portion of the data storage element; a second write

operation to write a second set of data bits into a second portion of the data storage element; and subsequently a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;”

- “the trigger condition occurs when the input voltage goes above both the first and second threshold voltages”
- “wherein: a specified operating voltage value is within a range of operating voltage values of the DIMM, the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value”
- “each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the system board”
- “wherein, when the system power is provided to the DIMM from the system board:

the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein: a specified operating voltage value is within a range of operating voltage values of the DIMM,
the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value

- "when the system power is provided to the DIMM from the system board:
the first and second groups of DDR SDRAM packages are configured to be **accessible and not in self-refresh mode**;

the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and

the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits."

The claimed specification fails to describe/support the limitation(s) as claimed.

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5. *Independent claims 44 and 58 recite similar limitations as presented above with respect to the independent claim 2.*

6. *Dependent claims 28-43, 45-67 and 68 are rejected at least by virtue their dependency from their respected claim.*

35 USC 112(b) or 35 USC 112 (pre-AIA), Second Paragraph

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 2, 44, and 58 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

7. *The independent claim 2, in part, recite the limitation:*

*“trigger condition occurs when the **input voltage goes above both the first and second threshold voltages**” (emphasis added).*

The specification appears to recite trigger condition occurs when input voltage is above a first threshold voltage and below a second threshold votage.

8. *The independent claim 2, in part, recite the limitation:*

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“the first and second groups of DDR SDRAM packages are configured to be **accessible and not in self-refresh mode**” (*emphasis added*).

It is not clear what is meant by phrase “accessible and not self-refresh mode”.

Does that mean that both groups may be in self-refresh and no access to either group possible or only one group at a time can be in the self-refresh mode?

The independent claims 44 and 58 recites a similar limitations and are rejected on the same ground of limitations. The dependent claims 28-43, 45-57, and 59-68 are rejected at least based on their dependency from their respected independent claims.

Response to Remarks

Applicant arguments have been fully considered but they have not been persuasive. Page 22-27 of the Remarks recites the newly amended independent claim 2. Page 27-28 recite the status of claim rejection as related to previous office action. On page 30-31, the Remarks make references to ¶ 0002 of Harris. The ¶ 0002 of Harris is background portion of the specification (e.g., refers to related art prior to Harris invention). ¶ 0003 Harris discusses the solution is a power distributed system in which the DIMM modules (e.g., board) receives external input voltage and generate a plurality of on-board regulated voltages to be supplied or provided to memory devices (e.g., DIMMs, see paragraph ¶ 0009 of Harris). Contrary to the Applicant arguments, the external input voltage regulated by VRM module 102 and supplied to the DRAM devices 110-1 to 110-N (e.g., DIMM, see ¶ 0009 and Figs. 1A and 1b of Harris) . In addition Harris DIMM may comprise any type of Double Data Rate (DDR) devices. Page 33 of

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the Remarks recites ¶ 0019 of the Remarks and highlight some portions of the paragraph and concludes that Harris cannot read on claim 2. Applicant argues:

“As the Office Action acknowledges, the memory module 100A of Harris receives its power through an external source. (“Fig. 1B Harris shows a block diagram of memory module 100B comprising DRAM devices 110, inputs a plurality voltages 120 from external (sic) source and generates a plurality of regulated voltages 106 and 108,” Office Action, p. 5, 11. 3-5, emphasis added).

Applicant respectfully maintains that Harris expressly teaches away from providing power to any memory module assembly via the memory socket from the system board of the computer system.” (*Emphasis added, bolded portion*).

The Examiner respectfully disagrees. Fig. 1A or 1B Harris teaches receiving input voltage(s) from an external voltage source (line(s) of 104 or 120s) and converting to the regulated voltages Vdd 108 and Vcc 106 using the voltage regulator module 102. Fig. 1B of Harris shows receiving voltages from a plurality of external voltage sources (e.g., lines), and converting to the regulated voltages Vdd 108 and Vcc 104 using a plurality of voltage regulator modules 122 (and logic 124). Harris discloses:

“[0014] FIG. 1B is a block diagram of voltage distribution system for a memory module 100B according to another embodiment. It should be apparent to one skilled in the art that the voltage distribution system for memory module 100B is essentially similar to the system shown in FIG. 1A, but for the redundancy implementation of local voltage conversion. A plurality of on-board VRMs 122-K, K=1, 2, . . . , N, are provided as part of the memory assembly/module 100B wherein each VRM is operable with an independent voltage supply path for locally converting an external supply voltage into appropriate local voltage levels. Reference numerals 120-1 through 120-K refer to K supply voltage paths which may be coupled to various voltage sources provided within the electronic system (e.g., a computer system) (not shown). It should be readily recognized that the external voltage sources may comprise any combination of known or heretofore

unknown voltage supplies, either regulated or unregulated, and even including variable voltages.

[0015] A **suitable logic module 124 is provided for selecting among the plurality of like voltage outputs from the VRMs 122-K in order to energize the V.sub.dd and V.sub.cc paths 108, 106, respectively.** In one implementation, a wired-OR arrangement is operable to **select from K voltage outputs (e.g., from K V.sub.dd outputs or from K V.sub.cc outputs) to power the appropriate local voltage supply path, i.e., the V.sub.dd and V.sub.cc paths.** As described previously, the V.sub.dd path 108 is coupled to each of the DRAM devices 110-1 through 110-N and the V.sub.cc path 106 is coupled to the buffer/logic component 112 of the memory assembly module 100B.” (Pars. 0014-0015, *emphasis added*).

Fig. 3 of Harris shows a plurality of DIMM board or module with with a plurality edge pins to be attached or coupled to memory controller via DIMM connector (.e.g., Par. 0019 OF Harris). Fig. 4 of Harris shows a computer system 400 comprising a plurality of memory modules 410 connected to memory controller 408 via memory link 411. Fig. 4 also shows that each of memory module comprise a local voltage regulator 412 for receiving input voltage from the external memory controller and converting it to regulated voltages used by each DIMM module of the plurality of DIMM modules. As showed in Figs. 3-4 of Harris, all of the communication between the computer system and the memory modules including external power voltage source or lines provided via edge pins and the DIMM connectors. Applicant argues that Harris expressly teaches away, apparently assuming that that there is no socket (e.g., connector). Applicant make numerous references to ¶ 0002 of Harris, which, as noted above, is a portion of background information. The Examiner respectfully disagrees with the Applicant that Harris teaches away, even if power was not provided through connector (e.g., socket). Harris teaches external voltage(s) are received by DIMM boards or modules and converted to regulated voltage(s) required by operating DIMMs.

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Page 28 of the Remarks states “support for all of the below amendments has been provided in the attachment accompanying this paper. The support is provided by claim number in a table within the attachment.” The Examiner respectfully submits that no support provided for the newly added amendments. The support means that the claimed specification describes/supports by pointing to portion(s) of the specification that describes/supports the newly added limitations. There is no table within the provided applicant’s Remarks that relates the limitations to claimed specification. Applicant describes various limitations in the claims without referring to the portions of specification that describes or supports them. The rejections under 35 USC 112, indicate at least some of the added new limitation not described/supported as is required to comply with written description requirement.

The arguments regarding the limitations associated with self-refresh subject is a newly introduced subject matter is considered moot. Although, as rejected under 112, there are some ambiguities in claim limitations that need clarification or modification.

Regarding the plurality Buck converted and voltage values, as previously noted, Harris teaches one or more regulated supply (e.g., Figs. 1A and 1B of Harris) to provide regulated voltages used by DIMM. The information regarding the type of voltage regulators such as buck converters and specific voltage values, as referred to in the previous office action, has been provided in the document “Trial Termination or final written description” or “Trial-Term”. For example page 23 of trial teaches how to combine (e.g., map) the teaching of Harris and FBDIMM standards to achieve require voltage values. The Trial-Term documents also describes that single and dual package

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buck converters were available before the time the Harris' invention filed, one having ordinary in the art would have been motivated to modify the DIMM board or module taught by Harris to include buck converters as voltage regulators. Regarding triggering voltage variation or fault, the combination of references should be considered.

For example, Harris teaches a power system supply voltage of +12V with a wide tolerance +/-15%. For an input supply source voltage 12V with +/-15%, The voltage range for DIMM to properly operates is from 10V to 13.8V. It means the DIMM module with an operating voltage ≤ 10.2 or an operating voltage ≥ 13.8 may fail (e.g., causes triggering condition). Amidi teaches a variation in a power supply may be detected and means has been provided to switch to the backup power supply (e.g., see claim 19, Figs. 5-6 of Amidi). In summary, the Examiner respectfully submit that combination or reference relied on previous office action teach or render obvious the pre-amended limitations. The current set of claims (e.g., newly amended and added claims) recite some limitations not described/supported by the claim specification.

Conclusion

*Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).*

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer

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Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HASHEM FARROKH/

Primary Examiner, Art Unit 2138

<i>Notice of References Cited</i>	Application/Control No. 17/582,797		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2138	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	2012-10-25	Chen; Chi-She	G11C29/52	711/E12.008
*	B	US-20070136523-A1	2007-06-14	Bonella; Randy M.	G06F12/0866	711/E12.019
*	C	US-20080126624-A1	2008-05-29	Prete; Edoardo	G11C7/1006	710/53
*	D	US-20060174140-A1	2006-08-03	Harris; Shaun L.	G11C5/04	713/300
*	E	US-7724604-B2	2010-05-25	Amidi; Mike H.	G11C7/22	711/115
*	F	US-6856556-B1	2005-02-15	Hajack; Michael J.	G11C16/30	365/189.11
*	G	US-20060080515-A1	2006-04-13	Spiers; John	G06F11/1456	714/14
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	01/19/2025								
	1	-								
	2	✓								
	3	-								
	4	-								
	5	-								
	6	-								
	7	-								
	8	-								
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	39	✓								
	40	✓								
	41	✓								
	42	✓								

<i>Index of Claims</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CLAIM		DATE								
Final	Original	01/19/2025								
	43	✓								
	44	✓								
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	64	✓								
	65	✓								
	66	✓								
	67	✓								
	68	✓								

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF
711	110	11/01/2023	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF
Inventor/Assignee Search Updated	10/31/2023	HF
EAST Text, CPC, FACET Search Updated	10/31/2023	HF
Inventor/Assignee Search Updated	02/22/2024	HF
EAST Text, CPC, FACET Search Updated	02/22/2024	HF
Inventor/Assignee Search Updated	08/05/2024	HF
EAST Text, CPC, FACET Search Updated	08/05/2024	HF
Inventor/Assignee Search Updated	01/19/2025	HF
EAST Text, CPC, FACET Search Updated	01/19/2025	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	11/01/2023	HF
G06F13/28	G06F13/00	11/01/2023	HF
G06F2212/205	G06F2212/00	11/01/2023	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L92	51	("5563838" "6351827" "6418065" "6434044" "7034569" "7161820" "7166934" "7269764" "7426649" "7613880" "7644216" "7656735" "20090020608" "20090027844").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 01:19 PM
L93	2	92 AND (buck near5 (conver\$3 OR regulat\$3))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 01:25 PM
L94	16	("8301833").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 03:47 PM
L95	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/01 05:00 PM
L96	2	("8874831").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 09:00 AM
L97	2	("8874831").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2024/07/02 09:00 AM

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L99	2	("20060174140").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 02:16 PM
L100	10	("7724604" "20060080515" "6856556").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/02 02:24 PM
L101	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/03 10:03 AM
L102	4	("11232054").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/07/30 10:26 AM
L103	2	("20070136523").pn.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 01:59 PM
L104	532	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:30 PM
L105	2281	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:31 PM
L106	97	105 AND (trigger\$3 near5 signal)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:31 PM
L107	0	97 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:32 PM
L108	410	105 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:32 PM
L109	13	106 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 02:33 PM
L110	55	(backup OR replicat\$3 OR cop\$4 OR writ\$4	(US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	ON	2024/08/02 04:48 PM

		OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH milli\$1second	JPO; DERWENT; IBM_TDB)				
L111	24	110 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:48 PM
L112	530	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) near5 (milli\$1second OR ms)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:51 PM
L113	143	112 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:52 PM
L114	71	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) ADJ (milli\$1second OR ms)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:53 PM
L115	14	114 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:54 PM
L116	6	(backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3 OR stor\$2) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH (every OR each) ADJ (milli\$1second)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:55 PM
L117	0	116 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 04:55 PM
L118	0	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM

		WITH millisecond					
L119	145	periodic\$4 WITH (backup OR replicat\$3 OR cop\$4 OR writ\$4 OR sav\$3) WITH ((non\$volatile OR flash) near2 (memory OR storage OR SSD)) WITH second	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM
L120	26	119 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:09 PM
L121	0	120 AND dimm	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:19 PM
L122	19	"backup may occur every millisecond"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:39 PM
L123	0	122 AND (@ad<"20070601")	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:39 PM
L124	19	(backup near5 occur near5 every near5 millisecond)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:40 PM
L125	19	(backup near5 occur near5 (every OR each OR one) near5 millisecond)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/08/02 05:41 PM
L126	31	(buck ADJ converters) SAME (regulat\$3 near2 voltage) SAME ((printed ADJ circuit ADJ board) OR PCB)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/08/05 08:32 AM
L127	3	((buck ADJ converters) SAME (regulat\$3 near2 voltage) SAME ((printed ADJ circuit ADJ board) OR PCB)).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2024/08/05 08:32 AM
L128	3	127 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/08/05 08:33 AM
L129	5	126 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2024/08/05 08:34 AM
L130	6	126 AND ((G06F1/185	(US-PGPUB; USPAT;	OR	ON	ON	2024/08/05

		OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)	USOCR; IBM_TDB)				08:35 AM
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2138	
	Examiner Name	HASHEM FARROKH	
	Attorney Docket Number	022955-000090	

U.S.PATENTS							Remove	
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1							
If you wish to add additional U.S. Patent citation information please click the Add button.							Add	
U.S.PATENT APPLICATION PUBLICATIONS							Remove	
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1							
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17582797
	Filing Date		2022-01-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2138	
	Examiner Name	HASHEM FARROKH	
	Attorney Docket Number	022955-000090	

1	Exhibit 2031, IPR 2022-00996, Declaration of Dr. William Henry Mangione-Smith regarding US Patent No. 11,016,918, 144 pages, executed on March 28, 2023.	☐
2	Paper 25, IPR 2022-00996, Petitioner's Reply to Patent Owner's Response, filed June 20, 2023, 57 pages.	☐
3	Exhibit 1014, Petition for Inter Partes Review of US Patent No. 8,301,833, filed June 20, 2014, 69 pages.	☐
4	Exhibit 1018, Petition for Inter Partes Review of US Patent No. 8,301,833, filed January 13, 2017, 70 pages.	☐
5	Exhibit 2013, JEDEC Standard No. 21-C, Page 4.20.3-1; PC133 SDRAM Unbuffered SO-DIMM, Reference Design Specification, Revision 1.02, 26 pages. downloaded August 16, 2022.	☐
6	FBDIMM: Architecture and Protocol, JEDEC Standard JESD206.01, January 2007, 128 pages.	☐

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Name/Print	Khaled Shami	Registration Number	38745

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	1	5563838	A	1996-10-08	MART et al.	
	2	6031762	A	2000-02-29	SAITOH	
	3	6351827	B1	2002-02-26	CO et al.	
	4	6418065	B2	2002-07-09	SATO et al.	
	5	6434044	B1	2002-08-13	GONGWER et al.	
	6	6556475	B2	2003-04-29	YAMAZAKI et al.	
	7	6889335	B2	2005-05-03	HARGIS et al.	
	8	7034569	B1	2006-04-25	BALASUBRAMANIAN et al.	

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	9	7161820	B2	2007-01-09	FUNABA et al.	
	10	7166934	B2	2007-01-23	DEWEY	
	11	7221617	B2	2007-05-22	FLACH et al.	
	12	7269764	B2	2007-09-11	DART et al.	
	13	7280417	B2	2007-10-09	CHOI et al.	
	14	7426649	B2	2008-09-16	BRITTAİN et al.	
	15	7495943	B2	2009-02-24	TAKEMURA et al.	
	16	7587559	B2	2009-09-08	BRITTAİN et al.	
	17	7613880	B2	2009-11-03	MIURA et al.	
	18	7644216	B2	2010-01-05	FAHR et al.	
	19	7656735	B2	2010-02-02	KAGAN et al.	

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	20	7679967	B2	2010-03-16	CHUNG et al.	
	21	8089795	B2	2012-01-03	RAJAN et al.	
	22	5272677	A	1993-12-21	YAMAMURA	
	23	5675467	A	1997-10-07	NISHIMURA et al.	
	24	5831421	A	1998-11-03	TAIRA et al.	
	25	6493282	B2	2002-12-10	IIDA et al.	
	26	7016249	B2	2006-03-21	ROSS	
	27	7061821	B2	2006-06-13	COTEUS et al.	
	28	7117334	B2	2006-10-03	ALMEIDA et al.	
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	30	7224595	B2	2007-05-29	DREPS et al.	

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	31	7702928	B2	2010-04-20	SCHUMACHER et al.	
	32	7779285	B2	2010-08-17	DOBLAR et al.	

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	1	20060136765	A1	2006-06-22	POISNER et al.	
	2	20060294437	A1	2006-12-28	WASHBURN et al.	
	3	20080307240	A1	2008-12-11	DAHAN et al.	
	4	20090020608	A1	2009-01-22	BENNETT et al.	
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	1	10-2005-0073902	KR	A	2005-07-18	SAMSUNG ELECTRONICS CO LTD	Machine Translation attached	☒
	2	0 744 748	EP	B1	2000-10-18	SILICON GRAPHICS, INC.		☐
	3	0 908 892	EP	B1	2005-07-27	NEC CORPORATION; NEC Electronics Corporation		☐
	4	3707888	JP	B2	2005-10-19	HITACHI LTD	Machine Translation attached	☒
	5	9-34806	JP	A	1997-02-07	CANON INC	Machine Translation attached	☒
	6	2001-318741	JP	A	2001-11-16	IDEC IZUMI CORP	Machine Translation attached	☒
	7	2001-357669	JP	A	2001-12-26	TOSHIBA TEC	Machine Translation attached	☒
	8	2006-268419	JP	A	2006-10-05	NEC CORP	Machine Translation attached	☒
	9	01/37090	WO	A1	2001-05-25	SUN MICROSYSTEMS, INC.		☐
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	1	ISL6537 "ACPI Regulator/Controller for Dual Channel DDR Memory Systems", Renesas, 16 pages, 07/18/2007, taken from internet on 16 April 2024 from: https://www.renesas.com/us/en/document/dst/isl6537-datasheet	☐
	2	ISL6532 "ACPI Regulator/Controller for Dual Channel DDR Memory Systems", Intersil, 16 pages, 07/2004, taken from internet on 17 April 2024 from: https://www.mouser.com/datasheet/2/465/intersil_fn9120-342258.pdf	☐
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	7	TONG et al., "A System Level Implementation of Rijndael on a Memory-slot based FPGA Card", Tong, 8 pages, 01/2003, taken from internet on 06 June 2024 from: https://phwl.org/assets/papers/aes_fpt02.pdf	☐
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	10	Xilinx (July 2005). OPB Synchronous DRAM (SDRAM) Controller (v1.00e). Retrieved from https://datasheet.datasheetarchive.com/originals/library/Datasheet-080/DASF007621.pdf . See pages 9 & 11.	☐

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11	Phillips (2003). New multimarket products quarterly highlights. Retrieved from https://datasheet.datasheetarchive.com/originals/library/Datasheets-DAV2/DSADA0022280.pdf	☐
12	Maxim (September 2005). Buck Converters Proliferate in Handhelds as Features and Processing Power Increase. Retrieved from https://datasheet.datasheetarchive.com/originals/library/Datasheet-081/DASF0031684.pdf	☐
13	Keller, S. J. (2006). The Physical Design of the ILLIAC 6 Supercomputing Platform (Doctoral dissertation, University of Illinois at Urbana-Champaign). See pages 12, 13 & highlighted sections.	☐
14	Interstill (2004). Power Management Application Guide for Altera FPGAs. Retrieved from https://www.datasheetarchive.com/pdf/download.php?id=c9facaf95737a8327c7217bff086b497d4f48&type=P&term=buck%2520SDRAM . See pages 4 & 9.	☐
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18	Renesas (2005). ISL6531 Datasheet.	☐
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21	Intel (2002). Mobile Intel® Pentium® 4 Processor-M and Intel® 845MP/845MZ Chipset Platform Design Guide.	☐

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22	Asus (2001). A7V133 A7V133-C jumperfree PC133/VC133 200/266MHz FSB AGP Pro/4X Socket A Motherboard User's Manual.	☐
23	Trescases, O. (2007). Integrated Power-Supplies for Portable Applications.	☐
24	Van Schaik, C. F. (2002). The Development of a Node for a Hardware Reconfigurable Parallel Processor (Master's thesis, University of Cape Town).	☐
25	Boroyevich, D. (2004). Power Electronics Building Blocks, "Plug-and-Play", Hardware and Software Control Architectures. VIRGINIA POLYTECHNIC INST AND STATE UNIV BLACKSBURG OFFICE OF SPONSORED PROGRAMS.	☐
26	McPartland, R. J., Loeper, D. J., Higgins, F. P., Singh, R., MacDonald, G., Komoriya, G., ... & Leung, C. W. (2000, May). SRAM Embedded Memory with Low Cost, FLASH EEPROM-Switch-Controlled Redundancy. In Proceedings of the IEEE 2000 Custom Integrated Circuits Conference (Cat. No. 00CH37044) (pp. 287-289). IEEE.	☐
27	Crisp, R. (1997). Direct Rambus Technology: The New Main Memory Standard. IEEE Micro, 17(6), 18-28. PC2-4200/PC2-3200 DDR2	☐
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Telephonic
Khalid Shami	Attorney	

Date of Interview: 05 March 2025

Issues Discussed:

35 U.S.C. 112

An applicant initiated interview with (the Applicant Mr. Khaled Shami Reg. # 38,745), was conducted on 03/05/2025. The interview was in response to the previous office action mailed 01/24/2025. The purpose of the interview was to clarify the rejections under 35 USC 112(a) first paragraph, as not complying with written description requirement. During the interview, agreement reached that applicant would provide a table that map claims limitations to the portions of the claimed specification that describe/support them.

☒ Attachment

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

FOR DISCUSSION PURPOSES ONLY

Proposed Claim Amendments (Marked-Up)

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket~~ slot connector of a system board of a ~~host computer~~ system, the memory ~~socket~~ slot connector including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board;~~ a 1 bit wide double data rate (DDR) data signal ~~signals is electrically conducted between the system board~~ host system and the DIMM ~~are transferred via the memory socket by each data conduit of the set of data conduits, and address and control signals are delivered electrically conducted from the host system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:~~

~~a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:~~

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory socket~~ slot connector of the host system, the edge connections including:

~~a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits and operable to deliver to the DIMM power from power rails of at the host system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM;~~

~~a second set of edge connections each of which is configured to electrically conduct a 1 bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and~~

~~a third set of edge connections configured to be electrically connected to~~

respective address and control conduits of the set of address and control conduits[[:]]

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM, ~~the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;~~

a set of data lines electrically ~~coupled~~ connected to respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1 bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and

a set of address and control lines electrically ~~coupled~~ connected to respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and ~~electrically connected to the~~ to an input voltage supply line of the set of voltage supply lines, the controller configured to receive the input voltage via the input voltage supply line, the controller including:

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits;~~

~~a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and~~

~~a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to: (i) monitor and compare the input voltage supplied via the input voltage supply line to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and (ii) generate a trigger signal upon detecting~~ detect a trigger condition; and (iii) transmit the trigger signal to

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at least one other portion of the controller, wherein the controller is configured to perform, in response to the detected trigger condition signal, the logic element is configured to perform a plurality of write operations, including:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;

wherein, when the system power is provided to the DIMM from the system board:

the trigger condition occurs when the input voltage exceeds a first threshold voltage goes above both the first and second threshold voltages; and

wherein:

a specified operating voltage value is within a range of operating voltage values of the DIMM;

the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value;

the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value;

the first threshold voltage corresponds to the lower operating voltage value; and

the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (DDR-SDRAM) packages devices coupled to the set of address and control lines and to the set of data lines, each the DDR SDRAM package of the plurality of DDR-SDRAM packages having a bit width of m bits and configured devices being operable to receive or output m-1 bit wide DDR data signals via respective m data lines of the set of data lines in response to based on address and control signals

received ~~from the host system via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and~~

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line; ~~and each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the input voltage supply line and to deliver power system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively,~~

~~wherein the set of components includes~~including:

a first component ~~configured to receive power via at least electrically connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;~~

a second component ~~configured to receive power via at least electrically connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;~~

a third component ~~configured to receive power via at least electrically connected to the third regulated voltage line and powered by the third regulated voltage when the system power is provided to the DIMM from the system board; and~~

a fourth component electrically ~~configured to receive power via at least connected to the fourth regulated voltage line and powered by the fourth regulated voltage when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and~~

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) ~~receives power when the system power is provided to the DIMM from the system board, is powered;~~

only by one or more of the first, second, third, and fourth converter circuits; and only via the one or more of the first, second, third, and fourth regulated voltage lines;

~~wherein the DIMM is configurable to operate in any of a plurality of states including a first state and a second state, wherein the controller is further configured to control power delivery to one or more components of the set of components for the DIMM to operate in the first state or the second state, whereby power to the set of components via the first, second, third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first state, and whereby power to the first component via the first regulated voltage line is turned off and power to the second, third, and fourth components via the second, third and fourth regulated voltage lines is turned on when the DIMM is operating in the second state~~

~~wherein the plurality of DDR SDRAM packages includes:~~

~~a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;~~

~~a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and~~

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and~~
- ~~• the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data~~

signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

- wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.

3 – 27. (Canceled)

~~28.~~ (Currently Amended) The DIMM of claim 2, wherein the ~~second component is~~ further connected to the first regulated voltage line and is powered by the first and second regulated voltages when the system power is provided to the DIMM from the system board plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;
- the at least 40 1-bit wide DDR data signals include at least 32 1-bit

wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals; and

the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

29. (Currently Amended) The DIMM of claim 2, wherein the controller further includes:
data storage element is a buffer configured to store the plurality of data bits or a First In First Out (FIFO) buffer configured to store the plurality of data bits.

a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Currently Amended) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage, the PCB has a form factor with a width and a height dimension, and wherein the width dimension is 133.35 millimeters.

31. (Currently Amended) The DIMM of claim 30, wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages, the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

32. (Currently Amended) The DIMM of claim 312, wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based

~~on a combination of the first and second sets of data bits written during the first and second write operations, a specified operating voltage value is within a range of operating voltage values of the DIMM.~~

33. (Currently Amended) The DIMM of claim ~~32~~, wherein, ~~prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations, the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM.~~

34. (Currently Amended) The DIMM of claim 2, wherein, when power from the power rails of the host system is provided to the DIMM~~the system power is provided to the DIMM from the system board~~, the nonvolatile memory is not powered using the first regulated voltage.

35. (Currently Amended) The DIMM of claim ~~2~~ 29, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger ~~condition~~signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger ~~condition~~signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Currently Amended) The DIMM of claim ~~235~~, wherein, ~~prior to performing the third write operation, the logic element is further configured to perform (i) a first read operation of the plurality of read operations to read the first set of data bits from the first portion of the data storage element, and (ii) a second read operation of the plurality of read operations to read the second set of data bits from the second portion of the data storage element, and to generate the~~

~~third set of data bits based on the read first and second sets of data bits; each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.~~

38. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger ~~condition~~ signal, one or more read operations.

39. (Previously Presented) The DIMM of claim 38, wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM ~~packages~~ devices, respectively.

40. (Currently Amended) The DIMM of claim ~~33~~ 38, wherein the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value. ~~the logic element is further configured to perform (i) first and second read operations of the one or more read operations, and (ii) subsequently, the first and second write operations.~~

41. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits, ~~in response to the detected trigger condition,~~ the voltage monitor circuit is further configured to generate and transmit a trigger signal.

43. (Currently Amended) The DIMM of claim ~~24~~ 0, wherein, when power from the power rails of the host system is provided to the DIMM ~~the system power is provided to the DIMM from the system board~~, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket~~ slot connector of a host system board of a computer system, the memory ~~socket~~ slot connector including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1 bit wide double data rate (DDR) data signals are transferred is electrically conducted between the host system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are delivered from the host system electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:~~

a printed circuit board (PCB) having ~~a form factor with a width dimension and a height dimension, the PCB~~ edge connections configured to fit into the memory slot connector of the host system, the edge connections including:

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:~~

a first set of edge connections configured to be electrically connected to ~~respective power conduits~~ of the set of power conduits and operable to deliver the DIMM power from the power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM.

a second set of edge connections ~~each of which is~~ configured to ~~electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and~~

a third set of edge connections configured to be electrically connected to ~~respective address and control conduits of the set of address and control conduits~~[[:]],

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power

rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board[[:]]

a set of data lines electrically ~~coupled to respective edge connections of~~ connected to the second set of edge connections, ~~wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; and~~

a set of address and control lines electrically ~~coupled to respective edge connections of~~ electrically connected to the third set of edge connections and ~~configured to electrically conduct address and control signals;~~

a controller ~~including coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power and the input voltage via the input voltage supply line, the controller including:~~

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits; and~~

~~a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and~~

a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and electrically connected to the ~~an~~ input voltage supply line of the set of voltage supply lines, and coupled to the logic element; the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line; and
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller, wherein, compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and
~~detect a trigger condition;~~

~~wherein, in response to the detected trigger condition, the logic element₁~~
in response to the detected trigger condition, the controller is configured to
periodically perform₁ at regular intervals of time₁ a plurality of write operations,
including[[:]]

a first write operation to write a first set of data bits into a first portion of
the data storage element; and

a second write operation to write a second set of data bits into a second
portion of the data storage element;

~~wherein, when the system power is provided to the DIMM from the system board:~~

- ~~• the trigger condition occurs when the input voltage goes above both
the first and second threshold voltages; and~~

~~wherein:~~

- ~~• a specified operating voltage value is within a range of operating
voltage values of the DIMM;~~
- ~~• the range of operating voltage values is bounded by an upper operating
voltage value and a lower operating voltage value;~~
- ~~• the lower operating voltage value is below the specified operating
voltage value of the DIMM and the upper operating voltage value is
above the specified operating voltage value;~~
- ~~• the first threshold voltage corresponds to the lower operating voltage
value, and~~
- ~~• the second threshold voltage corresponds to the upper operating
voltage value;~~

a set of components coupled to the PCB, the set of components including a
plurality of double data rate (DDR) synchronous dynamic random access memory (DDR
SDRAM) packages-devices coupled to the set of address and control lines and to the set of
data lines, ~~the each DDR SDRAM package of the plurality of DDR SDRAM packages~~
~~having a bit width of m bits and configured devices being operable to receive or output m~~
~~1-bit wide DDR data signals via respective m data lines of the set of data lines based on~~
~~response to address and control signals received from the host system via the set of~~
address and control lines via address and control lines of the set of address and control
~~lines, wherein m is equal to four or eight; and~~

first, second, third, and fourth converters ~~circuits~~ coupled to the PCB and to the input voltage supply line, ~~each of the first, second, third, and fourth converter circuits and~~ configured to, ~~in response to the detected trigger condition,~~ receive power ~~supplied by~~ from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines, ~~when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, wherein the set of components includes including:~~

a first component ~~electrically connected to the first regulated voltage line and powered by~~ configured to receive power via the first regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a second component ~~electrically connected to the second regulated voltage line and powered by~~ configured to receive power via the second regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a third component ~~electrically connected to the third regulated voltage line and powered by~~ configured to receive power via the third regulated voltage line ~~when the system power is provided to the DIMM from the system board; and~~

a fourth component ~~electrically connected to the fourth regulated voltage line and powered by~~ configured to receive power via the fourth regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and~~

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) receives power ~~when the system power is provided to the DIMM from the system board, is powered:~~

- ~~only by one or more of the first, second, third, and fourth converter circuits; and~~

only via the one or more of the first, second, third, and fourth regulated voltage

lines;

wherein the plurality of DDR SDRAM packages includes:

~~a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;~~

~~a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and~~

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are configured to be accessible and not in self-refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and~~
- ~~• the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error correcting code (ECC) data signals; and~~

~~wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.~~

45. (Currently Amended) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

~~the plurality of write operations further includes a third write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic~~

~~element is further configured to periodically perform at regular intervals of time all of the first and second write operations followed by the third write operation.~~

46. (Currently Amended) The DIMM of claim 44~~5~~, ~~wherein, the logic element is further configured to perform the third write operation by writing a third set of data bits from the data~~

~~storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations~~ wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector ~~socket~~, and the total number of data conduits is at least 72 data conduits.

47. (Currently Amended) The DIMM of claim 45, ~~wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element~~ the logic element is further configured to perform a second write operation to write a second set of data bits into a second portion of the

data storage element.

48. (Previously Presented) The DIMM of claim 44, wherein, when ~~the system~~ power is provided to the DIMM from the power rails of the host system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Currently Amended) The DIMM of claim 474, wherein, ~~in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal~~ the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Currently Amended) The DIMM of claim 496, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Currently Amended) The DIMM of claim 474, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM ~~packages~~ devices, respectively.

52. (Currently Amended) The DIMM of claim 44, wherein, ~~prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time a plurality of read operations~~ when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including

a first threshold voltage and a second threshold voltage.

53. (Currently Amended) The DIMM of claim 5246, wherein, ~~prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read data bits from the data storage element and to generate the third set of data bits based on the read data bits~~ when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

54. (Currently Amended) The DIMM of claim 5344, wherein a specified operating voltage value is within a range of operating voltage values of the DIMM, ~~the width dimension is 133.35 millimeters.~~

55. (Currently Amended) The DIMM of claim 54, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM, ~~the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.~~

56. (Currently Amended) The DIMM of claim 495, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Currently Amended) The DIMM of claim 5544, wherein, ~~when the system power is provided to the DIMM from the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and~~

~~wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal~~ the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds

to the upper operating voltage value.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket slot connector~~ of a ~~host system board of a computer system~~, the memory ~~socket slot connector~~ including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board~~, a 1-bit wide double data rate (DDR) data signals are transferred ~~dis electrically conducted~~ between the ~~host system board~~ and the DIMM via ~~the memory socket by each data conduit of the set of data conduits~~, and ~~address and control signals are delivered from the host system electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits~~, the DIMM comprising:

a printed circuit board (PCB) having ~~a form factor with a width dimension and a height dimension~~, the PCB edge connections configured to fit into the memory slot connector of the host system, the edge connections including:

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:~~

a first set of edge connections configured to be electrically connected to ~~respective power conduits of the set of power conduits and operable to deliver the DIMM power from the power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM.~~

a second set of edge connections ~~each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and~~

a third set of edge connections configured to be electrically connected to ~~respective address and control conduits of the set of address and control conduits[;].~~

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the

first set of edge connections of the DIMM, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board[::]

a set of data lines electrically ~~coupled to respective edge connections of~~ connected to the second set of edge connections, ~~wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines~~[::] and

a set of address and control lines electrically ~~coupled to respective edge connections of~~ connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines and configured to electrically conduct address and control signals;

a controller ~~including coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power and the input voltage via the input voltage supply line, the controller including:~~

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits; and~~

~~a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and~~

a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and electrically connected to the input voltage supply line of the set of voltage supply lines and coupled to the logic element; the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line; and
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller, wherein the compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second

threshold voltage; and

detect a trigger condition;

wherein, in response to the detected trigger condition, the logic element controller is configured to ~~periodically perform every millisecond~~, in response to the trigger signal, a plurality of write operations, including:

a first write operation to write a first set of data bits into a first portion of the data storage element; and

a second write operation to write a second set of data bits into a second portion of the data storage element;

wherein, when the system power is provided to the DIMM from the system board:

- the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein:

- a specified operating voltage value is within a range of operating voltage values of the DIMM;
- the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value;
- the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value;
- the first threshold voltage corresponds to the lower operating voltage value; and
- the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, ~~the each DDR SDRAM package of the plurality of DDR SDRAM packages~~ having a bit width of m bits and ~~configured devices being operable~~ to receive or output m 1-bit wide DDR data signals via ~~respective m data lines of the set of data lines~~ based on ~~response to~~ address and control signals received from the host system via the set of address and control lines ~~via address and control lines of the set of address and control~~

lines, wherein m is equal to four or eight; and

first, second, third, and fourth ~~voltage reducing circuits~~ converter circuits coupled to the PCB and to the input voltage supply line, ~~each of the first, second, third, and fourth converter circuits~~ and configured to, ~~in response to the detected trigger condition,~~ receive power ~~supplied by~~ from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines, ~~when the system power is provided to the DIMM from the system board,~~ the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and

(ii) ~~output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively,~~ wherein the set of components ~~includes~~ including:

a first component ~~electrically connected to the first regulated voltage line and powered by~~ configured to receive power via the first regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a second component ~~electrically connected to the second regulated voltage line and powered by~~ configured to receive power via the second regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a third component ~~electrically connected to the third regulated voltage line and powered by~~ configured to receive power via the third regulated voltage line ~~when the system power is provided to the DIMM from the system board; and~~

a fourth component ~~electrically connected to the fourth regulated voltage line and powered by~~ configured to receive power via the fourth regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter,~~ and the fourth converter circuit is a voltage reducing circuit; and

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) receives power ~~when the system power is provided to the DIMM from the system board, is powered;~~

- ~~only by one or more of the first, second, third, and fourth~~

converter circuits; and
only via the one or more of the first, second, third, and fourth
regulated voltage lines;

wherein the plurality of DDR SDRAM ~~packages~~ devices includes:

a first group of at least five DDR SDRAM ~~packages~~ devices that are each
connected to a first chip select line configured to electrically conduct a first chip
select signal; and

a second group of at least four DDR SDRAM ~~packages~~ devices that are each
connected to a second chip select line configured to electrically conduct a second
chip select signal; and

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are is-~~
~~configured to be accessible and not in self refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be-~~
~~enabled, in response to the first chip select signal, to receive or output~~
~~40 1 bit wide DDR data signals in parallel via 40 data lines of the set~~
~~of data lines independently of whether the second group of DDR~~
~~SDRAM packages is enabled, in response to the second chip select~~
~~signal, to receive or output an integer number of 1 bit DDR data~~
~~signals in parallel via the integer number of data lines of the set of data~~
~~lines; and~~
- ~~• the 40 1 bit wide DDR data signals include 32 1 bit wide DDR data~~
~~signals and eight 1 bit wide DDR error correcting code (ECC) data~~
~~signals; and~~

~~wherein the sum of the 40 data lines and the integer number of data lines~~
~~matches a total number of data conduits of the set of data conduits of the memory~~
~~socket, and the total number of data conduits is at least 72 data conduits.~~

59. (Currently Amended) The DIMM of claim 586, ~~wherein the plurality of write~~
~~operations further includes a third write operation subsequent to the first and second write~~
~~operations; and wherein, in response to the detected trigger condition, the logic element is~~

~~further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.~~

wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector ~~socket~~, and the total number of data conduits is at least 72 data conduits

60. (Currently Amended) The DIMM of claim 527, wherein the logic element is further configured to perform ~~the a~~ third write operation by subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Currently Amended) The DIMM of claim 560, wherein, ~~in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal~~ the logic element is further configured to perform, prior to the first write operation and the second write operation, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Currently Amended) The DIMM of claim 586, wherein, ~~in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond a plurality of read operations prior to performing the first and second write~~

operations when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.

63. (Currently Amended) The DIMM of claim ~~586~~0, wherein ~~the logic element is further configured to perform a first and second read operations of the plurality of read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively~~ when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Currently Amended) The DIMM of claim ~~563~~, wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages. ~~the width dimension is 133.35 millimeters.~~

65. (Currently Amended) The DIMM of claim ~~642~~, wherein a specified operating voltage value is within a range of operating voltage values of the DIMM. ~~the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.~~

66. (Currently Amended) The DIMM of claim ~~586~~, wherein each of the first, second, and third ~~voltage reducing~~converter circuits is a buck converter.

67. (Currently Amended) The DIMM of claim ~~586~~, wherein, when ~~the system power is provided to the DIMM from the power rails of the host system board,~~ the nonvolatile memory is not powered using the first regulated voltage.

68. (Currently Amended) The DIMM of claim ~~6358~~, wherein, when ~~the system power is provided to the DIMM from the power rails of the host system board,~~ an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

Proposed Claim Amendments (Clean)

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals are delivered from the host system to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and

(iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the host system via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line; and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component electrically configured to receive power via at least the fourth regulated voltage line; and

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.

3– 27. (Canceled)

28. (Currently Amended) The DIMM of claim 2, wherein the plurality of DDR

SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;
- the at least 40 1-bit wide DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals; and

the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

29. (Currently Amended) The DIMM of claim 2, wherein the controller further includes:

a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically

communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Currently Amended) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Currently Amended) The DIMM of claim 30, wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

32. (Currently Amended) The DIMM of claim 31, wherein a specified operating voltage value is within a range of operating voltage values of the DIMM.

33. (Currently Amended) The DIMM of claim 32, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM.

34. (Currently Amended) The DIMM of claim 2, wherein, when power from the power rails of the host system is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Currently Amended) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Currently amended) The DIMM of claim 38, wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

40. (Currently Amended) The DIMM of claim 33, wherein the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value.

41. (Currently Amended) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Currently Amended) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Currently Amended) The DIMM of claim 40, wherein, when power from the power rails of the host system is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the host system ~~board~~ and the DIMM via the set of data conduits, and control signals are delivered from the host system to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,

and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller,

wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the host system via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;
a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line;

and

a fourth component electrically configured to receive power via the fourth regulated

voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.

45. (Currently Amended) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Currently Amended) The DIMM of claim 44, wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

47. (Currently Amended) The DIMM of claim 45, wherein the logic element is further configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Currently Amended) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.

49. (Currently Amended) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Currently Amended) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Currently Amended) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Currently Amended) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input

voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Currently Amended) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

54. (Currently Amended) The DIMM of claim 53, wherein a specified operating voltage value is within a range of operating voltage values of the DIMM.

55. (Currently Amended) The DIMM of claim 54, wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM.

56. (Currently Amended) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Currently Amended) The DIMM of claim 55, wherein the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the host system and the DIMM via the set of data conduits, and control signals are delivered from the host system set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the host

system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,
and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller,

wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM-devices being operable to receive or output data signals via the set of data lines based on-address and control signals received from the host system via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

- a first component configured to receive power via the first regulated voltage line;
- a second component configured to receive power via the second regulated voltage line;
- line;
- a third component configured to receive power via the third regulated voltage line;
- and
- a fourth component electrically configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

- a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; and
- a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal.

59. (Currently Amended) The DIMM of claim 58, wherein when power from the power rails of the host system is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of

whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and

- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

60. (Currently Amended) The DIMM of claim 59, wherein the logic element is further configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Currently Amended) The DIMM of claim 60, wherein the logic element is further configured to perform, prior to the first write operation and the second write operation, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Currently Amended) The DIMM of claim 58, wherein when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.

63. (Currently Amended) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Currently Amended) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

65. (Currently Amended) The DIMM of claim 64, wherein a specified operating

voltage value is within a range of operating voltage values of the DIMM.

66. (Currently Amended) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Currently Amended) The DIMM of claim 58, wherein, when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.

68. (Currently Amended) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the host system, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.

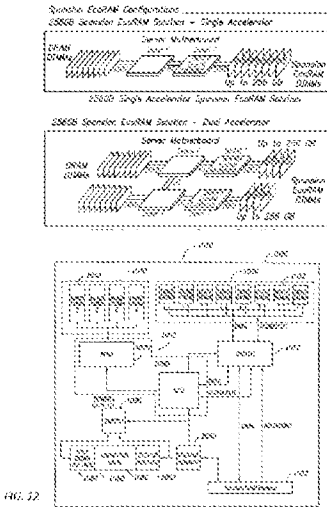
Support Table for Claim Amendments

Claim Element	Limitation	Support
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2.1

A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals are delivered from the host system to the DIMM via the set of address and control conduits

Para. [00123]
“For example, **the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system.** The interface 1022 of certain embodiments **provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system.** For example, the interface 1022 can comprise **a standard 240-pin DDR2 edge connector.**”
Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figure 2 and Fig.



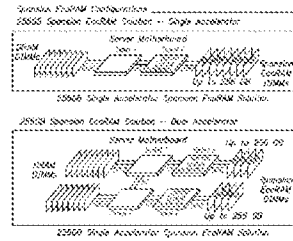
12.

The memory socket of a computer system includes pins soldered onto the system board that provide electrical conduits for system power, data, and address and control signals from the system board to be electrically connected to

		corresponding edge connections of the edge connector 1022 when it is inserted into the memory socket. Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”
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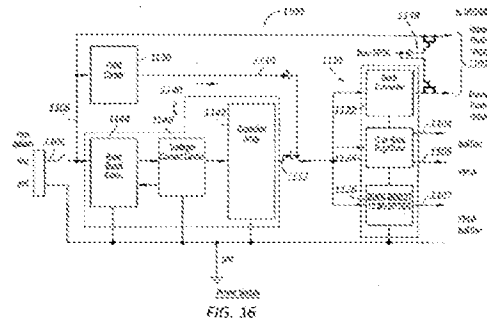
2.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including: a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits,	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector. When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB.
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Fig. 2 which shows the DIMM in the socket on the



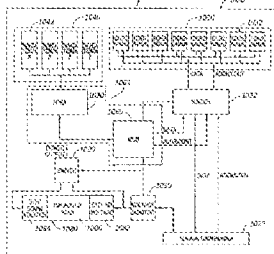
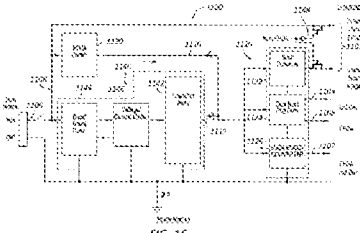
motherboard:

Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.



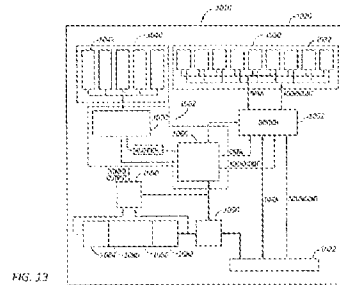
The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector.

See also limitation 2.1.

2.3	a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM a set of data lines electrically coupled to the second set of edge connections, and a set of address and control lines electrically to the third set of edge connections,	Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16. For example, in Fig. 12 the input voltage supply lines are shown electrically connected to the Voltage Monitor 1050. FIG. 12  FIG. 16  For example, respective power conduits (Vdd and gnd) of the set of power conduits of the memory socket are labeled “from system”, on left of Fig. 16, are electrically connected via a set of voltage supply lines to the PWM Controller 1130, and the ground supply line (gnd). Para. [00124]
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		“For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” Note also the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.
2.4	a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;	The current application specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178. Para. [00172] “In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).” Para. [00178] “For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA.”

For example, FIG. 13 shows controller 1062 comprising a buffer:



Para. [00133]

“The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.”

“In certain embodiments, **the voltage monitor circuit 1050 is part of the controller 1062** (e.g., part of the MCU 1060)”

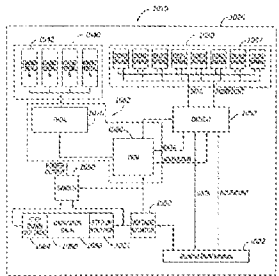
Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070

Para. [00133]

“In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and **the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.**”

Para. [00130]

“In certain embodiments, the trigger condition occurs when the memory system 1010 **detects** that the **system voltage** is below **a certain threshold voltage**. For example, in one embodiment, the threshold voltage is 10 percent below **a specified operating voltage**. In certain embodiments, **a trigger condition** occurs when the voltage goes above **a certain threshold value**, such as,

		for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
2.5	a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the host system via the set of address and control lines;	FIG. 12  FIG. 12 Para. [00124] “Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits

		of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” And Para. [00122] “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.” And Para. [00124] “In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”
2.6	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines	FIG. 16

	On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck convertor 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. Para. [00148] “The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g.,
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		flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
2.7	wherein the set of components includes: a first component configured to receive power via at least the first regulated voltage line; a second component configured to receive power via at least the second regulated voltage line; a third component configured to receive power via at least the third regulated voltage line; and a fourth component electrically configured to receive power via at least the fourth regulated voltage line	FIG. 16 FIG. 12 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the

		regulated voltage lines 1102, 1104, 1105, and 1107, respectively. On the right, each of the DRAM are shown as being connected to the first regulated voltage, for example. On the right, the isolation device is shown as being connected to the second regulated voltage, for example. On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example. On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
2.8	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and	FIG. 16

fourth regulated voltage lines	On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the system power is received, couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
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28.1	wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and	FIG. 21 Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above. Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
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	Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126].
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		If x4 are used, the number of DRAM are correspondingly more.
28.2	when power from the power rails of the host system is provided to the DIMM: • each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; • the at least 40 1-bit wide DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182]

		“In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
28.3	the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
29.1	a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage	The current application specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU)

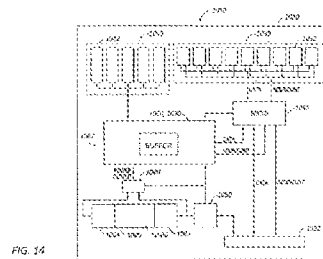
element and configured to electrically communicate with the nonvolatile memory and the data storage element,

1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16.

Para. [00127]

“FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.”

FIG. 14



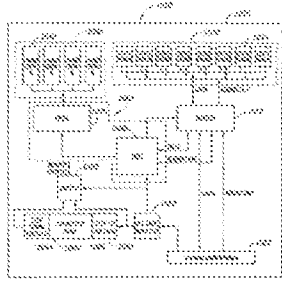
A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178.

Para. [00172]

“In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).”

Para. [00178]

	“For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA.” For example, FIG. 13 shows controller 1062 comprising a buffer: Para. [00127] “Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one embodiment, an FPGA without an internal flash is used
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		and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.” FIG. 12  FIG. 12
29.2	a first write operation to write a first set of data bits into a first portion of the data storage element; a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called “slicing” the volatile memory subsystem 1030.”

		Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above. Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
30	when power is provided to the DIMM from the power rails of the host system the	Para. [00133]

	voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage	“In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
31	when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition

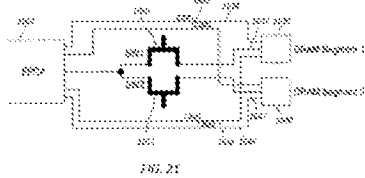
		occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
32	wherein a specified operating voltage value is within a range of operating voltage values of the DIMM	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
33	wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For

	operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM.	example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold." The first threshold and second threshold are the same as above, supported by "a threshold" and "another threshold" in the last line. The "10 percent" in the "the threshold voltage is 10 percent below a specified operating voltage" and the "10 percent" in the "above a specified operating voltage" support the lower operating voltage value, and the upper operating voltage value.
34	wherein, when power from the power rails of the host system is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] "In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102."
35	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read	Para. [00178] "The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash)." Para. [00178] "The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash)." Para. [00179]

	first and second sets of data bits.	“In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.”
35	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Difference from Claim 35 is the read from the buffer is a plurality of steps in Claim 35, and specifically two steps here. Same sentence in Para. [00178] supports both scenarios. Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040.” Reading a data word or a block of words to be written into the NVM.
37	wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.	Para. [00148] “The conversion element 1120 may comprise a plurality of sub- blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for

		providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system.”
38	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger signal, one or more read operations.	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.”
39	wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00174]

		“In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called “slicing” the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
40	wherein the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating

		voltage” support the lower operating voltage value, and the upper operating voltage value.
41	wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.	FIG. 21  Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.” Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” FIG. 13



As can be seen in FIG. 13, the transfer from the DRAM occurs on a connection on the right side of the buffer, and the transfer to NVM will occur on a connection on the left side of the buffer.

42

wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178.

Para. [00172]

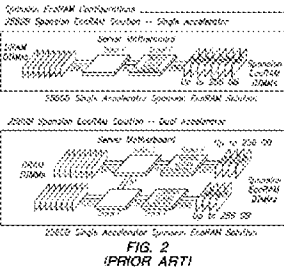
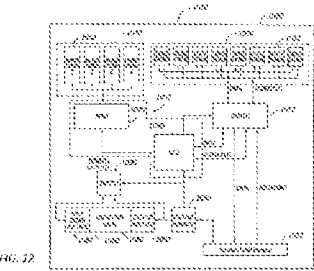
“In certain embodiments, **the controller 1062 comprises a buffer (not shown) which is configured to store data** from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).”

Para. [00178]

“For example, in one embodiment, the **buffer is a first-in, first-out buffer implemented in the controller 1062** which comprises an FPGA.”

For example, FIG. 13 shows controller 1062 comprising a buffer:

43.1	wherein, when power from the power rails of the host system is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.”
44.1	A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the host system and the DIMM are	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

	transferred via the set of data conduits, and control signals are delivered from the host system to the DIMM via the set of address and control conduits	Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figure 2 and Fig. 12.  FIG. 2 PRIOR ART The memory socket of a computer system includes pins soldered onto the system board that provide electrical conduits for system power, data, and address and control signals from the system board to be electrically connected to  FIG. 12 corresponding edge connections of the edge connector 1022 when it is inserted into the memory socket.
		Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown).

		For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system . The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector. ”
44.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including: a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector. When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB.

Fig. 2 which shows the DIMM in the socket on the

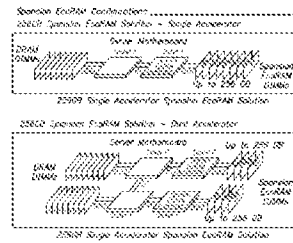
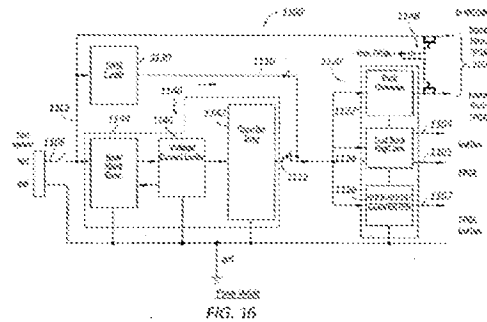


FIG. 2
(PRIOR ART)

motherboard:

Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.



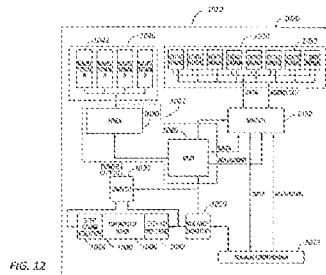
The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector.

		See also limitation 2.1.
44.3	a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM a set of data lines electrically coupled to the second set of edge connections, and a set of address and control lines electrically to the third set of edge connections ,	Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16. For example, in Fig. 12 the input voltage supply lines is shown electrically connected to the Voltage Monitor 1050. FIG. 12

		Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” Note also the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.
44.4	a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;	The current application specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178. Para. [00172] “In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).” Para. [00133] “The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.”

		“In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060)” Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070 Para. [00133] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
44.5	a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous	FIG. 12

dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the host system via the set of address and control lines;



Para. [00124]
“Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).”

Para. [00172]
 “In one embodiment, for example, the volatile memory subsystem 1030 comprises **a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)**, the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

And

Para. [00122]
 “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

And

Para. [00124]

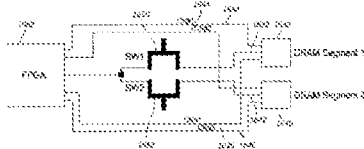
		“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”
44.6	first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines	FIG. 16

	sub- blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck- converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and
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On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the

		embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
44.8	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.	FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the system power is received, couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.”

		Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
45	wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.	Para. [00127] “FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.” FIG. 14

46.1	wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;	FIG. 21  FIG. 22 Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above. Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire

	data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or “slice”) may comprise 40 bits of the data word, and the second portion (or “slice”) may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as “slicing” the volatile memory during a backup operation. In the example embodiment, the process of “slicing” the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM
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		(5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more
46.2	wherein when power from the power rails of the host system is provided to the DIMM: - each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode; - the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182]

	integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	“In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
46.3	wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178]

		“In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
47	wherein the logic element is further configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.	Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above.
48	wherein, when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”

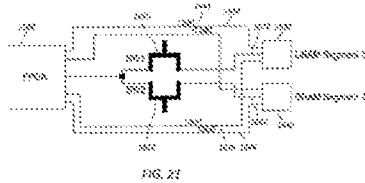
49	wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”
50	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00178] “The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice

	data bits based on the read first and second sets of data bits, respectively.	out of self-refresh, reading a 32-bit block from the slice into the buffer , and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. ”
51	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time, one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
52	wherein when power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130]

		“In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
53	wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.
54	wherein a specified operating voltage value is within a range of operating voltage values of the DIMM.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the

		voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold." The first threshold and second threshold are the same as above, supported by "a threshold" and "another threshold" in the last line. The "10 percent" in the "the threshold voltage is 10 percent below a specified operating voltage" and the "10 percent" in the "above a specified operating voltage" support the lower operating voltage value, and the upper operating voltage value.
55	wherein the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value, the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value of the DIMM.	Para. [00130] "In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold." The first threshold and second threshold are the same as above, supported by "a threshold" and "another threshold" in the last line. The "10 percent" in the "the threshold voltage is 10 percent below a specified operating voltage" and the "10 percent" in the "above a specified operating voltage" support the lower operating voltage value, and the upper operating voltage value.
56	wherein the third write operation comprises a transfer of the first and second set of data bits	FIG. 21

from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.



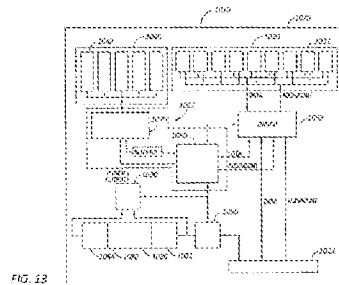
Para. [00172]

“FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.”

Para. [00177]

“The memory system 1010 may enter the second mode of operation, for example, **when a trigger condition (e.g., a power failure) occurs**. In certain embodiments, transferring each data word comprises **storing a first portion** (also referred to as a slice) **of the data word in a buffer** in an operational block 1622, **storing a second portion of the data word in the buffer** in an operational block 1624, and **writing the entire data word from the buffer to the non-volatile memory subsystem 1040** in an operational block 1626.”

FIG. 13

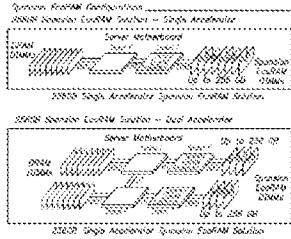


As can be seen in FIG. 13, the transfer from the DRAM occurs on a connection on the right side of the buffer, and

		the transfer to NVM will occur on a connection on the left side of the buffer.
57	wherein the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
58.1	A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a host system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the host system and the DIMM are transferred via the set of data conduits, and control signals are delivered from	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

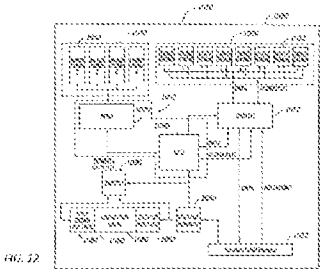
the host system to the DIMM via the set of address and control conduits

Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figure 2 and Fig.



12.

The memory socket of a computer system includes pins soldered onto the system board that provide electrical conduits for system power, data, and address and control signals from the system board to be electrically connected to corresponding edge connections of the edge connector 1022



when it is inserted into the memory socket.

Para. [00123]

“The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown).

For example, the interface 1022 can comprise a **plurality of edge connections which fit into a corresponding slot connector of the host system.** The interface 1022 of certain

		embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”
58.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the host system, the edge connections including: a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the host system via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the host system is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector. When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB.

be electrically connected to the set of address and control conduits,

Fig. 2 which shows the DIMM in the socket on the

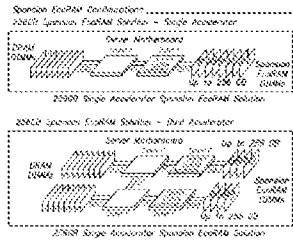
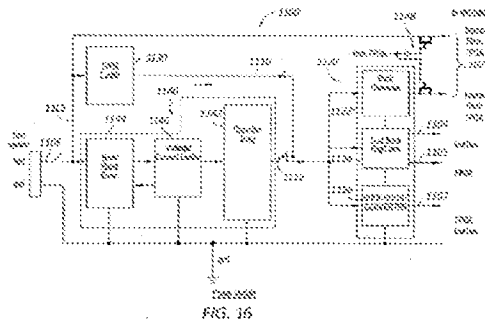


FIG. 2
(PRIOR ART)

motherboard:

Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.



configured to receive the power delivered to the DIMM from the power rails of the host system by way of the memory slot connector of the host system and the first set of edge connections of the DIMM

a set of data lines electrically coupled to the second set of edge connections, and

a set of address and control lines electrically to the third set of edge connections,

lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16. For example, in Fig. 12 the input voltage supply lines is shown electrically connected to the Voltage Monitor 1050.

FIG. 12

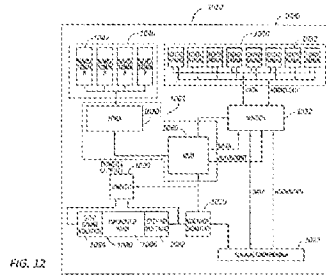
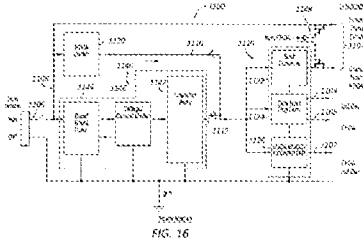


FIG. 16



For example, respective power conduits (Vdd and gnd) of the set of power conduits of the memory socket are labeled “from system”, on left of Fig. 16, are electrically connected via a set of voltage supply lines to the PWM Controller 1130, and the ground supply line (gnd).

Para. [00124]

“For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises **eight 64Mx8 DDR2 SDRAM** elements 1032.”

		Note also the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.
58.4	wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines;	Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.” And Para. [00172]

“In one embodiment, for example, the volatile memory subsystem 1030 comprises a **72-bit data bus** (e.g., each **data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC**), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

The embodiment of FIG. 13 shows nine x8 DRAM devices for a data bus bit width of 72:

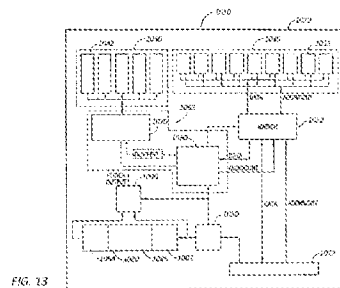


FIG. 13

And

Para. [00122]

*In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein."

58.5

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied

1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is

via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the data storage element;

found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16.

A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178.

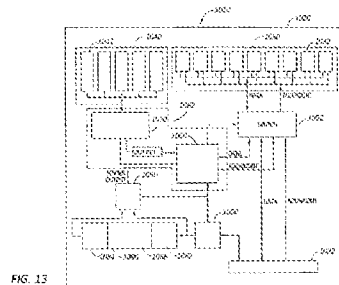
Para. [00172]

“In certain embodiments, **the controller 1062 comprises a buffer (not shown) which is configured to store data** from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).”

Para. [00178]

“For example, in one embodiment, the **buffer is a first-in, first-out buffer implemented in the controller 1062** which comprises an FPGA.”

For example, FIG. 13 shows controller 1062 comprising a buffer:



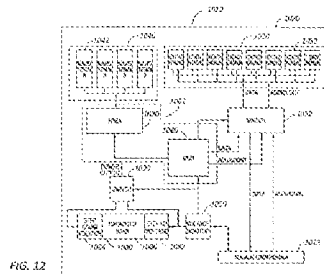
Para. [00133]

“The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.”

“In certain embodiments, **the voltage monitor circuit 1050 is part of the controller 1062** (e.g., part of the MCU 1060)”

		Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070 Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
58.6	a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data	FIG. 12

lines, the DDR SDRAM-
devices being operable to
receive or output data
signals via the set of data
lines based on-address and
control signals received
from the host system via
the set of address and
control lines;



Para. [00124]
“**Volatile memory elements 1032 compatible with certain embodiments described herein have packaging** which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).”

Para. [00172]
“In one embodiment, for example, the volatile memory subsystem 1030 comprises **a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)**, the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

And

Para. [00122]
“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

And

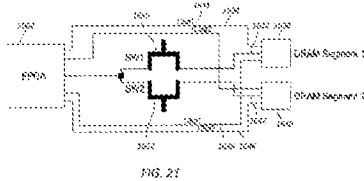
Para. [00124]

		“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”
58.7	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines	FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. Para. [00148] “The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters.

		The conversion element 1120 may comprise a plurality of sub- blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck- converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and
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On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the

		embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060
58.9	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines	FIG. 16

		“The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
58.10	wherein the plurality of DDR SDRAM devices includes: - a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; - a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal	FIG. 21  Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above. Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for

	the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062." Para. [00177] "In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626." Para. [00178] "In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the
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		32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
59.1	wherein when power from the power rails of the host system is provided to the DIMM: - each group of the first and second groups of DDR SDRAM is configured to be accessible and not in self-refresh mode; - the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the

	select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment." Para. [00182] "In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode."
59.2	wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits	Para. [00123] "The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector." Para. [00172] "In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the

	is at least 72 data conduits.	first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “ In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. ”
60	wherein the logic element is further configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
61	wherein the logic element is further configured to perform, prior to the first write operation and the second write operation, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	[00172] <u>FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.</u> <u>The controller read first and second data slices (sets of data bits) from first and second groups of DDR SDRAM packages, and then write the first and second sets of data bits into the controller 1062 (data storage element).</u>

62	wherein when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102. ”
63	when the power is provided to the DIMM from the power rails of the host system the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition. ” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage . For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage . In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value , such as, for example, 10 percent above a specified operating voltage . In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. ” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
64	wherein when power is provided to the DIMM from the power rails of the host system the trigger condition occurs when the input voltage goes above	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10

	both the first and the second threshold voltages.	percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
65	wherein a specified operating voltage value is within a range of operating voltage values of the DIMM.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.

66	wherein each of the first, second, and third converter circuits is a buck converter.	Para. [00148] “The conversion element 1120 may comprise a plurality of sub- blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system.”
67	wherein, when power is provided to the DIMM from the power rails of the host system, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”
68	wherein, when power is provided to the DIMM from the power rails of the host system, an additional trigger condition occurs when the input voltage is above the first threshold and the	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage.

	input voltage is below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.	In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.”
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Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

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PATENT CENTER # 69710469

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
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PAYMENT TRANSACTION ID
E20253NG10277610

PAYMENT AUTHORIZED BY
LISA MARGARET ELLIOTT

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1820	REQUEST FOR CONTINUED EXAMINATION (RCE)- 2ND AND SUBSEQUENT REQUEST (SEE 37 CFR 1.114)	2860.00	1	2860.00
TOTAL AMOUNT:				\$2,860.00

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	17582797	Filing Date	2022-01-24	Docket Number (if applicable)	022955-000090	Art Unit	2138
First Named Inventor	Chi-She Chen			Examiner Name	HASHEM FARROKH		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☒ Other

Appendix A - Claim Support Table; Appendix B - Clean Copy of Amended Claims

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 220585

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2025-03-24
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov**ELECTRONIC ACKNOWLEDGEMENT RECEIPT**APPLICATION #
17/582,797RECEIPT DATE / TIME
03/24/2025 04:10:05 PM Z ETATTORNEY DOCKET #
022955-000090**Title of Invention**

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application InformationAPPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 69710469

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She ChenCORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents**TOTAL DOCUMENTS: 2**

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
RCE - 022955-90.pdf	3	Request for Continued Examination (RCE)	1631 KB
Resp to 1-24-25 FOA w Appendix A and B - 022955- 90.pdf	99	Amendment Submitted/Entered with Filing of Continued Prosecution Application (CPA)/Request for Continued Examination(RCE)	2459 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
RCE - 022955-90.pdf	DB1FC70920753464AC76BD959DA2078658A2B02721B0211B9 107C08E09C817B1A56279487F8202F9CFE3E98A3234C9B137

DE433596459B9DE0E180ADDF57610B

Resp to 1-24-25 FOA w
Appendix A and B - 022955-
90.pdf

D7BC46D7161D95C3014355A0629AE46E079BDD4342FD77BE
3F65A401DD9DA0B804934A15DCEE10B16C6D80DCAB6D4349
AFBC12E13D8FB2D08EF95A514BDF5B76

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New International Application Filed with the USPTO as a Receiving Office

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

AMENDMENT

Commissioner:

In response to the final Office Action mailed January 24, 2025, Applicant submits the following Amendment and Remarks.

Listing of Claims begins on page **2**.

Remarks begin on page **29**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket~~ slot connector of a system board of a computer system, the memory ~~socket~~ slot connector including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board~~, a 1-bit wide double data rate (DDR) data signal signals are transferred is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are delivered electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:

~~a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB including:~~

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory sockets~~ slot connector of the system board, the edge connections including:

a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections ~~each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and~~

a third set of edge connections configured to be electrically connected to ~~respective address and control conduits of the set of address and control conduits,~~[[;]]

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM, ~~the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board;~~

a set of data lines electrically ~~coupled~~ connected to ~~respective edge connections of the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines;~~ and

a set of address and control lines electrically ~~coupled~~ connected to ~~respective edge connections of the third set of edge connections and configured to electrically conduct address and control signals;~~

a controller including a voltage monitor circuit and nonvolatile memory, ~~the controller coupled to the PCB and electrically connected to the~~ to an input voltage supply line of the set of voltage supply lines, ~~the controller configured to receive the input voltage via the input voltage supply line, the controller including:~~

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits;~~

~~a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and~~

~~a voltage monitor circuit electrically connected to the input voltage supply line and coupled to the logic element, the voltage monitor circuit configured to: (i) monitor and compare the input voltage supplied via the input voltage supply line to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and (ii) generate a trigger signal upon detecting detect a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the detected trigger conditionsignal, the logic element is configured to perform a plurality of write operations, including:~~

~~a first write operation to write a first set of data bits into a first portion of the data storage element;~~

~~a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently~~

~~a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations;~~

~~wherein, when the system power is provided to the DIMM from the system board:~~

~~the trigger condition occurs when the input voltage exceeds a first threshold voltagegoes above both the first and second threshold voltages; and~~

~~wherein:~~

~~a specified operating voltage value is within a range of operating voltage values of the DIMM, the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value;~~

~~the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value, the first threshold voltage corresponds to the lower operating voltage value, and the second threshold voltage corresponds to the upper operating voltage value;~~

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (DDR-SDRAM) ~~packages~~ devices coupled to the set of address and control lines and to the set of data lines, each the ~~the~~ DDR SDRAM package of the plurality of ~~DDR SDRAM packages~~ having a bit width of ~~m bits and configured devices being operable to receive or output m-1 bit wide DDR data signals via respective m data lines of the set of data lines in response to~~ based on address and control signals received from the system board via ~~address and control lines of the set of address and control lines, wherein m is equal to four or eight, and~~

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and each of the first, second, third, and fourth converter circuits configured to, in response to the detected trigger condition, receive power supplied by the input voltage supply line when the system power is provided to the DIMM from the input voltage supply line and to deliver power system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltage lines, respectively,

wherein the set of components includes~~including~~:

a first component configured to receive power via at least electrically ~~connected to the first regulated voltage line and powered by the first regulated voltage when the system power is provided to the DIMM from the system board;~~

a second component configured to receive power via at least electrically ~~connected to the second regulated voltage line and powered by the second regulated voltage when the system power is provided to the DIMM from the system board;~~

a third component configured to receive power via at least electrically
~~connected to the third regulated voltage line and powered by the third regulated~~
~~voltage when the system power is provided to the DIMM from the system board; and~~

a fourth component electrically configured to receive power via at least
~~connected to the fourth regulated voltage line and powered by the fourth regulated~~
~~voltage when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter,~~
~~and the fourth converter circuit is a voltage reducing circuit; and~~

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third,
and fourth regulated voltage lines; and

(ii) receives power ~~when the system power is provided to the DIMM~~
~~from the system board, is powered:~~

~~only by one or more of the first, second, third, and fourth converter~~
~~circuits; and only via the one or more of the first, second, third, and~~
~~fourth regulated voltage lines.[];~~

~~wherein the DIMM is configurable to operate in any of a plurality of states including a~~
~~first state and a second state, wherein the controller is further configured to control power~~
~~delivery to one or more components of the set of components for the DIMM to operate in the~~
~~first state or the second state, whereby power to the set of components via the first, second,~~
~~third, and fourth regulated voltage lines is turned on when the DIMM is operating in the first~~
~~state, and whereby power to the first component via the first regulated voltage line is turned off~~
~~and power to the second, third, and fourth components via the second, third and fourth~~
~~regulated voltage~~

~~lines is turned on when the DIMM is operating in the second state~~

~~wherein the plurality of DDR SDRAM packages includes:~~

~~a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;~~

~~a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and~~

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are configured to be accessible and not in self refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and~~
- ~~• the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and~~
- ~~• wherein the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.~~

3 – 27. (Canceled)

28. (Currently Amended) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second

~~regulated voltages when the system power is provided to the DIMM from the system board~~ plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;
- the at least 40 1-bit wide DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals; and

the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

29. (Currently Amended) The DIMM of claim 2, wherein the controller further includes: ~~data storage element is a buffer configured to store the plurality of data bits or a~~

~~First In First- Out (FIFO) buffer configured to store the plurality of data bits.~~

a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Currently Amended) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage. the PCB has a form factor with a width and a height dimension, and wherein the width dimension is 133.35 millimeters.

31. (Currently Amended) The DIMM of claim ~~30~~ 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM. the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

32. (Currently Amended) The DIMM of claim ~~2~~ 29, wherein, prior to performing the third write operation, the logic element is ~~further~~ configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Currently Amended) The DIMM of claim ~~2~~ 29, wherein, prior to performing the third write operation, the logic element is ~~further~~ configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Currently Amended) The DIMM of claim 2, wherein, when power from the power rails of the system board is provided to the DIMM~~the system power is provided to the DIMM from the system board~~, the nonvolatile memory is not powered using the first regulated voltage.

35. (Currently Amended) The DIMM of claim ~~2~~ 29, wherein, prior to performing the third write operation, the logic element is ~~further~~ configured to perform, in response to the detected trigger ~~conditions~~signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein, prior to performing the third write operation, the logic element is ~~further~~ configured to perform, in response to the detected trigger ~~conditions~~signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Currently Amended) The DIMM of claim ~~23~~ 5, wherein, ~~prior to performing the third write operation, the logic element is further configured to perform (i) a first read operation of the plurality of read operations to read the first set of data bits from the first portion of the data storage element, and (ii) a second read operation of the plurality of read operations to read the second set of data bits from the second portion of the data storage element, and to generate the third set of data bits based on the read first and second sets of data bits.~~ each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Currently Amended) The DIMM of claim ~~29~~ 2, wherein, prior to performing the first and second write operations, the logic element is ~~further~~ configured to perform, in

response to the detected trigger ~~condition~~signal, one or more read operations.

39. (Currently Amended) The DIMM of claim 38, wherein the logic element is ~~further~~ configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM ~~packages~~devices, respectively.

40. (Canceled)

41. (Currently Amended) The DIMM of claim 29 ~~2~~, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Currently Amended) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits, ~~in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal.~~

43. (Currently Amended) The DIMM of claim 23 ~~1~~, wherein, when power from the power rails of the system board is provided to the DIMM~~the system power is provided to the DIMM from the system board~~, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket~~ slot connector of a system board of a computer system, the memory ~~socket~~ slot connector including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power~~

~~conduits and the DIMM is configured to operate using only the provided system power from the system board, a 1-bit wide double data rate (DDR) data signals are transferred is electrically conducted between the system board and the DIMM via the memory socket by each data conduit of the set of data conduits, and address and control signals are delivered from the system board electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits, the DIMM comprising:~~

~~a printed circuit board (PCB) having a form factor with a width dimension and a height dimension, the PCB edge connections configured to fit into the memory slot connector of the system board, the edge connections including:~~

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory socket, the edge connections including:~~

~~a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,~~

~~a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits[[:]],~~

~~a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board[[:]]~~

~~a set of data lines electrically coupled to respective edge connections of connected to the second set of edge connections, wherein each data line of the set of~~

~~data lines is configured to electrically conduct a 1-bit wide DDR data signal, and
wherein the set of data lines includes at least 72 data lines; and~~

~~a set of address and control lines electrically coupled to respective edge
connections of electrically connected to the third set of edge connections and configured
to electrically conduct address and control signals;~~

~~a controller including coupled to the PCB and electrically connected to the input
voltage supply line, the controller configured to receive the system power and the input
voltage via the input voltage supply line, the controller including:~~

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits; and~~

~~a logic element coupled to the nonvolatile memory and the data storage element
and configured to electrically communicate with the nonvolatile memory and the data
storage element; and~~

~~a voltage monitor circuit and a data storage element configured to store a
plurality of data bits, the controller coupled to the PCB and electrically connected to
the an input voltage supply line of the set of voltage supply lines, and coupled to the
logic element, the voltage monitor circuit configured to:~~

- ~~(i) monitor an input voltage supplied via the input voltage supply
line; and~~
 - ~~(ii) generate a trigger signal upon detecting a trigger condition; and~~
 - ~~(iii) transmit the trigger signal to at least one other portion of the
controller, wherein, compare the input voltage to a plurality of
threshold voltages including a first threshold voltage and a
second threshold voltage; and~~
- ~~detect a trigger condition;~~

~~wherein, in response to the detected trigger condition, the logic element
-in response to the detected trigger condition, the controller is configured to
periodically perform, at regular intervals of time, a plurality of write operations,
including[[:]]~~

a first write operation to write a first set of data bits into a first portion of the data storage element; and

a second write operation to write a second set of data bits into a second portion of the data storage element;

wherein, when the system power is provided to the DIMM from the system board:

- the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and

wherein:

- a specified operating voltage value is within a range of operating voltage values of the DIMM;
- the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value;
- the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value;
- the first threshold voltage corresponds to the lower operating voltage value; and
- the second threshold voltage corresponds to the upper operating voltage value;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (DDR SDRAM) packages coupled to the set of address and control lines and to the set of data lines, the each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured devices being operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on in response to address and control signals received from the system board via the set of address and control lines via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and

first, second, third, and fourth converters ~~circuits~~ coupled to the PCB and to the input voltage supply line, ~~each of the first, second, third, and fourth converter circuits~~ and configured to, ~~in response to the detected trigger condition,~~ receive power ~~supplied by~~ from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines, ~~when the system power is provided to the DIMM from the system board,~~ the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and

(ii) ~~output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively,~~ wherein the set of components includes ~~including:~~

a first component ~~electrically connected to the first regulated voltage line and powered by~~ configured to receive power via the first regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a second component ~~electrically connected to the second regulated voltage line and powered by~~ configured to receive power via the second regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

a third component ~~electrically connected to the third regulated voltage line and powered by~~ configured to receive power via the third regulated voltage line ~~when the system power is provided to the DIMM from the system board; and~~

a fourth component ~~electrically connected to the fourth regulated voltage line and powered by~~ configured to receive power via the fourth regulated voltage line ~~when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and~~

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) receives power ~~when the system power is provided to the DIMM from the system board,~~ is powered:

- ~~only by one or more of the first, second, third, and~~

~~fourth converter circuits; and~~
only via the one or more of the first, second, third, and fourth regulated voltage lines.]]

~~wherein the plurality of DDR SDRAM packages includes:~~

~~a first group of at least five DDR SDRAM packages that are each connected to a first chip select line configured to electrically conduct a first chip select signal;~~

~~a second group of at least four DDR SDRAM packages that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and~~

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are configured to be accessible and not in self refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output 40 1-bit wide DDR data signals in parallel via 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and~~
- ~~• the 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and~~

~~wherein the sum of the 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory socket, and the total number of data conduits is at least 72 data conduits.~~

45. (Currently Amended) The DIMM of claim 44, wherein the controller further

includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

~~the plurality of write operations further includes a third write operation subsequent to the first and second write operations; and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform at regular intervals of time all of the first and second write operations followed by the third write operation.~~

46. (Currently Amended) The DIMM of claim 44~~5~~, ~~wherein, the logic element is further configured to perform the third write operation by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations~~ wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals

and eight 1-bit wide DDR error-correcting code (ECC) data signals; and
wherein the sum of the at least 40 data lines and the integer number of data lines matches
a total number of data conduits of the set of data conduits of the memory slot connector
socket, and the total number of data conduits is at least 72 data conduits.

47. (Currently Amended) The DIMM of claim 45, wherein, ~~prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element~~ the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Currently Amended) The DIMM of claim 44, wherein, when ~~the system~~ power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Currently Amended) The DIMM of claim 47, wherein, ~~in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a trigger signal~~ the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Currently Amended) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is ~~further~~ configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Currently Amended) The DIMM of claim 47~~4~~, wherein, prior to performing the first and second write operations, the logic element is ~~further~~ configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM ~~packages~~devices, respectively.

52. (Currently Amended) The DIMM of claim 44, wherein, ~~prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time a plurality of read operations~~ when power is provided to the DIMM from the power rails of the system board, the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Currently Amended) The DIMM of claim 52~~46~~, wherein, ~~prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read data bits from the data storage element and to generate the third set of data bits based on the read data bits~~ when power is provided to the DIMM from the power rails of the system board, the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

54. (Currently Amended) The DIMM of claim 49, wherein the logic element is ~~further~~ to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations ~~44, wherein the width dimension is 133.35 millimeters.~~

55. (Currently Amended) The DIMM of claim 49, wherein, ~~prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element~~ 54, wherein the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.

56. (Currently Amended) The DIMM of claim 49~~5~~, wherein the third write operation

comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Currently Amended) The DIMM of claim 5244, wherein ~~[[,]]~~ when ~~the system~~ power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and ~~the input voltage is~~ below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory ~~socket~~ slot connector of a system board of a computer system, the memory ~~socket~~ slot connector including a set of power conduits, a set of data conduits ~~including at least 72 data conduits~~, and a set of address and control conduits, wherein ~~system power from the system board is provided to the DIMM via the memory socket by the set of power conduits and the DIMM is configured to operate using only the provided system power from the system board~~, a 1-bit wide double data rate (DDR) data signals are transferred ~~is electrically conducted~~ between the system board and the DIMM via ~~the memory socket by each data conduit of the set of data conduits~~, and ~~address and control signals are delivered from the system board electrically conducted from the system board to the DIMM via the memory socket by the set of address and control conduits~~, the DIMM comprising:

a printed circuit board (PCB) having ~~a form factor with a width dimension and a height dimension~~, the PCB edge connections configured to fit into the memory slot connector, the edge connections including:

~~an interface having edge connections disposed along the width dimension and configured to fit into the memory socket~~, the edge connections including:

a first set of edge connections configured to be electrically connected to respective power conduits of the set of power conduits and operable to deliver to

the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

~~a second set of edge connections each of which is configured to electrically conduct a 1-bit wide DDR data signal and to be electrically connected to a respective data conduit of the set of data conduits, and~~

~~a third set of edge connections configured to be electrically connected to respective address and control conduits of the set of address and control conduits[[:]],~~

~~a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM, the set of voltage supply lines including an input voltage supply line to electrically conduct an input voltage when the system power is provided to the DIMM from the system board[[:]]~~

~~a set of data lines electrically coupled to respective edge connections of connected to the second set of edge connections, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines[[:]] and~~

~~a set of address and control lines electrically coupled to respective edge connections of connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines and configured to electrically conduct address and control signals;~~

~~a controller including coupled to the PCB and electrically connected to the input voltage supply line, the controller configured to receive the system power and the input voltage via the input voltage supply line, the controller including:~~

~~nonvolatile memory;~~

~~a data storage element configured to store a plurality of data bits; and~~

~~a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element; and~~

~~a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and electrically connected to the~~ an input voltage supply line of the set of voltage supply lines, and coupled to the logic element; the voltage monitor circuit configured to:

- ~~(i)~~ monitor an input voltage supplied via the input voltage supply line; and
- ~~(ii)~~ generate a trigger signal upon detecting a trigger condition; and
- ~~(iii)~~ transmit the trigger signal to at least one other portion of the controller, wherein the ~~compare the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage; and~~
detect a trigger condition;

~~wherein, in response to the detected trigger condition, the logic element controller is configured to periodically perform every millisecond, in response to the trigger signal, a plurality of write operations, including:~~

~~a first one or more write operations operation to write a first set of data bits into a first portion of the data storage element; and~~

~~a second write operation to write a second set of data bits into a second portion of the data storage element;~~

~~wherein, when the system power is provided to the DIMM from the system board:~~

- ~~• the trigger condition occurs when the input voltage goes above both the first and second threshold voltages; and~~

~~wherein:~~

- ~~• a specified operating voltage value is within a range of operating voltage values of the DIMM;~~

- ~~the range of operating voltage values is bounded by an upper operating voltage value and a lower operating voltage value;~~
- ~~the lower operating voltage value is below the specified operating voltage value of the DIMM and the upper operating voltage value is above the specified operating voltage value;~~
- ~~the first threshold voltage corresponds to the lower operating voltage value; and~~
- ~~the second threshold voltage corresponds to the upper operating voltage value;~~

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (DDR SDRAM) packages ~~devices~~ coupled to the set of address and control lines and to the set of data lines, ~~the each DDR SDRAM package of the plurality of DDR SDRAM packages having a bit width of m bits and configured~~ devices being operable to receive or output m 1-bit wide DDR data signals via respective m data lines of the set of data lines based on in response to address and control signals received from the system board via the set of address and control lines ~~via address and control lines of the set of address and control lines, wherein m is equal to four or eight; and~~

first, second, third, and fourth ~~voltage reducing circuits~~ converter circuits coupled to the PCB and to the input voltage supply line, ~~each of the first, second, third, and fourth converter circuits and configured to, in response to the detected trigger condition, receive power supplied by from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines, when the system power is provided to the DIMM from the system board, the first, second, third, and fourth converter circuits configured to (i) generate first, second, third, and fourth regulated voltages, respectively, from the power supplied by the input voltage supply line, and (ii) output the first, second, third, and fourth regulated voltages via first, second, third, and fourth regulated voltages lines, respectively, wherein the set of components includes including:~~

a first component ~~electrically connected to the first regulated voltage line and powered by~~configured to receive power via the first regulated voltage line~~when the system power is provided to the DIMM from the system board;~~

a second component ~~electrically connected to the second regulated voltage line and powered by~~configured to receive power via the second regulated voltage line~~when the system power is provided to the DIMM from the system board;~~

a third component ~~electrically connected to the third regulated voltage line and powered by~~configured to receive power via the third regulated voltage line~~when the system power is provided to the DIMM from the system board; and~~

a fourth component ~~electrically connected to the fourth regulated voltage line and powered by~~configured to receive power via the fourth regulated voltage line~~when the system power is provided to the DIMM from the system board;~~

~~wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit; and~~

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

(ii) receives power ~~when the system power is provided to the DIMM from the system board, is powered:~~

- ~~only by one or more of the first, second, third, and fourth converter circuits; and~~

~~only via the one or more of the first, second, third, and fourth regulated voltage lines;~~

wherein the plurality of DDR SDRAM ~~packages~~ devices includes:

a first group of at least five DDR SDRAM ~~packages~~ devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; and

a second group of at least four DDR SDRAM ~~packages~~ devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;~~;~~and

~~when the system power is provided to the DIMM from the system board:~~

- ~~• the first and second groups of DDR SDRAM packages are~~is
~~configured to be accessible and not in self-refresh mode;~~
- ~~• the first group of DDR SDRAM packages is configured to be~~
~~enabled, in response to the first chip select signal, to receive or~~
~~output 40 1-bit wide DDR data signals in parallel via 40 data lines~~
~~of the set of data lines independently of whether the second group of~~
~~DDR SDRAM packages is enabled, in response to the second chip~~
~~select signal, to receive or output an integer number of 1-bit DDR~~
~~data signals in parallel via the integer number of data lines of the set~~
~~of data lines; and~~
- ~~• the 40 1-bit wide DDR data signals include 32 1-bit wide DDR~~
~~data signals and eight 1-bit wide DDR error-correcting code~~
~~(ECC) data signals; and~~

~~wherein the sum of the 40 data lines and the integer number of data lines~~
~~matches a total number of data conduits of the set of data conduits of the memory~~
~~socket, and the total number of data conduits is at least 72 data conduits.~~

59. (Currently Amended) The DIMM of claim 586, ~~wherein the plurality of write~~
~~operations further includes a third write operation subsequent to the first and second write~~
~~operations; and wherein, in response to the detected trigger condition, the logic element is~~
~~further configured to periodically perform every millisecond all of the first and second write~~
~~operations followed by the third write operation.~~

wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to
be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response

- to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and
- wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector socket, and the total number of data conduits is at least 72 data conduits

60. (Currently Amended) The DIMM of claim 597, wherein the logic element is ~~further~~ configured to perform ~~the a~~ third write operation ~~by subsequent to the first and second write operations by~~ writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Currently Amended) The DIMM of claim 560, wherein, ~~in response to the detected trigger condition, the voltage monitor circuit is further configured to generate and transmit a~~ trigger signal the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Currently Amended) The DIMM of claim 586, wherein, ~~in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond a plurality of read operations prior to performing the first and second write operations~~ when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Currently Amended) The DIMM of claim 5860, wherein ~~the logic element is further configured to perform a first and second read operations of the plurality of read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM packages, respectively~~ when power is provided to the DIMM from the power rails of the system board, the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Currently Amended) The DIMM of claim 563, wherein when power is provided to the DIMM from the power rails of the system board, the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages. ~~the width dimension is 133.35 millimeters.~~

65. (Currently Amended) The DIMM of claim 58 62, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element ~~the height dimension is selected from one of a plurality of profiles ranging from 18.3 millimeters to 50 millimeters.~~

66. (Currently Amended) The DIMM of claim 586, wherein each of the first, second, and third ~~voltage reducing~~ converter circuits is a buck converter.

67. (Currently Amended) The DIMM of claim 586, wherein, when ~~the system~~ power is provided to the DIMM from the power rails of ~~from~~ the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Currently Amended) The DIMM of claim 6358, wherein, when ~~the system~~ power is provided to the DIMM from the power rails of ~~from~~ the system board, an additional trigger condition occurs when the input voltage is above the first threshold and ~~the input voltage is below the second threshold;~~ and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

REMARKS

The final Office Action mailed January 24, 2025, has been carefully considered.

Reconsideration in view of the following remarks is respectfully requested.

By this Amendment, Applicant has **amended claims 2 and 28-39 and 41-68; and canceled claim 40.**

Claims 2 and 28-39 and 41-68 are now pending and subject to examination.

For convenience, a Claim Support Table is provided below as Appendix A, as a well as a clean copy of the amended claims as Appendix B.

Interview Record

Applicant gratefully acknowledges the courtesy and consideration extended to Applicant's undersigned representative during the telephone interview with Examiner Farrokh on Wednesday March 5, 2025.

During the interview, Applicant explained that support for the instant amendment can be found in the specification, and that the accompanying Support Table detailed the location in the specification for said support by mapping each claim limitation to an example location in the written description and/or drawings at which it is discussed.

The Examiner specifically inquired about the support for “conduits” (plural), despite the discussion only of “conduit” (singular) in the specification. Applicant explained that the specification (para. [00123], FIGS. 2 and 12, etc.) describes a memory slot connector on the system board which the edge connectors of the DIMM plug into. The memory slot connector is further described as having a conduit that connects to an edge connector so that power can be transmitted from the system board to the DIMM. Other edge connectors of the DIMM serve to transmit data and control signals, while others still transmit address signals. These other edge connectors are also described as plugging into the memory slot connector on the system board, and the same reasoning—that the connections are made by way of corresponding conduits in the

memory slot connector—clearly applies. Therefore while not explicitly described, the presence of conduits for the transfer of data, address and control signals in addition to the data conduit is at least implicit as would be understood by a person of ordinary skill in the art.

The Examiner also inquired about the 1-bit wide DDR data signals, and for example the receipt or output of the claimed “40 1-bit wide DDR data signals.” The Examiner’s attention is respectfully directed to for example para. [00178] discussing the portions, or slices, into which a data word may be divided. It will be appreciated that a data word is the collection of bits that may for example be written into or read from volatile or nonvolatile memory, buffer, or other components. The word may comprise 72 individual bits, and may be sliced into two portions—for example a 32-bit portion, and a 40-bit portion. Other sized slices, or portions, as well as more than two portions, are also contemplated, and the portions may be handled by the inventive DIMM in unison or separately from one another.

Applicant also directed the Examiner’s attention to the added limitation “wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM” in the independent claims, which clarifies the intended scope of the limitation “the DIMM is configured to operate using only the provided system power from the system board” from the prior claim version. Applicant explained that this limitation is supported for example in FIG. 16, which shows the power module 1100 receiving power “from system” at nodes Vdd and gnd corresponding to the system’s power rails. It was explained that this contrasts starkly with the teachings of Harris et al. (US 2006/0174140, “Harris”), whose module derives power from an external power supply, and teaches away from supplying power to the memory module from the power rails of a system board, as further explained below.

Rejection(s) Under 35 U.S.C. § 112, First Paragraph

Claims 2 and 28-68 stand rejected under 35 U.S.C. § 112, first paragraph, as failing to comply with the written description requirement for allegedly containing subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C.

112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

Applicant has canceled claim 40, and has amended claims 2, 28-39 and 41-68 to re-write or cancel some of the limitations that are the basis of the instant 35 U.S.C. § 112, first paragraph rejection. Where other such limitations remain, attention is respectfully directed the Claim Support Table below, which maps all claim limitations to their support in the specification.

It is respectfully submitted that claims 2, 28-39 and 41-68 are now in compliance with 35 U.S.C. § 112, first paragraph, and the rejection thereunder should be withdrawn.

Amendment to Claims

The claims have been amended for improved clarity in particularly pointing out and distinctly claiming certain aspects of the invention. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references.

Rejection(s) Under 35 U.S.C. § 112(b) (or Pre-AIA 35 U.S.C. § 112, Second Paragraph)

Claims 2, 44 and 58 stand rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for allegedly failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

The limitation “trigger condition occurs when the input voltage goes above both the first and second threshold voltages” has been moved from the prior version of independent claim 2 to the current version of dependent claims 53 and 64. The support for this limitation can be found for example in para. [00130], which describes *inter alia* two thresholds, stating that “a trigger condition occurs when the voltage goes below a threshold or above another threshold.” According to this statement, “a threshold” can be considered the first, lower threshold, and

“another threshold” can be considered the second, upper threshold. If the input voltages goes above the second, upper threshold, then by definition it also goes above the first lower threshold. This therefore satisfies the language of the claims that “trigger condition occurs when the input voltage goes above both the first and second threshold voltages.”

The limitation relating to the DDR SDRAM devices being “configured to be accessible and not in self-refresh mode” has been moved from the prior version of independent claim 2 to the current version of dependent claims 28, 46 and 59. Support for this limitation can be found for example in para. [00178], which states that, *inter alia*, “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or “slice”) may comprise 40 bits of the data word, and the second portion (or “slice”) may comprise the remaining 32 bits of the data word.” (Emphasis added) Thus in this configuration the 72-bit data word comprises an accessible 40-bit slice and an accessible 32-bit slice. Para. [00178] then states that “the process of ‘slicing’ the volatile memory during a backup includes [1] bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.” In 1 above, the 32-bit slice becomes accessible and not in self-refresh; and in 2 above, the 40-bit slice becomes accessible and not in self-refresh mode. Both of these states satisfy the claim language of the DDR SDRAM devices being “configured to be accessible and not in self-refresh mode.”

It is respectfully submitted that the claims are in compliance with 35 U.S.C. § 112(b) (or pre-AIA 35 U.S.C. § 112, second paragraph), and the rejection thereunder should be withdrawn.

Harris Teaches Away from the Claimed Invention

In the Response to Remarks, the Office Action continues to cite Harris (U.S. Publication No. 2006/0174140) as relevant to the claimed invention. Harris, however, teaches away from the claimed invention. In particular, independent claims 2, 44 and 58, as amended, each recite “wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM.” (Emphasis added)

Contrary to the above claimed feature, Harris teaches “[t]o cope with power requirements, industry standard memory modules, e.g., Dual In-line Memory Modules (DIMMS) populated with dynamic random access memory (DRAM) devices, are provided with **power supply rails** (on a relatively large number of pins) that are **powered from system board or main board voltage sources**, and are specific to the memory technology. As the performance of the DRAM technology goes up, and timing margins shrink, **it is becoming increasingly more difficult for the system board sources to provide tightly regulated power for the DRAM cores** as well as input/output (I/O) interface buffers. Furthermore, each generation of DIMM/DRAM technology requires a different power supply which keeps getting lower (e.g., 3.3V, 2.5V, 1.8V, 1.5V and beyond), **thereby making it difficult to mix memory technologies on a system board**, or provide upgrades to next generation DRAM technology in a cost-effective manner.” (Harris at para. [0002], emphasis added.) Thus, Harris explicitly teaches away from using the power rails of the system board of the host computer system to supply power to the DIMM.

That the voltage powering the memory module of Harris does not come from the power rails and conduits of the memory slot connector of the system board is further evident from the term “external” used to describe it. This term is introduced in Harris’ Summary section (para. [0003]) as a counterpoint to the description of the conventional configuration in which DIMMs “provided with power supply rails . . . that are powered from system board or main board voltage sources, and are specific to the memory technology.” (Para. [0002]) This is problematic, Harris explains, because the specificity to the memory technology means DIMMs are customized to particular DRAMs due to the DRAMs’ specific power requirements, that impose specific conduit/memory slot connector constraints, and therefore, when improved DRAM technology emerges, DIMMs have to be completely redesigned rather than just adapted to the new improvements. (*Id.*) Harris’ external voltage source allows such adaptation because the conduit/memory slot connector power configuration constraints are removed when power is provided externally instead. The memory slot connector becomes independent of the power requirements of the DRAMs, which can be satisfied by provision of an external voltage, according to Harris.

This voltage independence of the memory slot connector is further explained in para. [0012], which states that “A voltage-independent FBD design incorporating the embodiment of FIG. 1A is capable of replacing these power supply interface pins with as few as six +12V pins (*from an external voltage source*).” (Emphasis added.) In addition, a further advantage of the voltage independence of the memory slot connector is described in para. [0014], which states that since voltage is provided externally, a variety of sources can be used, including “any combination of known or heretofore unknown voltage Supplies, either regulated or unregulated, and even including variable voltages.” In addition, Harris states at para. [0012] that due to this independence, “the overall power supply interface to the memory module 100A is improved to better support large memory capacity scaling, hot-pluggability, DRAM technology independence (e.g., DDR2, DDR3, DDR4, and other DRAM types), high DRAM device count (i.e., amenable to multi-ranking), as well as provide improved system-level cost sharing. Additionally, memory performance is also improved because of tighter voltage regulation.”

All these explanations inescapably point to the conclusion that the gist of Harris’ invention is to use an external power source that does not come from the power rails and conduits of the memory slot connector of the system board of a computer system, which directly contravenes the arrangement of “a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM” set forth in all the independent claims. Harris thus teaches away from the claimed invention, and cannot be relied upon as a basis for rejecting any of the pending claims. It is improper to combine references where the references teach away from their combination. *In re Grasselli*, 713 F.2d 731, 743 (Fed. Cir. 1983); see also MPEP 2145.X.D.2.

Canceled Claims

Claim 40 has been canceled without prejudice or disclaimer of the subject matter contained therein.

Conclusion

Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

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APPENDIX A
Claim Support Table

Claim Element	Limitation	Support
2.1	A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the system board and the DIMM are transferred via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figs. 2 and 12: FIG. 12 Based on the above, a POSITA would clearly understand that the memory socket of a computer system includes pins soldered onto the system board that provide <i>electrical conduits</i> for system power, data, and address and control signals from the system board to be electrically connected to corresponding edge connections of the edge connector 1022 when it is inserted into the memory socket. Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown).
2.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of

slot connector of the system board, the edge connections including: a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits.

certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector.

When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB.

Fig. 2 which shows the DIMM in the socket on the motherboard:

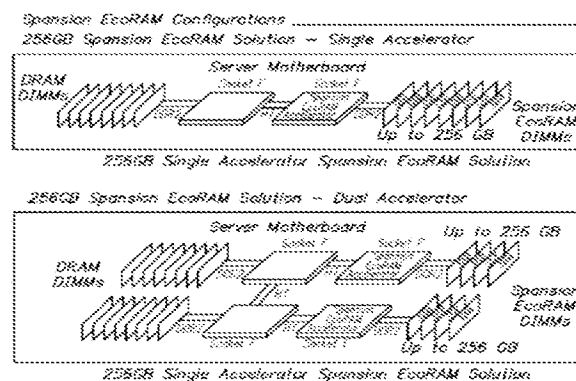
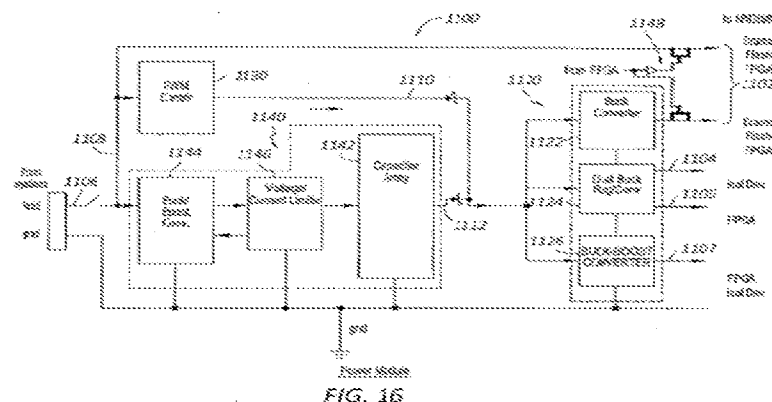


FIG. 2
(PRIOR ART)

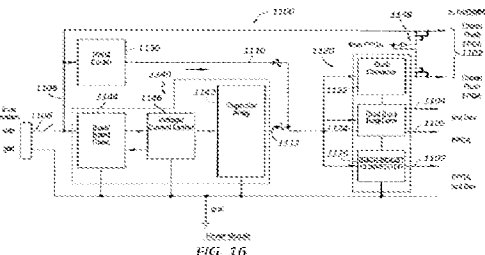
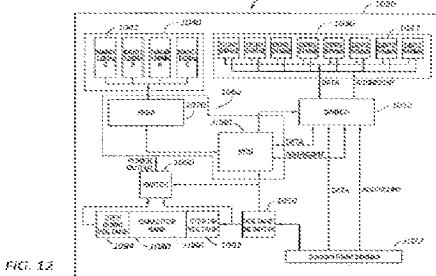
Note the power rails (Vdd and Gnd) of the system board coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.



		The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector.
2.3	a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM, a set of data lines electrically coupled to the second set of edge connections, and a set of address and control lines electrically connected to the third set of edge connections,	Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16. For example, in Fig. 12 the input voltage supply lines are shown electrically connected to the Voltage Monitor 1050. FIG. 12 is a block diagram of a memory system 1010. It shows a Standard DIMM Interface 1022 connected to a first memory bank 1030. The interface 1022 has multiple pins for data, address, and control signals. The memory bank 1030 contains eight 64Mx8 DDR2 SDRAM elements 1032. A Voltage Monitor 1050 is connected to the interface 1022 and a PWM Controller 1130. The PWM Controller 1130 is connected to the interface 1022 and a set of voltage supply lines 1130. The ground supply line (gnd) is also connected to the interface 1022. FIG. 16 is a detailed circuit diagram of the interface 1022. It shows the connection of the interface 1022 to the voltage supply lines 1130 and the ground supply line (gnd). The interface 1022 is connected to the voltage supply lines 1130 via a set of voltage supply lines 1130. The ground supply line (gnd) is connected to the interface 1022 via a set of ground supply lines 1130. The interface 1022 is also connected to the PWM Controller 1130 via a set of control lines 1130. The interface 1022 is connected to the first memory bank 1030 via a set of data lines 1130. The interface 1022 is connected to the Voltage Monitor 1050 via a set of control lines 1130. For example, the respective power rails (Vdd and gnd) are labeled “from system”, on left of Fig. 16, are electrically connected via a set of voltage supply lines to the PWM Controller 1130, and the ground supply line (gnd). Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” These are coupled to the data lines (by way of edge connections) of the Standard DIMM Interface 1022 by way of the switch 1052. Note also the address and control lines from edge connections of the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.
2.4	a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit	The specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example,

	configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;	support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. Para. [00133] “The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.” Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070 Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
2.5	a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the	FIG. 12

	set of address and control lines; location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” And Para. [00122] “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.” And Para. [00124] “In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein.”	
2.6	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines	FIG. 16

		converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
2.7	wherein the set of components includes: - a first component configured to receive power via at least the first regulated voltage line; - a second component configured to receive power via at least the second regulated voltage line; - a third component configured to receive power via at least the third regulated voltage line; and - a fourth component electrically configured to receive power via at	FIG. 16  FIG. 12  On the left of FIG. 16, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck convertor 1122, dual buck converter1124, and

	least the fourth regulated voltage line	buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. On the right, each of the DRAMs is shown as being connected to the first regulated voltage, for example. On the right, the isolation device is shown as being connected to the second regulated voltage, for example. On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example. On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
2.8	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines	FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the system power is received, couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010

		from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
28.1	wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and	FIG. 21

		embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash)." This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
28.2	when power from the power rails of the system board is provided to the DIMM: • each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second	Para. [00173] "In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode." Para. [00178] "In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into

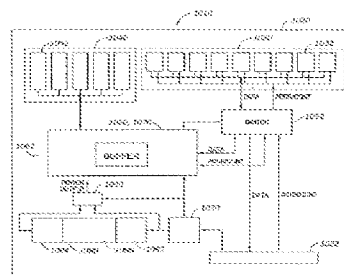
	group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; •the at least 40 1-bit wide DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals	self-refresh in the example embodiment.” Thus according to [00178], 72-bit data word comprises an accessible 40-bit slice and an accessible 32-bit slice. Para. [00178] also states that “the process of ‘slicing’ the volatile memory during a backup includes [1] bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.” In 1 above, the 32-bit slice is accessible and not in self-refresh; and in 2 above, the 40-bit slice is accessible and not in self-refresh mode. Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
28.3	the sum of the 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
29.1	a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and	The current application specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete

configured to electrically communicate with the nonvolatile memory and the data storage element.

electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. Para. [00127]

“FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.”

FIG. 14



A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178.

Para. [00172]

“In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).”

Para. [00178]

“For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA.”

For example, FIG. 13 shows controller 1062 comprising a buffer.

Para. [00127]

“Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the **logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash.** In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. **The internal flash can improve the speed of the memory system 1010 and save physical space.** Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, **the logic element 1070 is a custom device.** In certain embodiments, **the logic element 1070 comprises various discrete electrical elements,** while in certain other embodiments, **the logic element 1070 comprises one or more integrated circuits.** FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into

		a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.” FIG. 12 FIG. 12
29.2	a first write operation to write a first set of data bits into a first portion of the data storage element; a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called “slicing” the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above. Para. [00130]

		“In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The first threshold and second threshold are the same as above, supported by “a threshold” and “another threshold” in the last line. The “10 percent” in the “the threshold voltage is 10 percent below a specified operating voltage” and the “10 percent” in the “above a specified operating voltage” support the lower operating voltage value, and the upper operating voltage value.
30	when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
31	wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM	FIG. 16

		memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052. ”
32	wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
33	wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations	Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer.”
34	wherein, when power from the power rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102. ”
35	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash). ” Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash). ”

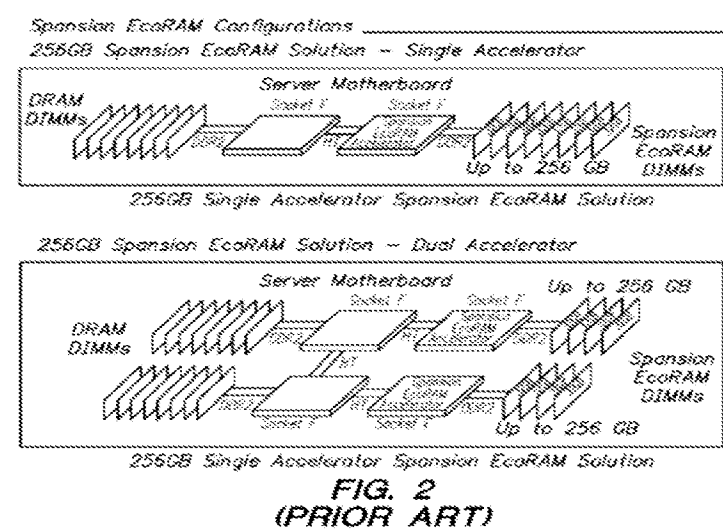
	data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.”
35	wherein, prior to performing the third write operation, the logic element is further configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” Difference from Claim 35 is the read from the buffer is a plurality of steps in Claim 35, and specifically two steps here. Same sentence in Para. [00178] supports both scenarios. Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040.” Reading a data word or a block of words to be written into the NVM.
37	wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.	Para. [00148] “The conversion element 1120 may comprise a plurality of sub blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system.”
38	wherein, prior to performing the first and second write operations, the logic element is further configured to	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”

	perform, in response to the detected trigger signal, one or more read operations.	Para. [00179] “In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.”
39	wherein the logic element is further configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called “slicing” the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
41	wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.	FIG. 21

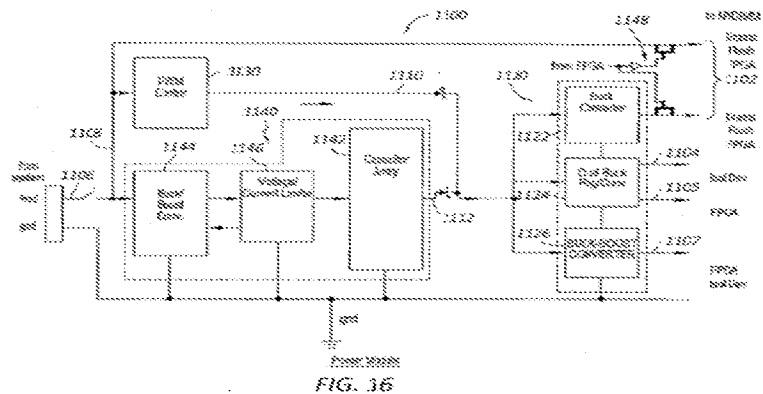
		(also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” FIG. 13 As can be seen in FIG. 13, the transfer from the DRAM occurs on a connection on the right side of the buffer, and the transfer to NVM will occur on a connection on the left side of the buffer.
42	wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.	A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178. Para. [00172] “In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).” Para. [00178] “For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA.” For example, FIG. 13 shows controller 1062 comprising a buffer:
43.1	wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In

		“The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”
44.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including: a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits,	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector. When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB.

Fig. 2 which shows the DIMM in the socket on the motherboard:



Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.



The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector.

44.3 a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge

Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 16. For example, in Fig. 12 the input voltage supply lines is shown electrically connected to the Voltage Monitor 1050.
FIG. 12

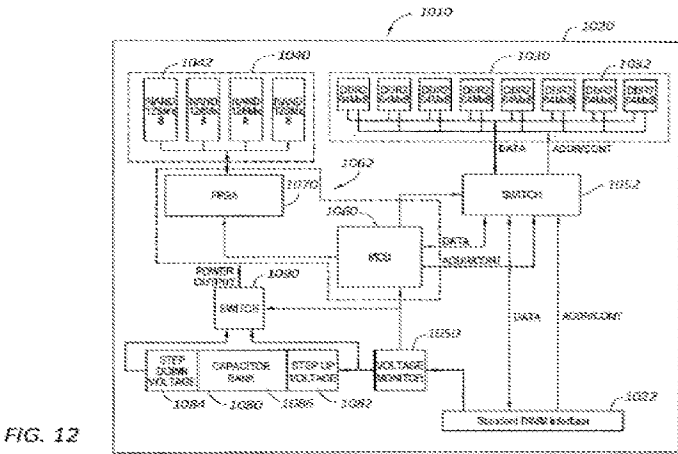
	connections of the DIMM a set of data lines electrically coupled to the second set of edge connections, and a set of address and control lines electrically coupled to the third set of edge connections ,	 FIG. 12
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FIG. 16

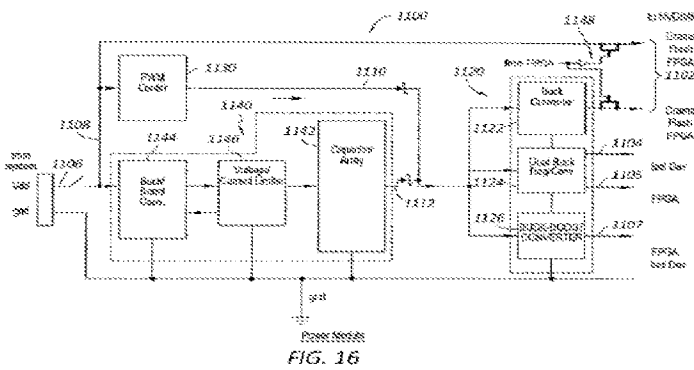


FIG. 16

For example, respective power conduits (Vdd and gnd) of the set of power conduits of the memory socket are labeled “from system”, on left of Fig. 16, are electrically connected via a set of voltage supply lines to the PWM Controller 1130, and the ground supply line (gnd).

Para. [00124]

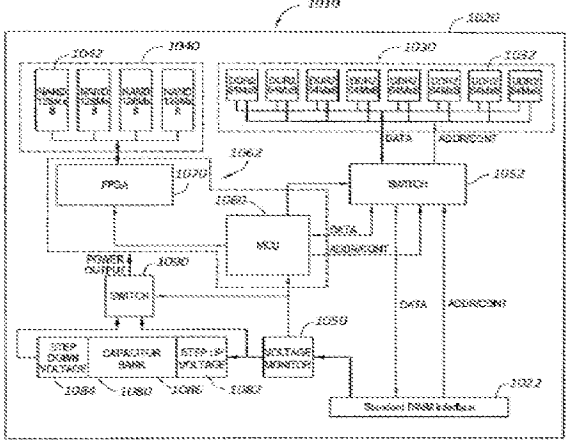
“For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises **eight 64Mx8 DDR2 SDRAM** elements 1032.”

Note also the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.

	supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;	A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178. Para. [00172] “In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).” Para. [00133] “The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.” “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060)” Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070 Para. [00133] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
44.5	a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM	FIG. 12

devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines;

FIG. 12



Para. [00124]
“Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).”

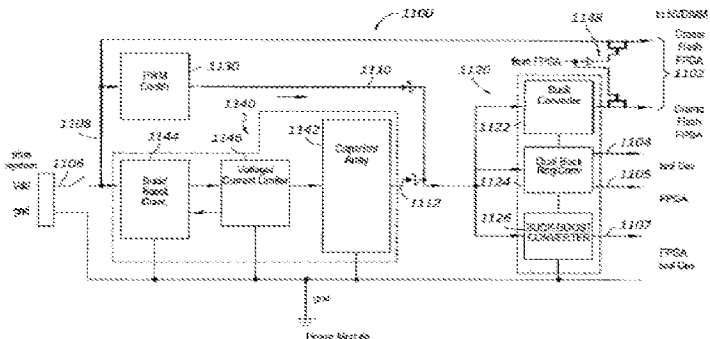
Para. [00172]
“In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

And

Para. [00122]
“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

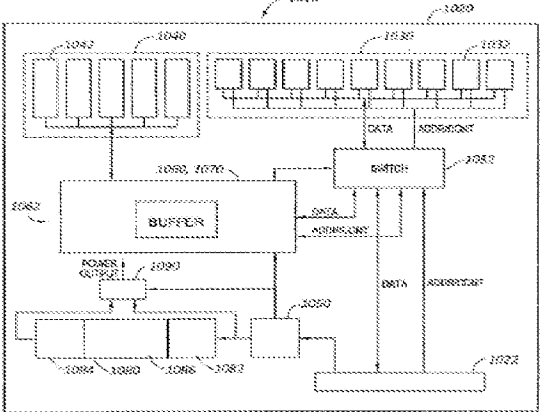
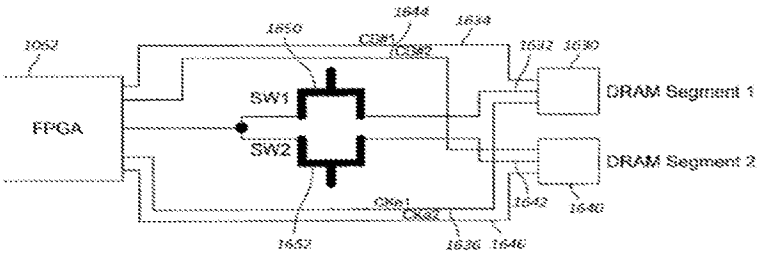
And

Para. [00124]
“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32,

		as well as other bit widths, are compatible with certain embodiments described herein.”
44.6	first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines	FIG. 16  FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. Para. [00148] “The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage

		1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”
44.7	wherein the set of components includes: - a first component configured to receive power via at least the first regulated voltage line; - a second component configured to receive power via at least the second regulated voltage line; - a third component configured to receive power via at least the third regulated voltage line; and - a fourth component electrically configured to receive power via at least the fourth regulated voltage line	FIG. 16

		As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060.
44.8	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.	FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the system power is received, couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
45	wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically	Para. [00127] “FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein.” FIG. 14

	communicate with the nonvolatile memory and the data storage element.	 FIG. 14
46.1	wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;	FIG. 21  FIG. 21 Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above. Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from

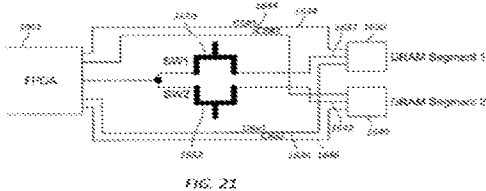
		the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more
46.2	wherein when power from the power rails of the system board is provided to the DIMM: • each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM packages is configured to be	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during

	enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Thus according to [00178], 72 bit data word comprises an accessible 40-bit slice and an accessible 32-bit slice. Para. [00178] also states that “the process of ‘slicing’ the volatile memory during a backup includes [1] bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.” In 1 above, the 32-bit slice is accessible and not in self-refresh; and in 2 above, the 40-bit slice is accessible and not in self-refresh mode. Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”
46.3	wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32

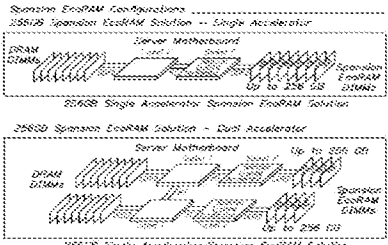
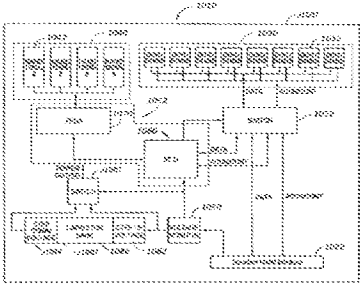
		bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
47	wherein the logic element is further configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.	Para. [00174] “In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” For the chain of system power electrical connection to the controller, see 2.1, 2.4, 2.5, and 2.6 above.
48	wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”
49	wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a

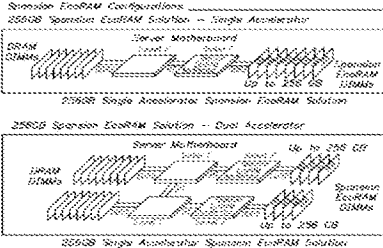
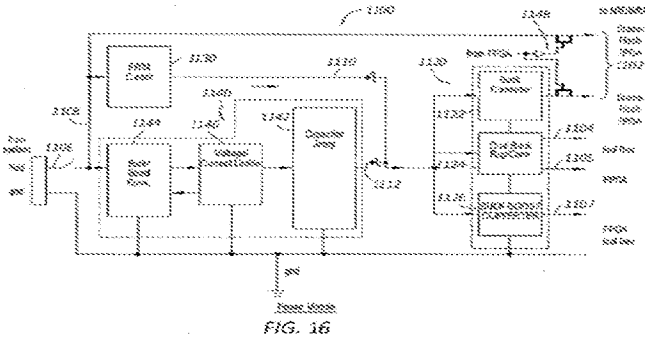
	the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is further configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.	backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.”
50	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits, respectively.	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00178] “The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.”
51	wherein, prior to performing the first and second write operations, the logic element is further configured to perform, in response to the detected trigger condition, periodically at regular intervals of time, one or more read operations to read the	Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion

	first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	(also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
52	wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
53	wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. Para. [00130] thus describes <i>inter alia</i> two thresholds, stating that “a trigger condition occurs when the voltage goes below a threshold or above another threshold.” From this statement, “a threshold” can be considered the first, lower threshold, and “another threshold” can be considered the second, upper threshold. If the input voltages goes above the second, upper threshold, then by definition it also goes above the first lower threshold.
54	wherein the logic element is further configured to perform the third write operation by writing the third set of data bits from the data	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data

	storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations	word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash). ”
55	wherein, prior to performing the third write operation, the logic element is further configured to generate the third set of data bits based on the groups of data bits stored in the data storage element	Para. [00178] “The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).”
56	wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.	FIG. 21  FIG. 21 Para. [00172] “FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.” Para. [00177] “The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” FIG. 13

		As can be seen in FIG. 13, the transfer from the DRAM occurs on a connection on the right side of the buffer, and the transfer to NVM will occur on a connection on the left side of the buffer.
57	wherein when power is provided to the DIMM from the power rails of the sys-tem board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.”
58.1	A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals between the system board and the	Para. [00123] “For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

	DIMM are transferred via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits	Dual in-line memory module (DIMM) has an edge connector 1022 that includes edge connections for power, data, and address and control signals designed to fit into a memory socket of a computer system. See Figure 2 and  Fig. 12. The memory socket of a computer system includes pins soldered onto the system board that provide electrical conduits for system power, data, and address and control signals from the system board to be electrically connected to corresponding edge connections of the edge connector 1022 when it is inserted into the memory socket. Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in  electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system . . . For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”
58.2	the DIMM comprising: a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including: a first set of edge connections configured	Para. [00123] “The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.”

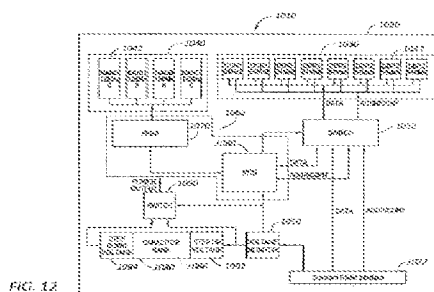
	to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM, a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits,	This paragraph describes the contact of the edge connections and the slot connector, as well as a JEDEC standard 240 pin DDR2 edge connector. The JEDEC standard specifies what voltages the system voltage regulator is to provide to which pins in the slot, which are then passed on to the edge connector. When the interface 1022 edge connector is inserted into the memory socket of the computer system, edge connections will mate with corresponding conduits and an electrical connection is established for system power, data, and address and control signals to be electrically conducted to the edge connections on the PCB, and PCB traces also called lines allow signals and power/voltage to be electrically conducted on the PCB. Fig. 2 which shows the DIMM in the socket on the motherboard:  FIG. 2 (PRIOR ART) Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 16.  FIG. 16 The DIMM operates using only the system power provided from the system board, for example as specified and is well known in the JEDEC standard 240-pin DDR2 edge connector.
58.3	a set of voltage supply lines electrically connected to the first set of edge connections and	Note the power rails (Vdd and Gnd) of the host system coupled to the Standard DIMM Interface 1022 and electrically connected to the input voltage and ground supply lines on the PCB, as shown in Fig. 12, Fig. 13, Fig. 14 and

	configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM a set of data lines electrically coupled to the second set of edge connections, and a set of address and control lines electrically to the third set of edge connections, FIG. 16. For example, in Fig. 12 the input voltage supply lines is shown electrically connected to the Voltage Monitor 1050. FIG. 12 FIG. 16 For example, respective power conduits (Vdd and gnd) of the set of power conduits of the memory socket are labeled “from system”, on left of Fig. 16, are electrically connected via a set of voltage supply lines to the PWM Controller 1130, and the ground supply line (gnd). Para. [00124] “For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.” Note also the address and control lines from the Standard DIMM Interface 1022 coupled to the DRAM 1030 through the Switch 1052 in FIG. 12.	
58.4	wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines; Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions	

		of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.” And Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” The embodiment of FIG. 13 shows nine x8 DRAM devices for a data bus bit width of 72: And Para. [00122] “In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”
58.5	a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i)	1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. A buffer (data storage element) is shown in Fig. 13 and Fig. 14, and disclosed to store data in at least 172, and 178. Para. [00172]

	monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the data storage element;	“In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO).” Para. [00178] “For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA.” For example, FIG. 13 shows controller 1062 comprising a buffer:
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including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines;



Para. [00124]

“Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).”

Para. [00172]

“In one embodiment, for example, the volatile memory subsystem 1030 comprises a **72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC)**, the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.”

And

Para. [00122]

“In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein.”

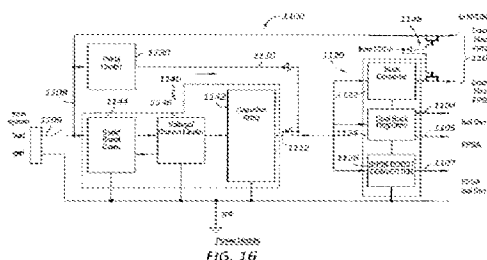
And

Para. [00124]

“In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. **Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM).** For example, in the block diagram of FIG. 12, **the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032.** The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, **volatile memory elements 1032 having bit widths of 4, 8, 16, 32,** as well as other bit widths, are compatible with certain embodiments described herein.”

58.7	first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines
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FIG. 16



On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck convertor 1122, dual buck converter 1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively.

Para. [00148]

“The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.”

58.8	wherein the set of components includes: a first component configured to receive power via the first regulated voltage line; a second component configured to receive power via the second regulated voltage line; a third component configured to receive power via the third regulated voltage line; and a fourth component electrically configured to receive power via the fourth regulated voltage line	FIG. 16 FIG. 12 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 which provides power to the four converter circuits. Buck convertor 1122, dual buck converter1124, and buck-boost converter 1126 provide regulated voltages on the regulated voltage lines 1102, 1104, 1105, and 1107, respectively. On the right, each of the DRAM are shown as being connected to the first regulated voltage, for example. On the right, the isolation device is shown as being connected to the second regulated voltage, for example. On the right, the FPGA 1070 is shown as being connected to the third regulated voltage, for example. On the right, the MCU 1060 is shown as also being connected to the fourth regulated voltage, as is shown in the embodiment of FIG. 12, the MCU 1060 is part of the controller 1062. This is further specified in the text of the disclosure: Para. [00148] “the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.” As shown in FIG. 12, above, the controller 1062 includes the components FPGA 1070, and the MCU 1060, and thus needs two different voltages for each of the components 1070, 1060
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58.9	wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines	FIG. 16 On the left of the figure, system power (Vdd) is provided on the input voltage supply line 1106 to the PWM controller 1130 that, when the system power is received, couples power to each of the four converter circuits via voltage 1110. Para. [00145] “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” Para. [00151] “The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.” The system power is provided to the PWM controller 1130 and from there to the four converter circuits (conversion element 1120) is the only power to the module. In the second state all components are powered from the power supplied from one or more of the four converter circuits via one or more of the four regulated voltages. There is no alternative power source claimed.
58.10	wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;	FIG. 21 Chip Select 1 and Chip Select 2 going to each group is shown in FIG. 21 above. Para. [00172]

	a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal	“FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062.” Para. [00177] “In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).” This claim is limited to x4 and x8 DRAM. When using x8, it takes five DRAM to receive or output 40 bits. The remaining bits in the word are 32, which means it would take four x8 DRAM to complete receiving or outputting
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		the word. Thus, you have a module with a total of nine DRAM (5+4) as shown in FIG. 13 and discussed in Para. [00126]. If x4 are used, the number of DRAM are correspondingly more.
59.1	wherein when power from the power rails of the system board is provided to the DIMM: • each group of the first and second groups of DDR SDRAM is configured to be accessible and not in self-refresh mode; • the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and • the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and	Para. [00173] “In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment.” Thus according to [00178], 72-bit data word comprises an accessible 40-bit slice and an accessible 32-bit slice. Para. [00178] also states that “the process of ‘slicing’ the volatile memory during a backup includes [1] bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer.” In 1 above, the 32-bit slice is accessible and not in self-refresh; and in 2 above, the 40-bit slice is accessible and not in self-refresh mode. Para. [00182] “In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.”

59.2	wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.	Para. [00123] “The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.” Para. [00172] “In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.” Para. [00178] “In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word.”
60	wherein the logic element is further configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.	Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
61	wherein the logic element is further configured to perform, prior to the first write operation and the second write operation, first and second read operations by reading the first and	<u>[00172]</u> <u>FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010.</u> <u>The controller read first and second data slices (sets of data bits) from first and second groups of DDR SDRAM packages, and then write the first and second sets of data bits into the controller 1062 (data storage element).</u>

	second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.	
62	wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.”
63	when the power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.	Para. [00133] “In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.” Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” The system voltage provides support for the input voltage. The “a threshold” in the last sentence provides support for “a first threshold” and the “another threshold,” also in the last sentence provides support for the “a second threshold.”
64	wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold.” Para. [00130] thus describes <i>inter alia</i> two thresholds, stating that “a trigger condition occurs when the voltage goes below a threshold or above another threshold.” From this statement, “a threshold” can be considered the first, lower threshold, and “another threshold” can be considered the second, upper threshold. If the input voltages goes above the second, upper threshold, then by definition it also goes above the first lower threshold.

65	wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to write a second set of data bits into a second portion of the data storage element	The specification discloses a controller 1062 designed to seamlessly integrate with various other components described within the memory system 1010 that enhances performance (speed), efficiency and reduces cost. The controller 1062 comprises a microcontroller unit (MCU) 1060, logic element 1070, buffer, voltage monitor circuit 1050, and one or more additional components. Logic element 1070 comprises internal nonvolatile memory, one or more integrated circuits and various discrete electrical elements. The controller 1062 receives system power from the standard DIMM interface 1022. For example, support is found in at least [00127], [00133], and [00172]; and Fig. 12, Fig. 13, Fig. 14, and Fig. 16. Para. [00133] “The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022.” Fig. 14 depicts the voltage monitoring circuit 1050 receiving the input voltage via interface 1022 and coupled to the logic element 1070. Para. [00130] “In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition.” Para. [00177] “In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.”
66	wherein each of the first, second, and third converter circuits is a buck converter.	Para. [00148] “The conversion element 1120 may comprise a plurality of sub blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010.” And Para. [00148] “In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the

		fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate amount for powering various components of the memory system. ”
67	wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.	Para. [00149] “In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102. ”
68	wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and the input voltage is below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.	Para. [00130] “In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage . For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage . In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value , such as, for example, 10 percent above a specified operating voltage . In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. ”

APPENDIX B

Clean Copy of Amended Claims

1. (Canceled)

2. (Currently amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM are transferred via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the

controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component electrically configured to receive power via at least the fourth regulated voltage line; and

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.

3– 27. (Canceled)

28. (Currently Amended) The DIMM of claim 2, wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;
- the at least 40 1-bit wide DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit wide DDR error-correcting code (ECC) data signals; and

the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

29. (Currently Amended) The DIMM of claim 2, wherein the controller further

includes:

a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Currently Amended) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Currently Amended) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

32. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based

on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Currently Amended) The DIMM of claim 2, wherein, when power from the power rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Currently Amended) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Currently Amended) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Currently amended) The DIMM of claim 38, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices,

respectively.

40. (Canceled)

41. (Currently Amended) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Currently Amended) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Currently Amended) The DIMM of claim 31, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the

system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller,

wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component electrically configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines.

45. (Currently Amended) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Currently Amended) The DIMM of claim 44, wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals and eight 1-bit wide DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

47. (Currently Amended) The DIMM of claim 45, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Currently Amended) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Currently Amended) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Currently Amended) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Currently Amended) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Currently Amended) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Currently Amended) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

54. (Currently Amended) The DIMM of claim 49, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

55. (Currently Amended) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on

the groups of data bits stored in the data storage element.

56. (Currently Amended) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Currently Amended) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit wide DDR data signal, and wherein the set of data lines includes at least 72 data lines;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller,

wherein the controller is configured to perform, in response to the trigger signal, one or more write operations to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component electrically configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; and

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal.

59. (Currently Amended) The DIMM of claim 58, wherein when power from the power rails of the system board is provided to the DIMM:

- each group of the first and second groups of DDR SDRAM devices is configured to be accessible and not in self-refresh mode;
- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit wide DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit wide DDR data signals include 32 1-bit wide DDR data signals

and eight 1-bit wide DDR error-correcting code (ECC) data signals; and
wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

60. (Currently Amended) The DIMM of claim 65, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Currently Amended) The DIMM of claim 65, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Currently Amended) The DIMM of claim 58, wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Currently Amended) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Currently Amended) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above both the first and the second threshold voltages.

65. (Currently Amended) The DIMM of claim 58, wherein the controller further

includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

66. (Currently Amended) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Currently Amended) The DIMM of claim 58, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Currently Amended) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.



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PETITION TO MAKE SPECIAL BASED ON AGE FOR ADVANCEMENT OF EXAMINATION UNDER 37 CFR 1.102(c)(1)

APPLICATION #	ATTORNEY DOCKET #	FILED ON
17582797	022955-000090	01/24/2022

In re Application of Chi-She Chen

This is a decision on the electronic petition under 37 CFR 1.102(c)(1), filed April 07, 2025 to make above-identified application special based on applicant's age as set forth in MPEP § 708.02, Section II.

The petition is **GRANTED**.

A grantable petition to make an application special under 37 CFR 1.102(c)(1), MPEP § 708.02, Section II:Applicant's

Age must include a statement by applicant or a registered practitioner having evidence that applicant is at least 65 years of age. No fee is required.

Accordingly, the above-identified application has been accorded "special" status and will be taken up for action by the examiner upon the completion of all pre-examination processing.

Telephone inquiries concerning this electronic decision should be directed to the Office of Petitions at 571-272-3282.

All other inquiries concerning either the examination or status of the application should be directed to the Technology Center.



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
04/07/2025 04:33:33 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Khaled Shami

PATENT CENTER # 69900618

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
petition-request.pdf	1	ePetition Request Form	29 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
petition-request.pdf	9B08C3FE87991DEF0EF284A91CDE4F7DE9663696B6FDA3D5 BA8CA884B367BB033E71A9E69F938200F112C0E1EFE3FD492 54251729346D2FE64FD17645BC57145

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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**PETITION TO MAKE SPECIAL BASED ON AGE FOR ADVANCEMENT OF EXAMINATION
UNDER 37 CFR 1.102(c)(1)**

APPLICATION #
17582797

ATTORNEY DOCKET #
022955-000090

FILING DATE
01/24/2022

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

Confirmation
Number **3546**

Filing Date **01/24/2022**

Art Unit **2138**

First Named
Inventor **Chi-She Chen**

Examiner **HASHEM FARROKH**

Attention: Office of Petitions

An application may be made special for advancement of examination upon filing of a petition showing that the applicant is 65 years of age, or more. No fee is required with such a petition. See 37 CFR 1.102(c)(1) and MPEP 708.02 (II).

APPLICANT HEREBY PETITIONS TO MAKE SPECIAL FOR ADVANCEMENT OF EXAMINATION IN THIS APPLICATION UNDER 37 CFR 1.102(c)(1) and MPEP 708.02 (II) ON THE BASIS OF THE APPLICANT'S AGE.

A grantable petition requires one of the following items:

- (1) Statement by one named inventor in the application that he/she is 65 years of age, or more; or
- (2) Certification by a registered attorney/agent having evidence such as a birth certificate, passport, driver's license, etc. showing one named inventor in the application is 65 years of age, or more.

Name of Inventor who is 65 years of age, or older

Given Name
Chi-She

Middle Name
—

Family Name
Chen

Suffix
—

Signature

I certify, in accordance with 37 CFR 1.4(d)(4) that I am an attorney or agent registered to practice before the Patent and Trademark Office, and I certify that I am in possession of evidence, and will retain such in the application file record, showing that the inventor listed above is 65 years of age, or more.

Signature
/Khaled Shami/

Name
Khaled Shami

Registration #
38745



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
04/17/2025 05:30:05 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 70056716

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Supplemental Amendment.pdf	18	Supplemental Response or Supplemental Amendment	166 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
Supplemental Amendment.pdf	6E7E89AE824138B12BC9995CACC9EC517E5FC3DDB74ED010 D618177C95410DB41D1468C7CF32C4DA231D8BEF2D1F0844 319154BEE2B9EE4D8BD2B43781C447B5

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

SUPPLEMENTAL AMENDMENT

Commissioner:

In further response to the final Office Action mailed January 24, 2025, Applicant submits the following Supplemental Amendment.

Listing of Claims begins on page **2**.

Remarks begin on page **29**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)
2. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM ~~are transferred~~ via the set of data conduits, and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component ~~electrically~~ configured to receive power via at least the fourth regulated voltage line; ~~and~~

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth

regulated voltage lines; and

- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines;
- the at least 40 1-bit DDR data signals include at least 32 1-bit ~~wide~~ DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and

the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

3– 28. (Canceled)

29. (Previously Presented) The DIMM of claim 2, wherein the controller further includes:

a data storage element configured to store a plurality of data bits and a logic element

coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Previously Presented) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Previously Presented) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

32. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Previously Presented) The DIMM of claim 2, wherein, when power from the power

rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Previously Presented) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Previously Presented) The DIMM of claim 38, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

40. (Canceled)

41. (Previously Presented) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a

transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Previously Presented) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Previously Presented) The DIMM of claim 31, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the

power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,
and

a set of address and control lines electrically~~electrically~~ connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
 - (ii) generate a trigger signal upon detecting a trigger condition; and
 - (iii) transmit the trigger signal to at least one other portion of the controller,
- wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component ~~electrically~~ configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; ~~and~~

(ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit DDR data signals include 32 1-bit ~~wide~~ DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

45. (Previously Presented) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Canceled)

47. (Previously Presented) The DIMM of claim 45, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Previously Presented) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Previously Presented) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Previously Presented) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Previously Presented) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Currently Amended) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above ~~both the first and the second threshold voltages~~ a specified operating voltage.

54. (Previously Presented) The DIMM of claim 49, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

55. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element.

56. (Previously Presented) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when

the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit ~~wide~~-DDR data signal, and

wherein the set of data lines includes at least 72 data lines;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
 - (ii) generate a trigger signal upon detecting a trigger condition; and
 - (iii) transmit the trigger signal to at least one other portion of the controller,
- wherein the controller is configured to perform, in response to the trigger signal, one or more write operations to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component ~~electrically~~ configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth

- regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; and

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal, and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and
- the at least 40 1-bit ~~wide~~ DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

59. (Canceled)

60. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein

the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Previously Presented) The DIMM of claim 58, wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Previously Presented) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Currently Amended) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above ~~both the first and the second threshold voltages~~ a specified operating voltage.

65. (Previously Presented) The DIMM of claim 58, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

66. (Previously Presented) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Previously Presented) The DIMM of claim 586, wherein, when power is provided to

the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Previously Presented) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

REMARKS

The final Office Action mailed January 24, 2025, has been carefully considered.

Reconsideration in view of the following remarks is respectfully requested.

By this Amendment, Applicant has **amended claims 2, 44, 53, 58, and 64; and canceled claims 28, 46, and 59.**

Claims 2, 29-39 and 41-45, 47-58, and 60-68 are now pending and subject to examination.

Interview Record

Applicant gratefully acknowledges the courtesy and consideration extended to Applicant's undersigned representative and Mr. Richard Kim of Netlist, Inc. during the telephone interviews with Examiner Farrokh on April 10, 16 and 17, 2025.

During the interviews, support for the claim limitations, as well as prior art reference Harris (U.S. Publication No. 2006/0174140), were discussed. The instant Supplemental Amendment, which includes adding the limitations from dependent claims 28, 46, and 59 to the respective independent claims was proposed. No agreement was reached.

Canceled Claims

Claims 28, 46, and 59 have been canceled without prejudice or disclaimer of the subject matter contained therein.

Conclusion

Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: April 17, 2025
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745



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NOTICE OF ALLOWANCE AND FEE(S) DUE

151145 7590 04/28/2025
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

EXAMINER	
FARROKH, HASHEM	
ART UNIT	PAPER NUMBER
2138	

DATE MAILED: 04/28/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	07/28/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

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151145 7590 04/28/2025
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(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	07/28/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2138	711-110000

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☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

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2 _____

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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

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Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

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☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	04/28/2025	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW				
9th Floor				
Washington, DC 20006-1152			ART UNIT	PAPER NUMBER
			2138	
DATE MAILED: 04/28/2025				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

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- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 17/582,797	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (FITF) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 04/17/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2,29-39,41-45,47-58 and 60-68 . As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____ .

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____ .

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____ .

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☒ Interview Summary (PTO-413), Paper No./Mail Date. <u>04/18/2025</u> .	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

DETAIL ACTION

Notice of Pre-AIA or AIA Status

1. *The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.*

Continued Examination Under 37 CFR 1.114

2. *A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 03/24/2025 has been entered.*

Response to Remarks

3. *In the previous office action mailed 01/24/2025, the claims were rejected under the 35 USC 112, first paragraphs as not complying with the written description requirement. The Applicant has overcome the 112 rejection.*

Examiner's Amendment

4. *An Examiner initiated interview with Applicant, Mr. Khaled Shami (Reg. # 38,745), was conducted on 04/18/2025. The Examiner informed the Applicant*

that claims in Application 17/582,797 are in condition for allowance except for a minor claim dependency error. Examiner inquired whether the applicant is willing to amend (e.g., change the claim dependency) the dependent claim 67 depending from claim 586 to depend from claim 58 instead.

Applicant agreed and authorized the Examiner to make the amendment. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment. It must be submitted no later than the payment of the issue fee.

Please amend the claim 67 as follow:

Claim 67:

Line 1, replace "DIMM of claim 586" with --DIMM of claim 58--

Allowable Subject Matter

Claims 2, 29-39, 41-45, 47-58, and 60-67 are allowed.

6. The invention directed to a hybrid memory System comprising volatile memory sub-system (e.g., DDR SDRAM DIMM) and non-volatile memory system (e.g., NAND flash memory). The memory system interface to the host system via edge connectors having separate conduits assigned for data, address, and power lines. A first group of DDR SDRAM devices are selected/enabled to be access in parallel by a first chip select signal and a second group of DDR SDRAM devices are selected/enabled to be access in parallel by a second chip select signal. The

reason for allowance of the independent claim 2 are the following: in interpreting the pending claims, in light of the specification, the Examiner finds the claimed invention to be patentably distinct from the prior art of record. The prior art of record neither anticipates, nor renders obvious the recited combination as a whole, including the limitation of “wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal: a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines; and

- the at least 40 1-bit DDR data signals include 32 1-bit wide-DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals: and wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.”

7. *Independent claims **29-39, 41-43, 45, 47-57, and 60-67** are allowable at least for reasons recited above including all the limitations of the allowable independent base claim upon which they depend..*

8. ***Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."***

Conclusion

9. ***Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.***

11. ***If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.***

12. ***Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system,***

contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786- 9199 (IN USA OR CANADA) or 571-272-1000.

***/HASHEM FARROKH/
Primary Examiner, Art Unit 2138***

<i>Examiner-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Telephonic
Khaled Shami	Attorney	

Date of Interview: 18 April 2025

Issues Discussed:

Other

Request for an examiner amendment to correct a minor error in the claim numbering. Examiner made a telephone call to the applicant, Mr. Khaled Shami (Reg. # 38,745) to inform the Applicant that claims in Application 17/582,797 are in condition for allowance except for a minor claim dependency error. Examiner inquired whether the applicant is willing to amend (e.g., change the claim dependency) the dependent claim 67 depending from claim 586 to depend from claim 58 instead. Applicant agreed and authorized the Examiner to make the amendment (see accompanying notice of allowance).

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner) _____ (Date) 19 April 2025		O.G. Print Claim(s) 2	O.G. Print Figure 6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	13	/ 28
G06F	/	1	/ 18

NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	110

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	19 April 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	2	6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10		19		28	15	37		46	22	55	33	64
1	2		11		20	2	29	7	38	18	47	23	56	35	65
	3		12		21	11	30	8	39	25	48	28	57	38	66
	4		13		22	12	31		40	19	49	29	58	30	67
	5		14		23	3	32	9	41	20	50		59	34	68
	6		15		24	4	33	10	42	24	51	36	60		
	7		16		25	14	34	13	43	26	52	37	61		
	8		17		26	5	35	16	44	27	53	31	62		
	9		18		27	6	36	17	45	21	54	32	63		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner) _____ (Date) 19 April 2025		O.G. Print Claim(s) 2	O.G. Print Figure 6

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF
711	110	11/01/2023	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF
Inventor/Assignee Search Updated	10/31/2023	HF
EAST Text, CPC, FACET Search Updated	10/31/2023	HF
Inventor/Assignee Search Updated	02/22/2024	HF
EAST Text, CPC, FACET Search Updated	02/22/2024	HF
Inventor/Assignee Search Updated	08/05/2024	HF
EAST Text, CPC, FACET Search Updated	08/05/2024	HF
Inventor/Assignee Search Updated	01/19/2025	HF
EAST Text, CPC, FACET Search Updated	01/19/2025	HF
Inventor/Assignee Search Updated	04/18/2025	HF
EAST Text, CPC, FACET Search Updated	04/18/2025	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	04/18/2025	HF
G06F13/28	G06F13/00	04/18/2025	HF
G06F2212/205	G06F2212/00	04/18/2025	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L178	1	"17582797"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 01:25 PM
L179	1	"17582797"	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 02:45 PM
L180	8	((plurality OR multiple OR several) SAME (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM OR DDR\$1) near2 (devices OR chips OR dies)) SAME (coupled OR connected) SAME (regulated near2 voltage)))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 06:00 PM
L181	7	((((plurality OR multiple OR several) WITH (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM OR DDR\$1) near2 (devices OR chips OR dies)) WITH (coupled OR connected) SAME (regulated near2 voltage))))).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 06:01 PM
L182	5	181 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 06:01 PM
L183	5	180 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2025/04/18 06:03 PM
L184	8	180 AND ((G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 06:03 PM

		G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)					
L185	0	(group OR cluster OR multiple OR plurality OR pool) SAME ((double ADJ data ADJ rate) OR DDR) near2 (devices OR chips OR dies) SAME ((connected OR attached OR hav\$3 OR "same") near2 (chip ADJ select))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/18 06:12 PM
L186	0	(group OR cluster OR multiple OR plurality OR pool) SAME ((double ADJ data ADJ rate) OR DDR) near2 (devices OR chips OR dies) SAME ((connected OR attached OR hav\$3 OR "same") near2 (chip ADJ select))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2025/04/18 06:13 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 17/582,797

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /HASHEM FARROKH/

Examiner's Signature

Initials

Title:

MEMORY WITH ON-MODULE POWER MANAGEMENT

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2022	711	2138	022955-000090
RULE			

APPLICANTS

Netlist, Inc., Irvine, CA, UNITED STATES

INVENTORS

Chi-She Chen, Walnut, CA, UNITED STATES

Jeffrey C. Solomon, Irvine, CA, UNITED STATES

Scott H. Milton, Irvine, CA, UNITED STATES

Jayesh Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17328019 05/24/2021 PAT 11232054

17328019 is a CON of 17138766 12/30/2020 PAT 11016918

17138766 is a CON of 15934416 03/23/2018ABN

15934416 is a CON of 14840865 08/31/2015 PAT 9928186

14840865 is a CON of 14489269 09/17/2014 PAT 9158684

14489269 is a CON of 13559476 07/26/2012 PAT 8874831

13559476 has PRO of 61512871 07/28/2011

13559476 is a CIP of 12240916 09/29/2008 PAT 8301833

12240916 is a CON of 12131873 06/02/2008ABN

12131873 has PRO of 60941586 06/01/2007

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

02/01/2022

STATE OR COUNTRY

UNITED STATES

ADDRESS

Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152
UNITED STATES

FILING FEE RECEIVED

\$1,980



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/19/2025 04:45:58 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 70486857

AUTHORIZED BY Khaled Shami

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585

PAYMENT TRANSACTION ID
E20255IG46339020

PAYMENT AUTHORIZED BY
LISA MARGARET ELLIOTT

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1202	EACH CLAIM IN EXCESS OF 20	200.00	5	1000.00
1820	REQUEST FOR CONTINUED EXAMINATION (RCE)- 2ND AND SUBSEQUENT REQUEST (SEE 37 CFR 1.114)	2860.00	1	2860.00
TOTAL AMOUNT:				\$3,860.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL
(Submitted Only via EFS-Web)**

Application Number	17582797	Filing Date	2022-01-24	Docket Number (if applicable)	022955-000090	Art Unit	2138
First Named Inventor	CHEN			Examiner Name	FARROKH		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____☐ Other _____☒ Enclosed☒ Amendment/Reply☐ Information Disclosure Statement (IDS)☐ Affidavit(s)/ Declaration(s)☐ Other _____**MISCELLANEOUS**☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)☐ Other _____**FEES****The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.**☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No 220585**SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED**☒ Patent Practitioner Signature☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2025-05-19
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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PATENT AND TRADEMARK OFFICE

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APPLICATION #
17/582,797

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Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 70486857

FILING DATE 01/24/2022

CUSTOMER # 151145

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Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Amendment and RCE after Allowance.pdf	18	Amendment Submitted/Entered with Filing of Continued Prosecution Application (CPA)/Request for Continued Examination(RCE)	167 KB
RCE - 022955-90.pdf	3	Request for Continued Examination (RCE)	1631 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
Amendment and RCE after Allowance.pdf	4B69A59336605C9F08BEFEBBA2CD1A642035A0EC71827E79F B21DC7AE94FB3E81E0BD463ADC78E64EDD075C1E5485A440

D6F0705683BA3BFA5D03FF788CEA6ED

RCE - 022955-90.pdf

7AD10C37BF7A5B7C4DD8C85D198CF60561EE5A090DFAB9F
AC7B5AF126C8785CB2F1A5A996955CF228F28B2D2E16899AB
F28A6B3FC5F5DF8FB3DD3F67C93185E8

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

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If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop RCE

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Submission Under 37 CFR § 1.313 of Amendment and RCE

After Allowance and Prior to Payment of Issue Fee

Commissioner:

Applicant gratefully acknowledges the issuance of the Notice of Allowance of April 28, 2025. Applicant respectfully submits under 37 CFR § 1.313 the instant Amendment and Request for Continued Examination after allowance and prior to payment of the Issue Fee.

Listing of Claims begins on page **2**.

Remarks begin on page **18**.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)
2. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component configured to receive power via at least the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, ~~and~~

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each configured to receive ~~connected to a first chip select line configured to electrically conduct~~ a first chip select signal;

a second group of at least four DDR SDRAM devices that are each configured to receive ~~connected to a second chip select line configured to electrically conduct~~ a second chip select signal; ~~and~~

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[40]]~~32 1-bit DDR data signals in parallel via at least ~~[[40]]~~32 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via ~~[[the]]~~an integer number of data lines of the set of data lines;
- ~~the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error correcting code (ECC) data signals; and~~

wherein the sum of the at least ~~[[40]]~~32 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[72]]~~64 data conduits.

3– 28. (Canceled)

29. (Previously Presented) The DIMM of claim 2, wherein the controller further includes:

a data storage element configured to store a plurality of data bits and a logic element

coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Previously Presented) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Previously Presented) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

32. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Previously Presented) The DIMM of claim 2, wherein, when power from the power

rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Previously Presented) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Previously Presented) The DIMM of claim 38, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

40. (Canceled)

41. (Previously Presented) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a

transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Previously Presented) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Previously Presented) The DIMM of claim 31, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB edge connections including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the

power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,
and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
 - (ii) generate a trigger signal upon detecting a trigger condition; and
 - (iii) transmit the trigger signal to at least one other portion of the controller,
- wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines;

(ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each configured to receive ~~connected to a first chip select line configured to electrically conduct~~ a first chip select signal;

a second group of at least four DDR SDRAM devices that are each configured to receive ~~connected to a second chip select line configured to electrically conduct~~ a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[40]]~~32 1-bit DDR data signals in parallel via at least ~~[[40]]~~32 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via ~~[[the]]~~an integer number of data lines of the set of data lines; and

- ~~the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and~~

wherein the sum of the at least ~~[[40]]~~32 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[72]]~~64 data conduits.

45. (Previously Presented) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Canceled)

47. (Previously Presented) The DIMM of claim 45, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Previously Presented) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Previously Presented) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element,

respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Previously Presented) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Previously Presented) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

54. (Previously Presented) The DIMM of claim 49, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

55. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element.

56. (Previously Presented) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the PCB ~~edge connections~~ including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge

connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit DDR data signal, ~~and wherein the set of data lines includes at least 72 data lines;~~

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, one or more write operations to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each configured to receive ~~connected to a first chip select line configured to electrically conduct~~ a first chip select signal; and

a second group of at least four DDR SDRAM devices that are each configured to receive ~~connected to a second chip select line configured to electrically conduct~~ a second chip select signal, and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[40]]~~32 1-bit DDR data signals in parallel via at least ~~[[40]]~~32 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via ~~[[the]]~~an integer number of data lines of the set of data lines; and
- ~~• the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error correcting code (ECC) data signals; and~~

wherein the sum of the at least ~~[[40]]~~32 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[72]]~~64 data conduits.

59. (Canceled)

60. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Previously Presented) The DIMM of claim 58, wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Previously Presented) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Previously Presented) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

65. (Previously Presented) The DIMM of claim 58, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

66. (Previously Presented) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Previously Presented) The DIMM of claim 58, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Previously Presented) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

69. (New) The DIMM of claim 2, wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output the integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines

70. (New) The DIMM of claim 69, wherein the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals.

71. (New) The DIMM of claim 69, wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

72. (New) The DIMM of claim 44, wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the

first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output the integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines

73. (New) The DIMM of claim 72, wherein the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals.

74. (New) The DIMM of claim 72, wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

75. (New) The DIMM of claim 58, wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output the integer number of 1-bit DDR data signals in parallel via the integer number of data lines of the set of data lines

76. (New) The DIMM of claim 75, wherein the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals.

77. (New) The DIMM of claim 75, wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

REMARKS

Applicant respectfully submits that the claims of the present application define patentable subject matter and should be passed to allowance.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant submission, or credit any overpayment not otherwise credited, to our deposit account no. 22-0585.

Respectfully submitted,

Date: May 19, 2025
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/582,797		Filing Date 01/24/2022		☐ To be Mailed		
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO										
APPLICATION AS FILED - PART I										
		(Column 1)		(Column 2)						
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)		
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =				
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =				
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).								
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL				
APPLICATION AS AMENDED - PART II										
		(Column 1)		(Column 2)		(Column 3)				
AMENDMENT	05/19/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		
								ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	* 47	Minus	** 42	= 5	x \$200 =		1000		
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$600 =		0		
	☐ Application Size Fee (37 CFR 1.16(s))									
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		1000
		(Column 1)		(Column 2)		(Column 3)				
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		
								ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =				
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =				
	☐ Application Size Fee (37 CFR 1.16(s))									
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE				
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/TAWANA E CAMPBELL/				
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".										
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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UNITED STATES PATENT AND TRADEMARK OFFICE

USPTO Automated Interview Request (AIR)

May 20 2025

This paper requesting to schedule and/or conduct an interview is appropriate because:

I have previously submitted an authorization for internet communications in a patent application. See form SB/0439.

Name(s) :

Khaled Shami

S-signature:

/Khaled Shami/

Registration Number:

38745

U.S. Application Number:

17582797

Confirmation Number:

3546

E-mail Address:

kshami@vorys.com

Phone Number:

+1 6507455353

Proposed Time of Interview:

5-28-2025 2:00 PM ET

Alternative Proposed Time(s) of Interview:

5-29-2025 2:00 PM ET

Alternative Proposed Time(s) of Interview:

5-30-2025 2:00 PM ET

Preferred Interview Type:

Telephonic

I am the applicant or applicant's representative for this application.

Topic for Discussion:

Newly-submitted amendment with RCE, prior art and support



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APPLICATION #
17/582,797

RECEIPT DATE / TIME
05/29/2025 02:23:07 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY LISA MARGARET ELLIOTT

PATENT CENTER # 70637843

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Khaled Shami

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Summary of Telephone Interview and Examiners Amendment.pdf	16	Applicant summary of interview with examiner	152 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
Summary of Telephone Interview and Examiners Amendment.pdf	013DCA47EE1919E1FD16C8DF9B1E8BF86701D73F49E1DF0B A11F13B344818ED9C07F360C6CE06B338A4F464E71F658C74 558CAE3598EB2FE0D452A0D1BAE489E

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH, Hashem

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT

SUMMARY OF TELEPHONE INTERVIEW

On May 28, 2025, a telephone interview was conducted between Examiner Farrokh and Applicant's attorney of record, Mr. Khaled Shami, and Mr. Richard C. Kim, VP of IP strategy at Netlist, Inc. Applicant sincerely thanks Examiner Farrokh for his time and consideration during the interview.

During the interview, it was agreed that Examiner Farrokh would enter, by Examiner's Amendment, claim changes that were discussed in order to place the application in condition for allowance. The claim changes are shown below in a courtesy copy provided for the Examiner's convenience. Specifically, they include clarifying amendments to independent claims 2, 44 and 58, and cancellation, without prejudice or disclaimer, of dependent claims 69-77.

COURTESY COPY OF EXAMINER'S AMENDMENT FOR ALLOWANCE

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Previously Canceled)
2. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component configured to receive power via at least the fourth regulated voltage line;

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth

regulated voltage lines; and

- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, ~~and~~

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are ~~each~~ configured to simultaneously receive a first chip select signal;

a second group of at least four DDR SDRAM devices that are ~~each~~ configured to simultaneously receive a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 32 1-bit DDR data signals in parallel via at least 32 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and

wherein the sum of the at least 32 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 64 data conduits.

3– 28. (Previously Canceled)

29. (Previously Presented) The DIMM of claim 2, wherein the controller further includes:

a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Previously Presented) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Previously Presented) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

32. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Previously Presented) The DIMM of claim 2, wherein, when power from the power rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third

write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Previously Presented) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Previously Presented) The DIMM of claim 38, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

40. (Previously Canceled)

41. (Previously Presented) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Previously Presented) The DIMM of claim 29, wherein the data storage element is a

buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Previously Presented) The DIMM of claim 31, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,

and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and
- (iii) transmit the trigger signal to at least one other portion of the controller, wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage

line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines;
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are ~~each~~ configured to simultaneously receive a first chip select signal;

a second group of at least four DDR SDRAM devices that are ~~each~~ configured to simultaneously receive a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 32 1-bit DDR data signals in parallel via at least 32 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and

wherein the sum of the at least 32 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 64 data conduits.

45. (Previously Presented) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory

and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Previously Canceled)

47. (Previously Presented) The DIMM of claim 45, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Previously Presented) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Previously Presented) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Previously Presented) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Previously Presented) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

54. (Previously Presented) The DIMM of claim 49, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

55. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element.

56. (Previously Presented) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector

including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit DDR data signal;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition; and

- (iii) transmit the trigger signal to at least one other portion of the controller,
wherein the controller is configured to perform, in response to the trigger signal,
one or more write operations to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

- a first component configured to receive power via the first regulated voltage line;

- a second component configured to receive power via the second regulated voltage line;

- a third component configured to receive power via the third regulated voltage line; and

- a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

- a first group of at least five DDR SDRAM devices that are ~~each~~ configured to

simultaneously receive a first chip select signal; and

a second group of at least four DDR SDRAM devices that are ~~each~~ configured to simultaneously receive a second chip select signal, and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 32 1-bit DDR data signals in parallel via at least 32 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and

wherein the sum of the at least 32 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 64 data conduits.

59. (Previously Canceled)

60. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Previously Presented) The DIMM of claim 58, wherein when power is provided to the

DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Previously Presented) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Previously Presented) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

65. (Previously Presented) The DIMM of claim 58, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

66. (Previously Presented) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Previously Presented) The DIMM of claim 58, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Previously Presented) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and
wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

69-77. (Previously Canceled)

Respectfully submitted,

Date: May 29, 2025

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

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022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	3546	FILED BY	LISA MARGARET ELLIOTT
PATENT CENTER #	70715685	FILING DATE	01/24/2022
CUSTOMER #	151145	FIRST NAMED INVENTOR	Chi-She Chen
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Khaled Shami

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
022955-000090 - Supplemental Amendment - 2025-06-03.pdf	17	Supplemental Response or Supplemental Amendment	164 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
022955-000090 - Supplemental Amendment - 2025-06-03.pdf	0968D5C8B7FFFE9674143900E478404E6D5EE1BDA3C3369F8 D4FB387AA6A651D940B8FAF9F50C75BF252683E5EF27641F0 D5A2C2AC93C943145839BB9045C103

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 3546

APPLICATION NO: 17/582,797

ART UNIT: 2138

FILING DATE: January 24, 2022

EXAMINER: FARROKH

TITLE: MEMORY WITH ON-MODULE POWER MANAGEMENT (as amended)

Mail Stop RCE

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Supplemental Amendment and Examiner Interview Summary

Commissioner:

Applicant thanks Examiner Farrokh for his time and consideration extended to the undersigned representative and to Mr. Richard C. Kim, in-house counsel at Netlist, Inc., during the telephone interview conducted on June 2, 2025. During the interview, Examiner Farrokh indicated that two prior art references, 2004/0193777 (LaBerge) and 2004/0143773 (Chen) may be used to reject the currently pending claims, filed on May 19, 2025 (Submission Under 37 CFR § 1.313 of Amendment and RCE After Allowance and Prior to Payment of Issue Fee). Applicant argued there is no motivation to combine LaBerge or Chen with Harris to arrive at the claimed invention, and the combinations do not teach all limitations of the claims. No agreement was reached on these points.

However, Examiner Farrokh stated that if the claims were restored to their previously allowed form, as of April 17, 2025 (Supplemental Amendment), and allowed by the Examiner in the Notice of Allowance of April 28, 2025, then they would again be allowed. Without agreeing on the rejections, and solely to expedite prosecution of the present application to allowance, the claims are amended herein without prejudice or disclaimer, to restore the claims to their previously allowed form. Per the agreement reached during the telephone interview, Applicant respectfully requests the Examiner to promptly pass the application to allowance.

Listing of Claims begins on page 2.

Remarks begin on page 17.

AMENDMENT OF THE CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Previously Canceled)
2. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and nonvolatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to: (i) monitor an input voltage supplied via the input voltage supply line; (ii) generate a trigger signal upon detecting a trigger condition; and (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory, and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via at least the first regulated voltage line;

a second component configured to receive power via at least the second regulated voltage line;

a third component configured to receive power via at least the third regulated voltage line; and

a fourth component configured to receive power via at least the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and

- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, ~~and~~

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each ~~configured to receive~~ connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each ~~configured to receive~~ connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[32]]~~40 1-bit DDR data signals in parallel via at least ~~[[32]]~~40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines;
- the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least ~~[[32]]~~40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[64]]~~72 data conduits.

3– 28. (Previously Canceled)

29. (Previously Presented) The DIMM of claim 2, wherein the controller further includes:

a data storage element configured to store a plurality of data bits and a logic element

coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

a first write operation to write a first set of data bits into a first portion of the data storage element;

a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

30. (Previously Presented) The DIMM of claim 2, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

31. (Previously Presented) The DIMM of claim 2, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

32. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

33. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

34. (Previously Presented) The DIMM of claim 2, wherein, when power from the power

rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

35. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

36. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

37. (Previously Presented) The DIMM of claim 2, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

38. (Previously Presented) The DIMM of claim 29, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

39. (Previously Presented) The DIMM of claim 38, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

40. (Previously Canceled)

41. (Previously Presented) The DIMM of claim 29, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a

transfer of the first and second sets of data bits into the data storage element via a second connection.

42. (Previously Presented) The DIMM of claim 29, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

43. (Previously Presented) The DIMM of claim 31, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

44. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the

power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections,
and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
 - (ii) generate a trigger signal upon detecting a trigger condition; and
 - (iii) transmit the trigger signal to at least one other portion of the controller,
- wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines;

(ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each ~~configured to receive~~ connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each ~~configured to receive~~ connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[32]]~~40 1-bit DDR data signals in parallel via at least ~~[[32]]~~40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and

- the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least ~~[[32]]~~40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[64]]~~72 data conduits.

45. (Previously Presented) The DIMM of claim 44, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

46. (Previously Canceled)

47. (Previously Presented) The DIMM of claim 45, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

48. (Previously Presented) The DIMM of claim 44, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

49. (Previously Presented) The DIMM of claim 47, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

50. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element,

respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

51. (Previously Presented) The DIMM of claim 47, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

52. (Previously Presented) The DIMM of claim 44, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

53. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

54. (Previously Presented) The DIMM of claim 49, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

55. (Previously Presented) The DIMM of claim 49, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element.

56. (Previously Presented) The DIMM of claim 49, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

57. (Previously Presented) The DIMM of claim 52, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

58. (Currently Amended) A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge

connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit DDR data signal;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
 - (ii) generate a trigger signal upon detecting a trigger condition; and
 - (iii) transmit the trigger signal to at least one other portion of the controller,
- wherein the controller is configured to perform, in response to the trigger signal, one or more write operations to write data into the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each ~~configured to receive~~ connected to a first chip select line configured to electrically conduct a first chip select signal; and

a second group of at least four DDR SDRAM devices that are each ~~configured to receive~~ connected to a second chip select line configured to electrically conduct a second chip select signal, and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least ~~[[32]]~~40 1-bit DDR data signals in parallel via at least ~~[[32]]~~40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and
- the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

wherein the sum of the at least ~~[[32]]~~40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least ~~[[64]]~~72 data conduits.

59. (Previously Canceled)

60. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

61. (Previously Presented) The DIMM of claim 65, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

62. (Previously Presented) The DIMM of claim 58, wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

63. (Previously Presented) The DIMM of claim 58, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

64. (Previously Presented) The DIMM of claim 63, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

65. (Previously Presented) The DIMM of claim 58, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

66. (Previously Presented) The DIMM of claim 58, wherein each of the first, second, and third converter circuits is a buck converter.

67. (Previously Presented) The DIMM of claim 58, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

68. (Previously Presented) The DIMM of claim 63, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

69-77. (Canceled)

REMARKS

Independent claims 2, 44 and 58 are amended herein to restore them to their previously allowed form, and with a correction of minor typographical error (“edge connections” replaced with “PCB”) as previously discussed with the Examiner. The amendments herein are made without prejudice or disclaimer and solely for the purpose of expediting prosecution of the present application to allowance. Per the agreement reached during the telephone interview on June 2, 2025, Applicant respectfully submits that the claims of the present application define patentable subject matter and should be passed to allowance.

Applicant does not believe that extensions of time are necessary. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 22-0585. Respectfully submitted,

Date: June 3, 2025
Vorys, Sater, Seymour, and Pease LLP
1909 K Street, NW 9th Floor
Washington, DC 20006-1152
Customer No.: 151145
Tel: (202) 467-8800

/Khaled Shami/
Khaled Shami, Reg. No. 38,745

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/582,797	Filing Date 01/24/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	06/03/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 38	Minus	** 47	= 0	x \$200 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$600 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						Supervisory Prog. Sup. Spec.	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/LISA E FULTON/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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NOTICE OF ALLOWANCE AND FEE(S) DUE

151145 7590 06/18/2025
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

EXAMINER	
FARROKH, HASHEM	
ART UNIT	PAPER NUMBER
2138	

DATE MAILED: 06/18/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	09/18/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 06/18/2025
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	09/18/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2138	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546
151145	7590	06/18/2025	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW				
9th Floor				
Washington, DC 20006-1152			ART UNIT	PAPER NUMBER
			2138	

DATE MAILED: 06/18/2025

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 17/582,797	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 06/03/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2,29-39,41-45,47-58 and 60-68. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material _____. | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>06/02/2025</u> . | |

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

DETAIL ACTION

Notice of Pre-AIA or AIA Status

1. *The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.*

Continued Examination Under 37 CFR 1.114

2. *A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 05/19/2025 has been entered.*
3. *This office action is in response to claim amendment filed on 06/03/2025. There are a total 38 claims pending in the application. Claims 1, 3-27, 40, 46, 59, and 69-77 have been canceled; claims 2, 44, and 58 have been amended; and no new claims have been added.*

Response to Remarks

4. *Applicant's arguments have been fully considered and they are persuadable. Thus, pending claims 2, 29-39, 41-45, 47-58, and 60-67 are considered to be allowable (see also the attached interview summary).*

Allowable Subject Matter

5. Claims 2, 29-39, 41-45, 47-58, and 60-67 are allowed.

The invention directed to a hybrid memory System comprising volatile memory sub-system (e.g., DDR SDRAM DIMM) and non-volatile memory system (e.g., NAND flash memory). The memory system interface to the host system via edge connectors having separate conduits assigned for data, address, and power lines. A first group of DDR SDRAM devices are selected/enabled to be accessed in parallel by a first chip select signal and a second group of DDR SDRAM devices are selected/enabled to be accessed in parallel by a second chip select signal. The reason for allowance of the independent claim 2 are the following: in interpreting the pending claims, in light of the specification, the Examiner finds the claimed invention to be patentably distinct from the prior art of record. The prior art of record neither anticipates, nor renders obvious the recited combination as a whole, including the limitation of: "wherein each component of the set of components: (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and(ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and wherein the plurality of DDR SDRAM devices includes: a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal; and wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode: the first group of DDR SDRAM

devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 32 1-bit DDR data signals in parallel via at least 32 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and wherein the sum of the at least 32 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 64 data conduits.”

Independent claims 44 and 58 include similar limitations as claim 1 and are allowed for the reasons recited above with respect to claim 1.

6. *Independent claims **29-39, 41-43, 45, 47-57, and 60-67** are allowable at least for reasons recited above including all the limitations of the allowable independent base claim upon which they depend.*

7. *Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”*

Conclusion

Art Unit: 2138

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.
9. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Tim Vo can be reached on (571)272-3642. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.
10. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HASHEM FARROKH/
Primary Examiner, Art Unit 2138

<i>Applicant-Initiated Interview Summary</i>	Application No. 17/582,797	Applicant(s) Chen et al.		
	Examiner HASHEM FARROKH	Art Unit 2138	AIA (First Inventor to File) Status No	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
HASHEM FARROKH	Primary Examiner	Telephonic
Khaled Shami	Attorney	
Richard C. Kim	Applicant Counsel	

Date of Interview: 02 June 2025

Issues Discussed:

35 U.S.C. 103

An Applicant initiated interview with Applicant, Mr. Khaled Shami (reg. # 38,745), was conducted on 06/02/2025. The purpose of interview was to discuss the amendment of independent claims in the communication filed on 05/19/2025. During the interview, Examiner indicated the prior art references of LaBerge (US 2004/0193777) and Chen (US 2004/0143773) teach or render obvious the newly added limitations. During the interview an agreement reached that, except for some minor amendment, if the claim restored to the previously allowed version submitted on 04/17/2025, the claims would be in condition for allowance.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	05 June 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	2	6

Issue Classification 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2138	

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	13	/ 28
G06F	/	1	/ 18
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	110

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	05 June 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	2	6

<i>Issue Classification</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10		19		28	15	37		46	22	55	33	64
1	2		11		20	2	29	7	38	18	47	23	56	35	65
	3		12		21	11	30	8	39	25	48	28	57	38	66
	4		13		22	12	31		40	19	49	29	58	30	67
	5		14		23	3	32	9	41	20	50		59	34	68
	6		15		24	4	33	10	42	24	51	36	60		
	7		16		25	14	34	13	43	26	52	37	61		
	8		17		26	5	35	16	44	27	53	31	62		
	9		18		27	6	36	17	45	21	54	32	63		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 38	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2138 (Primary Examiner) _____ (Date) 05 June 2025		O.G. Print Claim(s) 2	O.G. Print Figure 6

<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

CPC - Searched*		
Symbol	Date	Examiner
G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/ 205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00	10/06/2022	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	10/06/2022	HF
711	110	11/01/2023	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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<i>Search Notes</i> 	Application/Control No. 17/582,797	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2138

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	10/06/2022	HF
EAST Text, CPC, FACET Search	10/06/2022	HF
IP Search	10/06/2022	HF
Google Search	10/06/2022	HF
Inventor/Assignee Search Updated	07/07/2023	HF
EAST Text, CPC, FACET Search Updated	07/07/2023	HF
Inventor/Assignee Search Updated	10/31/2023	HF
EAST Text, CPC, FACET Search Updated	10/31/2023	HF
Inventor/Assignee Search Updated	02/22/2024	HF
EAST Text, CPC, FACET Search Updated	02/22/2024	HF
Inventor/Assignee Search Updated	08/05/2024	HF
EAST Text, CPC, FACET Search Updated	08/05/2024	HF
Inventor/Assignee Search Updated	01/19/2025	HF
EAST Text, CPC, FACET Search Updated	01/19/2025	HF
Inventor/Assignee Search Updated	04/18/2025	HF
EAST Text, CPC, FACET Search Updated	04/18/2025	HF
Inventor/Assignee Search Updated	06/05/2025	HF
EAST Text, CPC, FACET Search Updated	06/05/2025	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	04/18/2025	HF
G06F13/28	G06F13/00	04/18/2025	HF
G06F2212/205	G06F2212/00	04/18/2025	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2138	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L200	0	(group OR pool OR cluster) SAME (least near5 five near5 DDR near5 SDRAM) near5 (devices OR chips OR dies) SAME (each near3 (connected OR attached)) near5 (chip ADJ select\$3)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2025/06/05 02:12 PM
L201	0	(group OR pool OR cluster) SAME ((plurality OR multiple OR pool) near5 DDR near5 SDRAM) near5 (devices OR chips OR dies) SAME (each near3 (connected OR attached)) near5 (chip ADJ select\$3)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2025/06/05 02:14 PM
L202	0	(group OR pool OR cluster) SAME ((rank OR bank) near5 DDR near5 SDRAM) near5 (devices OR chips OR dies) SAME (each near3 (connected OR attached)) near5 (chip ADJ select\$3)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2025/06/05 02:15 PM
L203	0	(five) SAME ((rank OR bank) near5 DDR near5 SDRAM) near5 (devices OR chips OR dies) SAME (each near3 (connected OR attached)) near5 (chip ADJ select\$3)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	OFF	ON	2025/06/05 02:37 PM
L204	17	(group OR pool OR cluster OR plurality OR two OR five) SAME ((double\$1data\$1rate OR DDR) near3 ((synchronous near3 dynamic near3 random near3 access near3 memory) OR SDRAM)) SAME (devices OR chips OR dies OR ASICs) SAME (simultaneously OR concurrently OR parallel) SAME (chip	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:39 PM

		near3 (select\$3 OR enabl\$3))					
L205	0	((group OR pool OR cluster OR plurality OR two OR five) SAME ((double\$1data\$1rate OR DDR) near3 ((synchronous near3 dynamic near3 random near3 access near3 memory) OR SDRAM)) SAME (devices OR chips OR dies OR ASICs) SAME (simultaneously OR concurrently OR parallel) SAME (chip near3 (select\$3 OR enabl\$3))).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:39 PM
L206	0	((first near5 second near5 third near5 fourth) OR (plurality OR multiple)) SAME (voltage near2 (converter OR regulator)) SAME ((coupled OR connected) near5 ((printed ADJ circuit ADJ board) OR PCB)) SAME (input near5 (voltage near5 supply near5 line))	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:53 PM
L207	7	((plurality OR multiple OR several OR pool) SAME (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM OR DDR\$1) near2 (devices OR chips OR dies)) SAME (coupled OR connected) SAME (regulated near2 voltage))).clm.	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:56 PM
L208	8	((plurality OR multiple OR several OR pool) SAME (((synchronous ADJ dynamic ADJ random ADJ access ADJ memory) OR SDRAM OR DDR\$1) near2 (devices OR chips OR dies)) SAME (coupled OR	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:56 PM

		connected) SAME (regulated near2 voltage)))					
L209	5	207 AND ("BHAKTA; JAYESH".ine. OR "CHEN; CHI-SHE".ine. OR "MILTON; SCOTT H.".ine. OR "SOLOMON; JEFFREY C.".ine.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 02:57 PM
L210	0	204 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2025/06/05 02:59 PM
L211	5	208 AND ("NETLIST, INC.".ase.)	(US-PGPUB; USPAT; USOCR; IBM_TDB)	OR	ON	ON	2025/06/05 03:01 PM
L212	8	208 AND ((G06F1/185 OR G06F12/0246 OR G06F12/0638 OR G06F13/1694 OR G06F13/28 OR G06F13/4027 OR G06F13/4243 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0613 OR G06F3/0659 OR G06F3/0685 OR G11C11/005 OR G11C11/409 OR G11C14/0018 OR G11C16/10 OR G11C7/1072 OR Y02D10/00).cpc. OR (711/110).ccls. OR (G06F1/18 OR G06F12/02 OR G06F12/06 OR G06F13/16 OR G06F13/28 OR G06F13/40 OR G06F13/42 OR G06F3/06 OR G11C14/00 OR G11C7/10).ipc.)	(US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/06/05 03:01 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 17/582,797

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /HASHEM FARROKH/

Examiner's Signature

Initials

Title:

MEMORY WITH ON-MODULE POWER MANAGEMENT

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2022	711	2138	022955-000090
RULE			

APPLICANTS

Netlist, Inc., Irvine, CA, UNITED STATES

INVENTORS

Chi-She Chen, Walnut, CA, UNITED STATES

Jeffrey C. Solomon, Irvine, CA, UNITED STATES

Scott H. Milton, Irvine, CA, UNITED STATES

Jayesh Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17328019 05/24/2021 PAT 11232054

17328019 is a CON of 17138766 12/30/2020 PAT 11016918

17138766 is a CON of 15934416 03/23/2018ABN

15934416 is a CON of 14840865 08/31/2015 PAT 9928186

14840865 is a CON of 14489269 09/17/2014 PAT 9158684

14489269 is a CON of 13559476 07/26/2012 PAT 8874831

13559476 has PRO of 61512871 07/28/2011

13559476 is a CIP of 12240916 09/29/2008 PAT 8301833

12240916 is a CON of 12131873 06/02/2008ABN

12131873 has PRO of 60941586 06/01/2007

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

02/01/2022

STATE OR COUNTRY

UNITED STATES

ADDRESS

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9th Floor
Washington, DC 20006-1152
UNITED STATES

FILING FEE RECEIVED

\$1,980



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
06/24/2025 04:13:28 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lauren Siefke

PATENT CENTER # 71055701

FILING DATE 01/24/2022

CUSTOMER # 151145

FIRST NAMED
INVENTOR Chi-She Chen

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Michael Messinger

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-06-24- Issue Fee - 022955-90.pdf	1	Issue Fee Payment (PTO-85B)	124 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-06-24- Issue Fee - 022955-90.pdf	193428BFD4E0867A974792662B9120D236478374F4BB5C929F C7F4AD3511B045665F1B197C6A651E46BCC1FE2F671777793 A998232A5BC77EF3CD4C547D7DA59

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
06/24/2025 04:13:28 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3546

FILED BY Lauren Siefke

PATENT CENTER # 71055701

AUTHORIZED BY Michael Messinger

CUSTOMER # 151145

FILING DATE 01/24/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Chi-She Chen

Payment Information

PAYMENT METHOD
DA / 220585

PAYMENT TRANSACTION ID
E20256NG14266729

PAYMENT AUTHORIZED BY
Lauren Siefke

PRE-AUTHORIZED ACCOUNT
220585

PRE-AUTHORIZED CATEGORY
37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement

Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 06/18/2025
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	01/24/2022	Chi-She Chen	022955-000090	3546

TITLE OF INVENTION: MEMORY WITH ON-MODULE POWER MANAGEMENT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	09/18/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2138	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Vorys, Sater, Seymour and Pease LLP

2 Khaled Shami

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE
NETLIST, INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)
Irvine, California, UNITED STATES OF AMERICA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 220585

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Michael Messinger/

Date June 24, 2025

Typed or printed name Michael Messinger

Registration No. 37575

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135 2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

U.S.PATENTS

[Remove](#)

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
Change(s) applied to document, P.H./ 11/21/2023	1	10949339		2021-03-16	Netlist, Inc. Lee et al.	
	2	10860506		2020-12-08	Netlist, Inc. Lee et al.	
	3	10217523		2019-02-26	Netlist, Inc. Lee et al.	
	4	10474595		2019-11-12	Netlist, Inc. Lee	
	5	9858218		2018-01-02	Netlist, Inc. Lee	
	6	7619912		2009-11-17	Netlist, Inc. Bhakta et al.	
	7	11232054		2022-01-25	Netlist, Inc. Chen et al.	
	8	8026485				

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
Change(s) applied to document, /P.H./ 11/21/2023	1	20190004985		2019-01-03	Netlist, Inc. Lee et al.	
	2	20060174140		2006-08-03	Shaun Harris, et al.	
	3	20060080515		2006-04-13	LEFTHAND NETWORKS, INC. Spiers et al.	
	4	20080238536		2008-10-02	ELPIDA MEMORY, INC. Hayashi et al.	
	5	20100257304		2010-10-07	GOOGLE INC. Rajan et al.	
	6	20100205470		2010-08-12	STEC, Inc. Moshayedi et al.	
	7	20080101147		2008-05-01	Hosseini Amidi	
	8	20070136523		2007-06-14	Randy Bonella, et al.	
	9	20090034354		2009-02-05	MICRON TECHNOLOGY, INC. Resnick	

If you wish to add additional U.S. Published Application citation information please click the Add button.

FOREIGN PATENT DOCUMENTS

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

Change(s) applied to document	9	6798709	2004-09-28	Sim et al. Samsung Electronics Co., Ltd.
P.H./				
1/21/2023	10	6856556	2005-02-15	Hajeck Siliconsystems, Inc.
	11	7023187	2006-04-04	Shearon et al. Interoil Americas Inc.
	12	7692938	2010-04-06	Petter Northern Power Systems, Inc.
	13	5563839	1996-10-18	Herd et al. Sintek Corporation
	14	6693840	2004-02-17	Matsushita Electric Industrial Co., Ltd. Shimada et al.
	15	7061214	2006-06-13	Mayega et al. Texas Instruments Incorporated
	16	5630096	1997-05-13	Microunity Systems Engineering, Inc. Zuravleff et al.
	17	10672458	2020-06-02	Shaeffer et al. Rambus Inc.
	18	6067241	2000-05-23	Lu International Power Devices, Inc.

If you wish to add additional U.S. Patent citation information please click the Add button.

Add

U.S.PATENT APPLICATION PUBLICATIONS

Remove

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17582797
Filing Date	2022-01-24
First Named Inventor	CHEN
Art Unit	2135 2138
Examiner Name	FARROKH
Attorney Docket Number	022955-000090

U.S. PATENTS

[Remove](#)

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
Change(s) applied to document, P.H./ 11/21/2023	1	11016918		2021-05-25	Chen et al. Netlist, Inc.	
	2	8301833		2012-10-10	Chen et al. Netlist, Inc.	
	3	8874831		2014-10-28	Lee et al. Netlist, Inc.	
	4	9158684		2015-10-13	Lee et al. Netlist, Inc.	
	5	9928186		2018-03-27	Lee et al. Netlist, Inc.	
	6	7724604		2010-05-25	SMART Modular Technologies, Inc. Amidi et al.	
	7	7719866		2010-05-18	Boldo Agence Spatiale Europeenne	
	8	7721130		2010-05-18	Prete et al. Qimonda AG	

**POWER OF ATTORNEY TO PROSECUTE APPLICATIONS
BEFORE THE USPTO**

I hereby revoke all previous powers of attorney given in the application identified with this filing or in the attached statement under 37 CFR 3.73(c):

I hereby appoint: Practitioner(s) associated with the Customer Number: **207012**

Please change the correspondence address for the application identified with this filing or in the attached statement under 37 CFR 3.73(c) to:

The address associated with the above-mentioned Customer Number: **207012**

Applicant Name and Address:

NETLIST, INC
111 Academy Way, Suite 100
Irvine, California 92617
UNITED STATES OF AMERICA

Signature of Applicant

The individual whose signature and title is supplied below is authorized to act on behalf of the Applicant

Signature Rich Kim Date 6-10-25

Name Richard Kim Telephone 949-679-0124

Title Vice President of Intellectual Property Strategy



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
17/582,797

RECEIPT DATE / TIME
06/30/2025 05:37:59 PM Z ET

ATTORNEY DOCKET #
022955-000090

Title of Invention

MEMORY WITH ON-MODULE POWER MANAGEMENT

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	3546	FILED BY	LISA MARGARET ELLIOTT
PATENT CENTER #	71153942	FILING DATE	01/24/2022
CUSTOMER #	151145	FIRST NAMED INVENTOR	Chi-She Chen
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Khaled Shami

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
POA Netlist 207012_Signed 6-10-2025.pdf	1	Power of Attorney	34 KB
022955-90 - X - 17582797 - 3546 - aia373c.pdf	3	Assignee showing of ownership per 37 CFR 3.73	123 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
POA Netlist 207012_Signed 6-10-2025.pdf	794914ECA68B16D47E940FE5C960FC376FED75D1292CB1BD108E048FFBCB9D8F03676352EA6A89A9CEC04C3AF466E9DAF8C4095511687C78D450D575127E6109
022955-90 - X - 17582797 -	9092C7E5045AA0F5194AA6BE68E3148937A705274261951705

3546 - aia373c.pdf

2D2EA9FE612DDFFB27AB3EE1557BEE2E75A874A07DEB3FF3
CF607FCE16DFDB00563C934BFFB356

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

STATEMENT UNDER 37 CFR 3.73(c)Applicant/Patent Owner: NETLIST, INC.Application No./Patent No.: 17/582,797 Filed/Issue Date: 01-24-2022Titled: MEMORY WITH ON-MODULE POWER MANAGEMENTNETLIST, INC., a corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that, for the patent application/patent identified above, it is (choose **one** of options 1, 2, 3 or 4 below):

1. ☒ The assignee of the entire right, title, and interest.
2. ☐ An assignee of less than the entire right, title, and interest (check applicable box):
- ☐ The extent (by percentage) of its ownership interest is _____%. Additional Statement(s) by the owners holding the balance of the interest must be submitted to account for 100% of the ownership interest.
- ☐ There are unspecified percentages of ownership. The other parties, including inventors, who together own the entire right, title and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

3. ☐ The assignee of an undivided interest in the entirety (a complete assignment from one of the joint inventors was made). The other parties, including inventors, who together own the entire right, title, and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

4. ☐ The recipient, via a court proceeding or the like (e.g., bankruptcy, probate), of an undivided interest in the entirety (a complete transfer of ownership interest was made). The certified document(s) showing the transfer is attached.

The interest identified in option 1, 2 or 3 above (not option 4) is evidenced by either (choose **one** of options A or B below):

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 034531, Frame 0385, or for which a copy thereof is attached.
- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

[Page 1 of 2]

This collection of information is required by 37CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(c)

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

4. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

5. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

6. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).



As required by 37 CFR 3.73(c)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

Khaled Shami

Printed or Typed Name

June 30, 2025

Date

38745

Title or Registration Number

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/582,797	01/24/2022	Chi-She Chen	022955-000090

CONFIRMATION NO. 3546

POWER OF ATTORNEY NOTICE



151145
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Date Mailed: 07/08/2025

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 06/30/2025.

- The Power of Attorney to you in this application has been revoked by the applicant. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ytdemisse/



UNITED STATES PATENT AND TRADEMARK OFFICE

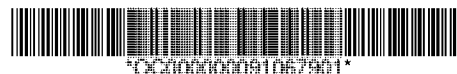
UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/582,797	01/24/2022	Chi-She Chen	022955-000090

CONFIRMATION NO. 3546

207012
Netlist Inc.
1909 K St., NW
9th Floor
Washington, DC 20006-1152

POA ACCEPTANCE LETTER



000000091067901

Date Mailed: 07/08/2025

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 06/30/2025.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ytdemisse/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/582,797	07/29/2025	12373366	022955-000090	3546

207012 7590 07/16/2025
Netlist Inc.
1909 K St., NW
9th Floor
Washington, DC 20006-1152

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

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To Promote the Progress

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The Director

*of the United States Patent and Trademark Office has received
an application for a patent for a new and useful invention. The title
and description of the invention are enclosed. The requirements
of law have been complied with, and it has been determined that
a patent on the invention shall be granted under the law.*

Therefore, this United States

Patent

grants to the person(s) having title to this patent the right to exclude others from making, using, offering for sale, or selling the invention throughout the United States of America or importing the invention into the United States of America, and if the invention is a process, of the right to exclude others from using, offering for sale or selling throughout the United States of America, products made by that process, for the term set forth in 35 U.S.C. 154(a)(2) or (c)(1), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b). See the Maintenance Fee Notice on the inside of the cover.

Coke Moya Smead

ACTING DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

Maintenance Fee Notice

If the application for this patent was filed on or after December 12, 1980, maintenance fees are due three years and six months, seven years and six months, and eleven years and six months after the date of this grant, or within a grace period of six months thereafter upon payment of a surcharge as provided by law. The amount, number and timing of the maintenance fees required may be changed by law or regulation. Unless payment of the applicable maintenance fee is received in the United States Patent and Trademark Office on or before the date the fee is due or within a grace period of six months thereafter, the patent will expire as of the end of such grace period.

Patent Term Notice

If the application for this patent was filed on or after June 8, 1995, the term of this patent begins on the date on which this patent issues and ends twenty years from the filing date of the application or, if the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121, 365(c), or 386(c), twenty years from the filing date of the earliest such application (“the twenty-year term”), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b), and any extension as provided by 35 U.S.C. 154(b) or 156 or any disclaimer under 35 U.S.C. 253.

If this application was filed prior to June 8, 1995, the term of this patent begins on the date on which this patent issues and ends on the later of seventeen years from the date of the grant of this patent or the twenty-year term set forth above for patents resulting from applications filed on or after June 8, 1995, subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b) and any extension as provided by 35 U.S.C. 156 or any disclaimer under 35 U.S.C. 253.



US012373366B2

(12) **United States Patent**
Chen et al.

(10) **Patent No.:** **US 12,373,366 B2**
(45) **Date of Patent:** **Jul. 29, 2025**

(54) **MEMORY WITH ON-MODULE POWER MANAGEMENT**

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(Continued)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Related U.S. Application Data

(63) Continuation of application No. 17/328,019, filed on May 24, 2021, now Pat. No. 11,232,054, which is a (Continued)

(51) **Int. Cl.**
G06F 13/28 (2006.01)
G06F 1/18 (2006.01)
(Continued)

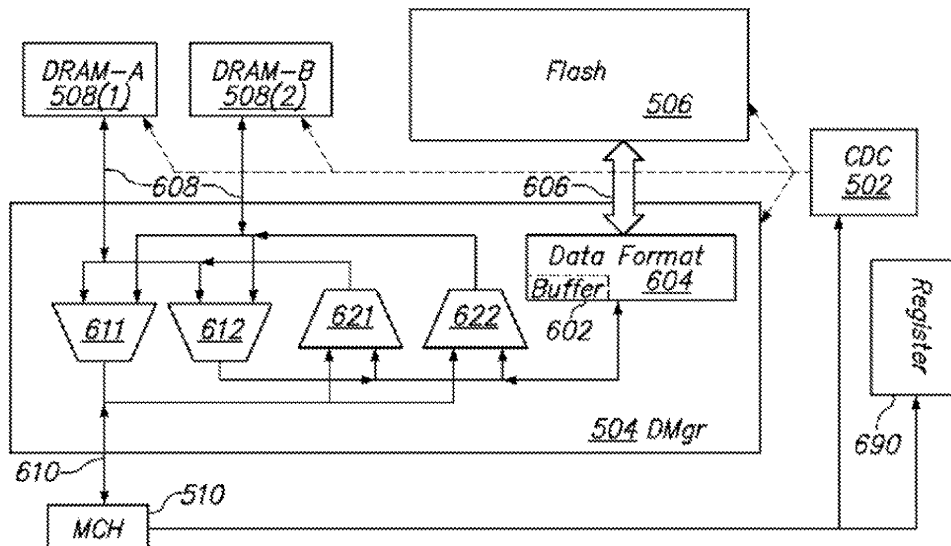
(52) **U.S. Cl.**
CPC **G06F 13/28** (2013.01); **G06F 1/185** (2013.01); **G06F 3/0613** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC .. G06F 1/185; G06F 12/0246; G06F 12/0638;
G06F 13/1694; G06F 13/28;
(Continued)

(57) **ABSTRACT**

In certain embodiments, a memory module includes a printed circuit board (PCB) having an interface that couples it to a host system for provision of power, data, address and control signals. First, second, and third buck converters receive a pre-regulated input voltage and produce first, second and third regulated voltages. A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage. Synchronous dynamic random access memory (SDRAM) devices are coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, and a voltage monitor circuit monitors an input voltage and produces a signal in response to the input voltage having a voltage amplitude that is greater than a threshold voltage.

38 Claims, 22 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/138,766, filed on Dec. 30, 2020, now Pat. No. 11,016,918, which is a continuation of application No. 15/934,416, filed on Mar. 23, 2018, now abandoned, which is a continuation of application No. 14/840,865, filed on Aug. 31, 2015, now Pat. No. 9,928,186, which is a continuation of application No. 14/489,269, filed on Sep. 17, 2014, now Pat. No. 9,158,684, which is a continuation of application No. 13/559,476, filed on Jul. 26, 2012, now Pat. No. 8,874,831, which is a continuation-in-part of application No. 12/240,916, filed on Sep. 29, 2008, now Pat. No. 8,301,833, which is a continuation of application No. 12/131,873, filed on Jun. 2, 2008, now abandoned.

- (60) Provisional application No. 61/512,871, filed on Jul. 28, 2011, provisional application No. 60/941,586, filed on Jun. 1, 2007.

(51) Int. Cl.

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G06F 12/02 (2006.01)
G06F 12/06 (2006.01)
G06F 13/16 (2006.01)
G06F 13/40 (2006.01)
G06F 13/42 (2006.01)
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G11C 14/00 (2006.01)

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CPC **G06F 3/0659** (2013.01); **G06F 3/0685** (2013.01); **G06F 12/0246** (2013.01); **G06F 12/0638** (2013.01); **G06F 13/1694** (2013.01); **G06F 13/4027** (2013.01); **G06F 13/4243** (2013.01); **G11C 7/1072** (2013.01); **G11C 14/0018** (2013.01); **G06F 2212/205** (2013.01); **G06F 2212/7208** (2013.01)

(58) Field of Classification Search

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USPC 711/110
 See application file for complete search history.

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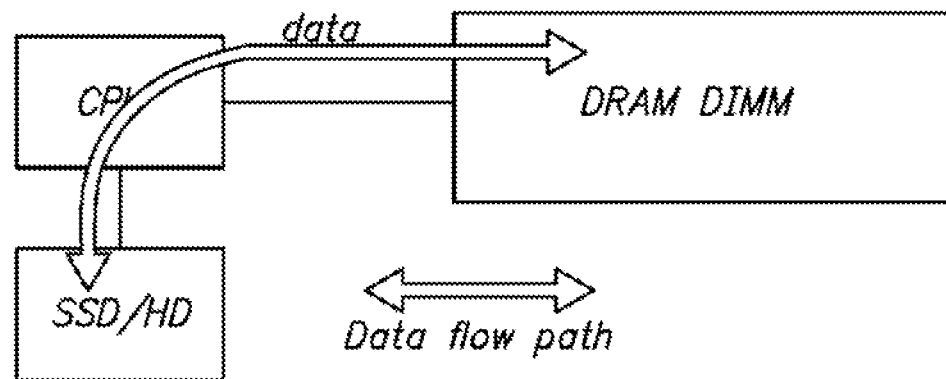


FIG. 1
(PRIOR ART)

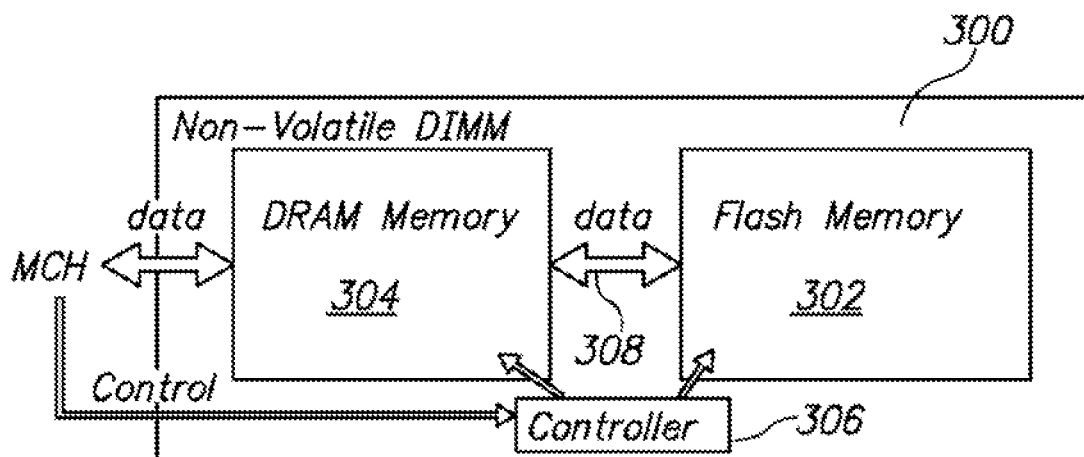
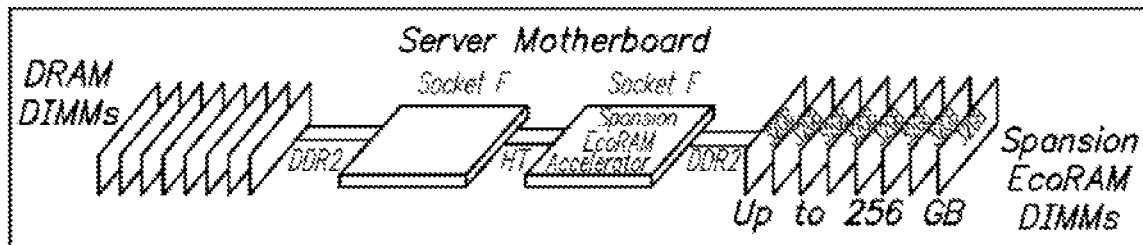
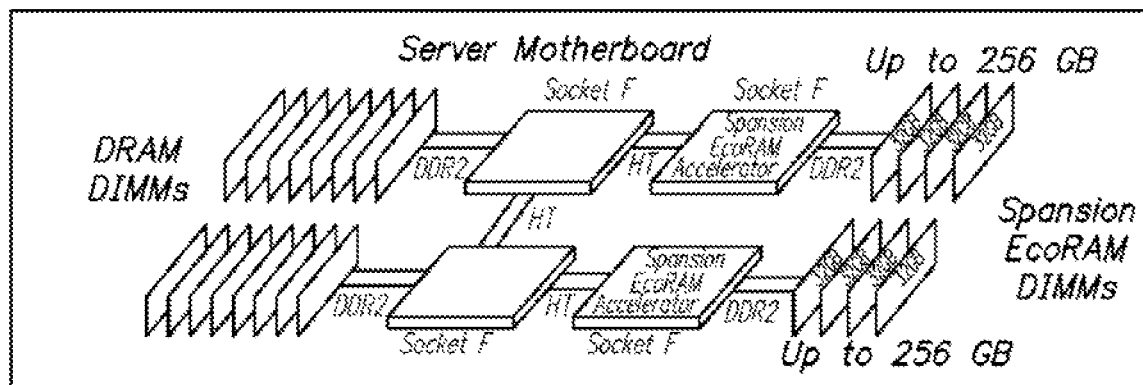
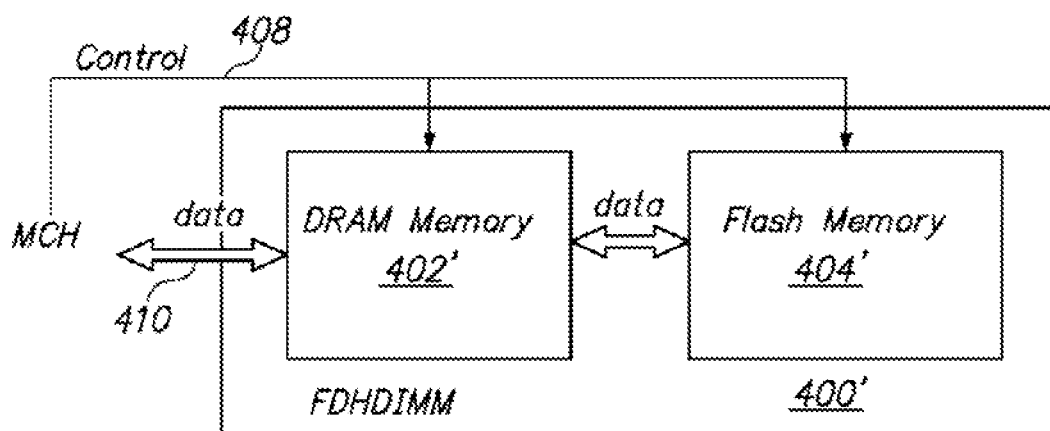
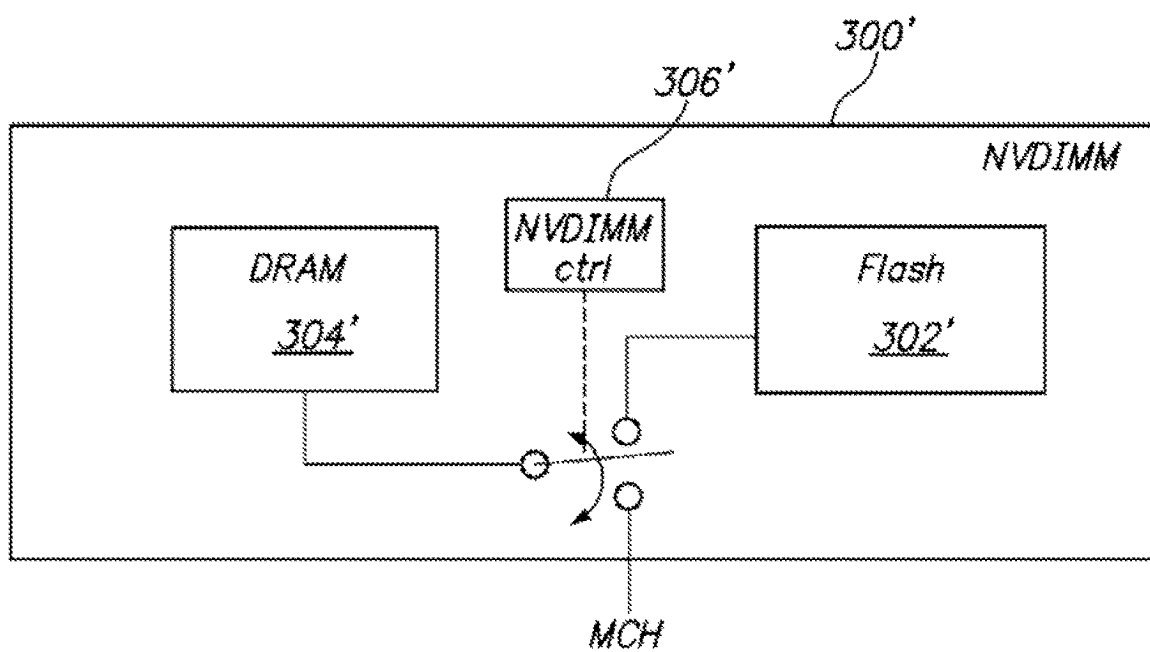
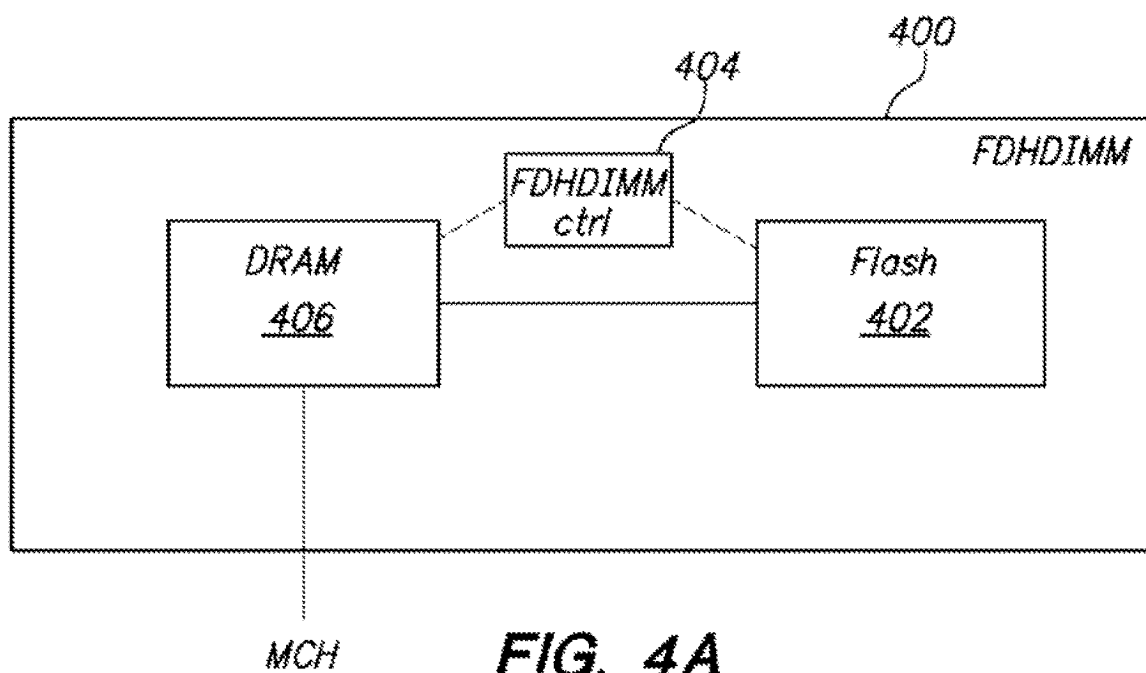
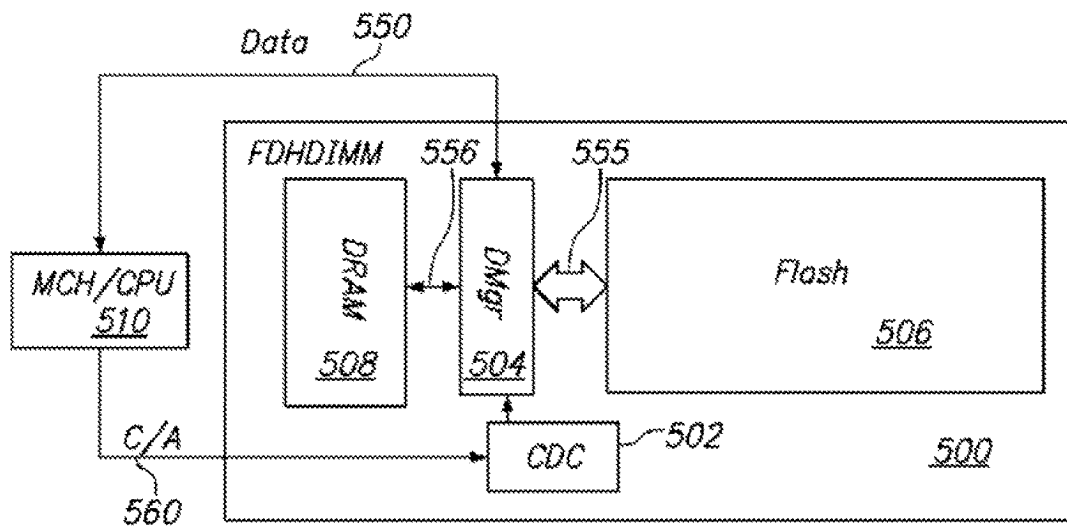
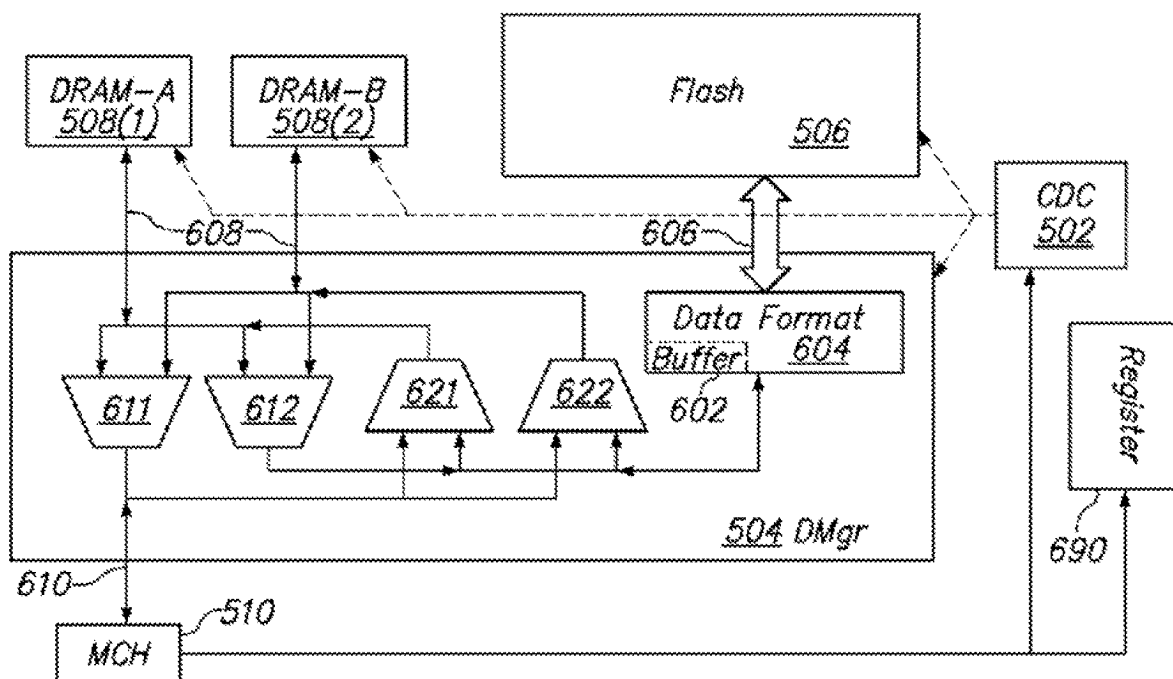
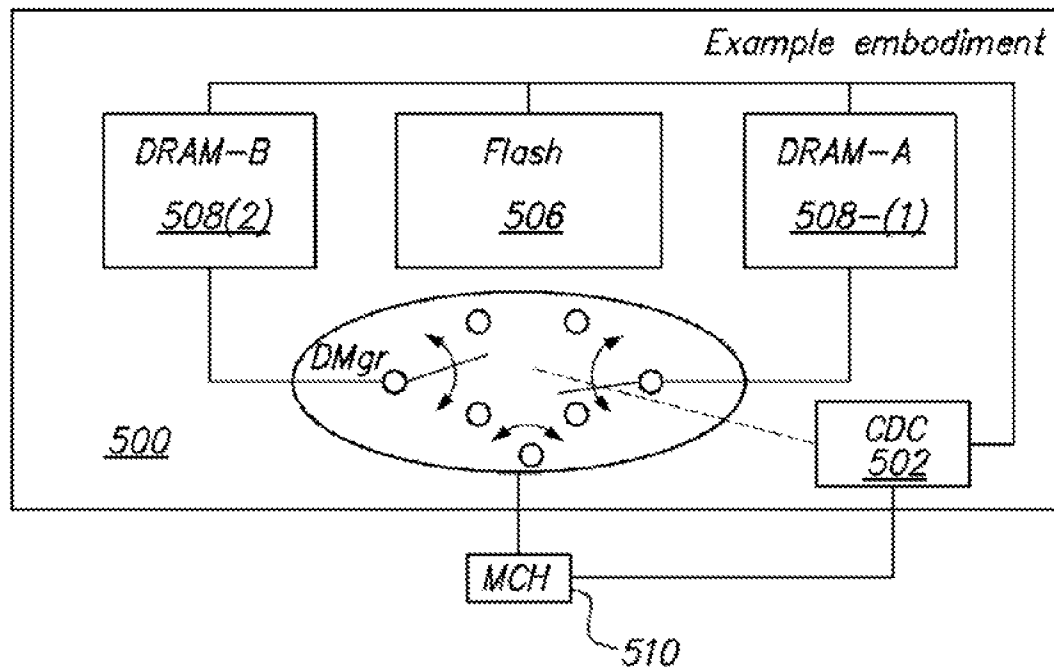


FIG. 3A

*Spansion EcoRAM Configurations**256GB Spansion EcoRAM Solution – Single Accelerator**256GB Single Accelerator Spansion EcoRAM Solution**256GB Spansion EcoRAM Solution – Dual Accelerator**256GB Single Accelerator Spansion EcoRAM Solution***FIG. 2
(PRIOR ART)****FIG. 4B**

**FIG. 3B****FIG. 4A**

**FIG. 5A****FIG. 6**

**FIG. 5B**

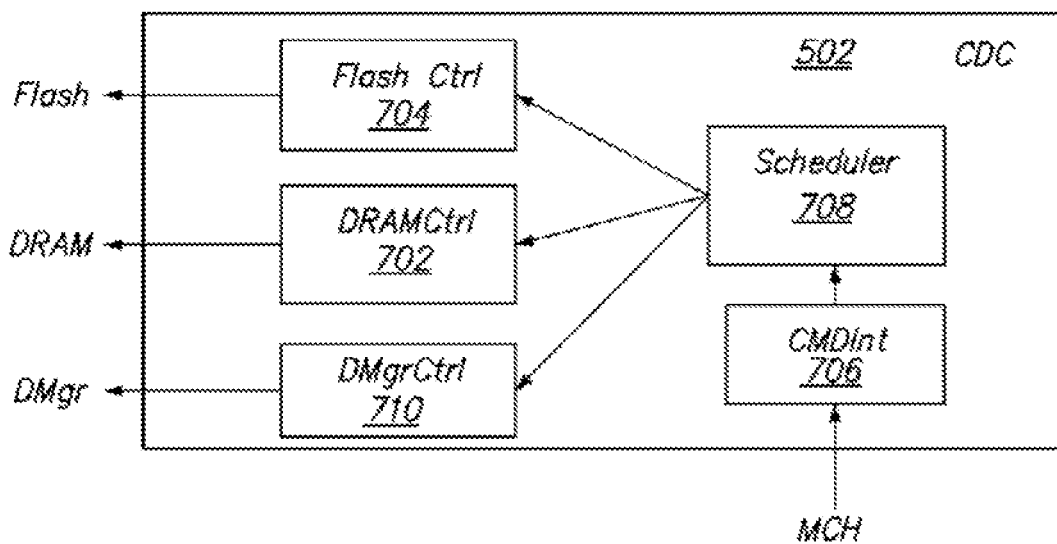


FIG. 7

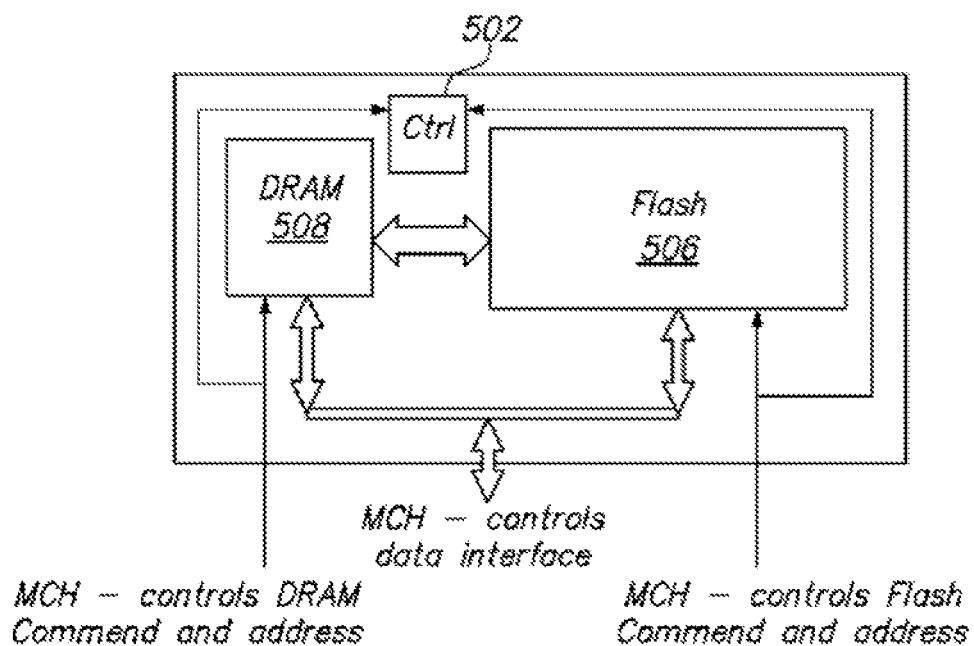


FIG. 8A

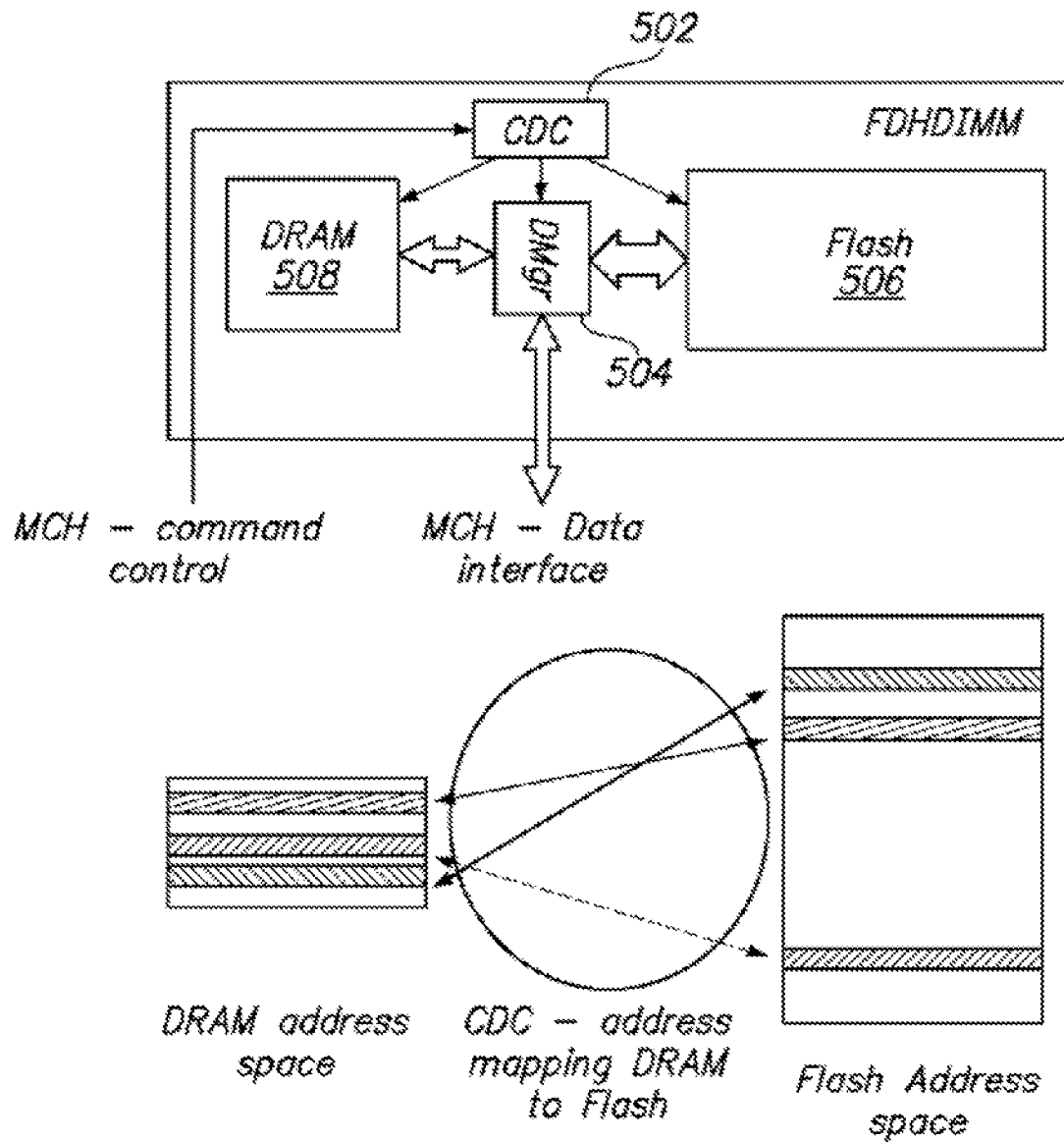
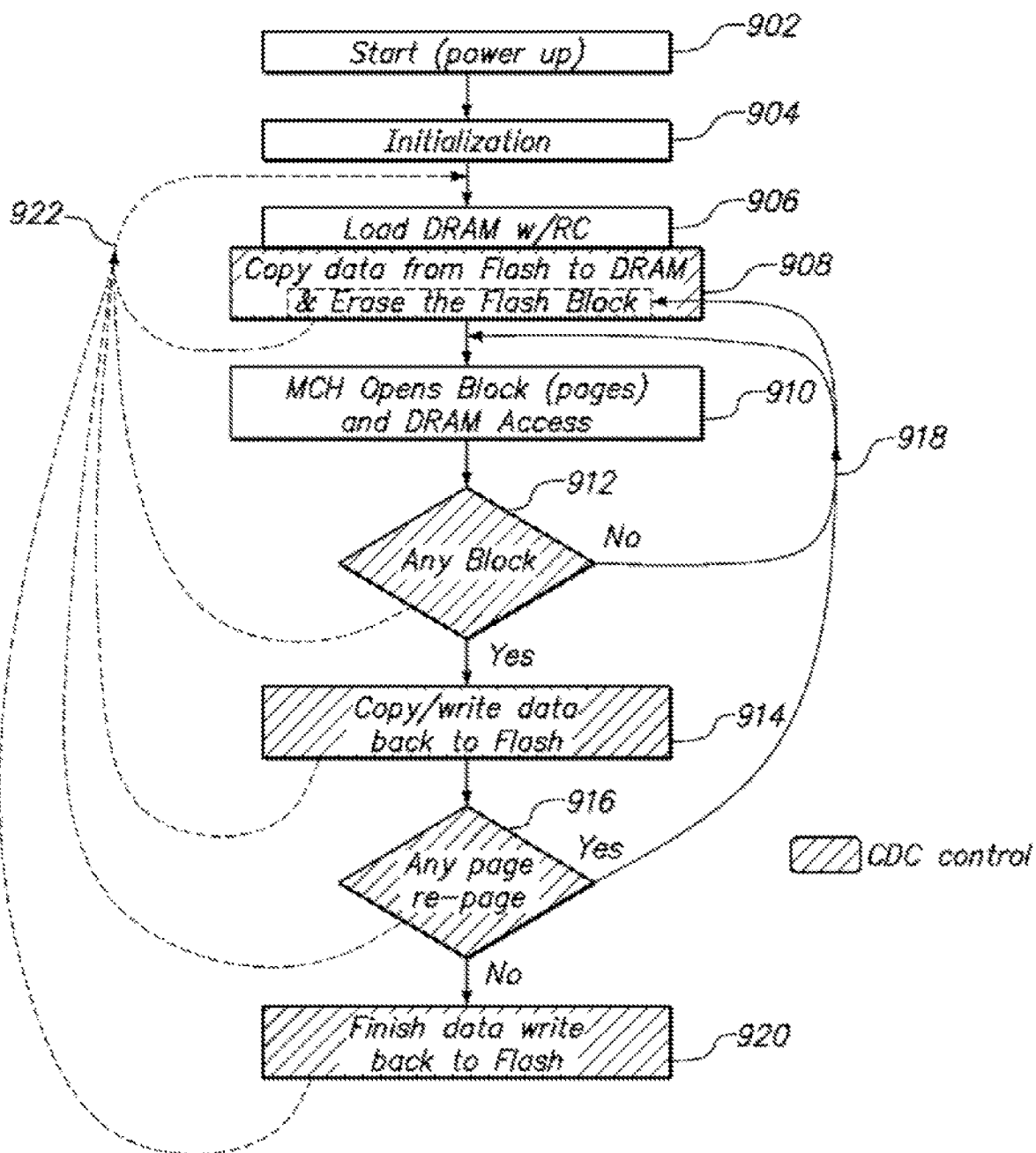
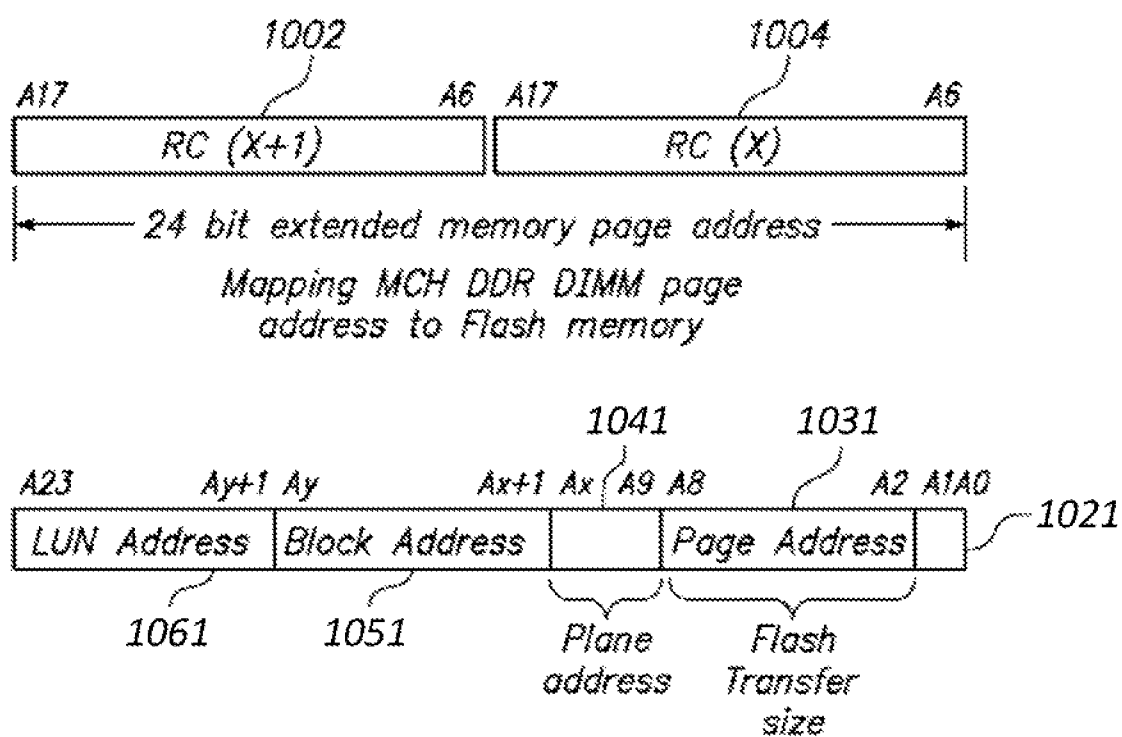


FIG. 8B

**FIG. 9**

**FIG. 10**

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

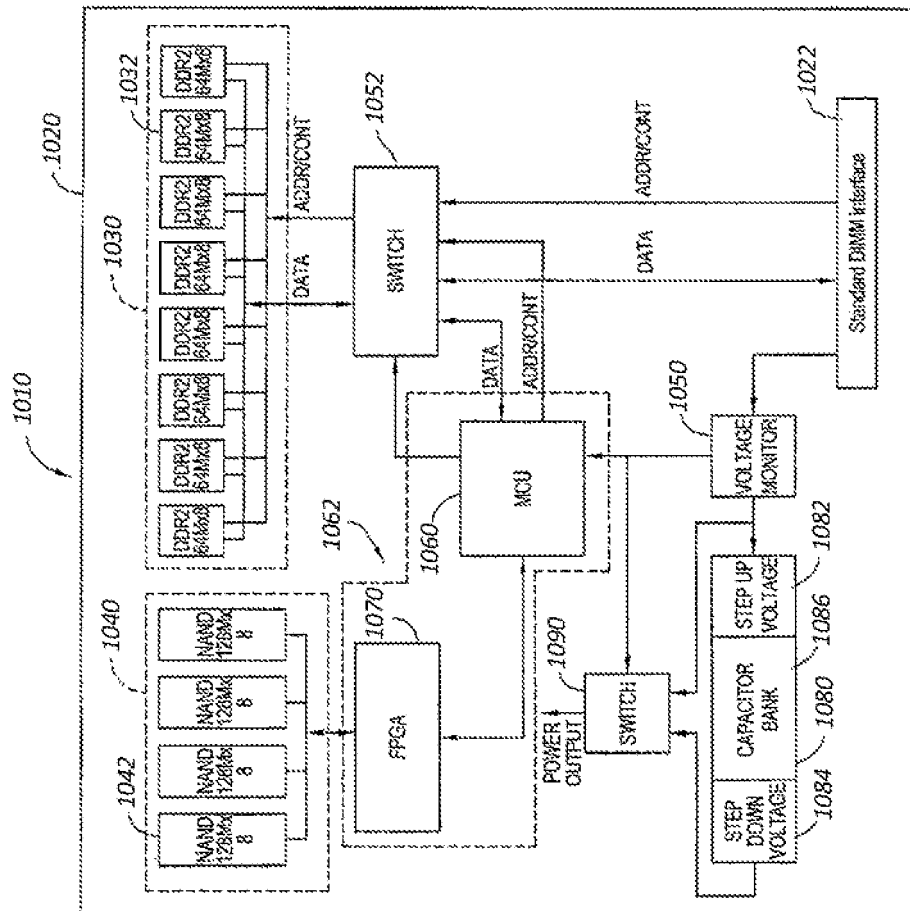


FIG. 12

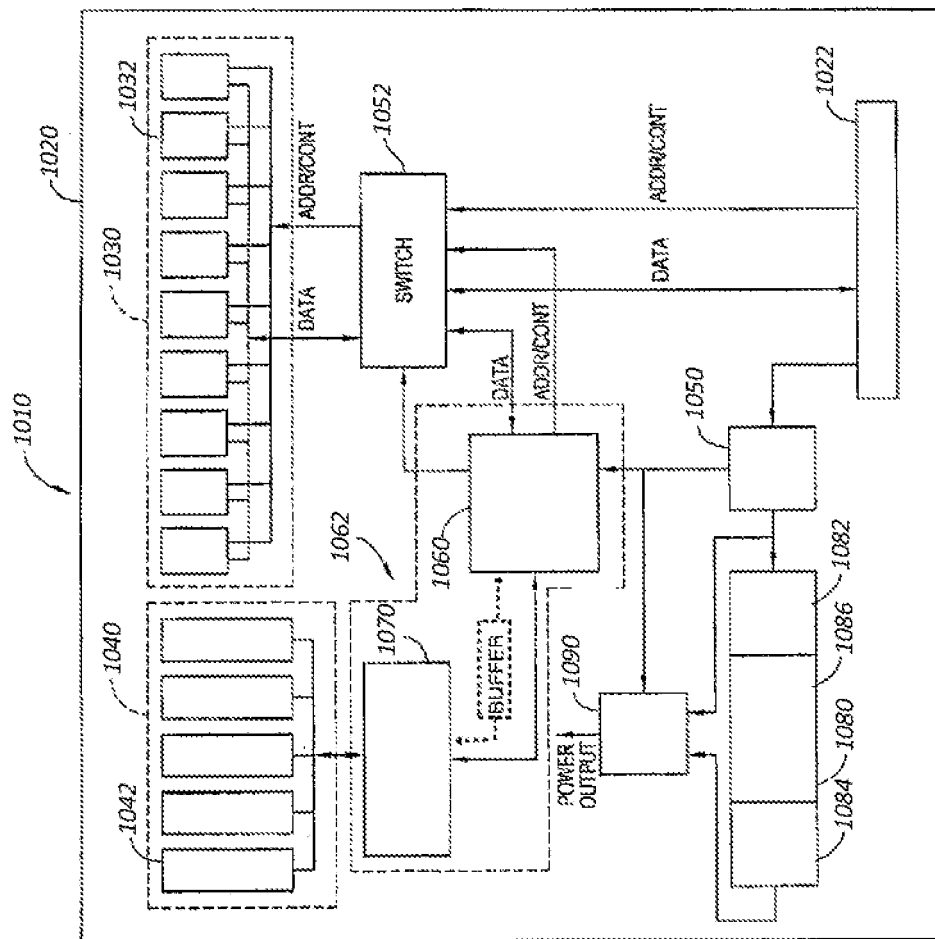


FIG. 13

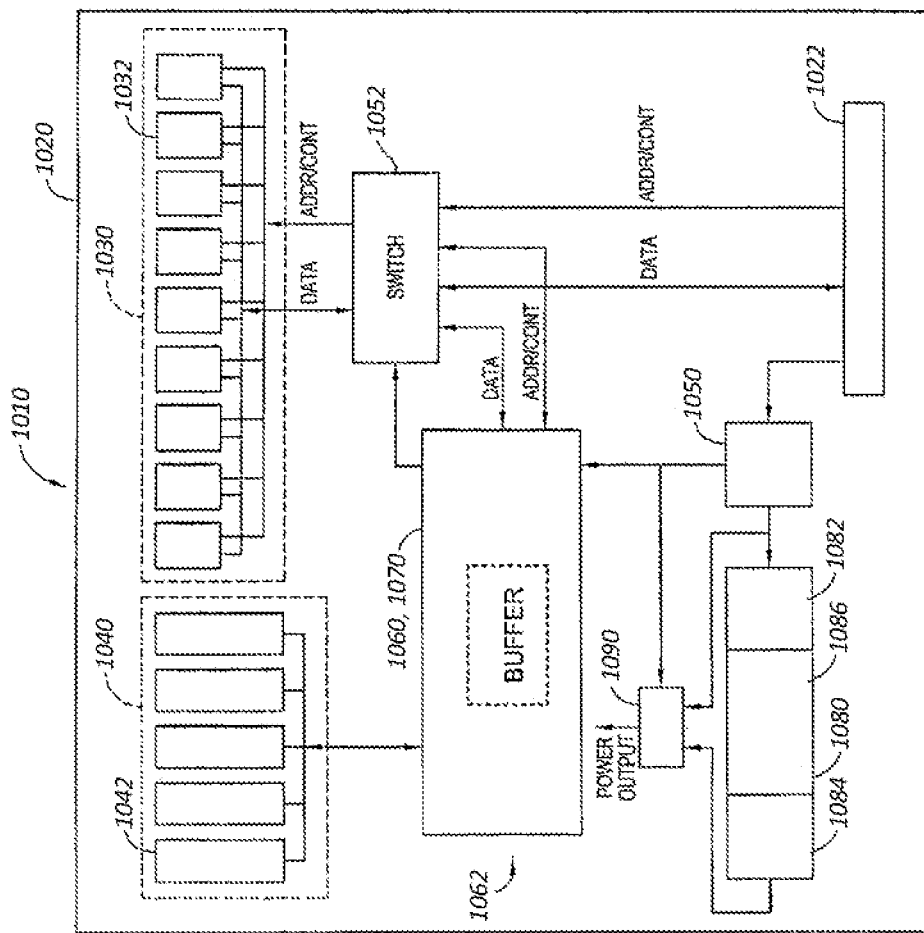


FIG. 14

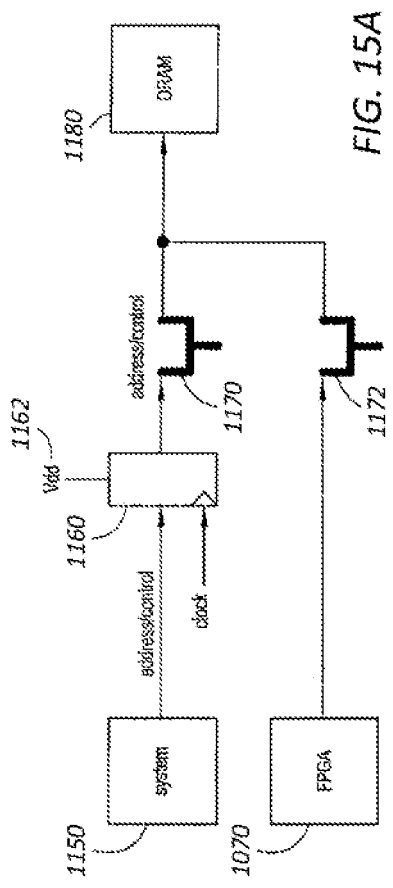


FIG. 15A

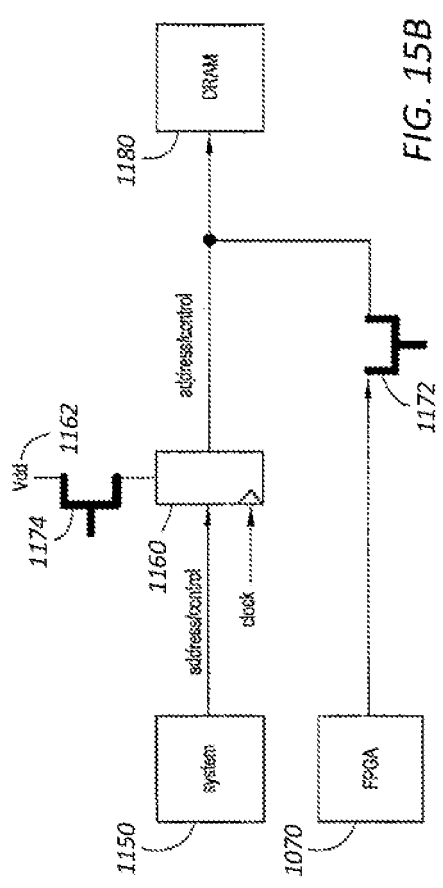


FIG. 15B

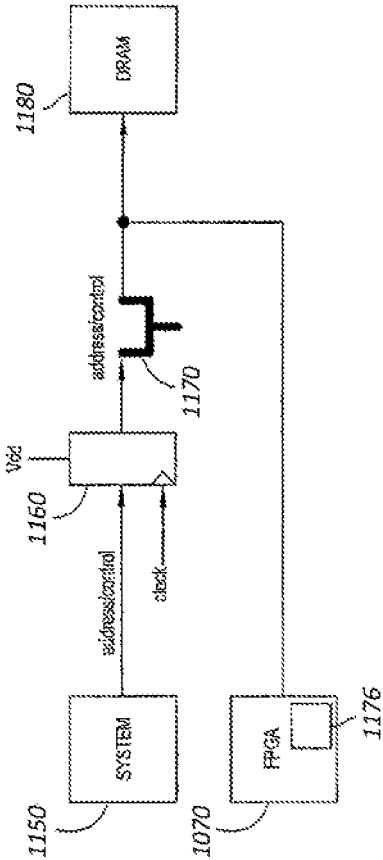


FIG. 15C

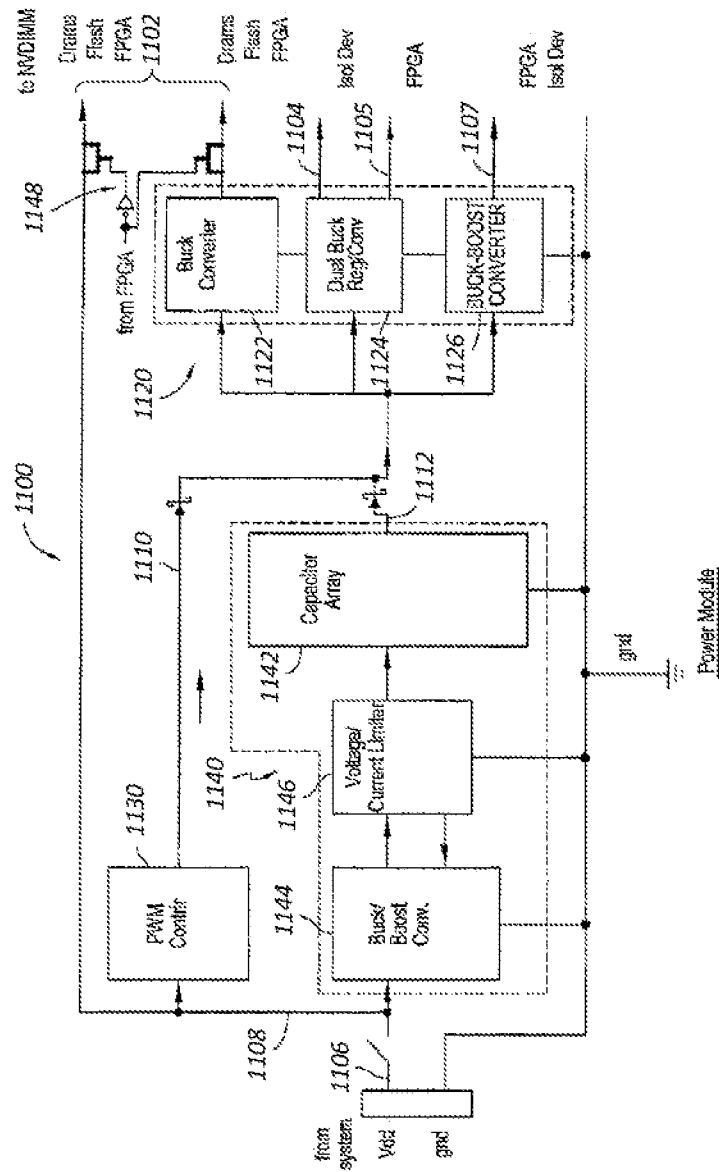


FIG. 16

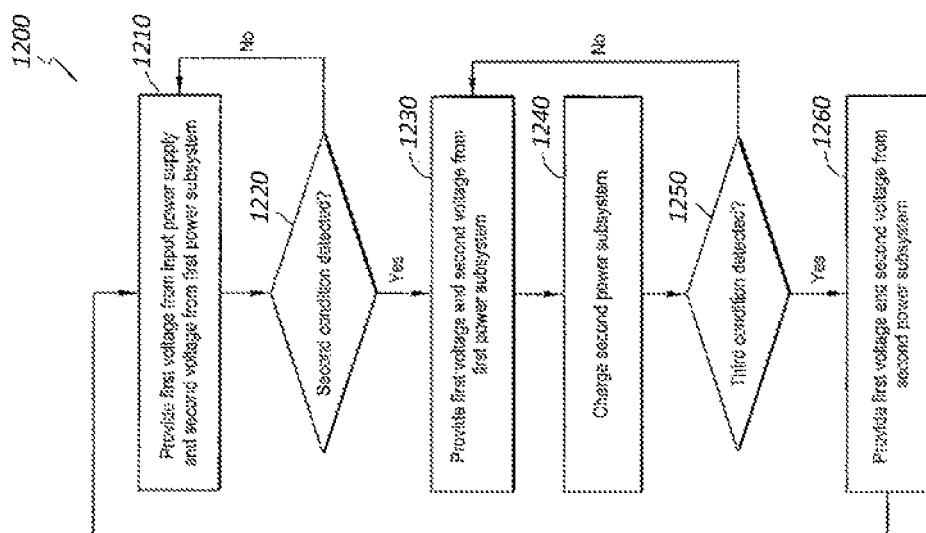


FIG. 17

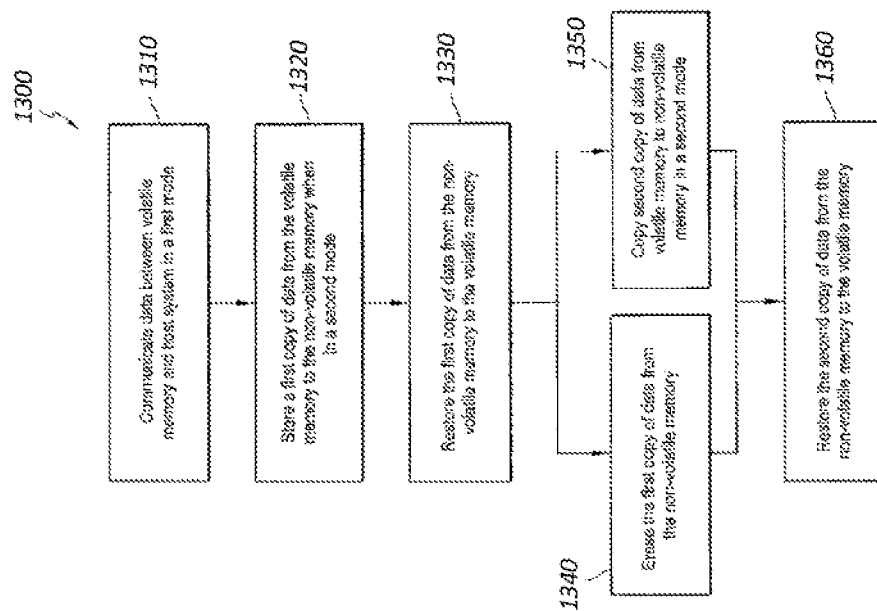


FIG. 18

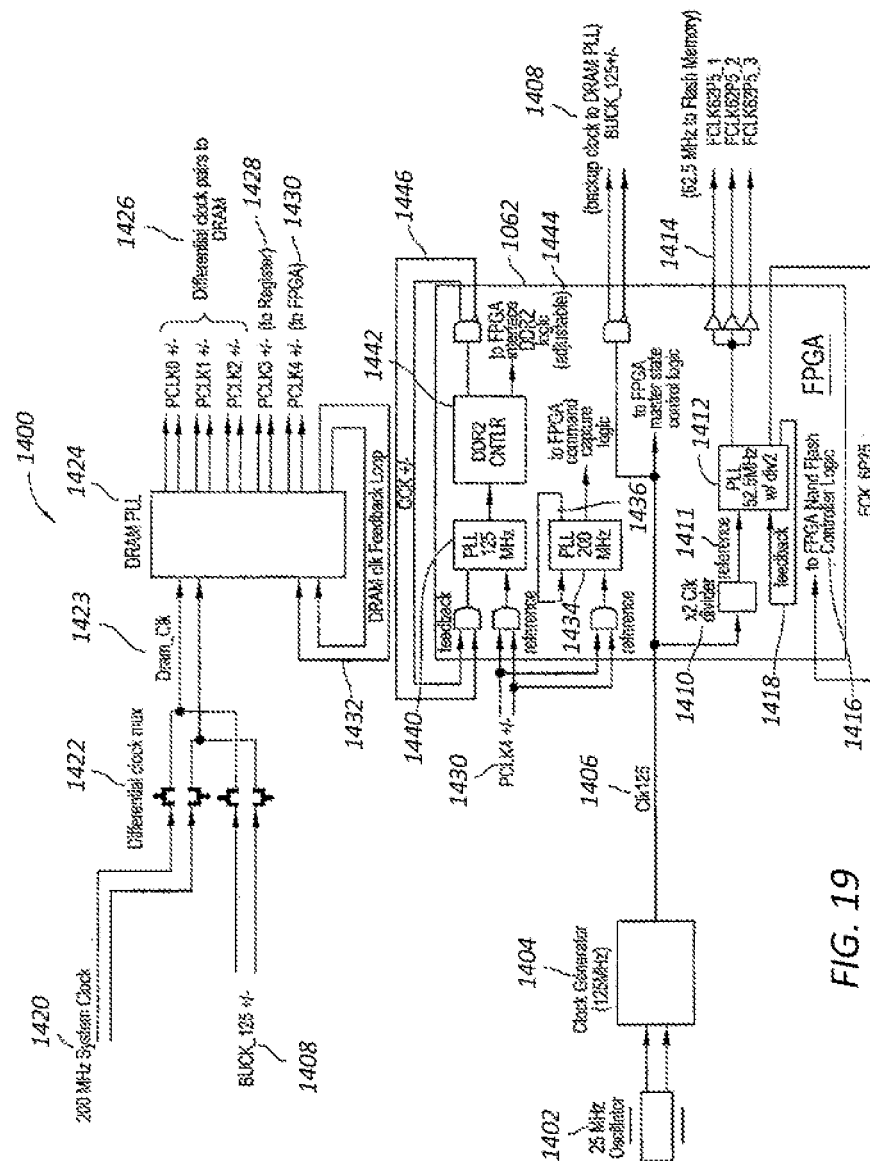


FIG. 19

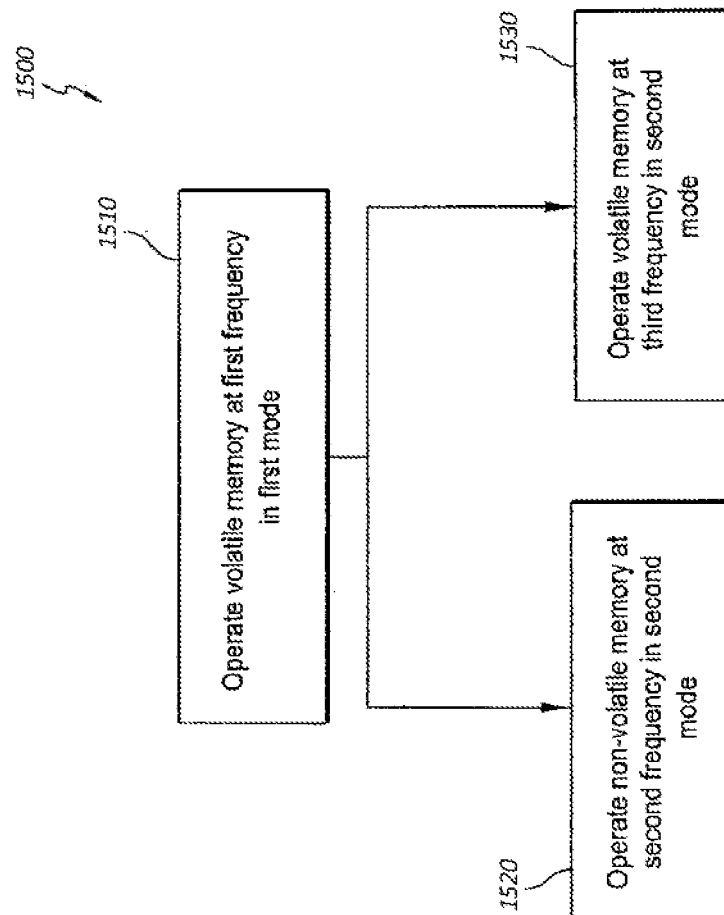


FIG. 20

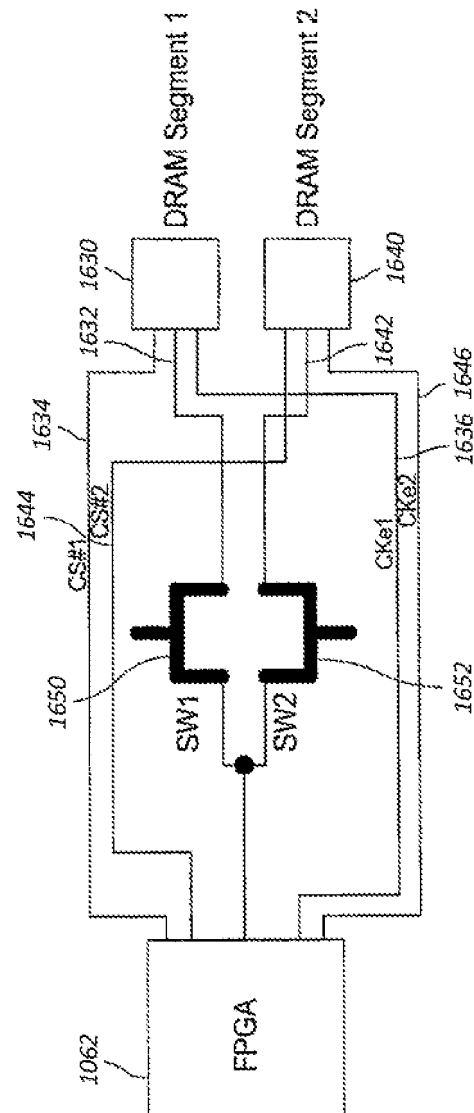


FIG. 21

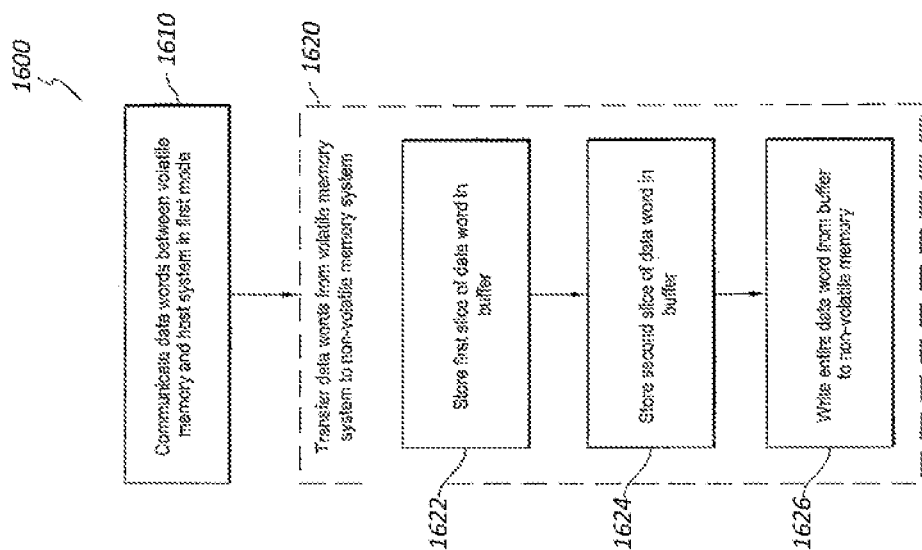


FIG. 22

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**MEMORY WITH ON-MODULE POWER
MANAGEMENT****PRIORITY CLAIM**

This application is a continuation of U.S. patent application Ser. No. 17/328,019, filed May 24, 2021, titled "FLASH-DRAM HYBRID MEMORY MODULE", now U.S. Pat. No. 11,232,054, which is a continuation of U.S. patent application Ser. No. 17/138,766, filed Dec. 30, 2020, titled "Flash-Dram Hybrid Memory", now U.S. Pat. No. 11,016,918, which is a continuation of U.S. patent application Ser. No. 15/934,416, filed Mar. 23, 2018, titled "Flash-Dram Hybrid Memory Module," which is a continuation of U.S. patent application Ser. No. 14/840,865, filed Aug. 31, 2015, titled "Flash-Dram Hybrid Memory Module," now U.S. Pat. No. 9,928,186, which is a continuation of U.S. patent application Ser. No. 14/489,269, filed Sep. 17, 2014, titled "Flash-Dram Hybrid Memory Module," now U.S. Pat. No. 9,158,684, which is a continuation of U.S. patent application Ser. No. 13/559,476, filed Jul. 26, 2012, titled "Flash-Dram Hybrid Memory Module," now U.S. Pat. No. 8,874,831, which claims the benefit of U.S. Provisional Patent Application No. 61/512,871, filed Jul. 28, 2011, and is a continuation-in-part of U.S. patent application Ser. No. 12/240,916, filed Sep. 29, 2008, titled "Non-Volatile Memory Module," now U.S. Pat. No. 8,301,833, which is a continuation of U.S. patent application Ser. No. 12/131,873, filed Jun. 2, 2008, which claims the benefit of U.S. Provisional Patent Application No. 60/941,586, filed Jun. 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

This application may be considered related to U.S. patent application Ser. No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed Feb. 5, 2014, which is a continuation of U.S. patent application Ser. No. 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Pat. No. 8,677,060, issued Mar. 18, 2014, which is a continuation of U.S. patent application Ser. No. 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed Jun. 28, 2012, now U.S. Pat. No. 8,516,187, issued Aug. 20, 2013, which is a divisional of U.S. patent application Ser. No. 12/240,916, titled "Non-Volatile Memory Module," filed Sep. 29, 2008, now U.S. Pat. No. 8,301,833, issued Oct. 30, 2012, which is a continuation of U.S. patent application Ser. No. 12/131,873, filed Jun. 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed Jun. 1, 2007, the contents of which are incorporated by reference herein in their entirety.

This application may also be considered related to U.S. patent application Ser. No. 15/000,834, filed Jan. 19, 2016, which is a continuation of U.S. patent application Ser. No. 14/489,332, filed Sep. 17, 2014, now U.S. Pat. No. 9,269,437, which is a continuation of U.S. patent application Ser. No. 14/173,219, filed Feb. 5, 2014, now U.S. Pat. No. 8,904,099, which is a continuation of U.S. patent application Ser. No. 13/905,048, filed May 29, 2013, now U.S. Pat. No. 6,671,243, which is a continuation U.S. patent application Ser. No. 13/536,173 above.

This application may also be considered related to U.S. patent application Ser. No. 15/924,866, which is a continuation of U.S. patent application Ser. No. 14/489,281, filed Sep. 17, 2014, now U.S. Pat. No. 9,921,762, which is a continuation of U.S. patent application Ser. No. 13/625,563, filed Sep. 24, 2012, now U.S. Pat. No. 8,904,098, which

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claims the benefit of U.S. Provisional Application No. 61/583,775, filed Sep. 23, 2011.

TECHNICAL FIELD

The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as 'finance transactions' (for example, point-of-service, inter-bank transaction, stock market transaction), 'scientific computation' (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), 'medical diagnostics' (for example, diagnostics based on the fuzzy logic, medical data processing), 'simple information sharing and searching' (for example, web search, retail store website, company home page), 'email' (information distribution and archive), 'security service', 'entertainment' (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes

a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3 GB/s compared with 400 MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5× performance improvement. However, the performance improvement for write random access throughput rate is less than 2× (197 MBs for the EcoRAM vs. 104 MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. FIG. 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™, which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400 MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

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Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

Also described herein is a memory module wherein the controller includes a volatile memory control module, a

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non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

Also described herein is a method for managing a memory module further comprising operating the data ports

to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the

description of example embodiments, serve to explain the principles and implementations of the embodiments.

In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system,

the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

The term "exemplary" where used herein is intended to mean "serving as an example, instance or illustration." Any embodiment described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments.

Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio)

between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406

at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1 KB or 2 KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host

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system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term “read/write access” means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

In certain embodiments, memory module **500** is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module **500** is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. **4A** and **4B**. Memory module **500** includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) **502** and an on-DIMM data manager (DMgr) **504**. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC **502** and data manager DMgr **504** are operative to manage the interface between a non-volatile memory subsystem such as a Flash **506**, a volatile memory subsystem such as a DRAM **508**, and a host system represented by MCH **510**.

In certain embodiments, CDC **502** controls the read/write access to/from Flash memory **506** from/to DRAM memory **508**, and to/from DRAM memory from/to MCH **510**. Read/write access between DRAM **508**, Flash **506** and MCH **510** may be referred to herein generally as communication, wherein control and address information C/A **560** is sent from MCH **510** to CDC **502**, and possible data transfers follow as indicated by Data **550**, Data **555**, and/or Data **556**. In certain embodiments, the CDC **502** performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC **502** ensures that memory module **500** provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In the arrangement shown in FIGS. **5A** and **5B**, there is no direct access from the MCH **510** to the Flash **506** memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. **5B** and in accordance with one embodiment, the CDC controller **502** receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing

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amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

An exemplary role of DMgr **504** is described with reference to FIG. **6**. In certain embodiments and in response to communication from CDC **502**, DMgr **504** provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr **504**) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr **504** is to provide interoperability among various memory subsystems or components and/or MCH **510**.

In one embodiment, an exemplary host system operation begins with initialization. The CDC **502** receives a first command from the MCH **510** to initialize FDHDIMM **500** using a certain memory space. The memory space as would be controlled by MCH **510** can be configured or programmed during initialization or after initialization has completed. The MCH **510** can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC **502** maps the actual physical Flash **506** and DRAM **508** memory space using the information sent by MCH **510** via the first command. In one embodiment, the CDC **502** maps the memory address space of any one of the Flash **506** and DRAM **508** memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM **500**, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash **506** or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM **500**.

In one embodiment, MCH **510** may send a command to restore a certain amount of data information from Flash **506** to DRAM **508**. The CDC **502** provides control information to DMgr **504** to appropriately copy the necessary information from Flash **506** to the DRAM **508**. This operation can provide support for various host system booting operations and/or a special host system power up operation.

In one embodiment, MCH **510** sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC **502** receives and interprets the command and provides control signals to DMgr **504** to control the data traffic between the Flash **506**, the DRAM **508**, and the MCH **510**. For example, DMgr **504** receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as **611**, **612**, **621**, **622** as detailed below), and the starting time of the actual data transfer from CDC **502**. DMgr **504** may also receive additional control information from the CDC **502** to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr **504** also functions as a bi-directional data transfer fabric. For example, DMgr **504** may have more than 2 sets of data ports facing the Flash **506** and the DRAM **508**. Multiplexers **611** and **612** provide controllable data paths from any one of the DRAMs **508(1)** and **508(2)** (DRAM-A and DRAM-B) to any one of the MCH **510** and the Flash **506**. Similarly multiplexers **621** and **622** provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs **508(1)** and **508(2)** (DRAM-A and DRAM-B). In one embodiment, DRAM **508(1)** is a segment of DRAM **508**, while in other

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embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC 502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors

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commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due

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to the difference in the data access time between the Flash memory and the DRAM memory.

In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHIMM is shown in FIG. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be

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optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In one embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908-922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The

CDC 502 monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in FIG. 5 and FIG. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4 TB of Flash memory storage space. If a memory module has 1 GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1 G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in FIG. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061 field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A₀A₁ 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μs, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22 ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22 ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11 ms instead of 22 ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than

9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100 ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$\left(\frac{\text{(\#of blocks per bank)}}{\text{(ratio of 'Flash_block_write_time' to 'Flash_read_time')}} \right) * \left(\frac{\text{(Block usage time)}}{\text{('Flash_block_write_time')}} \right)$$

In order to maintain less than 100 ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22 ms write access time per Block would result in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88 ms (=22 ms*4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2 GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor array). In the event of a power failure or other

trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

FIG. 12 is a block diagram of an example memory system **1010** compatible with certain embodiments described herein. The memory system **1010** can be coupled to a host computer system and can include a volatile memory subsystem **1030**, a non-volatile memory subsystem **1040**, and a controller **1062** operatively coupled to the non-volatile memory subsystem **1040**. In certain embodiments, the memory system **1010** includes at least one circuit **1052** configured to selectively operatively decouple the controller **1062** from the volatile memory subsystem **1030**.

In certain embodiments, the memory system **1010** comprises a memory module. The memory system **1010** may comprise a printed-circuit board (PCB) **1020**. In certain embodiments, the memory system **1010** has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system **1010** has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems **1010** having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB **1020** has an industry-standard form factor. For example, the PCB **1020** can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB **1020** has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB **1020** has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FB-DIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible

such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

In certain preferred embodiments, the memory system **1010** is in electrical communication with the host system. In other embodiments, the memory system **1010** may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, I U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system **1010** can be in communication with a disk controller of a computer system, for example. The PCB **1020** can comprise an interface **1022** that is configured to be in electrical communication with the host system (not shown). For example, the interface **1022** can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface **1022** of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system **1010** and the host system. For example, the interface **1022** can comprise a standard 240-pin DDR2 edge connector.

The volatile memory subsystem **1030** comprises a plurality of volatile memory elements **1032** and the non-volatile memory subsystem **1040** comprises a plurality of non-volatile memory elements **1042**. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem **1040** in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem **1030**. In certain embodiments, the first plurality of volatile memory elements **1032** comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements **1032** compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank **1030** comprises eight 64M×8 DDR2 SDRAM elements **1032**. The volatile memory elements **1032** may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements **1032** having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements **1032** compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

In certain embodiments, the second plurality of non-volatile memory elements **1042** comprises one or more flash memory elements. Types of flash memory elements **1042** compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank **1040** comprises 512 MB of flash memory organized as four 128 Mb×8 NAND flash memory elements **1042**. In addition, nonvolatile memory elements **1042** having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements **1042** compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

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FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements 1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of

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DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash "mirror" to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, 10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodi-

ments, the logic element **1070** is also programmed to perform address/address translation between the volatile memory subsystem **1030** and the non-volatile memory subsystem **1040**. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements **1042** because of their superior read speed and compact structure.

The memory system **1010** of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller **1062** and the non-volatile memory subsystem **1040** are operatively decoupled (e.g., isolated) from the volatile memory subsystem **1030** by the at least one circuit **1052** and a second state in which the volatile memory subsystem **1030** is operatively coupled to the controller **1062** to allow data to be communicated between the volatile memory subsystem **1030** and the nonvolatile memory subsystem **1040** via the controller **1062**. The memory system **1010** may transition from the first state to the second state in response to a trigger condition, such as when the memory system **1010** detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

The memory system **1010** may further comprise a voltage monitor **1050**. The voltage monitor circuit **1050** monitors the voltage supplied by the host system via the interface **1022**. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit **1050** may transmit a signal to the controller **1062** indicative of the detected condition. The controller **1062** of certain embodiments responds to the signal from the voltage monitor circuit **1050** by transmitting a signal to the at least one circuit **1052** to operatively couple the controller to the volatile memory system **1030**, such that the memory system **1010** enters the second state. For example, the voltage monitor **1050** may send a signal to the MCU **1060** which responds by accessing the data on the volatile memory system **1030** and by executing a write cycle on the nonvolatile memory subsystem **1040**. During this write cycle, data is read from the volatile memory subsystem **1030** and is transferred to the non-volatile memory subsystem **1040** via the MCU **1060**. In certain embodiments, the voltage monitor circuit **1050** is part of the controller **1062** (e.g., part of the MCU **1060**) and the voltage monitor circuit **1050** transmits a signal to the other portions of the controller **1062** upon detecting a power threshold condition.

The isolation or operational decoupling of the volatile memory subsystem **1030** from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system **1010** during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem **1030**. For example, in one embodiment during such periods of operation, the controller **1062** and the nonvolatile memory subsystem **1040** do not add a significant capacitive load to the volatile memory system **1030** when the memory system **1010** is in the first state. In certain such embodiments, the capacitive load of the controller **1062** and the non-volatile memory subsystem **1040** do not significantly affect the signals propagating between the volatile memory subsystem **1030** and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit **1052** comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit **1052** comprises other types of isolation devices.

Power may be supplied to the volatile memory subsystem **1030** from a first power supply (e.g., a system power supply) when the memory system **1010** is in the first state and from a second power supply **1080** when the memory system **1010** is in the second state. In certain embodiments, the memory system **1010** is in the first state when no trigger condition (e.g., a power failure) is present and the memory system **1010** enters the second state in response to a trigger condition. In certain embodiments, the memory system **1010** has a third state in which the controller **1062** is operatively decoupled from the volatile memory subsystem **1030** and power is supplied to the volatile memory subsystem **1030** from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory subsystem **1030** when the memory system **1010** detects that a trigger condition is likely to occur but has not yet occurred.

In certain embodiments, the second power supply **1080** does not comprise a battery. Because a battery is not used, the second power supply **1080** of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. **12-14**, the second power supply **1080** comprises a step-up transformer **1082**, a step-down transformer **1084**, and a capacitor bank **1086** comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply **1080** is configured to power the volatile memory subsystem **1030** for less than thirty minutes. In certain embodiments, the second power supply **1080** may comprise a battery. For example, in certain embodiments, the second power supply **1080** comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem **1030** for no more than thirty minutes.

In certain embodiments, the capacitor bank **1086** of the second power supply **1080** is charged by the first power supply while the memory system **1010** is in the first state. As a result, the second power supply **1080** is fully charged when the memory system **1010** enters the second state. The memory system **1010** and the second power supply **1080** may be located on the same printed circuit board **1020**. In other embodiments, the second power supply **1080** may not be on the same printed circuit board **1020** and may be tethered to the printed circuit board **1020**, for example.

When operating in the first state, in certain embodiments, the step-up transformer **1082** keeps the capacitor bank **1086** charged at a peak value. In certain embodiments, the step-down transformer **1084** acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements **1032** and 3.0V to the non-volatile flash memory elements **1042**) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. **12-14**, the memory module **1010** further comprises a switch **1090** (e.g., FET switch) that switches power provided to the controller **1062**, the volatile memory subsystem **1030**, and the non-volatile memory subsystem **1040**, between the power from the second power supply **1080** and the power from the first power supply (e.g., system power) received via the interface **1022**. For example, the switch **1090** may switch from the first power supply to the second power supply **1080** when the voltage monitor **1050** detects a low voltage con-

dition. The switch **1090** of certain embodiments advantageously ensures that the volatile memory elements **1032** and non-volatile memory elements **1042** are powered long enough for the data to be transferred from the volatile memory elements **1032** and stored in the non-volatile memory elements **1042**. In certain embodiments, after the data transfer is complete, the switch **1090** then switches back to the first power supply and the controller **1062** transmits a signal to the at least one circuit **1052** to operatively decouple the controller **1062** from the volatile memory subsystem **1030**, such that the memory system **1010** reenters the first state.

When the memory system **1010** re-enters the first state, data may be transferred back from the non-volatile memory subsystem **1040** to the volatile memory subsystem **1030** via the controller **1062**. The host system can then resume accessing the volatile memory subsystem **1030** of the memory module **1010**. In certain embodiments, after the memory system **1010** enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem **1030** rather than the non-volatile memory subsystem **1040** because the volatile memory elements **1032** have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank **1030** to the nonvolatile memory bank **1040**, or from the non-volatile memory bank **1040** to the volatile memory bank **1030**, takes less than one minute per GB.

In certain embodiments, the memory system **1010** protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system **1010** copies the entire contents of the volatile memory subsystem **1030** into the nonvolatile memory subsystem **1040** on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem **1040** are copied back into the volatile memory subsystem **1030** on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem **1040** are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem **1040** (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system **1010**. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system **1010** and can improve the performance of the memory system **1010**. In certain other embodiments, the entire contents of the volatile memory subsystem **1030** are not copied into the non-volatile memory subsystem **1040** on each backup operation, but only a partial copy is performed. In certain embodiments, other management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem **1040** are performed in the controller **1062**.

The memory system **1010** generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem **1030** which then writes the data to non-volatile storage which is not part of the memory system **1010**, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system **1010** indicating that the data has been written to the hard disk or is otherwise secure. The memory system **1010** of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system **1010** will still

be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system **1010** may be operated as a write-through cache or as some other type of cache.

FIG. **16** schematically illustrates an example power module **1100** of the memory system **1010** in accordance with certain embodiments described herein. The power module **1100** provides power to the various components of the memory system **1010** using different elements based on a state of the memory system **1010** in relation to a trigger condition. In certain embodiments, the power module **1100** comprises one or more of the components described above with respect to FIG. **12**. For example, in certain embodiments, the power module **1100** includes the second power supply **1080** and the switch **1090**.

The power module **1100** provides a plurality of voltages to the memory system **1010** comprising non-volatile and volatile memory subsystems **1030**, **1040**. The plurality of voltages comprises at least a first voltage **1102** and a second voltage **1104**. The power module **1100** comprises an input **1106** providing a third voltage **1108** to the power module **1100** and a voltage conversion element **1120** configured to provide the second voltage **1104** to the memory system **1010**. The power module **1100** further comprises a first power element **1130** configured to selectively provide a fourth voltage **1110** to the conversion element **1120**. In certain embodiments, the first power element **1130** comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element **1130** is configured to receive a 1.8V input system voltage as the third voltage **1108** and to output a modulated 5V output as the fourth voltage **1110**.

The power module **1100** further comprises a second power element **1140** can be configured to selectively provide a fifth voltage **1112** to the conversion element **1120**. The power module **1100** can be configured to selectively provide the first voltage **1102** to the memory system **1010** either from the conversion element **1120** or from the input **1106**.

The power module **1100** can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage **1102** is provided to the memory system **1010** from the input **1106** and the fourth voltage **1110** is provided to the conversion element **1120** from the first power element **1130**. In a second state, the fourth voltage **1110** is provided to the conversion element **1120** from the first power element **1130** and the first voltage **1102** is provided to the memory system **1010** from the conversion element **1120**. In the third state, the fifth voltage **1112** is provided to the conversion element **1120** from the second power element **1140** and the first voltage **1104** is provided to the memory system **1010** from the conversion element **1120**.

In certain embodiments, the power module **1100** transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module **1100** may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage **1102** in the second state from the first power element **1130** rather than from the input **1106** allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage **1102** from the first power element **1130** has capacitive and other smoothing effects. In addition, switching the point of power transition

to be between the conversion element **1120** and the first and second power elements **1130**, **1140** (e.g., the sources of the pre-regulated fourth voltage **1110** in the second state and the pre-regulated fifth voltage **1112** in the third state) can smooth out potential voltage spikes.

In certain embodiments, the second power element **1140** does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. **16**, the second power element **1140** comprises a capacitor array **1142**, a buck-boost converter **1144** which adjusts the voltage for charging the capacitor array and a voltage/current limiter **1146** which limits the charge current to the capacitor array **1142** and stops charging the capacitor array **1142** when it has reached a certain charge voltage. In one example embodiment, the capacitor array **1142** comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For example, in one example embodiment, the buck-boost converter **1144** receives a 1.8V system voltage (first voltage **1108**) and boosts the voltage to 4.3V which is outputted to the voltage current limiter **1146**. The voltage/current limiter **1146** limits the current going to the capacitor array **1142** to 1 A and stops charging the array **1142** when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element **1140** may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array **1142**.

The conversion element **1120** can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element **1120** may comprise a plurality of sub-blocks **1122**, **1124**, **1126** as schematically illustrated by FIG. **16**, which can provide more voltages in addition to the second voltage **1104** to the memory system **1010**. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system **1010**. For example, in one embodiment, sub-block **1122** comprises a buck converter, sub-block **1124** comprises a dual buck converter, and sub-block **1126** comprises a buck-boost converter as schematically illustrated by FIG. **16**. Various other components for the sub-blocks **1122**, **1124**, **1126** of the conversion element **1120** are also compatible with certain embodiments described herein. In certain embodiments, the conversion element **1120** receives as input either the fourth voltage **1110** from the first power element **1130** or the fifth voltage **1112** from the second power element **1140**, depending on the state of the power module **1100**, and reduces the input to an appropriate amount for powering various components of the memory system. For example, the buck-converter of sub-block **1122** can provide 1.8V at 2 A for about 60 seconds to the volatile memory elements **1032** (e.g., DRAM), the non-volatile memory elements **1042** (e.g., flash), and the controller **1062** (e.g., an FPGA) in one embodiment. The sub-block **1124** can provide the second voltage **1104** as well as another reduced voltage **1105** to the memory system **1010**. In one example embodiment, the second voltage **1104** is 2.5V and is used to power the at least one circuit **1052** (e.g., isolation device) and the other reduced voltage **1105** is 1.2V and is used to power the controller **1062** (e.g., FPGA). The subblock **1126** can provide yet another voltage **1107** to the memory system **1010**.

For example, the voltage **1107** may be 3.3V and may be used to power both the controller **1062** and the at least one circuit **1052**.

Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element **1120** may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements **1032** and nonvolatile memory elements **1042** are powered using independent voltages and are not both powered using the first voltage **1102**.

FIG. **17** is a flowchart of an example method **1200** of providing a first voltage **1102** and a second voltage **1104** to a memory system **1010** including volatile and nonvolatile memory subsystems **1030**, **1040**. While the method **1200** is described herein by reference to the memory system **1010** schematically illustrated by FIGS. **12-15**, other memory systems are also compatible with embodiments of the method **1200**. During a first condition, the method **1200** comprises providing the first voltage **1102** to the memory system **1010** from an input power supply **1106** and providing the second voltage **1104** to the memory system **1010** from a first power subsystem in operational block **1210**. For example, in one embodiment, the first power subsystem comprises the first power element **1130** and the voltage conversion element **1120** described above with respect to FIG. **16**. In other embodiments, other first power subsystems are used.

The method **1200** further comprises detecting a second condition in operational block **1220**. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method **1200** comprises providing the first voltage **1102** and the second voltage **1104** to the memory system **1010** from the first power subsystem in an operational block **1230**. For example, referring to FIG. **16**, a switch **1148** can be toggled to provide the first voltage **1102** from the conversion element **1120** rather than from the input power supply.

The method **1200** further comprises charging a second power subsystem in operational block **1240**. In certain embodiments, the second power subsystem comprises the second power element **1140** or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element **1140** and the voltage conversion element **1120** described above with respect to FIG. **16**. In other embodiments, some other second power subsystem is used.

The method **1200** further comprises detecting a third condition in an operational block **1250** and during the third condition, providing the first voltage **1102** and the second voltage **1104** to the memory system **1010** from the second power subsystem **1140** in an operational block **1260**. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method **1200** may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem **1140** is charged before detecting the second condition.

In certain embodiments, the memory system **1010** comprises a volatile memory subsystem **1030** and a non-volatile

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memory subsystem **1040** comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system **1010** also comprises a controller **1062** operatively coupled to the volatile memory subsystem **1030** and operatively coupled to the non-volatile memory subsystem **1040**. The controller **1062** can be configured to allow data to be communicated between the volatile memory subsystem **1030** and the host system when the memory system **1010** is operating in a first state and to allow data to be communicated between the volatile memory subsystem **1030** and the non-volatile memory subsystem **1040** when the memory system **1010** is operating in a second state.

Although the memory system **1010** having extra storage capacity of the non-volatile memory subsystem **1040** has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem **1040** than in the volatile memory subsystem **1030**. In various embodiments, there may be at least 200, 300, or 400 percent more storage capacity in the non-volatile memory subsystem **1040** than in the volatile memory subsystem **1030**. In other embodiments, the non-volatile memory subsystem **1040** includes at least some other integer multiples of the storage capacity of the volatile memory subsystem **1030**. In some embodiments, the non-volatile memory subsystem **1040** includes a non-integer multiple of the storage capacity of the volatile memory subsystem **1030**. In one embodiment, the non-volatile memory subsystem **1040** includes less than 100 percent more storage capacity than does the volatile memory subsystem **1030**.

The extra storage capacity of the non-volatile memory subsystem **1040** can be used to improve the backup capability of the memory system **1010**. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem **1040** which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem **1040** allows the volatile memory subsystem **1030** to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem **1040** may allow the memory system **1010** to backup the volatile memory subsystem **1030** efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system **1030** is copied to a first, previously erased portion of the nonvolatile memory subsystem **1040** via the controller **1062**. Since the non-volatile memory subsystem **1040** has more storage capacity than does the volatile memory subsystem **1030**, there is a second portion of the non-volatile memory subsystem **1040** which does not have data from the volatile memory subsystem **1030** copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller **1062** of the memory system **1010** restores the data to the volatile memory subsystem **1030** by copying the backed-up data from the non-volatile memory subsystem **1040** back to the volatile memory subsystem **1030**. After the data is restored, the memory system **1010** erases the non-volatile memory subsystem **1040**. While the first portion of the non-volatile memory subsystem **1040** is being erased, it may be temporarily inaccessible.

If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem **1040** is completely erased, the volatile memory subsystem **1030** can be backed-up or stored again in the second portion of the

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non-volatile memory subsystem **1040** as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem **1040** may allow the memory system **1010** to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem **1040**, the memory system **1010** can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem **1040**.

FIG. **18** is a flowchart of an example method **1300** of controlling a memory system **1010** operatively coupled to a host system and which includes a volatile memory subsystem **1030** and a non-volatile memory subsystem **1040**. In certain embodiments, the non-volatile memory subsystem **1040** comprises at least 100 percent more storage capacity than does the volatile memory subsystem **1030** as described herein. While the method **1300** is described herein by reference to the memory system **1010** schematically illustrated by FIGS. **12-14**, the method **1300** can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block **1310**, the method **1300** comprises communicating data between the volatile memory subsystem **1030** and the host system when the memory system **1010** is in a first mode of operation. The method **1300** further comprises storing a first copy of data from the volatile memory subsystem **1030** to the non-volatile memory subsystem **1040** at a first time when the memory system **1010** is in a second mode of operation in an operational block **1320**.

In an operational block **1330**, the method **1300** comprises restoring the first copy of data from the non-volatile memory subsystem **1040** to the volatile memory subsystem **1030**. The method **1300** further comprises erasing the first copy of data from the non-volatile memory subsystem **1040** in an operational block **1340**. The method further comprises storing a second copy of data from the volatile memory subsystem **1030** to the non-volatile memory subsystem **1040** at a second time when the memory system **1010** is in the second mode of operation in an operational block **1350**. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem **1040**.

In some embodiments, the memory system **1010** enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem **1040**. The method **1300** can also include restoring the second copy of data from the non-volatile memory subsystem **1040** to the volatile memory subsystem **1030** in an operational block **1360**. The operational blocks of method **1300** referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem **1030** at operational block **1360** before the first copy of data is completely erased in the operational block **1340**.

FIG. **19** schematically illustrates an example clock distribution topology **1400** of a memory system **1010** in accordance with certain embodiments described herein. The clock distribution topology **1400** generally illustrates the creation and routing of the clock signals provided to the various components of the memory system **1010**. A clock source **1402** such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source **1402** may feed a clock generator **1404** which provides a clock signal **1406** to the controller **1062**, which may be an FPGA. In one embodiment,

ment, the clock generator **1404** generates a 125 MHz clock signal **1406**. The controller **1062** receives the clock signal **1406** and uses it to clock the controller **1062** master state control logic. For example, the master state control logic may control the general operation of an FPGA controller **1062**.

The clock signal **1406** can also be input into a clock divider **1410** which produces a frequency-divided version of the clock signal **1406**. In an example embodiment, the clock divider **1410** is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal **1406**. A non-volatile memory phase-locked loop (PLL) block **1412** can be included (e.g., in the controller **1062**) which distributes a series of clock signals to the non-volatile memory subsystem **1040** and to associated control logic. For example, a series of clock signals **1414** can be sent from the controller **1062** to the non-volatile memory subsystem **1040**. Another clock signal **1416** can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem **1040**. For example, the clock signal **1416** may clock the portion of the controller **1062** which is dedicated to generating address and/or control lines for the non-volatile memory subsystem **1040**. A feedback clock signal **1418** is fed back into the non-volatile memory PLL block **1412**. In one embodiment, the PLL block **1412** compares the feedback clock **1418** to the reference clock **1411** and varies the phase and frequency of its output until the reference **1411** and feedback **1418** clocks are phase and frequency matched.

A version of the clock signal **1406** such as the backup clock signal **1408** may be sent from the controller to the volatile memory subsystem **1030**. The clock signal **1408** may be, for example, a differential version of the clock signal **1406**. As described herein, the backup clock signal **1408** may be used to clock the volatile memory subsystem **1030** when the memory system **1010** is backing up the data from the volatile memory subsystem **1030** into the non-volatile memory subsystem **1040**. In certain embodiments, the backup clock signal **1408** may also be used to clock the volatile memory subsystem **1030** when the memory system **1010** is copying the backed-up data back into the volatile memory subsystem **1030** from the nonvolatile memory subsystem **1040** (also referred to as restoring the volatile memory subsystem **1030**). The volatile memory subsystem **1030** may normally be run at a higher frequency (e.g., DRAM running at 400 MHz) than the nonvolatile memory subsystem **1040** (e.g., flash memory running at 62.5 MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem **1030** may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem **1040**) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem **1030** at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system **1010**.

In one embodiment, the backup clock **1408** and the volatile memory system clock signal **1420** are received by a multiplexer **1422**, as schematically illustrated by FIG. **19**. The multiplexer **1422** can output either the volatile memory system clock signal **1420** or the backup clock signal **1408** depending on the backup state of the memory system **1010**. For example, when the memory system **1010** is not performing a backup or restore operation and is communicating with the host system (e.g., normal operation), the volatile memory system clock signal **1420** may be provided by the

multiplexer **422** to the volatile memory PLL block **1424**. When the memory system **1010** is performing a backup (or restore) operation, the backup clock signal **1408** may be provided.

The volatile memory PLL block **1424** receives the volatile memory reference clock signal **1423** from the multiplexer **1422** and can generate a series of clock signals which are distributed to the volatile memory subsystem **1030** and associated control logic. For example, in one embodiment, the PLL block **1424** generates a series of clock signals **1426** which clock the volatile memory elements **1032**. A clock signal **1428** may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal **1430** may be sent to the controller **1062**. A feedback clock signal **1432** is fed back into the volatile memory PLL block **1424**. In one embodiment, the PLL block **1424** compares the feedback clock signal **1432** to the reference clock signal **1423** and varies the phase and frequency of its output until the reference clock signal **1423** and the feedback clock signal **1432** clocks are phase and frequency matched.

The clock signal **1430** may be used by the controller **1062** to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem **1030**. For example, control logic in the controller **1062** may be used to control the volatile memory subsystem **1030** during a backup or restore operation. The clock signal **1430** may be used as a reference clock signal for the PLL block **1434** which can generate one or more clocks **1438** used by logic in the controller **1062**. For example, the PLL block **1434** may generate one or more clock signals **1438** used to drive logic circuitry associated with controlling the volatile memory subsystem **1030**. In certain embodiments, the PLL block **1434** includes a feedback clock signal **1436** and operates in a similar manner to other PLL blocks described herein.

The clock signal **1430** may be used as a reference clock signal for the PLL block **1440** which may generate one or more clock signals used by a sub-block **1442** to generate one or more other clock signals **1444**. In one embodiment, for example, the volatile memory subsystem **1030** comprises DDR2 SDRAM elements and the sub-block **1442** generates one or more DDR2 compatible clock signals **1444**. A feedback clock signal **1446** is fed back into the PLL block **1440**. In certain embodiments, the PLL block **1440** operates in a similar manner to other PLL blocks described herein.

While described with respect to the example embodiment of FIG. **19**, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem **1030** operates on the volatile memory clock signal **1420** and there is no backup clock signal **1408**. In some embodiments, the volatile memory subsystem **1030** is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem **1030** is operated at a reduced frequency during a restore operation and not during a backup operation.

FIG. **20** is a flowchart of an example method **1500** of controlling a memory system **1010** operatively coupled to a host system. Although described with respect to the memory system **1010** described herein, the method **1500** is compatible with other memory systems. The memory system **1010** may include a clock distribution topology **1400** similar to

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the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 1040.

In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice from the second DRAM segment 1640 may comprise the remaining 32 bits

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of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630 into self-refresh mode by sending a signal via the CKe line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKe line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system

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when the memory system **1010** is in a first mode of operation in an operational block **1610**. For example, the memory system **1010** may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

In an operational block **1620**, the method further comprises transferring data words from the volatile memory subsystem **1030** to the non-volatile memory subsystem **1040** when the memory system **1010** is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system **1010**. The memory system **1010** may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block **1622**, storing a second portion of the data word in the buffer in an operational block **1624**, and writing the entire data word from the buffer to the non-volatile memory subsystem **1040** in an operational block **1626**.

In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller **1062**. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller **1062** which comprises an FPGA. The method **1600** may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The 40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller **1062** and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements **1040** write each backed-up data word to the controller **1062** which writes a first portion of the data word to the volatile memory subsystem **1030** and then a second portion of the data word to the volatile memory subsystem **1030**. In certain embodiments, slicing the volatile memory subsystem **1030** during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem **1030** during a backup operation.

The method **1600** can advantageously provide significant power savings and can lead to other advantages. For

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example, in one embodiment where the volatile memory subsystem **1030** comprises DRAM elements, only the slice of the DRAM which is currently being accessed (e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method **1600** can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKE) signal which allows each slice to be accessed independently.

In addition, the connection between the DRAM elements and the controller **1062** may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller **1062** and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller **1062** may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem **1030**. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem **1030** and the controller **1062** but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

What is claimed is:

1. A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM;

a second set of edge connections configured to be electrically connected to the set of data conduits, and a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

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a controller including a voltage monitor circuit and non-volatile memory, the controller coupled to the PCB and to an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

- (i) monitor an input voltage supplied via the input voltage supply line;
- (ii) generate a trigger signal upon detecting a trigger condition;
- (iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal, a write operation to write data into the nonvolatile memory; and wherein the trigger condition occurs when the input voltage exceeds a first threshold voltage;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line, and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

- a first component configured to receive power via at least the first regulated voltage line;
- a second component configured to receive power via at least the second regulated voltage line;
- a third component configured to receive power via at least the third regulated voltage line; and
- a fourth component configured to receive power via at least the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines,

wherein the plurality of DDR SDRAM devices includes:

- a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;
- a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

- the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines;
- the at least 40 1-bit DDR data signals include at least 32 1-bit wide DDR data signals and at least eight 1-bit DDR error-correcting code (ECC) data signals; and

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wherein the sum of the at least 40 data lines and the integer number of data lines equals a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

2. The DIMM of claim 1, wherein the controller further includes:

- a data storage element configured to store a plurality of data bits and a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and wherein the write operation conducted by the controller in response to the trigger signal includes:

- a first write operation to write a first set of data bits into a first portion of the data storage element;

- a second write operation to write a second set of data bits into a second portion of the data storage element; and subsequently

- a third write operation to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

3. The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on a combination of the first and second sets of data bits written during the first and second write operations.

4. The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on groups of data bits stored in the data storage element including the first and second sets of data bits written during the first and second write operations.

5. The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a plurality of read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

6. The DIMM of claim 2, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger signal, a read operation to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

7. The DIMM of claim 2, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger signal, one or more read operations.

8. The DIMM of claim 7, wherein the logic element is configured to perform a first read operation of the one or more read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

9. The DIMM of claim 2, wherein the third write operation comprises a transfer of the third set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second sets of data bits into the data storage element via a second connection.

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10. The DIMM of claim 2, wherein the data storage element is a buffer configured to store a plurality of data bits or a First-In First-Out (FIFO) buffer configured to store the plurality of data bits.

11. The DIMM of claim 1, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including the first threshold voltage and a second threshold voltage.

12. The DIMM of claim 1, wherein the second component is further connected to the first regulated voltage line and is powered by the first and second regulated voltages when power from the power rails of the system board is provided to the DIMM.

13. The DIMM of claim 12, wherein, when power from the power rails of the system board is provided to the DIMM, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

14. The DIMM of claim 1, wherein, when power from the power rails of the system board is provided to the DIMM, the nonvolatile memory is not powered using the first regulated voltage.

15. The DIMM of claim 1, wherein each of the first, second, and third converter circuits is a buck converter, and the fourth converter circuit is a voltage reducing circuit.

16. A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector of the system board, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and to an input

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voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

(i) monitor an input voltage supplied via the input voltage supply line;

(ii) generate a trigger signal upon detecting a trigger condition; and

(iii) transmit the trigger signal to at least one other portion of the controller,

wherein, in response to the detected trigger condition, the controller is configured to periodically perform, at regular intervals of time, a plurality of write operations, including a first write operation to write a first set of data bits into a first portion of the data storage element;

a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converters coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

a first component configured to receive power via the first regulated voltage line;

a second component configured to receive power via the second regulated voltage line;

a third component configured to receive power via the third regulated voltage line; and

a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

(i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines;

(ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines, and

wherein the plurality of DDR SDRAM devices includes:

a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal;

a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal;

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

the first group of DDR SDRAM packages is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM packages is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

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wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

17. The DIMM of claim 16, wherein the controller further includes a nonvolatile memory and a logic element, the logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element.

18. The DIMM of claim 17, wherein the logic element is configured to perform a second write operation to write a second set of data bits into a second portion of the data storage element.

19. The DIMM of claim 18, wherein the plurality of write operations further includes a third write operation subsequent to the first and second write operations to write a third set of data bits from the data storage element into the nonvolatile memory, wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations, and wherein, in response to the detected trigger condition, the logic element is configured to periodically perform every millisecond all of the first and second write operations followed by the third write operation.

20. The DIMM of claim 19, wherein, prior to performing the third write operation, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second portions of the data storage element, respectively, and to generate the third set of data bits based on the read first and second sets of data bits.

21. The DIMM of claim 19, wherein the logic element is configured to perform the third write operation by writing the third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second sets of data bits written during the first and second write operations.

22. The DIMM of claim 19, wherein, prior to performing the third write operation, the logic element is configured to generate the third set of data bits based on the groups of data bits stored in the data storage element.

23. The DIMM of claim 19, wherein the third write operation comprises a transfer of the first and second set of data bits from the data storage element to the nonvolatile memory via a first connection, and wherein the first and second write operations comprise a transfer of the first and second set of data bits into the data storage element via a second connection.

24. The DIMM of claim 18, wherein, prior to performing the first and second write operations, the logic element is configured to perform, in response to the detected trigger condition, periodically at regular intervals of time one or more read operations to read the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

25. The DIMM of claim 16, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

26. The DIMM of claim 16, wherein when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

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27. The DIMM of claim 26, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

28. The DIMM of claim 26, wherein when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

29. A dual in-line memory module (DIMM) configured to fit into a memory slot connector of a system board of a computer system, the memory slot connector including a set of power conduits, a set of data conduits, and a set of address and control conduits, wherein data signals are transferred between the system board and the DIMM via the set of data conduits, and address and control signals are delivered from the system board to the DIMM via the set of address and control conduits, the DIMM comprising:

a printed circuit board (PCB) having edge connections configured to fit into the memory slot connector, the PCB including:

a first set of edge connections configured to be electrically connected to the set of power conduits and operable to deliver to the DIMM power from the power rails of the system board via the set of power conduits, wherein the power delivered to the DIMM from the power rails of the system board is the only power received by the DIMM,

a second set of edge connections configured to be electrically connected to the set of data conduits,

a third set of edge connections configured to be electrically connected to the set of address and control conduits,

a set of voltage supply lines electrically connected to the first set of edge connections and configured to receive the power delivered to the DIMM from the power rails of the system board by way of the memory slot connector of the system board and the first set of edge connections of the DIMM,

a set of data lines electrically connected to the second set of edge connections, and

a set of address and control lines electrically connected to the third set of edge connections, wherein the set of data lines are electrically coupled to respective edge connections of the second set of edge connection, wherein each data line of the set of data lines is configured to electrically conduct a 1-bit DDR data signal;

a controller including a voltage monitor circuit and a data storage element configured to store a plurality of data bits, the controller coupled to the PCB and an input voltage supply line of the set of voltage supply lines, the voltage monitor circuit configured to:

(i) monitor an input voltage supplied via the input voltage supply line;

(ii) generate a trigger signal upon detecting a trigger condition; and

(iii) transmit the trigger signal to at least one other portion of the controller, wherein the controller is configured to perform, in response to the trigger signal,

one or more write operations to write data into the data storage element;

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a set of components coupled to the PCB, the set of components including a plurality of double data rate (DDR) synchronous dynamic random access memory (SDRAM) devices coupled to the set of address and control lines and to the set of data lines, the DDR SDRAM devices being operable to receive or output data signals via the set of data lines based on address and control signals received from the system board via the set of address and control lines; and

first, second, third, and fourth converter circuits coupled to the PCB and to the input voltage supply line and configured to receive power from the input voltage supply line and to deliver power via first, second, third, and fourth regulated voltage lines,

wherein the set of components includes:

- a first component configured to receive power via the first regulated voltage line;
- a second component configured to receive power via the second regulated voltage line;
- a third component configured to receive power via the third regulated voltage line; and
- a fourth component configured to receive power via the fourth regulated voltage line;

wherein each component of the set of components:

- (i) is electrically connected to one or more of the first, second, third, and fourth regulated voltage lines; and
- (ii) receives power only via the one or more of the first, second, third, and fourth regulated voltage lines;

wherein the plurality of DDR SDRAM devices includes:

- a first group of at least five DDR SDRAM devices that are each connected to a first chip select line configured to electrically conduct a first chip select signal; and
- a second group of at least four DDR SDRAM devices that are each connected to a second chip select line configured to electrically conduct a second chip select signal, and

wherein when power from the power rails of the system board is provided to the DIMM and each group of the first and second groups of DDR SDRAM devices is not in self-refresh mode:

the first group of DDR SDRAM devices is configured to be enabled, in response to the first chip select signal, to receive or output at least 40 1-bit DDR data signals in parallel via at least 40 data lines of the set of data lines independently of whether the second group of DDR SDRAM devices is enabled, in response to the second chip select signal, to receive or output an integer number of 1-bit DDR data signals in parallel via an integer number of data lines of the set of data lines; and

the at least 40 1-bit DDR data signals include 32 1-bit DDR data signals and eight 1-bit DDR error-correcting code (ECC) data signals; and

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wherein the sum of the at least 40 data lines and the integer number of data lines matches a total number of data conduits of the set of data conduits of the memory slot connector, and the total number of data conduits is at least 72 data conduits.

30. The DIMM of claim 29, wherein, when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

31. The DIMM of claim 29, wherein when power is provided to the DIMM from the power rails of the system board, the nonvolatile memory is not powered using the first regulated voltage.

32. The DIMM of claim 29, when power is provided to the DIMM from the power rails of the system board the voltage monitor circuit compares the input voltage to a plurality of threshold voltages including a first threshold voltage and a second threshold voltage.

33. The DIMM of claim 32, wherein when power is provided to the DIMM from the power rails of the system board the trigger condition occurs when the input voltage goes above a specified operating voltage.

34. The DIMM of claim 32, wherein, when power is provided to the DIMM from the power rails of the system board, an additional trigger condition occurs when the input voltage is above the first threshold and below the second threshold; and

wherein, in response to the additional trigger condition, the voltage monitor circuit is further configured to generate and transmit a second trigger signal.

35. The DIMM of claim 29, wherein the controller further includes a logic element coupled to the nonvolatile memory and the data storage element and configured to electrically communicate with the nonvolatile memory and the data storage element, and the one or more write operations include a first write operation to write a first set of data bits into a first portion of the data storage element and a second write operation to wire a second set of data bits into a second portion of the data storage element.

36. The DIMM of claim 35, wherein the logic element is configured to perform a third write operation subsequent to the first and second write operations by writing a third set of data bits from the data storage element into the nonvolatile memory, and wherein the third set of data bits includes the first and second set of data bits written during the first and second write operations.

37. The DIMM of claim 35, wherein the logic element is configured to perform, prior to the first and second write operations, first and second read operations by reading the first and second sets of data bits from the first and second groups of DDR SDRAM devices, respectively.

38. The DIMM of claim 29, wherein each of the first, second, and third converter circuits is a buck converter.

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